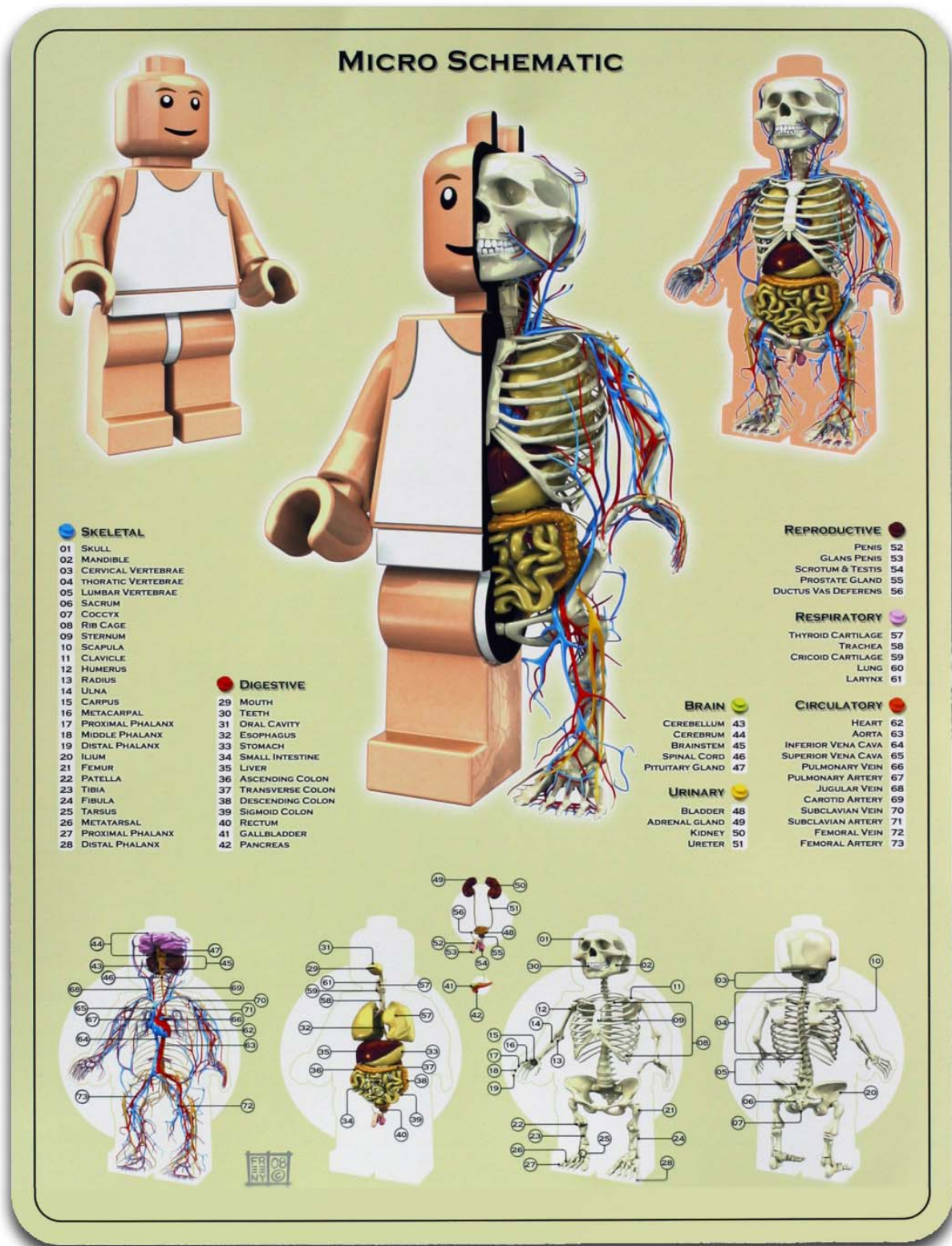


# Anatomy notes collection



***Dear Students,***

In those pages you will find a collection of notes about the gastrointestinal & respiratory tracts, the heart, as well as the topography of the thoracic and abdominal walls together with the retroperitoneal cavity and mediastinum. I have added also the topography of the neck section including fasciae, triangles and Nuchal region. at the end of the file you will find an appendix containing hand made drawings that can be useful for the preparation for the midterms, semifinal or final exams. (in the printed version only)

These notes were summarized and collected from different sources that I list here:

- *Moore: Clinically oriented anatomy*
- *Thieme: Internal organs (vol. 2)*
- *Rohen Yokochi color atlas of anatomy*
- *BRS anatomy*
- *Kovacs anatomy*
- *Instant anatomy website*
- *Grey's anatomy*
- *Netter Atlas*
- *Anatomy lectures at the department of Anatomy, Histology & Embryology*
- *Dr. Madarasz's notes & drawings*
- *Dr. Somogyi's & Dr. Altdorfer's file about peritoneum & anatomical cartoons*
- *Dr. Altdorfer's & Dr. Hanics's file about the branchial apparatus*

The different chapters of this file were revised by some of the teachers of the department of Anatomy, Histology & Embryology (listed in alphabetical order) in order to meet with the demands and requirements of the department:

- *Dr. Altdorfer, Károly*: Esophagus, stomach, small intestines, large intestines, peritoneum, heart
- *Dr. Barna, János*: Liver & gallbladder, porto-caval anastomosis, pancreas, spleen
- *Dr. Gallatz Katalin*: Kidneys, adrenals, ureters, aorta, IVC, external carotid & maxillary arteries
- *Dr. Madarász, Magdolna*: Trachea, pleura, lungs, diaphragm. heart
- *Dr. Somogyi, György*: Abdominal wall, inguinal region, mediastinum, peritoneum
- *Dr. Székelyi, Andrea*: oral cavity, teeth, tongue, palate, pharynx
- *Dr. Tótpál, Géza*: External nose, nasal cavity, larynx

Please understand that these notes do not guarantee the passing grade in the exams or replace the original atlases or books suggested by the department. The main purpose of these notes is to orientate you around the important and main points & aspects required by the department of Anatomy, Histology & Embryology during the course of your preparation toward the final exam at the end of the 2nd year.

If you do encounter any type of mistake or you have any ideas or suggestions please contact me at: haythamtb@gmail.com

I would like to thank the teachers who participated in revising these notes for their time & effort and continues support not only for me, but also for the medical and dental students at the department of Anatomy, Histology & Embryology. I would like also to thank Yoav Litani (2<sup>nd</sup> year) for his aid in the graphic design of most of the illustrations.

Wish you all the best & lots of luck  
Haytham Bayadsi

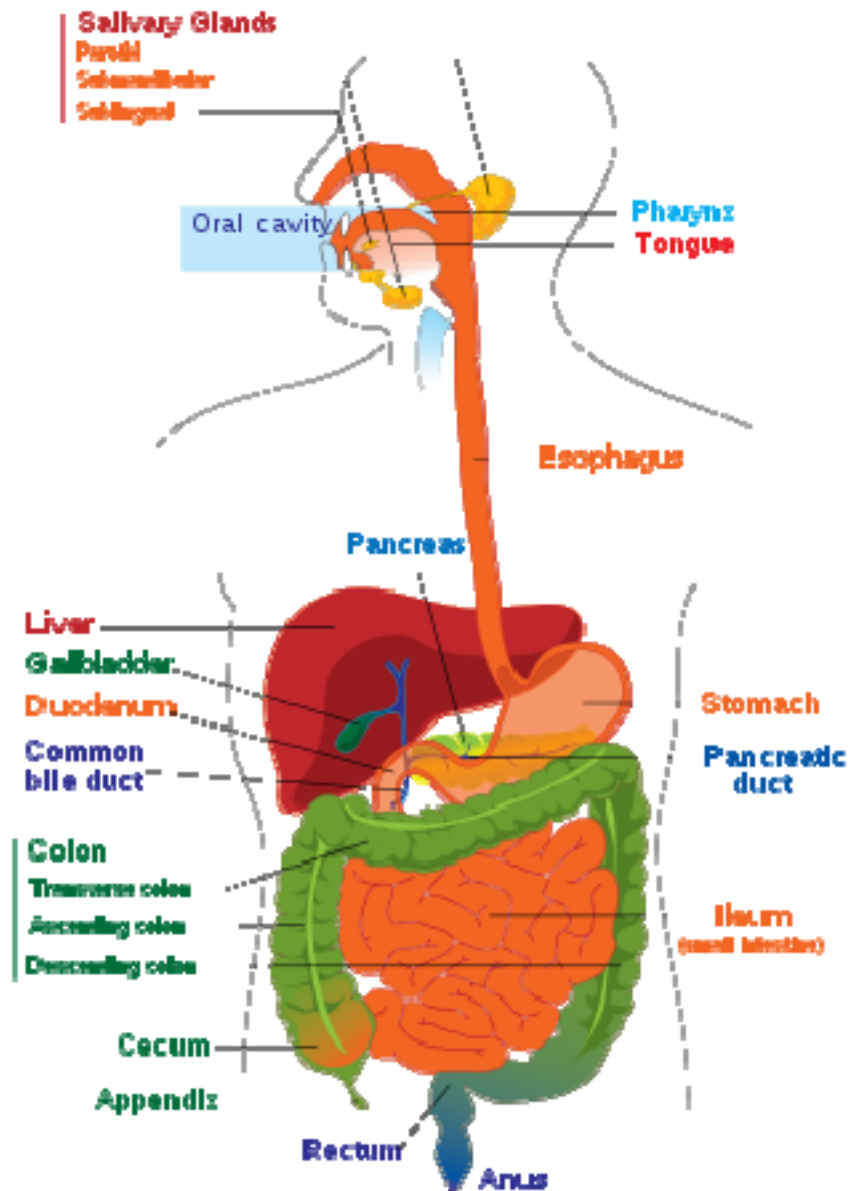
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# *The Gastrointestinal tract*



## ***The oral cavity***

### **→ Different parts of the oral cavity:**

- ***The vestibule:*** that part of the mouth bounded anteriorly and laterally by the lips and the cheeks, posteriorly and medially by the teeth and/or gums, and above and below by the reflections of the mucosa from the lips and cheeks to the gums.
- ***The oral cavity proper:*** the space between the dental arches, limited posteriorly by the isthmus of the fauces.
  - if the jaws are totally closed and dentition is normal then there is no connection between the 2 parts of the oral cavity, except behind the last molar tooth

### **→ Borders of the oral cavity:**

- ***Superior border (Roof):*** *Hard* (anterior) & *Soft* (posterior) *palates*
- ***Inferior border (Floor):*** the muscular floor or the diaphragm of oral cavity, formed by the *Mylohyoid muscles* that arise from the mylohyoid line to the medial raphe and to the hyoid bone. Above this floor and near the midline there are the 2 *Geniohyoid* muscles that also can be considered as part of the inferior border (Floor)
- ***Lateral border:*** teeth arches (cavity proper) or the cheeks formed by the *Buccinator* muscle, *Bichat's fat pad* and *mucosa* (the whole oral cavity)
- ***Anterior border:*** the lips
- ***Posterior border:*** open, the ***Isthmus of Fauces***, bordered by:
  - Superior: ***soft palate***
  - Inferior: ***root of the tongue***
  - lateral: ***palatoglossal and palatopharyngeal arches*** (*Anterior & Posterior Palatine Arches, inside them muscles with the same names*)

### **→ The vestibule of the oral cavity:**

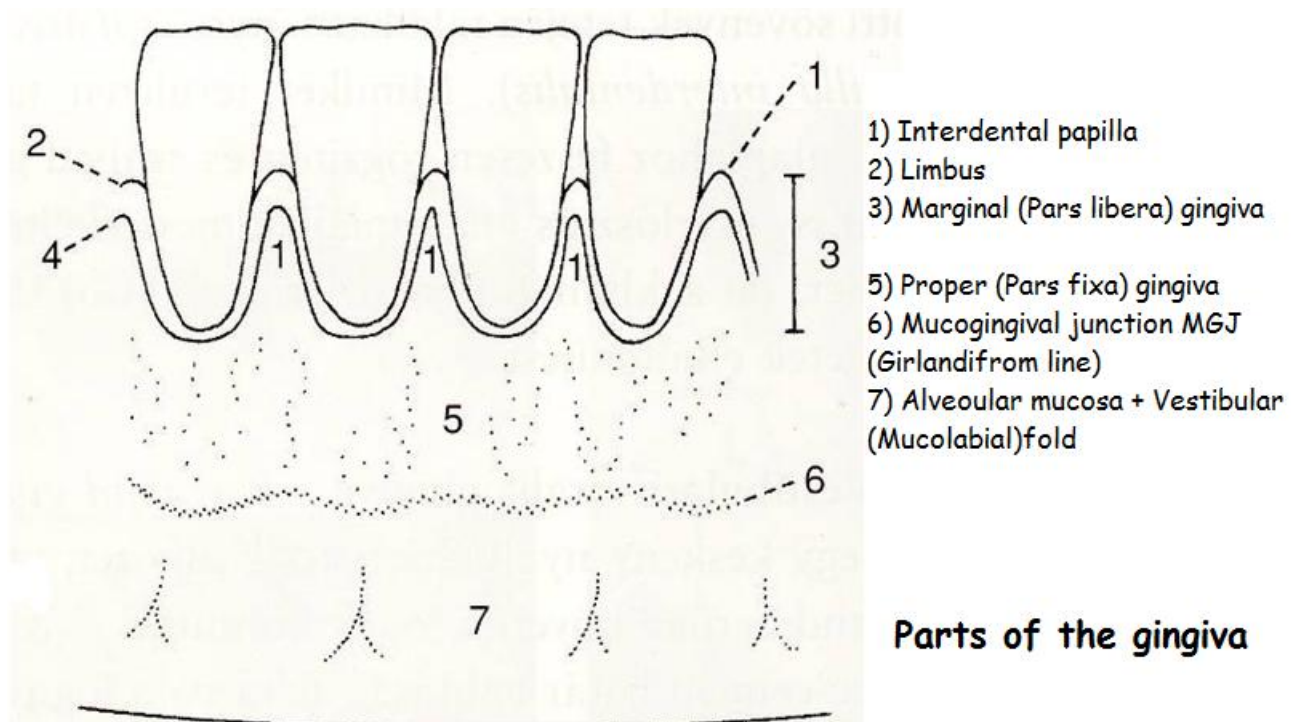
- The *lips* are found anteriorly, guarding the entrance to the mouth. Both lips are connected laterally at the corners of the mouth
  - The *upper lip* reaches the base of external nose and to the ***Nasolabial fold***
  - The *lower lip* reaches the chin-lip groove (***mentolabial fold***)
  - The ***Philtrum*** is the groove that extends downward from the nose over the midline of the upper lip
  - ***Each lip has 2 parts:***
    - External cutaneous part (covered by stratified squamous *keratinized* hairy epithelium)
    - Internal mucous part (covered by stratified squamous *non-keratinized* epithelium)
- \*\* The Red area of the lip (*Vermilion border*)** demarcates the border between the 2 parts, it has a thinly keratinized type of stratified squamous epithelium deeply penetrated by well-vascularized dermal papillae
- The *mucobuccal* or *mucolabial* folds are the line of flexure of the mucous membrane as it passes from the mandible or maxillae to the cheek or lips respectively
  - The anterior upper and lower mucosal folds in the vestibule are called the ***Frenulum of lips***

→ **Gums (gingivae):**

The dense fibrous tissue and overlying mucous membrane, which envelops the alveolar processes of the upper and lower jaws and surrounds the necks of the teeth, has 2 parts:

- ***Pars fixa gingivae:*** or the *proper* gingiva, the *attached* part of the gums that is tightly connected to the periosteum of the neck of teeth and alveolar processes of jaws, it is demarcated from lining mucosa on the facial aspect by a clearly defined line which marks the *mucogingival junction (Girlandiform line)*, and, in contrast to the lining mucosa, is keratinized and lighter in color; on the palatal surface, the gingiva blends imperceptibly with the palatal mucosa.
- ***Pars libera (free) gingivae:*** or the *marginal* gingiva, the most coronal part that extends beyond the alveolar bone by 1-2 mm surrounding the tooth, it is supported and stabilized by the *gingival fibers* and composed of:
  - **Interdental papilla:** thickening (seen as an elevation) of the gingiva that fills the interproximal space between two adjacent teeth.
  - **Gingival limbus:** the tip of the marginal gingiva, encircling the neck of teeth as a belt at the *dentinoenamel* junction of tooth

The **gingival sulcus:** the space or furrow between the surface of the tooth and the free soft tissue gingiva



## ***The Tongue***

A mobile muscular organ that is covered by a mucous membrane (stratified squamous non-keratinized epithelium), partly found in the *oral cavity* (2/3) as ***pars papillaris*** (*Presulcal*), partly in the *isthmus of fauces* (1/3) as ***pars follicularis*** (*Postsulcal*) with the lingual tonsil (the border between the 2 parts is the ***V-shaped terminal sulcus*** lined anteriorly by the group of the *Circumvallate papillae*, in the middle of the terminal sulcus we can find the ***Foramen Cecum***, which is the remnant of the *Thyroglossal* duct in thyroid gland development.

At rest, the tongue occupies the entire *oral cavity proper*

Main functions:

- a. Forming words during speech
- b. Mastication
- c. Squeezing food into the pharynx
- d. Taste

### **→ *Parts of the tongue:***

- *Body (corpus)*
- *Apex* (tip)
- *Root*
- *Dorsum* (back): has the median sulcus of tongue
- *Inferior surface*: has the frenulum, deep lingual vein and fimbriated fold

### **→ *Connections of the base of tongue:***

- a. ***Hyoid bone*** (by the *Hyoglossus* muscle)
- b. ***Epiglottis***: by,
  - 1 median glossoepiglottic fold
  - 2 lateral glossoepiglottic folds
    - Between these 2 fold folds we find the *Epiglottic Valleculae*
- c. ***Soft Palate***: by the *Palatoglossal* arch
- d. ***Pharynx***: by part of the superior constrictor muscle (4<sup>th</sup> part, Glossopharyngeal)
- e. ***Mandible***:
  - by the *Frenulum lingua*: a fold under the surface of tongue, formed by the mucus membrane, on both sides of this fold we find the *Lingual Valleculae*
    - The *Sublingual caruncula* is a papilla on each side of the base of the lingual frenulum, in the Valleculae, which is the location of the opening of the *Submandibular duct* + main sublingual duct
  - by the *Genioglossus* muscle (from mental spine)

### **→ *Lingual papillae:***

Numerous variously shaped projections of the mucous membrane of the dorsum of the tongue; that have mechanical and sensory (taste) functions, there are 4:

- a. ***Circumvallate papillae***: anterior to the terminal sulcus. Surrounded by circular trenches (fossa), and having a slightly raised walls (vallum) of which studded by taste buds.
- b. ***Foliate papillae***: on the lateral sides of tongue, poorly developed in humans, has taste buds.

- c. **Filiform papillae:** arranged V-shaped rows, parallel to terminal sulcus, keratinized on the top and contain afferent nerve endings; sensitive to touch. At the apex of tongue they are arranged transversally.
- d. **Fungiform papillae:** scattered among the Filiform papillae, and most numerous at the apex and sides of the tongue. Mushroom shaped and appears as pink or red spots. Have taste buds.

→ **Extrinsic muscles of the tongue:**

Arise from structure nearby the tongue, and their fibers insert into the tongue in a 3-D network. Mainly control the position of the tongue inside the oral cavity  
The first 3 muscles are innervated by the **Hypoglossal nerve (XII)** while the *Palatoglossus* is innervated by the *Pharyngeal plexus* (Glossopharyngeal nerve IX & Vagus nerve X)

- 1. **Genioglossus:** *Depression of tongue* (more anterior fibers) and **protracting the tongue** outside the oral cavity (posterior fibers, only muscle that protracts tongue!)  
*Origin:* Mental spine (superior part)  
*Insertion:* fans out from apex to the root of tongue
- 2. **Hyoglossus:** *depresses and retracts tongue backwards*  
*Origin:* greater horn of hyoid + body of hyoid  
*Insertion:* lateral side and inferior surface of tongue
- 3. **Styloglossus:** pulls the tip backward, and pulls the whole tongue upwards & backwards  
*Origin:* styloid process + stylohyoid ligament  
*Insertion:* side up to the anterior aspect of tongue
- 4. **Palatoglossus:** *Elevates posterior part of tongue*  
*Origin:* palatine aponeurosis (soft palate)  
*Insertion:* side of tongue

→ **Intrinsic muscles of tongue:**

group of fibers running in 3 principal planes and attached to the connective tissue framework of tongue that is composed of the *Lingual Septum* (fibrous, dividing the tongue in 2 halves) and the *Lingual aponeurosis* (tough sheet, between the mucous membrane and muscles)  
Also all of these muscles innervated by the **Hypoglossal nerve (XII)**

- 1. **Superior longitudinal:** curls tip and side superiorly
- 2. **Inferior longitudinal:** curls tip inferiorly
- 3. **Transversus:** narrows and elongates tongue
- 4. **Verticalis:** flattens and broadens tongue

**\*\* An important note:** In case of hypoglossal nerve half paralysis (*Hypoglossal paresis*), the tongue will protrude, but the *Genioglossus*, *Transversus* and *Verticalis* muscles of the *healthy* side will push the *ipsilateral* (same side) *longitudinal* muscles forward, and thus the *tongue*

will **turn (point) towards the diseased side (paralyzed)** because it offers no resistance to the healthy muscle.

E.g. if you ask the patients to protrude his tongue, and his tongue will *deviate* to the *left* it means that the *Left Hypoglossal* is damaged, which means that it could be a problem in the *Right hemisphere of brain* (because of the decussating or crossing nerve fibers in brain)

→ **Innervation of tongue:**

- **Motor innervation: Hypoglossal nerve (XII):** for all muscles of tongue, except the *Palatoglossus* (innervated by the *pharyngeal plexus*)
  - **General sensory innervation** (touch, temperature):
    - **Anterior 2/3 (Presulcal): Lingual nerve**, branch of *mandibular nerve (CN-V3)*
    - **Posterior 1/3 (Postsulcal): Glossopharyngeal nerve** (CN-IX)
  - **Taste sensation:**
    - **Anterior 2/3 (Presulcal): Chorda tympani nerve**, branch of *Facial nerve (CN-VII)*
      - \*\* But the **Circumvallate papillae**, located at the border, are innervated by the **glossopharyngeal nerve (CN-IX)**
    - **Posterior 1/3 (Postsulcal): Glossopharyngeal nerve** (CN-IX)
- \*\***Vagus nerve (CN-X):** innervates the tongue from its base, until the epiglottis (folds and Valleculae)

→ **Taste sensation:**

1. *Sweetness*: at apex
2. *Sourness and bitterness*: posterior part
3. *Saltiness*: lateral margin

→ **Vasculature of tongue:**

**External carotid artery → lingual artery → branches:**

1. **Dorsal lingual arteries:** supplies *posterior* part and *palatine tonsils*
  2. **Deep lingual artery:** supplies the *anterior* part, the anastomosis between the right and the left end-branch of the deep lingual artery is called the *Beclard* or *Ranine anastomosis*
  3. **Sublingual artery:** supplies *floor of mouth* and *sublingual glands*
- All veins run parallel to the arteries and drain into *internal jugular vein*, the *Apical veins of the tongue* are seen very well on both sides of the frenulum on the inferior surface of the tongue forming the *fimbriated folds*

\*\* The **lingual artery** is the 2<sup>nd</sup> branch of the External carotid artery (or commonly also branches from the facial artery, then its known as the **Linguofacial trunk**), and it can be recognized as the artery immediately found above the greater horn of the Hyoid bone. It passes behind 2 different triangles which are important in surgery (control of tongue bleeding) before it ends up in the *medial* lingual sulcus:



**1) Pirogoff triangle, borders:**

- Superior → *Hypoglossal nerve (CN-XII)*
- Anterior → *Posterior margin of mylohyoid muscle*
- Inferior → *Tendon of digastric muscle/ stylohyoid muscle*

**2) Beclard triangle, borders:**

- Superior → Tendon of *digastric muscle*
- Posterior → *free margin of hyoglossus muscle*
- Inferior → *Greater horn of hyoid bone*

**\*\* Content of both triangles: *lingual artery***

**→ Lingual grooves (Sulci)**

We have to distinguish different spaces or specifically called Lingula grooves between the tongue and the nearby muscles and structures, we have 2 lingual grooves with different contents inside them:

**1) Lateral groove (sulcus) of the tongue:**

*Borders:*

- Medial: *Hyoglossus muscle*
- lateral: *Mylohyoid muscle*
- Superior: *Sublingual mucous membrane*

*Contents:*

- *Lingual nerve*
- *Submandibular duct*
- *Hypoglossal nerve (XII)* (more posteriorly)
- *Sublingual gland* (only *anteriorly*)

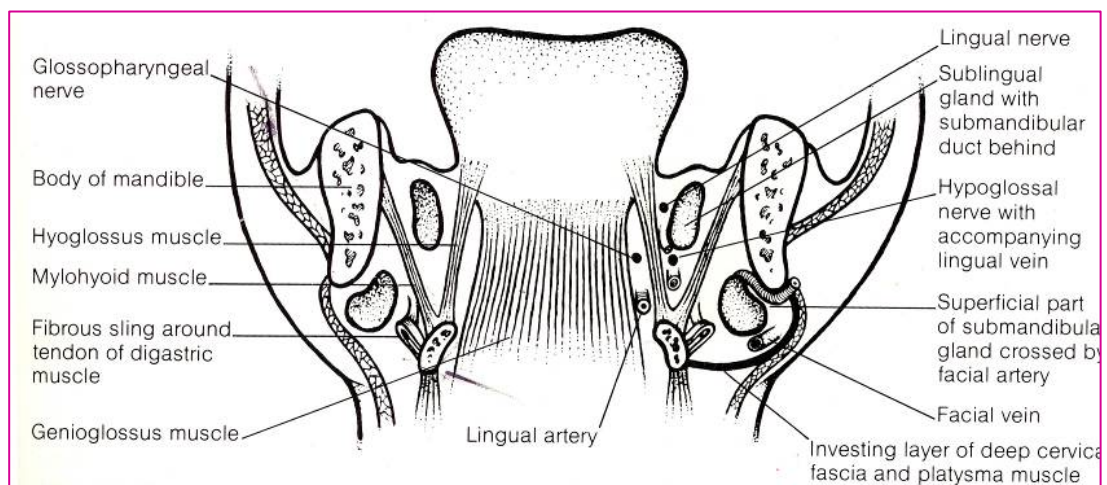
**2) Medial groove (sulcus) of the tongue:**

*Borders:*

- Medial: *Genioglossus muscle*
- lateral: *Hyoglossus muscle*

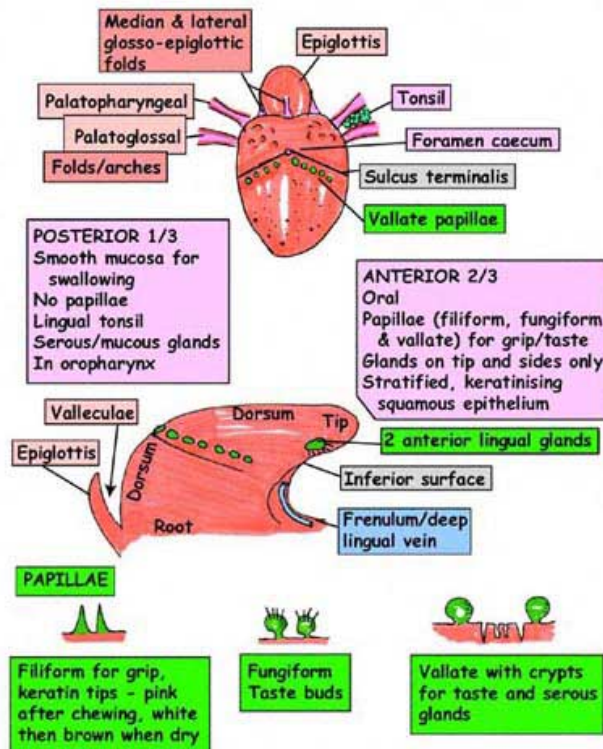
*Contents:*

- *Lingual artery*
- *Glossopharyngeal nerve (IX)* (not totally)

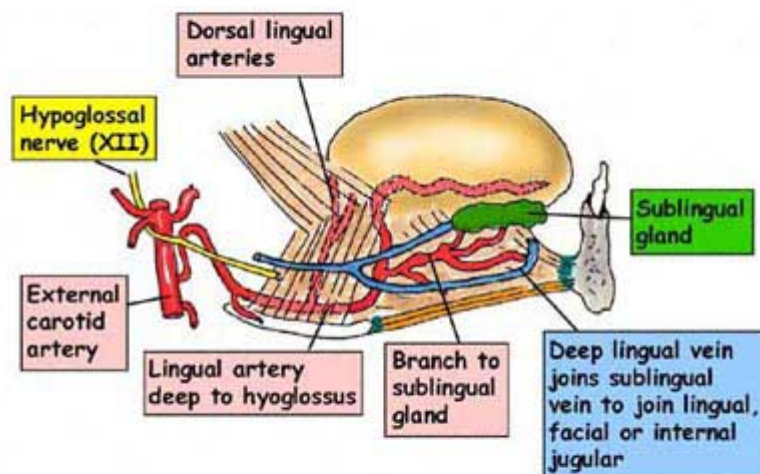


### TONGUE - GENERAL TOPOGRAPHY

The tongue is a mass of skeletal muscle covered by mucous membrane. It is divided functionally and embryologically into an anterior 2/3 and a posterior 1/3 by the sulcus terminalis



### TONGUE - LINGUAL ARTERY

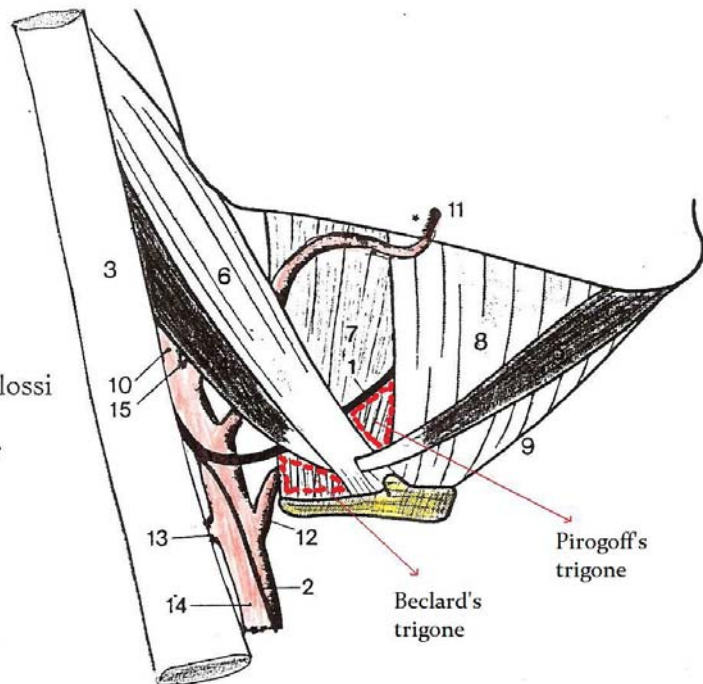


#### LYMPH

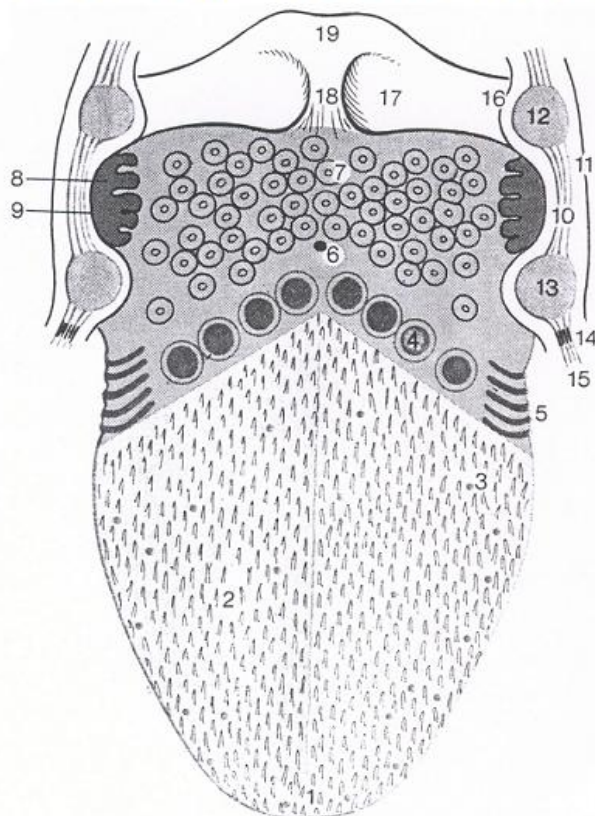
Tip to submental glands bilaterally  
Dorsum to submandibular mostly unilaterally  
Posterior to jugulo-omohyoid & deep cervical

### Triangles of Pirogoff & Beclard

1. n.hypoglossus
2. ramus descendens nervi hypoglossi
3. m. sternocleidomastoideus
4. m. digastricus, venter posterior
5. m. digastricus, venter anterior
6. m. stylohyoideus
7. m. hyoglossus
8. m. mylohyoideus
9. raphe mylohyoidea
10. art. temporalis superficialis
11. art. facialis
12. art. lingualis
13. art. occipitalis
14. art. carotis externa
15. art. maxillaris



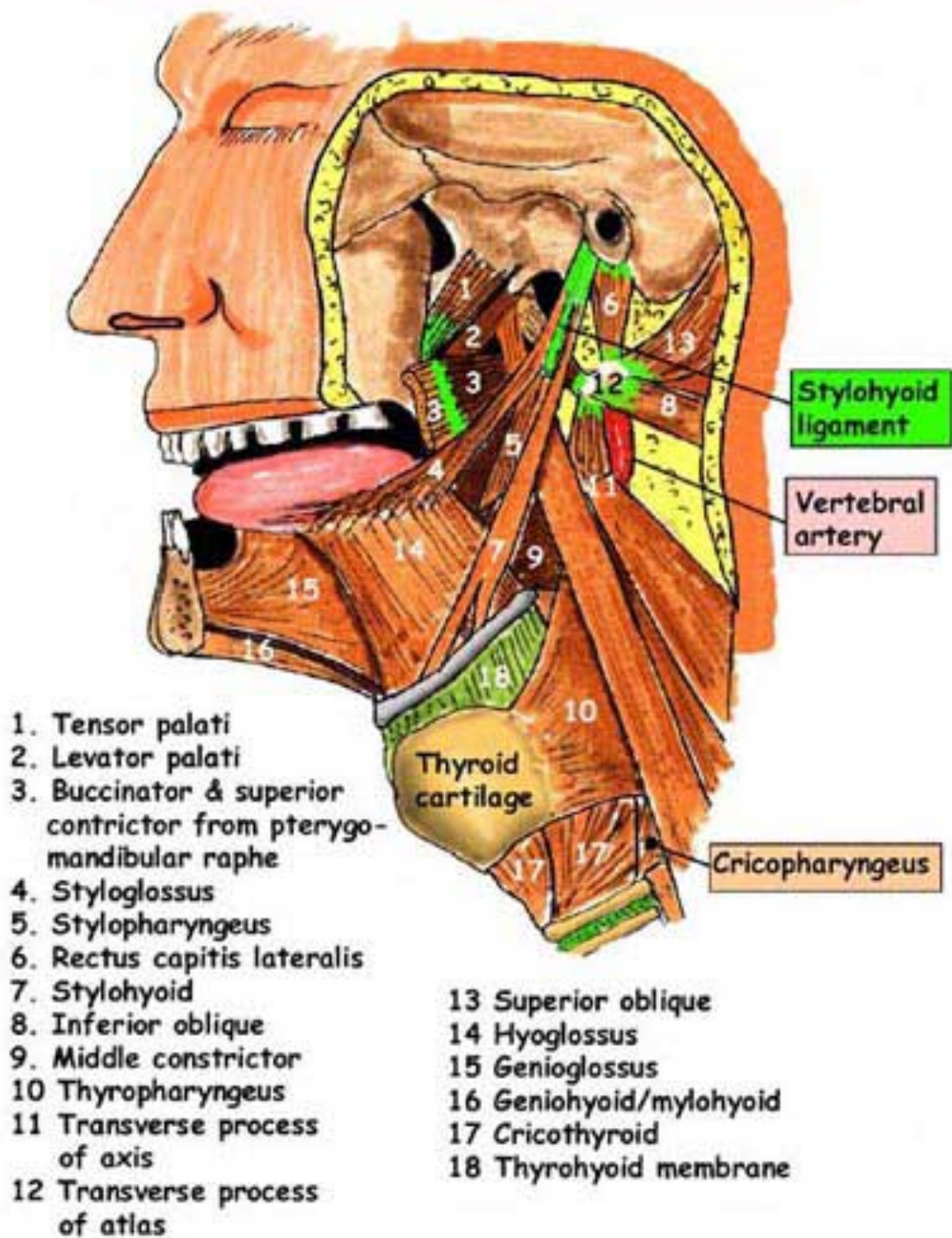
### Isthmus of fauces (horizontal section) & the tongue



- 1) Apex of tongue
- 2) Body of tongue
- 3) Filiform papillae
- 4) Vallate papillae
- 5) Foliate papillae
- 6) Foramen cecum
- 7) Lingual tonsil (lymphatic follicles)
- 8) Palatine tonsil
- 9) Capsule of palatine tonsil
- 10) Superior constrictor muscle of pharynx
- 11) Pharyngeal fascia
- 12) Palatopharyngeus muscle
- 13) Palatoglossus muscle
- 14) Pterygomandibular raphe
- 15) Buccinator muscle
- 16) Palatopharyngeal fold
- 17) Epiglottic valleculae
- 18) Median glossoepiglottic fold
- 19) Epiglottis



## MUSCLES OF TONGUE, MOUTH & NECK



## The Palate

Is the arched roof of mouth and floor of nasal cavity, Separates oral, nasal and nasopharynx cavities, and has:

- Superior part covered by *respiratory* mucosa (pseudostratified columnar epith.)
- Inferior part covered by *oral* mucosa (Stratified squamous non-keratinized epith.)

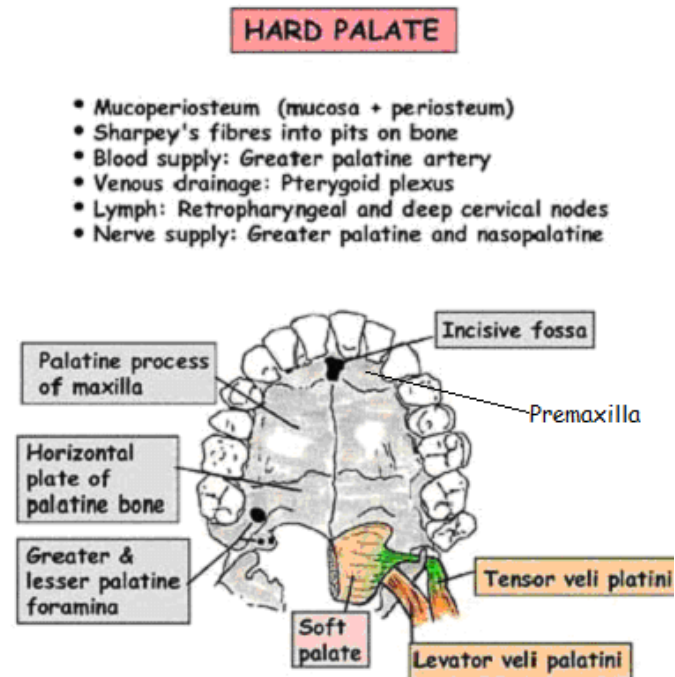
Composed of 2 parts:

- The *Hard palate* anteriorly
- The *Soft palate* posteriorly

### → Hard palate:

The anterior 2/3 of palate, Composed of:

- *Premaxilla or incisive bone*: the anterior and inner portion of the maxilla, which in the fetus and sometimes in the adult is a separate bone.
- *Palatine process of maxilla*
- *Horizontal plates of palatine bones*



In the middle of the hard palate's mucosa there is the Rapih palate, and further between the periosteum and the mucosa we can find numerous *Palatine glands*. It has 3 foramina:

- Incisive fossa (with 2 foramina)**: contains Nasopalatine nerves (of Scarpa) → incisive papilla
- Greater palatine foramen**: medial to 3<sup>rd</sup> molar, greater palatine vessels and nerves
- Lesser palatine foramina**: lesser palatine vessels and nerves.

### → Soft palate:

The posterior 1/3 of palate, motile, has 4 parts:

- **Anterior**:
  - Aponeurotic palate, attaches to hard palate formed by an expanded tendon of *tensor veli palatini*
- **Posterior**:
  - Muscular palate
- **Midline**: Uvula
- **Lateral**:
  - Continuous with the wall of the pharynx by Palatoglossal and palatopharyngeal arches

### → Role in Swallowing:

The palate is initially tensed to allow the tongue to press against it, squeezing bolus to the back of the mouth. The soft palate is then elevated posteriorly and superiorly against the wall of the pharynx, preventing passage of food into the nasal cavity.

→ **Muscles of the soft palate:**

- a. **Tensor veli palatini:** tenses the palate & opens auditory tube  
*Origin:* Scaphoid fossa & membranous part of the Eustachian tube  
*Insertion:* Palatine aponeurosis, the muscles tendon forms a hook around the Pterygoid hamulus, so radiating horizontally into the soft palate
  - b. **Levator veli palatini:** elevates the soft palate up and backward  
*Origin:* petros part of temporal bone + cartilaginous part of the Eustachian tube  
*Insertion:* Palatine aponeurosis
  - c. **Palatoglossus:** elevates posterior part of the tongue + draws the palate to the tongue, closing the isthmus of Fauces  
*Origin:* Palatine aponeurosis  
*Insertion:* side of the tongue
  - d. **Palatopharyngeus:** tenses the palatine + pulls the walls of pharynx during swallowing  
*Origin:* palatine aponeurosis and hard palate  
*Insertion:* lateral wall of pharynx
  - e. **Musculus uvulae:** shortens the uvula and pulls it superiorly  
*Origin:* palatine aponeurosis and posterior nasal spine  
*Insertion:* mucosa of uvula
- All muscles are innervated by *pharyngeal plexus (Glossopharyngeal IX, Vagus X)*, except to the **tensor veli palatine**, innervated by the a nerve branch of the Mandibular division of the *Trigeminal nerve (CN-V3)* which loops around the Pterygoid hamulus

**\*\* Note:** To test glossopharyngeal and Vagus nerve function, examine the position of the uvula and its movement by asking the patient to say "Ah." The soft palate should elevate symmetrically and the uvula should remain in the midline. The gag reflex can be tested by touching the pharyngeal wall on each side with a cotton tip applicator. This reflex relies on an intact sensory arc, as mediated by sensory fibers of the glossopharyngeal nerve to the soft palate, and an intact motor arc, as mediated by the motor fibers of the Vagus nerve to the soft palate and pharynx. Deviation of the uvula to one side implies a **lower motor** lesion of the Vagus nerve **contralateral** to the side the uvula is deviating to. An **upper motor** neuron Vagus nerve lesion will present with the uvula deviating **toward** the **side** of the lesion.

→ **Vasculature and innervation of the palate:**

**Vasculature:**

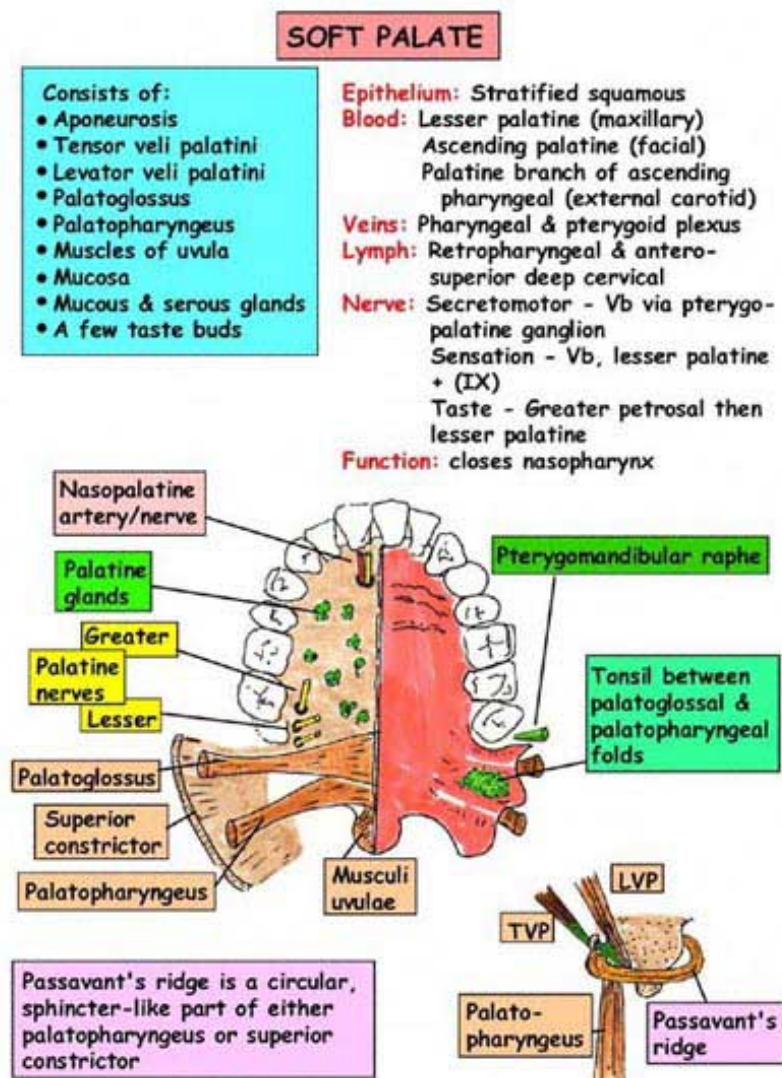
- Greater palatine artery
- Branch of descending palatine artery (From Maxillary artery)
- Lesser palatine artery anastomosis with ascending palatine (From facial artery)

**Innervation**

- Greater palatine nerve: supplies the gingivae, mucous membrane and glands of the hard palate
- Nasopalatine nerve: mucous membrane of anterior part of the hard palate (Premaxilla)
- Lesser palatine nerve: soft palate

→ *Waldeyer's lymphatic pharyngeal ring, formed by:*

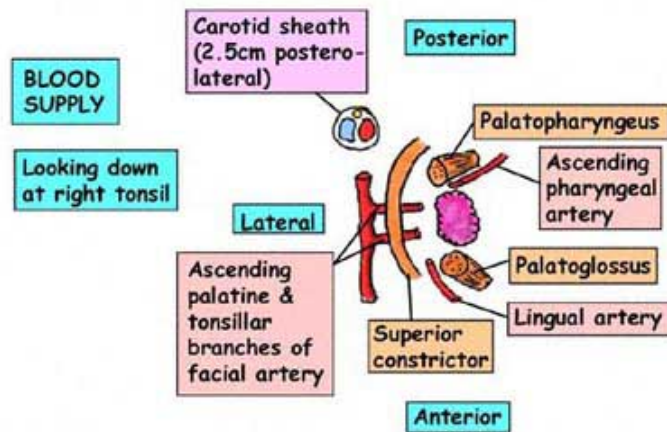
- Palatine tonsils*, most frequently inflamed, in the tonsillar fossa between the pillars of fauces (2).
- Lingual tonsil*, made by the lingual follicles, which are collections of lymphoid tissue in the mucosa of the pharyngeal part of the tongue posterior to the terminal sulcus.
- Pharyngeal tonsil*, adenoid, in the nasopharynx.
- Tubal tonsils*, nasopharynx, close to the opening of each auditory tube (2).





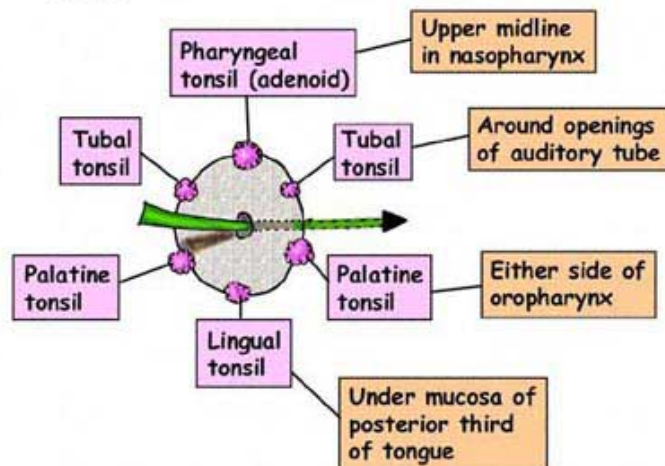
## PALATINE TONSIL

- Lymphoid tissue in tonsillar fossa
- Anterior/posterior: palatoglossal/palatopharyngeal arches
- Superior: soft palate
- Inferior: tongue
- Medial: mucosa & 20 tonsillar crypts, intratonsillar cleft (this is a large crypt from 2nd pharyngeal pouch)
- Bed: submucosa (capsule), superior constrictor, facial artery & its branches
- Lymph: to deep cervical & jugulodigastric
- Veins: plexus in capsule to pharyngeal venous plexus. Also external palatine (paratonsillar) from soft palate
- Nerves: tonsillar branch of glossopharyngeal (IX) - hence referred pain to the middle ear. Also lesser palatine (maxillary via pterygopalatine ganglion)
- Development: 2nd pharyngeal pouch endoderm gives mucosa & crypts  
Surrounding mesenchyme gives lymphoid tissue
- Surface marking: medial to lower masseter



## WALDEYER'S RING

An interrupted circle of protective lymphoid tissue at the upper ends of the respiratory and alimentary tracts





## The Teeth

Hard conical structures set in the alveoli of the upper and lower jaws, used in mastication and assisting in articulation, forming an arch that separates the vestibule from the oral cavity proper. Humans usually have *20 primary teeth* (also called *deciduous*, baby, or milk teeth) and *32 permanent teeth*, which means 2 sets of teeth or **Diphyodont**

Among primary teeth, 10 are found in the (upper) *maxilla* and the other 10 in the (lower) *mandible*.

In the primary set of teeth, there are two types of incisors, centrals and laterals, and two types of molars, first and second. All primary teeth exfoliate and are replaced with permanent counterparts except for molars, which are replaced by permanent premolars.

Among permanent teeth, 16 are found in the maxilla with the other 16 in the mandible, lying closely to each other with no space between them (ideally!) (If there is space between the teeth it's called: **Diastema**).

→ Humans have **heterodont** dentition, which means the teeth have 4 different shapes (types) for different functions:

- **Incisors**: chisel shaped, for biting (cutting edge), has a *tubercle* on the oral side and a single conical tapering *root*; the maxillary incisors are more broad
- **Canines**: firmly rooted & long, for tearing and gripping. Got an acute angle, a chewing point and a long, slightly flattened conical root
- **Premolars**: broad crown, for grinding & crushing, has 2 tubercles or cusps on the occlusal surface of crown
  - Maxillary *1st premolar* got divided roots (furrowed)
  - Mandibular premolars, and 2<sup>nd</sup> maxillary premolar got simple roots
- **Molars**: broad surface, for chewing. Got 4 tubercles on the chewing surface
  - Maxillary molars got 2 *vestibular root and 1 oral*
  - Mandibular molars got 1 *vestibular root and 1 oral*

The teeth form 2 arcades or arches, the *upper teeth arch (maxillary)* is **ellipsoid** or *half eclipse shaped*, while the *lower (Mandibular)* is **parabola shaped**

→ Each tooth has different *surfaces*:

- **Masticatory surface (occlusal surface)**: contacts the teeth of the opposing dental arcade,
- **Vestibular surface**: facing the lips or cheeks
- **Lingual or Palatal surface**: the inner surface of teeth, facing the oral cavity proper
- **Approximal surface**: facing the adjacent tooth, subdivided into,
  - Mesial surface: facing anteriorly or medially
  - Distal surface: facing posteriorly or laterally

The normal healthy bite (*Dentition*) is called the **Scissors bite (Psaliodontia)**, in which the teeth don't meet exactly, but the maxillary incisors overlap over the Mandibular incisors)

*The lingual nerve passes close to the 3<sup>rd</sup> Mandibular molar*

→ **Parts of the tooth:**

- **Crown:** the portion of the tooth covered by the hard acellular *enamel*, which is the hardest substance of the human body, made of enamel prisms (ground substance 90% hydroxyapatite and 6-8% calcium carbonate, calcium fluoride, and magnesium carbonate, the remainder comprising an organic matrix of protein and glycoprotein)  
We can distinguish:
  - **Anatomical crown:** the whole crown of the tooth covered by the enamel
  - **Clinical crown:** the portion of the crown visible above the gingiva in the oral cavity
- **Dentine:** the avascular ivory forming mass of the tooth, under the enamel. About 20% is organic matrix, mostly collagen, with some elastin and mucopolysaccharide; the inorganic fraction (70%) is mainly hydroxyapatite, with some carbonate, magnesium, and fluoride  
On its inner surface we find *odontoblasts*, sending projections called *Odontoblastic processes (Tomes fibers)* into the dental canaliculi, which extend up to the dentinoenamel or cementodentinal junction
- **Cementum:** bonelike mineralized tissue covering the dentine of the root and neck of a tooth that serves to anchor the fibers of the periodontal ligament
- **Pulp cavity:** contains loose connective tissue, blood vessels and nerves known as **Dental pulp**, subdivided into:
  - *Pulp cavity* of the crown: the space within the crown of a tooth
  - *Root canal:* the chamber of the dental pulp lying within the root portion of a tooth
  - *Apical foramen:* the opening at the apex of the root of a tooth that gives passage to the nerve and blood vessels
- **Neck of tooth (Collum):** the slightly constricted part, between the crown and the root
- **Root:** that part of a tooth below the neck, covered by cementum rather than enamel, and attached by the *periodontal ligament* to the alveolar bone

**\*\* Note:** The histological structure of the teeth should be studied with more details from the lectures provided by the department, or from the atlases of histology recommended by the department, as well as the embryological development and eruption of the teeth that can be found in the histological manual of the department or in the Thieme Internal organ book, especially for dental students.

→ **Supporting structures of teeth:**

- **Periodontal ligament:** springy suspension of teeth in the bony alveoli, *perforation fibers of Sharpey's* (collagen) are found anchored in the cementum and reaching the periosteum of the bony alveolus, permits slight mobility
- **Periodontium:** periodontal ligament + cement + gingiva + alveolus
- **Tooth sockets (Alveoli dentales):** small pockets of the maxillary or mandibular alveolar processes housing the teeth, separated by the *interalveolar septa*
- **Interdental septa:** separates the different roots of teeth where there is more than 1 root

→ **Teeth formula:**

4 groups of teeth in 2 arches, mirror image of each side of median plane

From middle line to the distal end we have:

- **2** incisors, **1** canine, **2** premolars, **3** molars (*Third molars are commonly called "Wisdom teeth" and may never erupt into the mouth or form at all*)

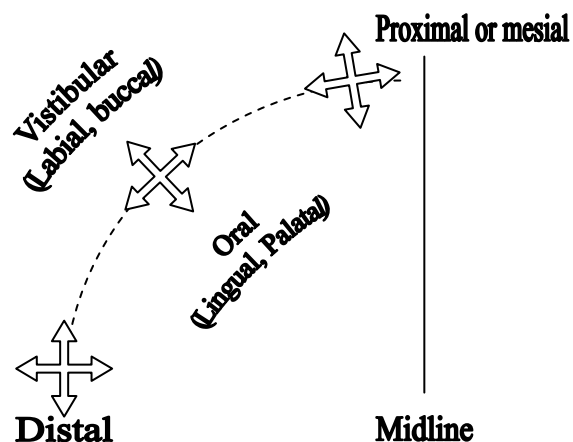
This gives a formula of **2123** on each quadrant of mouth according to this table:

| FDI Two-Digit Notation |    |    |    |    |    |    |    |            |    |    |    |    |    |    |    |
|------------------------|----|----|----|----|----|----|----|------------|----|----|----|----|----|----|----|
| Permanent teeth        |    |    |    |    |    |    |    |            |    |    |    |    |    |    |    |
| upper right            |    |    |    |    |    |    |    | upper left |    |    |    |    |    |    |    |
| 18                     | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 21         | 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 48                     | 47 | 46 | 45 | 44 | 43 | 42 | 41 | 31         | 32 | 33 | 34 | 35 | 36 | 37 | 38 |
| lower right            |    |    |    |    |    |    |    | lower left |    |    |    |    |    |    |    |

**Examples:**

- "13" = permanent upper right canine
- "32" = permanent lower left lateral incisor

→ **Orientation in the oral cavity:**



→ **Arterial supply of teeth:**

- 1) **Lower jaw:** *Inferior alveolar artery* (branch of *maxillary*) inside the *Mandibular canal*
- 2) **Upper jaw:** anastomosis between,
  - a. *Superior-Posterior alveolar artery* (branch of *maxillary*) → through the *maxillary tuberosity* to supply the posterior teeth and gums
  - b. *Superior-Anterior alveolar artery* (branch of *infraorbital*) supplies the anterior teeth and gums

→ **Nerve supply of teeth:**

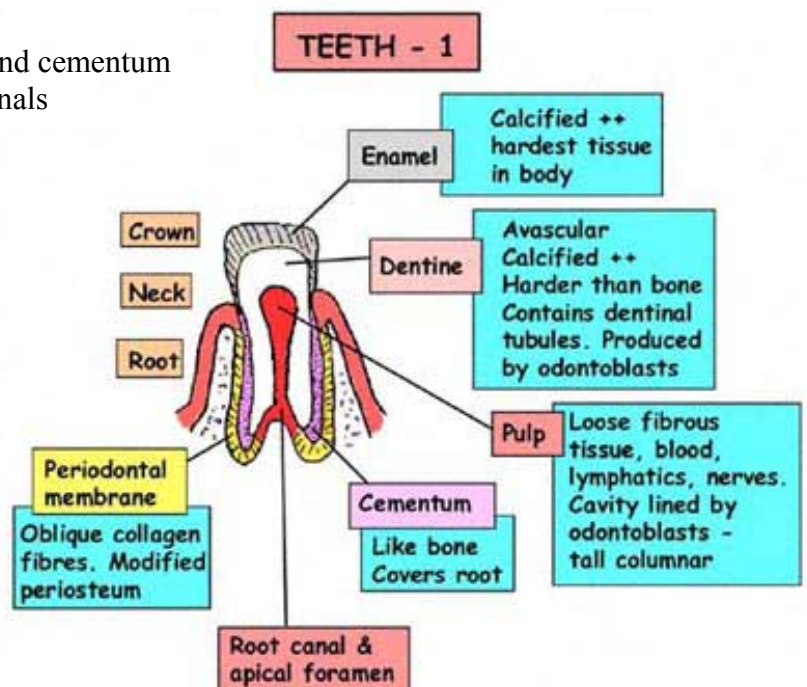
- 1) **Lower jaw:** *Mandibular branch of the trigeminal nerve (CNV3): Inferior alveolar (Dental) nerve & plexus*
- 2) **Upper jaw:** the *superior dental plexus* formed by the *Maxillary branch* of the *trigeminal nerve (CNV2)* and the *Infraorbital nerve*
  - a. 876 teeth by the *Posterior superior alveolar* (*Maxillary branch*)
  - b. 543 teeth by the *Middle superior alveolar* (*Infraorbital branch*)
  - c. 21 teeth by the *Anterior superior alveolar* (*Infraorbital branch*)

→ **Deciduous Teeth features:**

- 20 in number
- 2 incisors, 1 canine, no premolars and 2 molars
- Smaller size, globular crown, short and wide
- Their shapes are more stable
- Bluish enamel
- Sharp border between enamel and cementum
- Wide pulp chamber and root canals
- Thin dentin, thick enamel
- Serious abrasion

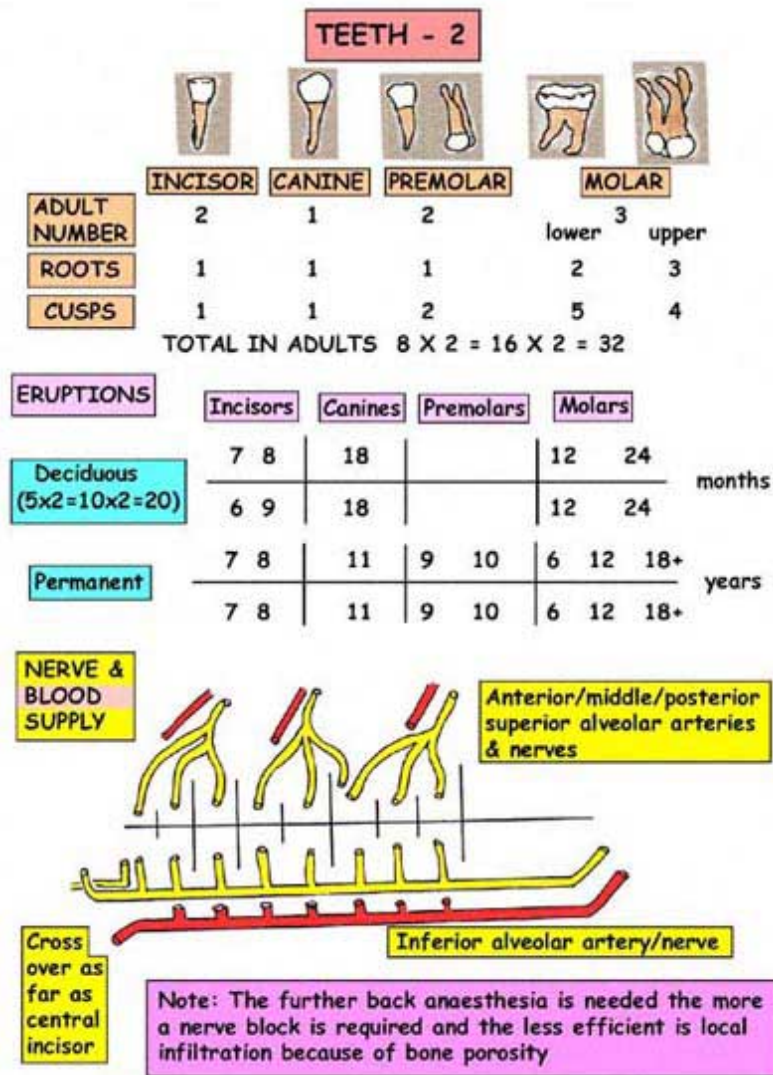
→ **Order of Eruption:**

- A) **Primary dentition (deciduous):**
  - 1) Incisors (1<sup>st</sup> then 2<sup>nd</sup>)
  - 2) 1<sup>st</sup> primary molars
  - 3) Canine teeth
  - 4) 2<sup>nd</sup> primary molar
- B) **Permanent dentition:**
  - 1) 1<sup>st</sup> molar
  - 2) Incisors (1<sup>st</sup> then 2<sup>nd</sup>)
  - 3) 1<sup>st</sup> premolars
  - 4) Canine teeth, 2<sup>nd</sup> premolar
  - 5) 2<sup>nd</sup> molar
  - 6) 3<sup>rd</sup> molar



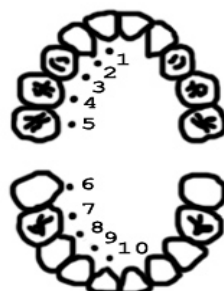
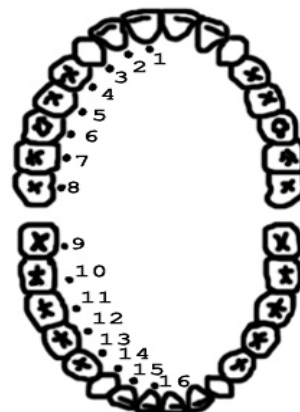
**Development**

- Budding of mouth ectoderm gives enamel
- Mesoderm is evoked to produce dentine & cementum
- Ameloblasts give enamel
- Mesoderm gives dental papillae which give odontoblasts which give dentine
- Dental papilla also gives pulp



### Adult Teeth

- Upper Teeth
1. Central Incisor
  2. Lateral Incisor
  3. Canine (cuspid)
  4. First Premolar (first bicuspid)
  5. Second Premolar (second bicuspid)
  6. First Molar
  7. Second Molar
  8. Third Molar (wisdom tooth)
- Lower Teeth
9. Third Molar (wisdom tooth)
  10. Second Molar
  11. First Molar
  12. Second Premolar (second bicuspid)
  13. First Premolar (first bicuspid)
  14. Canine (cuspid)
  15. Lateral Incisor
  16. Central Incisor



### Baby Teeth

- Upper Teeth
1. Central Incisor
  2. Lateral Incisor
  3. Canine (cuspid)
  4. First Molar
  5. Second Molar
- Lower Teeth
6. Second Molar
  7. First Molar
  8. Canine (cuspid)
  9. Lateral Incisor
  10. Central Incisor



## ***The Salivary glands***

- Parotid gland
- Submandibular gland
- Sublingual gland
- Accessory minor glands scattered over palate, lips, cheeks and tongue

Secrete the tasteless, clear, odorless *Saliva*, which has functions in:

- 1) Keep mucous membrane of mouth moist (intrinsic mouth wash), which is the main role of the minor salivary glands
- 2) Lubrication of food & beginning of starch digestion
- 3) Prevention of teeth decay

→ ***Parotid gland: pure serous*** (check drawing of the *Parotid nest*)

Largest, enclosed with a tough fascial capsule, the *parotid sheath* (part of the *investing layer* of the cervical fascia). Has an irregular shape parotid bed or nest, in which the gland lays anteroinferiorly to the external acoustic meatus, *behind the ramus of the mandible and in front of the styloid process*

Between the lobes there is numerous *fat* that increase the flexibility and accommodates the gland to the movement of the mandible

The *parotid duct* of ***Stensen*** passes *horizontally* and at the anterior border of the masseter muscle it pierces the buccinator muscle and opens behind the 2<sup>nd</sup> *maxillary molar* into the oral cavity

The accessory parotid gland is an occasional islet of parotid tissue separate from the mass of the primary gland, lying anteriorly just above the commencement of the parotid duct

\*\* Structures embedded within the gland:

- *Parotid plexus of fascial nerve branches*
- *Retromandibular vein*
- *External carotid artery and branches (supply the gland also)*
- *Auriculotemporal nerve (of CN-V3)*
- *Parotid lymph nodes*

→ ***Submandibular gland: mostly serous***

Found along the *body of mandible*. The duct is ~5cm long and arises between the *mylohyoid* and *hyoglossus* muscles, passing from *lateral* to *medial*, opening on a *sublingual papilla*, found both sides *lateral* to the *lingual frenulum*

Supplied by the *submental artery/vein*

→ ***Sublingual gland: mostly mucous***

The smallest and most deep gland, almond shaped, lying in the floor of the mouth between the *mandible* and the *Genioglossus muscle*. Both glands unite to form a *horseshoe* shape under the lingual frenulum. It has many small ducts that open to the mouth

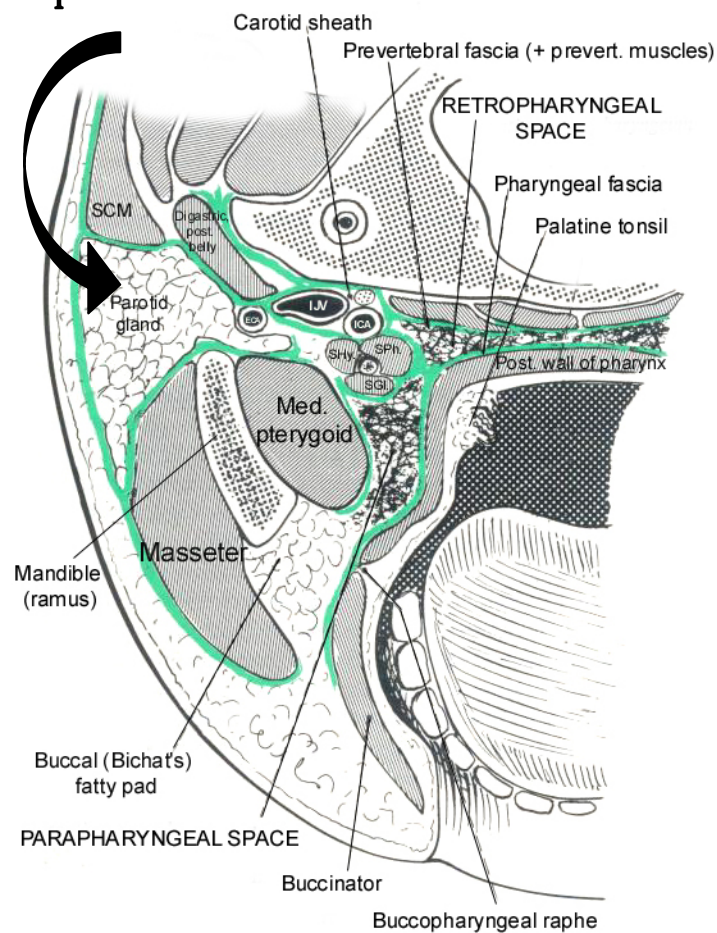
Supplied by the *Sublingual artery* (branch of lingual artery) and the *submental artery* (branch of fascial artery)

→ **Borders of the Parotid nest (Nidus parotideus):**

- *Anterior:*
  - Masseter muscle
  - Ramus of the mandible
  - Medial pterygoid muscle
- *Posterior:*
  - Sternocleidomastoid
  - Posterior belly of the digastric
- *Medial:*
  - Stylohyoid muscle
  - Stylopharyngeus muscle
  - Styloglossus muscle

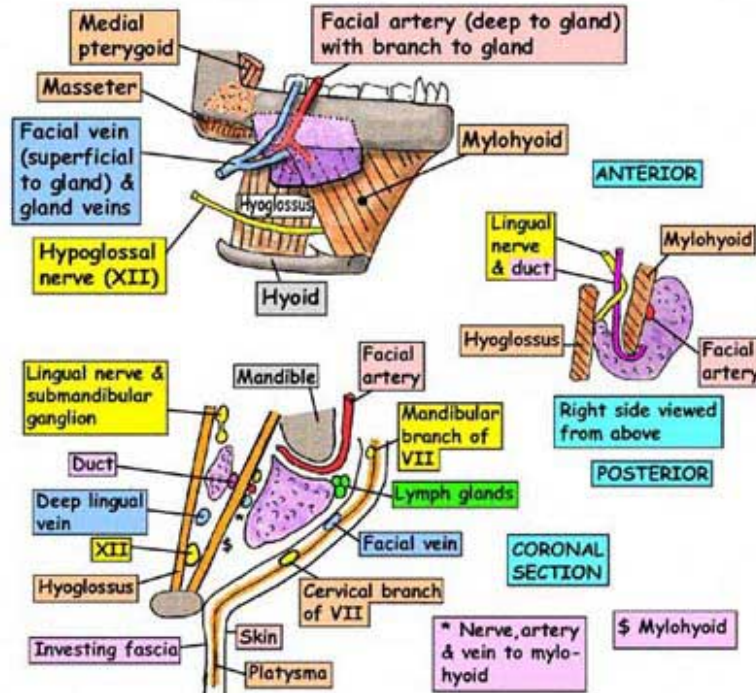
PARA- and RETROPHARYNGEAL SPACES

& parotid nest

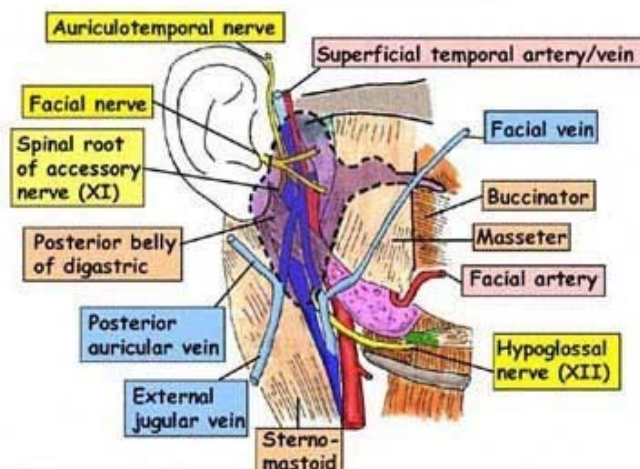


## SUBMANDIBULAR GLAND

- Mixed - mucous and serous
- 2 parts - Superficial: larger. Deep: smaller. Join behind posterior edge of mylohyoid
- Duct: (Wharton's) 5cm long. First between mylohyoid & hyoglossus, then between sublingual gland and genioglossus. Opens in floor of mouth beside frenulum. Develops in ectoderm from a groove in the floor of mouth
- Produces 70% of the saliva
- Lymph nodes in it and on it. Drain to submandibular glands
- Histology: See parotid



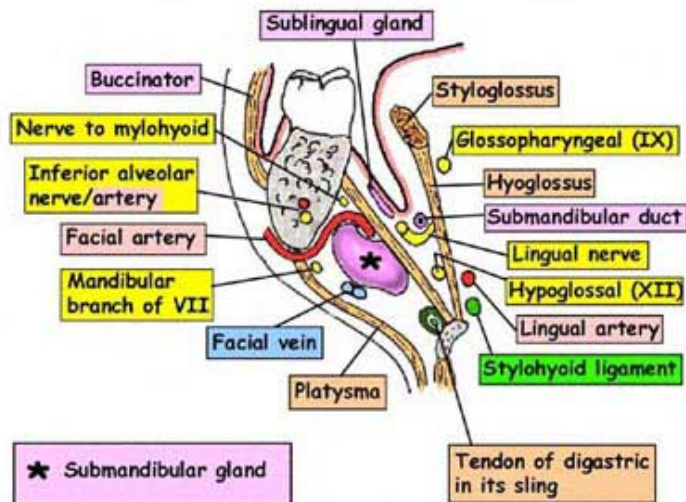
## PAROTID GLAND 1



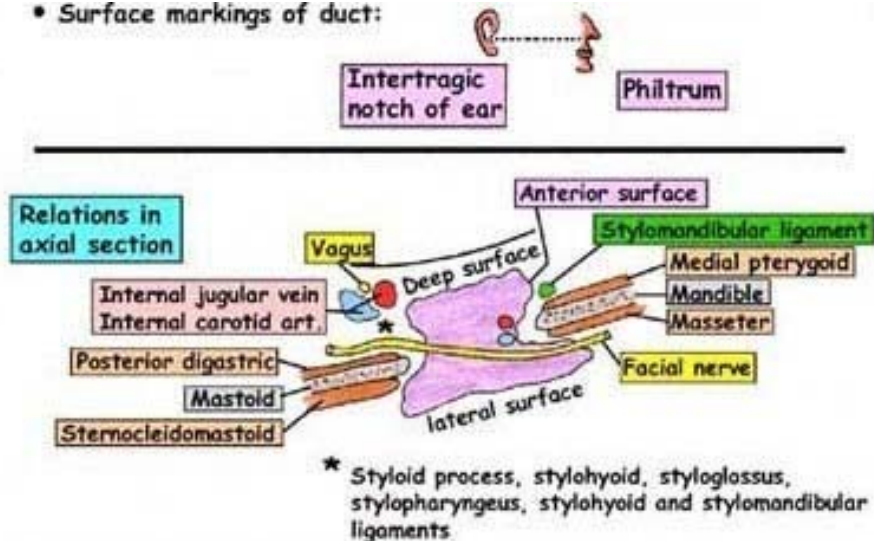
- Serous secretions
- Produces amylase, water, Ig factors (lubricates & oral hygiene)
- Lies between mastoid, styloid process, ramus of mandible
- Has an upper & lower pole, lateral, anterior & deep surfaces
- Surrounded by parotid fascia (investing layer of deep fascia)

## SUBLINGUAL GLAND

- Mucous gland
- Between mylohyoid and genioglossus
- 15 ducts - 1/2 into submandibular duct  
1/2 into sublingual fold
- Nerve supply - secretomotor via submandibular ganglion  
general sensation via lingual (Vc)
- Blood supply - Lingual artery & branches of submental artery
- Develops from a groove in floor of mouth that becomes a tunnel  
Blind end proliferates (ectodermal) to give secreting acini
- (Note: all salivary glands develop from epithelial lining of mouth)



- Surface markings of duct:



## ***The Pharynx***

The pharynx is the continuation of the GI tract after the oral cavity. It's a *fibromuscular* tube that is used as a pathway for both food and air.

The Pharynx is divided to 3 parts:

- An upper ***Nasopharynx*** that receives the posterior openings of the nasal cavity, the 2 Choana.
- Middle ***Oropharynx***, behind the oral cavity (after the isthmus of fauces)
- A lower ***Laryngopharynx***, behind the Larynx.

The Pharynx is 15cm and it extends from the base of the skull down to the lower border of the Cricoid cartilage anteriorly, and to the lower border of C-6 *posteriorly*, from this point the pharynx becomes the Esophagus

### **→ The Pharyngobasilar fascia:**

Is the *fibrous* coat of the pharyngeal wall situated between the mucous and muscular coats; it is attached above to the basilar part of the occipital bone, and the petrous part of the temporal bone. This layer and the mucosa which lines it forms the wall of the non-muscular pharynx (pharyngeal vault) above the superior pharyngeal constrictor muscle (so basically the whole pharynx is hanging down from the base of skull by this fascia)

### **A) The Nasopharynx (Epipharynx):**

This part is located above the soft palate, and to it the *Choanae* (also known as internal nares) opens. The roof and the posterior walls are formed by the Sphenoid bone and the basilar part of the Occipital bone. In the mucous membrane we can find the "*Pharyngeal tonsils*".

On the lateral wall of the Nasopharynx we find the "*Orifice for the auditory tube*" ("*Eustachian tube*"), above this orifice we find the "***Torus Tubarius***" a small ridge caused by the projection of the cartilaginous part of the tube. Under this Torus, and posteriorly, we find the "***Salpingopharyngeal fold***" that covers a muscle with the same name which is in charge of opening the tube during chewing. Anterior to the Torus, we can find a small fold called "*Salpingopalatal fold*". Under the hiatus for the Eustachian tube we find the so called small "***Torus Levatorius***" made by the underlying Levator veli palatini muscle

Behind the orifice of the auditory tube, we can find the ***Tubarian tonsils***, and above the Torus we can see a "*pharyngeal recess*"

***Passavant Ridge:*** a prominence on the posterior wall of the nasopharynx formed by contraction of the superior constrictor muscle of the pharynx during swallowing

### **B) The Oropharynx (Mesopharynx):**

To this area the oral cavity opens via the "***Oropharyngeal Isthmus***" or "Isthmus of Fauces". The superior border of this part is the soft palate, and the inferior border is the base of the tongue and the superior border of the Epiglottis.

The *Palatine tonsils* are located within the Oropharynx in an area called the "***Tonsillar bed***" located between the palatine arches: the "*Palatoglossal arch*" anteriorly and the "*palatopharyngeal arch*" posteriorly.



### **C) Laryngopharynx (Hypopharynx):**

This part runs from the superior border of the epiglottis till the inferior border of the cricoid cartilage, and from this area, the Pharynx continues as the esophagus.

The Laryngopharynx is in contact with the larynx by the "**Inlet or Aditus of the larynx**", and at the sides of this inlet we find the bilateral **Piriform recess** which is a depression of the *laryngoharengal* cavity. Between these Piriform recess and the inlet of the larynx we find the "**Aryepiglottic fold**"

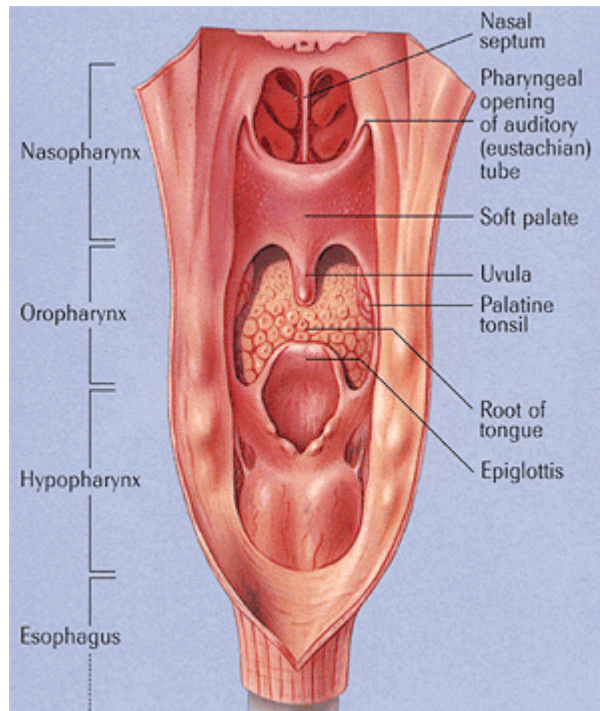


Image made available by a generous grant from Bristol-Myers Squibb

### **→ Muscles of the pharynx:**

#### **A) The Constrictors of pharynx:**

The constrictors are organized in a way that the superior constrictors are the innermost and the inferior constrictors are the outermost (like the roof tiles, but opposite). Their job is to involuntarily propelling the food downward towards the esophagus.

*Origin of the 3 Constrictors of Pharynx:*

##### **1) Superior constrictor:**

- *Pterygopharyngeal part* (Pterygoid process)
- *Buccopharyngeal part* (Pterygomandibular raphe)
- *Mylopharyngeal part* (Mylohyoid line of mandible)
- *Glossopharyngeal part* (root of tongue)

##### **2) Middle constrictor:**

- *Chondropharyngeal part* (Greater & lesser horns of Hyoid bone)
- *Ceratopharyngeal part* (Stylohyoid ligament)

**3) Inferior constrictor:**

- *Thyropharyngeal part* (Thyroid cartilage)
- *Cricopharyngeal part* (Cricoid cartilage): has a sphincteric function for the esophagus, forming the upper esophageal sphincter (UES) allowing some voluntary control of eructation and reflux. Can be a separate muscle

**\*\* All the constrictors of pharynx insert posteriorly in the midline on the Median raphe of pharynx and the pharyngeal tubercle**

**B) Elevators of pharynx:**

The pharynx has 3 important intrinsic muscles that lay deep to the constrictors that their job is to elevate the Pharynx and the Larynx during speaking or swallowing

**5) Stylopharyngeus muscle:**

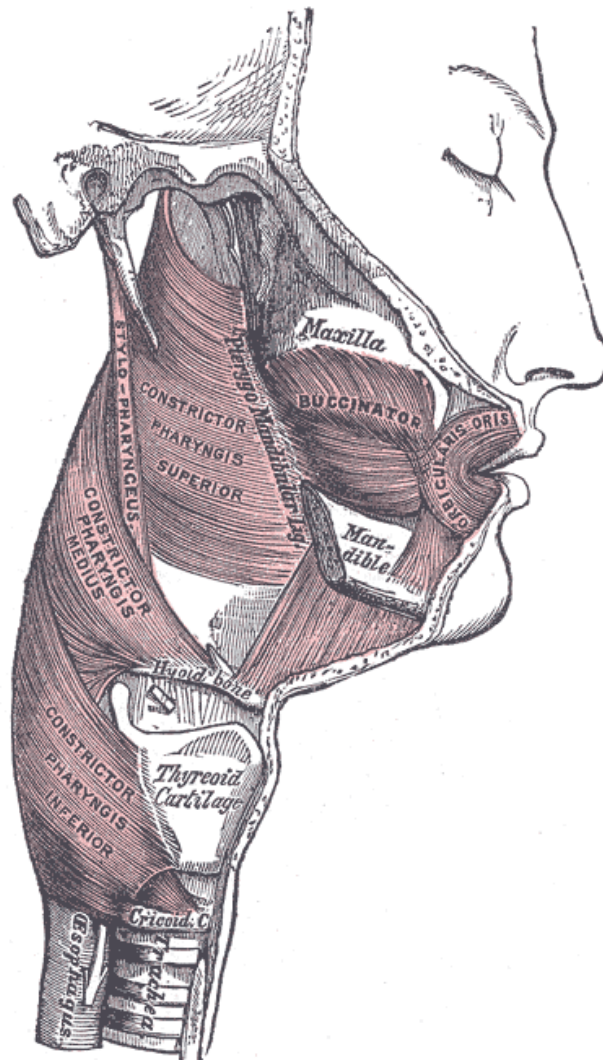
- *origin*: styloid process
- *Insertion*: the anterior and posterior parts of the thyroid cartilage

**6) Palatopharyngeus muscle:**

- *origin*: Hard palate and the palatine aponeurosis
- *Insertion*: posterior border of the Thyroid cartilage lamina

**7) Salpingopharyngeal muscle:**

- *origin*: cartilaginous part of the auditory tube
- *Insertion*: Palatopharyngeus muscle



### → **Peripharyngeal space**

See the discussion of the peripharyngeal space (retrovisceral, retropharyngeal, danger & parapharyngeal spaces in the neck fascia section)

### → **Pharyngeal mucosa:**

The nasal portion is covered by the nasal respiratory epithelium, and the oral and laryngeal portion has the same epithelium as the oral cavity (Stratified squamous non-keratinized).

Mucous glands, the pharyngeal glands, provide the lubricating mucous. The subepithelial connective tissue contains abundant elastic fibers, allowing the pharyngeal wall to stretch and recoil. In the laryngeal portion at the transition to the esophagus, the mucosa of the anterior and posterior wall of the pharynx is separated from the skeleton of the larynx and the cervical spinal column by a venous plexus.

**The buccopharyngeal fascia:** the fascia that covers the muscular layer of the pharynx and is continued forward onto the buccinator muscle

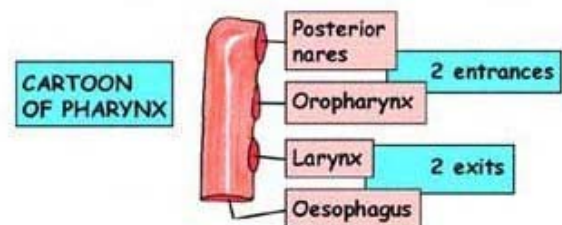
### → **Arterial supply of the pharynx:**

- The upper parts of the pharynx receive their blood supply from branches of the *facial* and *maxillary* arteries
- The *ascending pharyngeal* artery arises from the external carotid artery to supply the lower pharynx and larynx
- The *superior thyroid* artery arises from the external carotid artery to give the pharyngeal branches that supply the lowest part of the pharynx and part of the esophagus
- **Venous drainage:** into the pharyngeal plexus, posterior to the pharynx

### → **Innervation of the pharynx:**

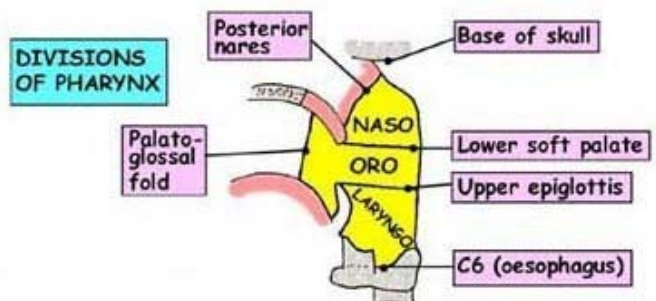
- **Pharyngeal plexus:** branches of Glossopharyngeal, Vagus and Sympathetic nerves
- **Motor supply:** cranial part of accessory n. which indirectly via the Vagus, branch to pharyngeal plexus supplies all muscles of pharynx except Stylopharyngeus, superior constrictor of pharynx & the upper 1/2 of the middle constrictor of pharynx! (Supplied by glossopharyngeal n.)
- **Sensory supply:** mucous membrane of *nasopharynx* is supplied mainly by *maxillary nerve* (CN-V2) via Pharyngeal branch of sphenopalatine ganglion.  
*The Oropharynx* mainly by glossopharyngeal n. (Posterior 3<sup>rd</sup> of tongue and Oropharynx).

Around the laryngeal inlet,  
Laryngopharynx: internal laryngeal Branch of Vagus, (branch of superior laryngeal)



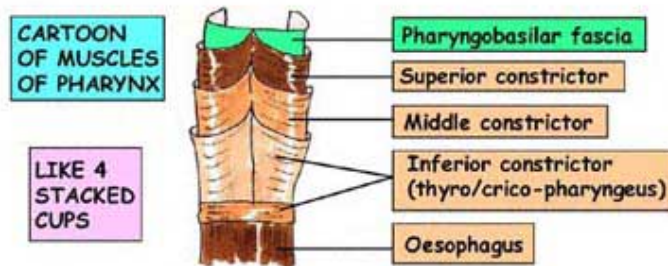
### → **Lymphatic drainage of the pharynx:**

- Retropharyngeal lymph nodes





## PHARYNX - CARTOON OF MUSCLES & PHARYNGBASILAR FASCIA



### PHARYNGBASILAR FASCIA

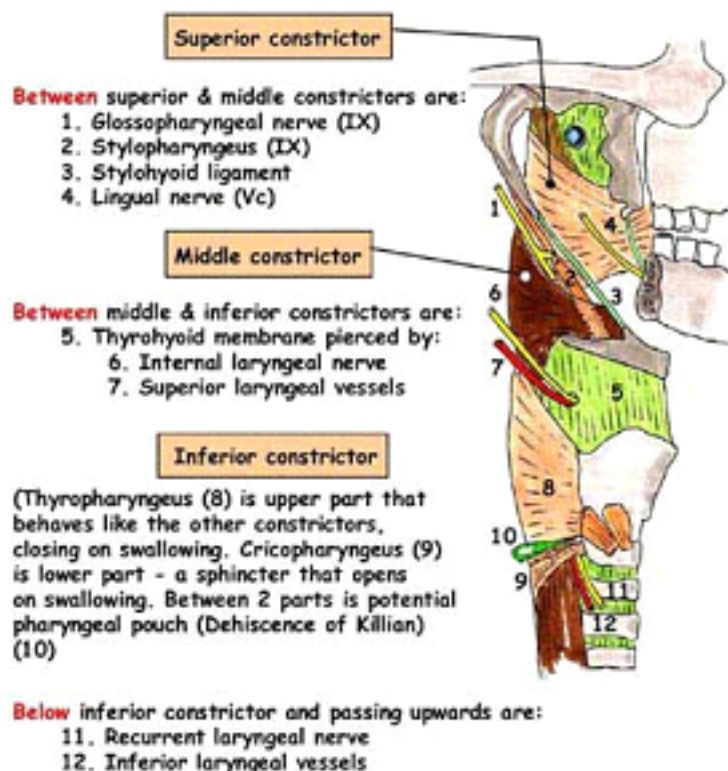
(a rigid membrane with cartilage of auditory tube passing just above it. Shown in yellow)

**Arises:** Pharyngeal tubercle, back of foramen lacerum, petrous temporal anterior to carotid foramen, cartilage of auditory tube, medial pterygoid plate, pterygoid hamulus then across to opposite side

**Lower border:** Junction of hard and soft palates (Passavant's ridge)

**Note:** Levator palati arises within pharynx

## PHARYNX - MUSCLES & STRUCTURES ENTERING IT



## PHARYNX - VESSELS & NERVES

### ARTERIAL SUPPLY

Ascending pharyngeal

Internal carotid



Common carotid

Greater palatine (maxillary)

Artery to pterygoid canal (maxillary)

Tonsillar (facial)

Ascending palatine (facial)

Lingual

Superior laryngeal (superior thyroid)

### VENOUS DRAINAGE

Plexus on middle constrictor draining to:  
Pterygoid plexus  
Internal jugular vein  
Lower pharynx to inferior thyroid veins

### NERVE SUPPLY

Levator palati  
Salpingopharyngeus  
Palatopharyngeus  
Palatoglossus  
3 constrictors  
Striated oesophagus

All from PHARYNGEAL PLEXUS on posterior wall of pharynx. It is:  
IX (glossopharyngeal) sensory only  
X (pharyngeal branch of vagus)  
Branchiomotor fibres from nucleus ambiguus via cranial accessory (XI)  
Sympathetic - vasoconstrictor only

Stylopharyngeus

Glossopharyngeal (IX) only

Cricopharyngeus (+/- some oesophagus)

Recurrent laryngeal (X)

Sensation/taste

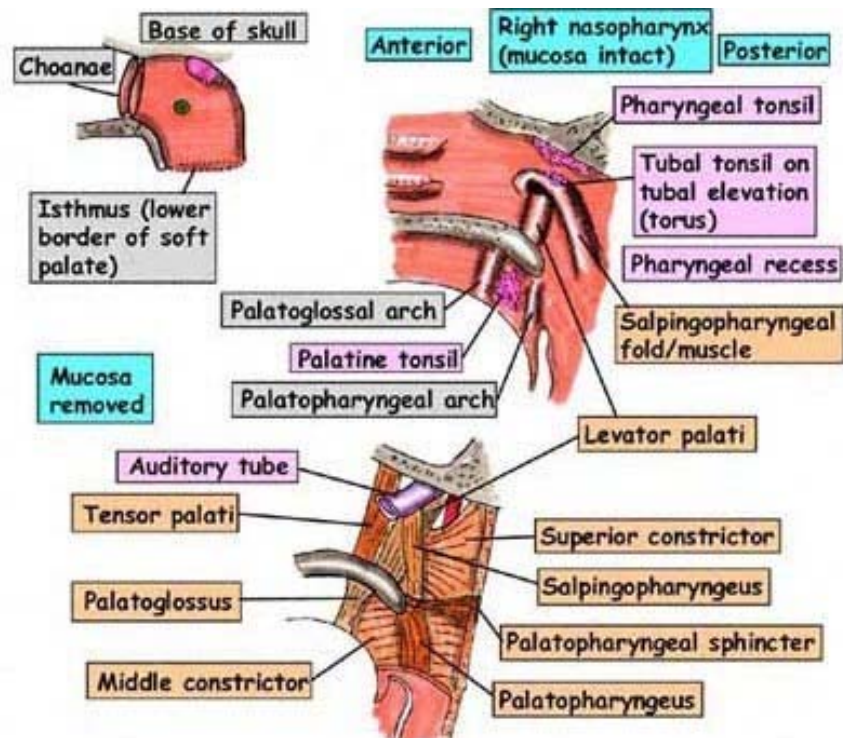
Nasopharynx: pharyngeal branch of maxillary (Vb) via pterygopalatine ganglion and palatovaginal canal  
Oropharynx/valleculae: IX and X

Sensation only

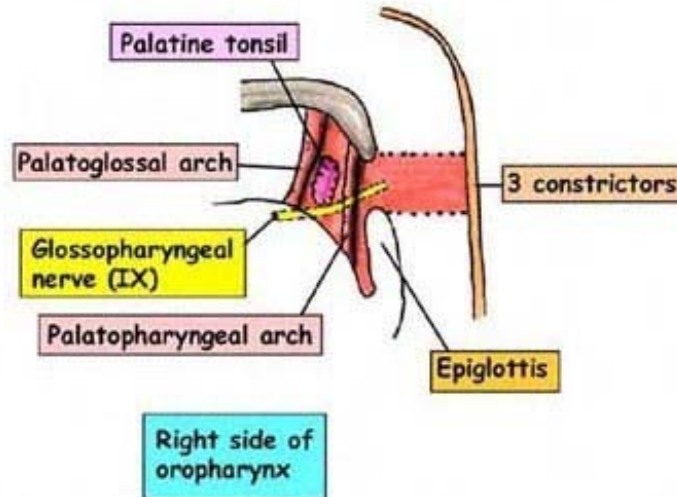
Laryngopharynx: Internal branch of superior laryngeal (X) & recurrent laryngeal (X)

LYMPH

Retropharyngeal to upper/lower deep cervical

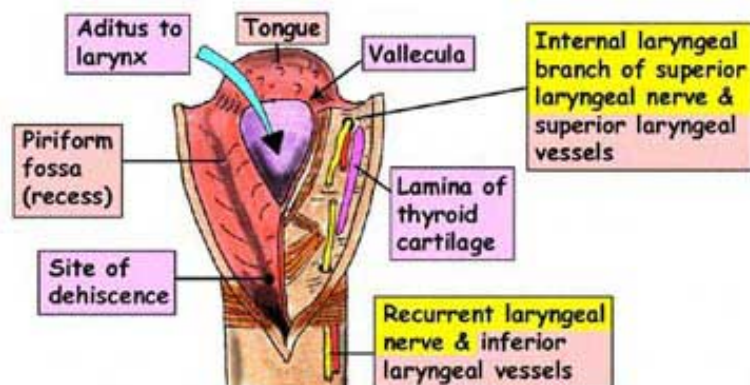






## LARYNGOPHARYNX

- Extends from: tip of epiglottis - C3
- To: start of oesophagus - C6
- Anterior: larynx, aditus, epiglottis
- Posterior: 3 overlapping constrictors, dehiscence of Killian, cricopharyngeus, vertebrae C4,5,6
- Nerve supply: internal laryngeal branch of superior laryngeal nerve (X) & recurrent laryngeal nerve (X).  
Note that there is some overlapping of supply in the laryngopharynx unlike in the larynx
- Lining: squamous non-keratinising epithelium
- Features: aditus to larynx & piriform fossa



**Piriform fossa**  
Medial: quadrangular membrane  
Lateral: thyrohyoid membrane & lamina of thyroid cartilage

**Hypopharynx**  
A clinical term for that part of the laryngopharynx below the aditus  
Anterior: arytenoid cartilages  
Posterior: dehiscence of Killian

## ***The Esophagus***

A muscular tube that is 25cm long, with an average diameter of 2 cm, conveying the food bolus from the pharynx into the stomach, it consists of three parts: the ***cervical part***, from the *cricoid* cartilage to the *thoracic inlet*; the ***thoracic part***, from the *thoracic inlet* to the *diaphragm*; and the ***abdominal part***, below the *diaphragm* to the *cardiac opening* of the stomach.

**\*\*** *The cervical and thoracic parts are fused with the membranous wall of the trachea*

The esophageal wall has an internal circular and an external longitudinal layer of muscle, the ***Upper 1/3*** – striated muscle, ***middle 1/3*** is mixed fibers of smooth and skeletal muscles, ***lower 1/3*** – smooth muscle.

It has also 3 constrictions where adjacent structures produce impressions on it:

- ***Cervical constriction***: at *Pharyngo-esophageal* junction (C5 level), caused by the *cricoid* cartilage (*Cricoid angustia*). At this point, the *Cricopharyngeous* part of the lower constrictor of pharynx muscle forms the **Upper Esophageal Sphincter (UES)**.
- ***Thoracic (Aortic) constriction***: is a compound constriction, 1<sup>st</sup> crossed by the arch of aorta then by the left main bronchus (in some sources, *the Broncho-aortic angustia*)
- ***Diaphragmatic constriction***: at the esophageal hiatus in the diaphragm, *Th-10* level, formed by the *Right Crus* of diaphragm (although the esophagus is on the left side!) and the ***phrenico-esophageal ligament*** attach the esophagus to diaphragm at hiatus. The **Lower Esophageal Sphincter (LES)** is formed here by the right crus and the ligament, and this sphincter is very important for preventing the acidic contents of the stomach to be regurgitated back into the esophagus (*Gastro-esophageal reflux*)

**\*\* *Laimer trigone***: a triangular (or V-shaped) area on the posterior aspect of the proximal esophagus, with its apex directed inferiorly in the midline and the cricopharyngeous muscle forming its base, which is an area of weakness due to the *absence of outer longitudinal muscle layer* of the esophagus. It's a potential site of herniation of pharyngeal or esophageal mucosa known as ***Zenker Diverticulum*** (Pharyngo-esophageal diverticulum)

Following the curve of vertebral column as it descends through neck and posterior mediastinum in a slight arch towards the right but reaching the left side at the diaphragm, contacting the *Left atrium* of the heart posteriorly in the middle of the thorax.

After it pierces the diaphragm through the esophageal hiatus, it has the ***abdominal part*** which is 1.5cm long (that has a groove on the posterior surface of the liver). It terminates entering stomach at the *Cardial* orifice (Cardia) (Esophago-gastric junction), *Left* To midline, at the level of the ***7<sup>th</sup> costal cartilage or Th-11*** vertebral body

**\*\*** weakness in the connective tissue of the esophageal hiatus of the diaphragm can result in ***Hiatal hernia***, in which the abdominal part of the esophagus as well as parts of the stomach protrude into the thoracic cavity

→ ***Z-line***: where the mucosa abruptly changes from Esophageal (Stratified non-keratinized squamous) to Gastric (Simple columnar, glandular) at the cardia

→ The ***Arterial supply*** of the esophagus:

- Esophageal branch of Inferior thyroidal artery (cervical part)
- Esophageal branch of Left & Right bronchial arteries (thoracic part)

- Esophageal branches of Thoracic aorta (segmental) (thoracic part)
- L. Gastric art. (From Celiac trunk) (Abdominal part)
- L. Inf. Phrenic art. (Abdominal part)

→ **Venous drainage** is from the sub-mucosal veins as part of the Porto-Caval anastomosis  
(See the liver for more details):

- into the Portal system through L. Gastric v. of stomach (coronary vein)
- into the Systemic venous system (SVC by the Azygos and hemiazygos veins) through the Esophageal veins

→ **Lymphatic drainage:**

- L. gastric lymph nodes into celiac lymph nodes.
- Inferior Deep cervical lymph nodes
- Para-tracheal lymph nodes
- Posterior mediastinal & Intercostal lymph nodes

→ **Innervation:**

It has an esophageal nerve plexus, formed by the 2 Vagal nerves (running parallel) as the Parasympathetic portion + Thoracic sympathetic trunks via the greater Splanchnic nerve and the peri-arterial plexus around L. gastric and inf. Phrenic arteries.

The parasympathetic nerve supply comes from the nucleus ambiguus and dorsal motor nucleus of the vagus nerve and provides motor innervation to the esophageal muscular coat and secretomotor innervation to the glands. The sympathetic nerve supply comes from the cervical and the thoracic sympathetic chain (spinal segments T1–T10) and regulates blood vessel constriction, esophageal sphincters contractions, relaxation of the muscular wall, and increases in glandular and peristaltic activity.

The thin nerve fibers and numerous ganglia of the intramural myenteric and the submucosal plexi provide the intrinsic innervation of the esophagus. The ganglia that lie between the longitudinal and the circular layers of the tunica muscularis form the myenteric or Auerbach's plexus, whereas those that lie in the submucosa form the submucous or Meissner's plexus. Auerbach's plexus regulates contraction of the outer muscle layers, whereas Meissner's plexus regulates secretion and the peristaltic contractions of the muscularis mucosae.

**OESOPHAGUS**

27cm long (45cm from teeth to cardia of stomach)  
 Nerves: Sensation and motor via vagus nerves  
 Lining: • Stratified squamous (non-keratinising) becoming columnar at stomach  
 • Thick muscularis mucosae ++  
 • Mucous glands in mucosa and submucosa  
 Endoscopic narrowings: From mouth at 17cm, 28cm, 45cm

**Slight compression from:**

- Cricoid cartilage
- Aorta
- Left bronchus
- Diaphragmatic hiatus



**Relations**

**Posterior:** vertebrae, thoracic duct crosses to left at T5, hemiazygos/ accessory hemiazygos cross to right at T8/9, descending aorta, first 2 intercostal arteries from aorta

**Anterior:** trachea to T4/5, recurrent laryngeal nerves, left bronchus, left atrium, diaphragm

**Left:** thoracic duct, aorta, left subclavian artery, lung

**Right:** lung, azygos vein (hence good side to approach the oesophagus)

|        | 1/3rds | MUSCLE           | ARTERY           | VEIN             | LYMPH         | LENGTH |
|--------|--------|------------------|------------------|------------------|---------------|--------|
| Upper  |        | Striated         | Inferior thyroid | Inferior thyroid | Deep cervical | 9cm    |
| Middle |        | Striated/ smooth | Aortic branches  | Azygos branches  | Mediastinal   | 9cm    |
| Lower  |        | Smooth           | Left gastric     | Left gastric     | Gastric       | 9cm    |

## The Stomach

A **J-shaped** (Usually) **hollow organ**, but its shape varies and depends on the body eating habitus or position. (Can be Elongated or Transverse shaped also)

In supine position, it rests on the so called **Stomach bed**, made of the following organs:

→ **Pancreas, spleen, L. kidney, L. suprarenal gland, transverse colon and Mesocolon**

Totally **intra-peritoneal organ**, located in **L. Hypochondrium and Epigastric** regions of the Abdomen, and as a J-shaped organ it has 2 margins or **Curvatures**:

- **Lesser curvature:** forms the shorter **concave** border of stomach, the **Angular Incisure** (notch) is the sharper indentation at the 2/3 distance along the lesser curvature, that indicates the junction of the body with pylorus. The **Hepatogastric ligament** is a very thin duplication of the peritoneum (Part of the Lesser omentum) reaches the lesser curvature.
- **Greater curvature:** forms the longer **convex** border of stomach, less fixed part, or not at all. **2 double Layers** of the peritoneum (**4 total layers**) meet each other at **greater curvature** forming together the **Greater Omentum** (and the **Gastrocolic ligament** fused together with **2 layers** of the **Transverse Mesocolon**, so total 6 layers).

### → Parts of the stomach:

- **Cardia:** is the funnel shaped part surrounding cardiac orifice, left side of **Th-11** vertebra, related to ribs (7<sup>th</sup>, 2cm left of midline). It's the **most fixed** point of stomach, attached to the diaphragm by the **Gastrophrenic ligament**.
- **Fundus:** is the dilated superior part related to left dome of diaphragm, it can reach the level of the **5<sup>th</sup> IC space**. **The Cardial notch** is a deep incisure on the greater curvature between the **Esophagus and the Fundus**, forming an acute **angle of His**, which functions as an additional valve besides the LES that **prevents reflux of gastric contents back into the esophagus**. It can be dilated by different substances (gas, H<sub>2</sub>O, food) forming the **Stomach Bubble**. Located in **Hypochondrium** and related also to spleen.
- **Body: (corpus)** major part of the stomach that has the **lesser and greater curvatures**.
- **Pyloric part:** is a funnel shaped **outflow** region of the stomach divided into 2 parts:
  - **Pyloric Antrum:** wide part, leads into the
  - **Pyloric Canal:** narrow part, that has the **pylorus** at its end

→ **The Pylorus:** is a distal **sphincteric** region of the pyloric part of stomach, which is marked by the **thickening of the circular layer of smooth muscle**, named the **pyloric sphincter muscle**: controls the **discharge of stomach's contents through pyloric orifice into duodenum**.

\*\* It is the **Second** most fixed point of stomach, Right side of **L-1** vertebra, **1.5cm** from Midline (**Trans-pyloric line**)

### → The Interior of stomach:

**Reddish brown gastric mucosa**, **pink pylorus**,

When contracted, gastric mucosa is thrown into longitudinal ridges called **Gastric folds or Gastric Rugae**, and when stomach is dilated folds are decreased



**A Gastric canal (furrow)** (food canal) (**Magan Strasse**) forms temporarily during swallowing between the longitudinal gastric folds of mucosa along lesser curvature to direct the food into pylorus. The folds become irregular at the fundic region of stomach

**The muscular layer** of the stomach consists of bundles of smooth muscle and divided into:

- *Inner circular layer*
- *Outer longitudinal layer*
- *Innermost 3<sup>rd</sup> Oblique layer (from cardia to the angular notch only)*

→ **Ligaments of the stomach (Duplications of peritoneum):**

- **Hepatogastric ligament:** from the liver (fissure of ligamentum Venosum) into the lesser curvature, part of the Lesser Omentum. Contains the L. and R. gastric art. /v.
- **Gastrocolic ligament:** from the *greater omentum* (hanging from the greater curvature) to the *transverse Mesocolon*, fusing with it forming the 6 layers Gastrocolic ligament. Contains the anastomosis between the L. and R. Gastroepiploic arteries
- **Gastrosplenic ligament:** the continuation of the *Gastrocolic ligament* on the left side. Passes to the spleen and continues further to the left as the *Phrenico-splenic ligament*. Contains the short gastric arteries and the beginning of L. Gastroepiploic art.

→ **Arterial Supply of stomach:**

Rich arterial supply arising from celiac trunk and its branches:

- **Left Gastric art.** (Branch of *celiac trunk*): *ascends* to esophageal hiatus in a semi-intraperitoneal position first giving cardiac branches, forming a fold of peritoneum called *Gastropancreatic fold*, and then *descends* along lesser curvature to anastomose with *R. gastric art.*
- **Right gastric art.** (Branch of *proper hepatic art.* or *common hepatic art.*): runs along the lesser curvature of stomach to anastomose with *L. gastric art.*
- **Posterior gastric art.** (Branch of *Splenic art.*): behind the stomach, ascends along the posterior wall of the *Omental bursa* to enter the *Gastrophrenic ligament*. (for the posterior wall of stomach + cardia)
- **Short gastric arteries** (4-5 branches of *splenic art.* at the hilum of spleen): passes between the layers of **Gastrosplenic ligament** to the fundus of stomach
- **Left Gastro-omental (Epiploic) art.** (Branch of *splenic art.* at the hilum of spleen): passes into the **greater omentum (or the Gastrocolic lig.)** to the stomach, then along the greater curvature to anastomose with *R. gastro-omental (Epiploic) art.*
- **Right Gastro-omental (Epiploic) art.** (Branch of *Gastro-duodenal art.*): passes into the greater omentum, behind duodenum along the greater curvature of the stomach to anastomose with the *Left Gastro-omental (Epiploic) art.*

→ **Venous Drainage of stomach:**

- **Left Gastric vein: (Coronary v. of stomach):** made by **esophageal branches** and veins from surface of cardia of stomach. Runs in the lesser omentum and drains into **portal vein**. (participates in the porto-caval anastomosis)
- **R. gastric vein:** receives veins from both surfaces of upper portion of stomach, runs along the lesser curvature and drains into **portal vein**.
- **Short gastric veins:** drain Fundus + L. portion of stomach → drain into **splenic vein**.



- **L. Gastroepiploic vein:** runs with the artery → empties into **splenic v.**
- **R. Gastroepiploic vein:** runs with the artery → empties into **Sup. Mesenteric v.**
- **Prepyloric vein:** ascends over pylorus to the R. gastric v, used to identify the **pylorus** by surgeons.

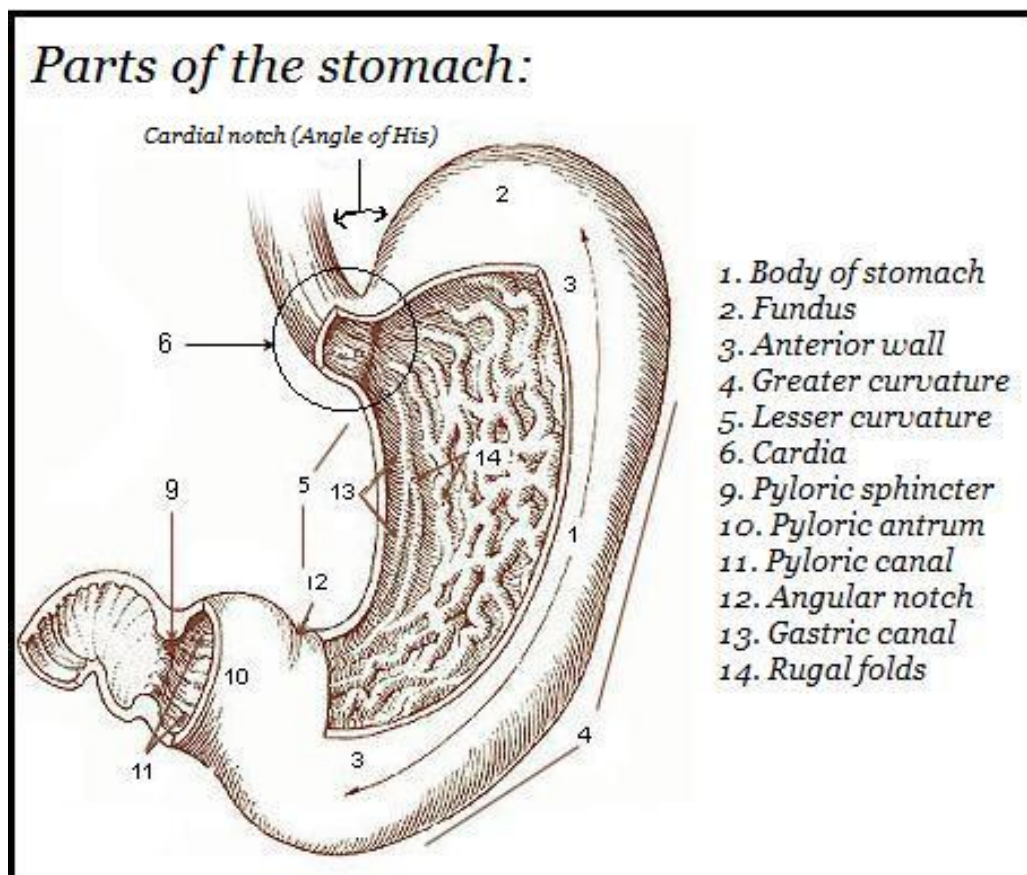
→ **Lymphatic drainage of stomach:**

- **Lymph from sup 1/3 of stomach:** along the R. and L. gastric vessels into → Gastric lymph nodes which goes to → **Cisterna Chyli**
- **Lymph from fundus & superior body:** along short gastric art. And Left gastroepiploic artery → **Pancreatico-splenic lymph nodes**
- **Lymph from R. 2/3 of inf. 1/3 of stomach:** **Pyloric & gastrointestinal lymph** nodes (connected to hepatic lymph nodes, can cause metastasis to liver in case of gastric cancer)
- **Lymph from L. 1/3 of greater curvature** → **Pancreatico-duodenal lymph nodes.**

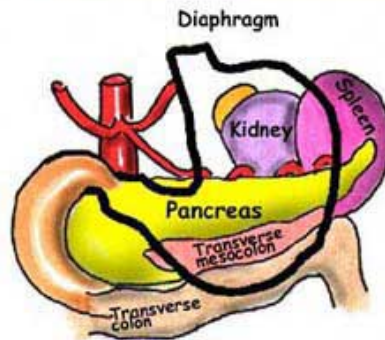
**\*\*Note:** Lymph from the stomach can also drain into the **mediastinal lymph nodes** and then to the **Left Supraclavicular Lymph node of Virchow** (Signal lymph node), this is very important clinically, since it can be a sign of metastatic gastric cancer. Another sign of gastric cancer can be the **Sister Mary Joseph's nodule**, which is a palpable lymph node bulging into the umbilicus.

→ **Parasympathetic** innervation: from **Anterior (L.) / Posterior (R.) Vagal trunks + celiac plexus**

→ **Sympathetic** innervation: **Th6 – Th9** → **celiac plexus through greater splanchnic nerves**



## STOMACH - RELATIONS



### ANTERIOR

Abdominal wall  
Left costal margin  
Diaphragm  
Left lobe of liver

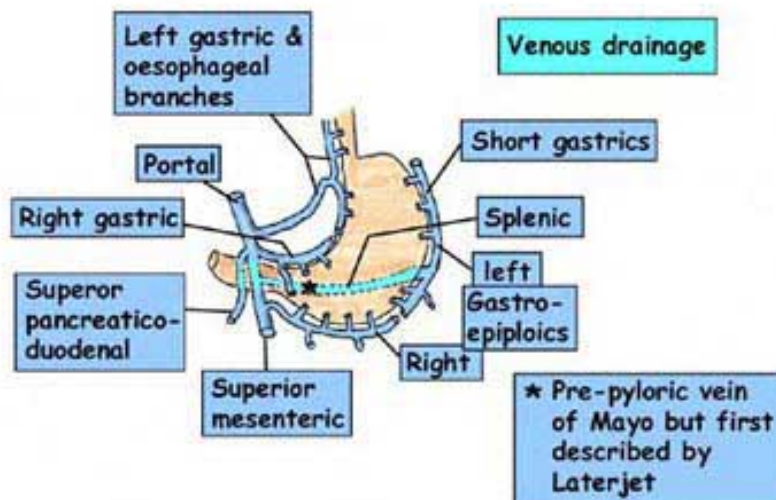
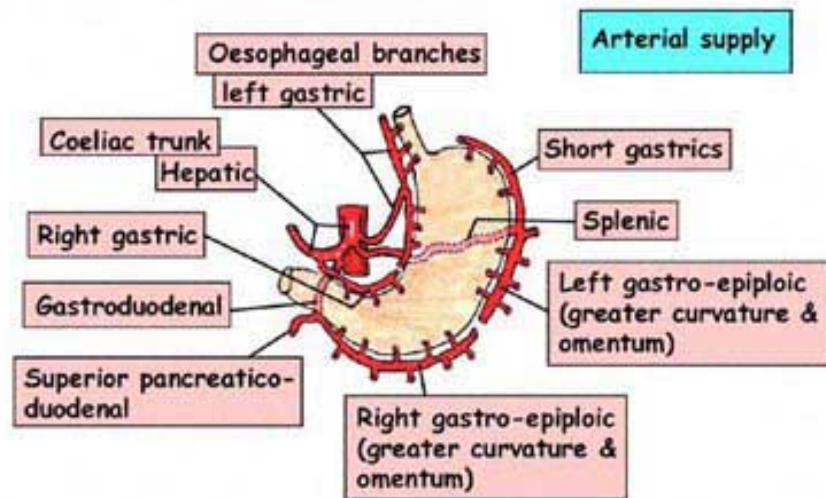
### SUPERIOR

Left dome of diaphragm

### POSTERIOR

Lesser sac  
Pancreas  
Transverse mesocolon  
Transverse colon  
Left kidney/suprarenal gland  
Spleen/splenic artery

## STOMACH - BLOOD SUPPLY & VENOUS DRAINAGE



## The Small intestine

About 5 meters long, Consists of **Duodenum, Jejunum and Ileum**

The primary site for absorption of nutrition from ingested materials, extends from the *Pylorus* to the **Ileocecal** junction, where the ileum joins the cecum

### A) Duodenum:

C- Shaped hollow organ surrounding the head of the pancreas, **widest but shortest** part of the small intestines (Duodenum means 12 fingers ~25cm). Starts at the Pylorus and ends at the **Duodenojejunal flexure**

→ **The duodenojejunal junction:** occurs at the level of L2, 2-3 cm left of midline, makes an acute angle, the *duodenojejunal flexure*. It's a fixed point by the **Suspensory ligament of Treitz**, which is attached to the **crus of the diaphragm** and has some skeletal muscle fibers in it (in some sources, it says it's attached to the celiac trunk of the abdominal aorta).

**\*\* The superior & inferior duodenal folds cover the superior & inferior duodenal recesses respectively**

**\*\* The paraduodenal fold** a sickle-shaped fold of peritoneum sometimes found arching between the left side of the duodenojejunal flexure and the **medial border of the left kidney**; its right free edge contains the ascending branch of the left colic artery and inferior mesenteric vein; forms anterior boundary of the **paraduodenal recess**.

They are typical places of *internal herniation (Treitz Hernias)*

The duodenum consists of 4 parts:

#### 1) **Superior Horizontal 1<sup>st</sup> part:**

Short (~5cm), lies anterolateral to the **body of L1 vertebra**, continuous at the **superior duodenal flexure** with the 2<sup>nd</sup> descending part of the duodenum. Has a dilated portion called duodenal **Ampulla** in which the pylorus connects (Called **duodenal cap** or **Bulb on radiographs**) → it's a mobile part and attaches the **Hepatoduodenal ligament** (part of the lesser omentum). This part of the duodenum is **Intraperitoneal**

*Relations:*

- Anterior: Peritoneum, gallbladder, caudate lobe of the liver
- Posterior: Common bile duct, gastroduodenal artery, IVC, portal vein
- Inferior: Neck of pancreas
- Superior: Neck of gallbladder, Epiploic foramen

#### 2) **Descending 2<sup>nd</sup> part:**

7-10 cm descends along the **R. side of L1-L3 vertebrae** until the **inferior duodenal flexure**, parallel to IVC, contains the **junction the foregut with midgut**.

The common bile duct and the main pancreatic duct unite to form **Hepatopancreatic ampulla**, which is a dilation that opens into the posteromedial wall of this part of duodenum as **Vater's papilla (major duodenal papilla)**, guarded by the muscular sphincter of **Oddi**, located 10 cm from the pylorus.

Another **minor duodenal papilla** (of **Schneider or Santorini**) is located 2-3 cm proximal to Vater's; it's the opening of the **accessory pancreatic duct of Santorini**

This part of the duodenum is totally **Retroperitoneal (Secondary)**

*Relations:*

- Anterior: Transverse colon, Mesocolon, coils of small intestine
- Posterior: Hilum of R. kidney, renal vessels, Ureter
- Medial: head of pancreas

**3) Inferior Horizontal 3<sup>rd</sup> part:**

6-8 cm, crosses L3 vertebra to the left side transversally, passing over IVC, aorta and L3 vertebral body, *crossed anteriorly by the SMA/SMV* and the root of mesentery. Together with the *left renal vein & uncinat process*, this part of the duodenum is located between the abdominal aorta & SMA.

(In extreme cases, if we eat too much, the SMA & Aorta can compress the filled 3<sup>rd</sup> part of the duodenum and passage of the food will stop causing **Ileus**).

This part of duodenum is **Retroperitoneal (secondary)**.

*Relations:*

- Anterior: SMA, SMV, coils of small intestines
- Posterior: R. Psoas major, IVC, Aorta, R. Ureter
- Superior: Head of pancreas + Uncinate process + SMA/ SMV

**4) Ascending 4<sup>th</sup> part:**

~5cm, begins at the L. side *L3 level and ascends superiorly until L2*, reaching the inferior border of the body of the pancreas, then curves again to join the jejunum, becoming again Intraperitoneal, forming the *Duodenojejunal flexure (junction)*, this part of duodenum is **Retroperitoneal (secondary)**

*Relations:*

- Anterior: Beginning of root of mesentery, coils of jejunum
- Posterior: L. Psoas major
- Medially: head of pancreas
- Superior: body of pancreas

**→ Arterial supply of the duodenum:**

*Supplied by the celiac trunk and Superior mesenteric artery*

- **Superior Pancreaticoduodenal artery** (branch of *Gastroduodenal*): divides into Anterior & Posterior arteries that descend around the pancreatic head, anastomosing with the branches of the Inferior PD artery of SMA. *Supplies the Proximal duodenum and the Superior part of head pancreas*

→ The duodenum develops from *both the foregut & midgut*, which are supplied by the celiac trunk and SMA respectively

The *anastomosis between the Superior & Inferior Pancreaticoduodenal arteries* around the head of pancreases and duodenum is the so called **First Riolan arch (or the Minor/Lesser Riolan anastomosis)**, which marks the border between foregut and midgut embryologically

**\*\* See surgical significance of the Pancreaticoduodenal arteries in the description of pancreas**

- **Gastroduodenal artery** (branch of the *common hepatic artery*): descends posteriorly to the gastroduodenal junction (pylorus) and supplies the stomach, pancreas, 1<sup>st</sup> part of duodenum and distal part of the bile duct. Gives also the *Supraduodenal artery*

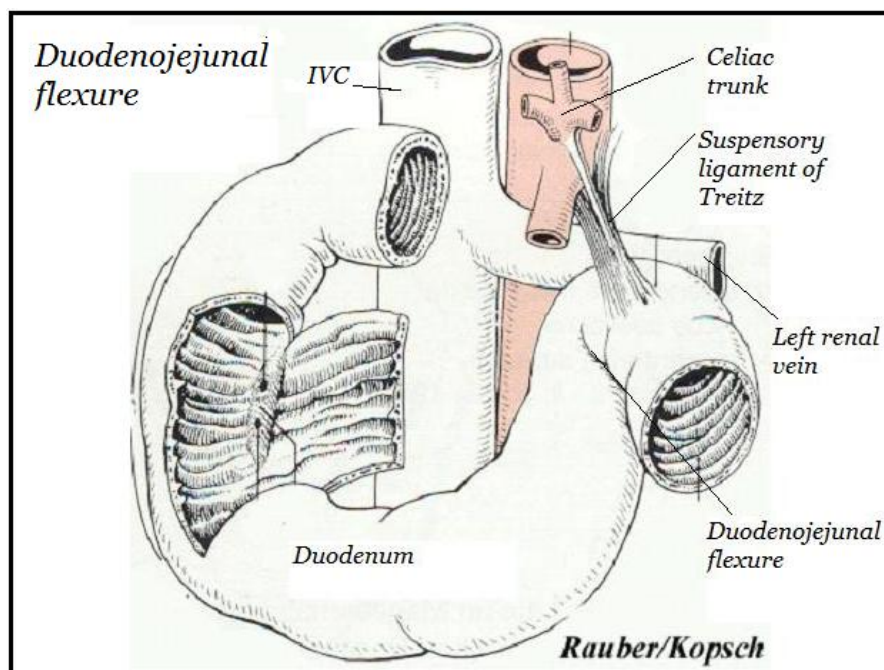
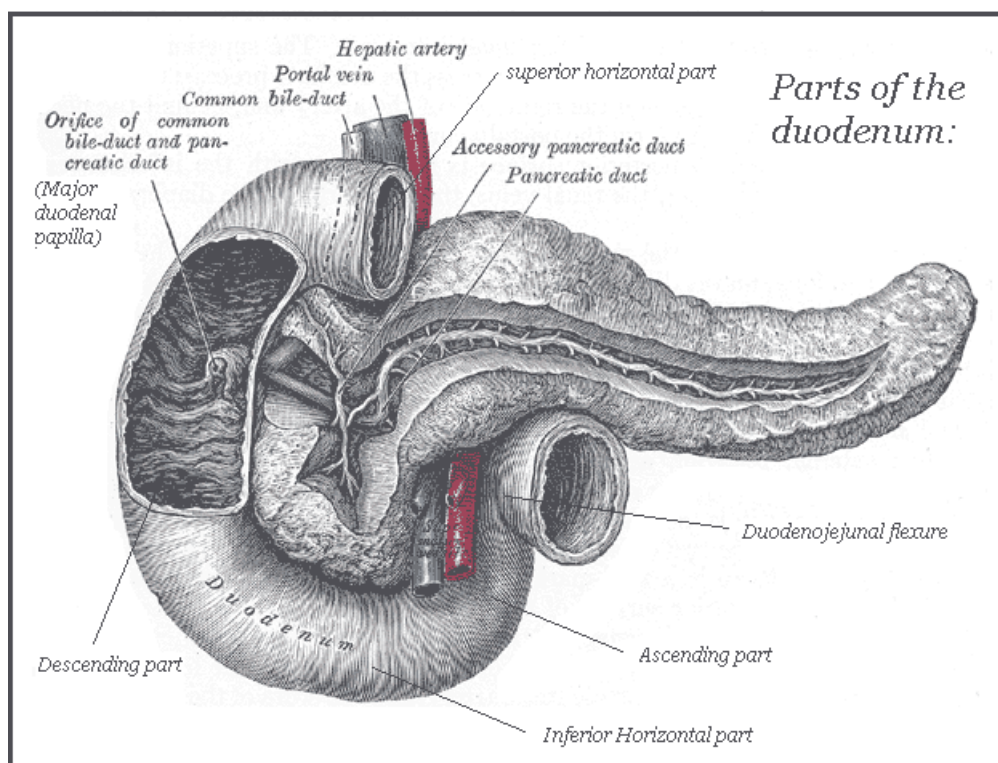


- **Inferior Pancreaticoduodenal artery** (branch of the superior mesenteric artery): divides into Anterior & Posterior arteries that ascend on each side of the pancreatic head, anastomosing with the similar branches from the superior PD artery. *Supplies the Distal parts of duodenum and the inferior part of head of pancreas*

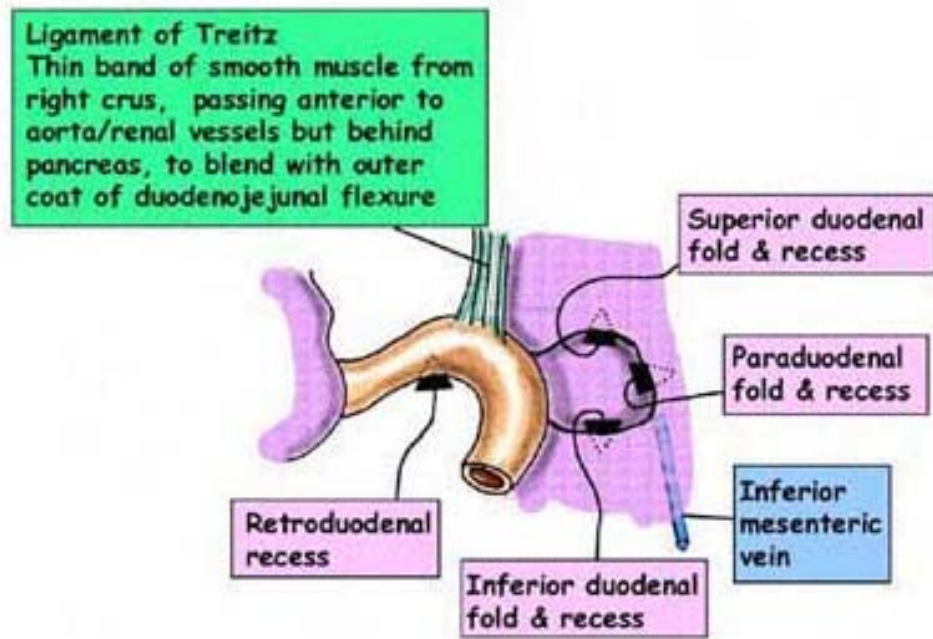
→ **Venous drainage of the duodenum:** by the **Pancreaticoduodenal veins** → into the SMV

→ **Lymphatics:** - Anterior lymph vessels → into Pancreaticoduodenal and pyloric lymph nodes  
 - Posterior lymph vessels → into the SMLN (superior mesenteric lymph nodes)

→ **Innervation:** - **Parasympathetic:** Vagus  
 - **Sympathetic:** Lesser/Greater Splanchnic nerves







### B) Jejunum & Ileum:

→ **The jejunum** is the second part of the small intestine that begins at the *Duodenojejunal flexure* located *intraperitoneally*. Makes about 2/5 of the length of small intestines. Its wider, emptier and thicker walled than the ileum, most of it lays in the *upper left quadrant* of the abdominal cavity. Grossly with the naked eye, the *plica circularis* can be seen (circular folds of Kerckring) in the lumen. Microscopically it contains **no Peyer patches**! It has more translucent windows between the vessels in its mesentery and *less fat, less prominent arterial arcades compared with the ileum, but has longer Vasa recta*.

→ **The ileum** is longer than the jejunum, 3/5 of the small intestines, and occupies the *false pelvis in the Right lower quadrant of the abdomen*, it is darker, has *more vertical loops* than the jejunum. Histologically, it **contains Peyer patches** (on the anti-mesenterial side, bulging into the lumen in some cases), *shorter plica circularis, shorter vasa recta, but more mesenteric fat and more prominent arcades compared with jejunum*

\* **Meckel's diverticulum**: the remains of the yolk stalk of the embryo, which, when persisting abnormally as a blind sac or pouch in the adult, is located *60-90 cm orally* from the Ileocecal junction; it may be free in the abdominal cavity or attached to the umbilicus. Inside we can find Ileal tissue, with frequent islets of *gastric or pancreatic* tissues!

### → The Root of mesentery:

~15 cm directed obliquely, inferiorly and to the right extends from *Duodenojejunal flexure* on the *left side of L2* to the *Ileocecal junction in the right iliac fossa, at the level of right sacroiliac joint*

Crosses part 2 & 3 of duodenum, abdominal aorta, IVC, R. Ureter, R. Psoas major, R. Testicular artery/ ovarian artery and contains the superior mesenteric vessels, lymphatics and nerves

→ **Arterial supply of Jejunum & Ileum:**

**Superior mesenteric artery**, which arises at the level of L1, 1cm under the celiac trunk, and goes between the head of pancreas, uncinate process and the inferior horizontal part of duodenum into the root of mesentery to give 15-18 Jejunal & Ileal (4-5 Jejunal, 11-13 ileal) branches to the intestines

Those branches anastomose together forming the *Intestinal arterial arcades*, which become more prominent towards the terminal ileum. From the arcades, the straight branches or *Arteria recta* or *Vasa recta* reach the intestines as terminal branches (not to be confused with the vasa recta in the kidney)

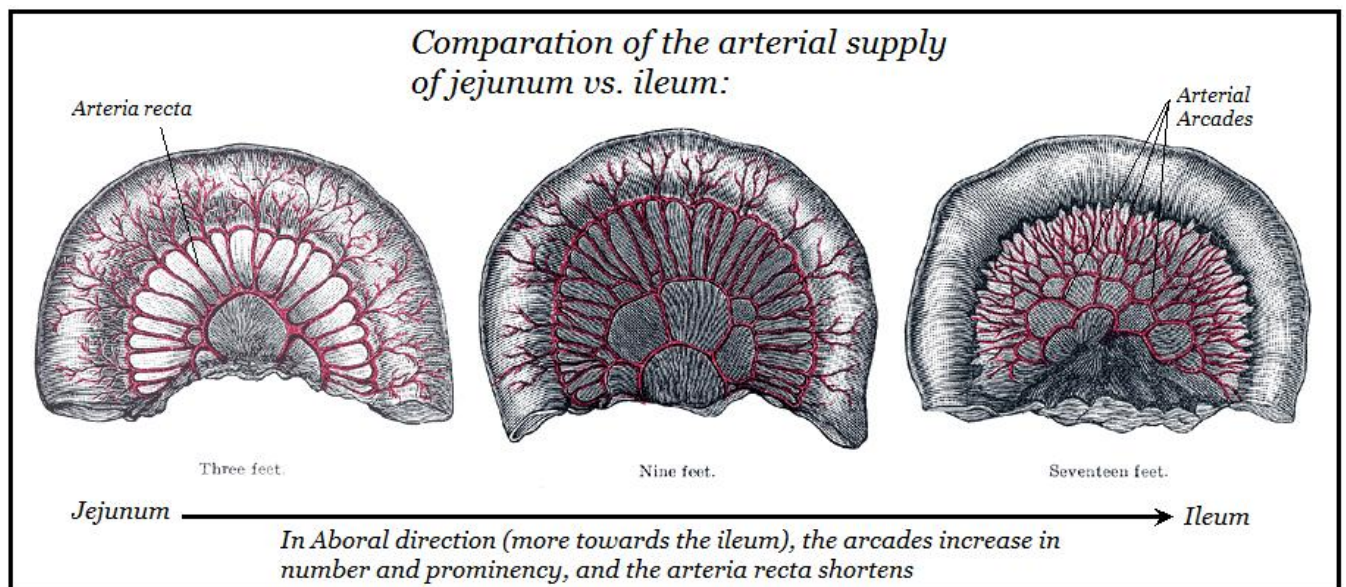
→ **Venous drainage** is by the **superior mesenteric vein**, which ends behind the neck of pancreas to join the splenic vein to form the portal vein

→ **Lymphatics:**

- *Juxta-intestinal lymph* nodes: close to the intestinal wall
- *Mesenteric lymph* nodes: among the arterial arcades
- *Superior central* nodes: along the proximal part of the SMA
- they drain into the celiac lymph nodes then into Cisterna Chyli and thoracic duct

→ **Innervation:**

- **Sympathetic:** Th8-Th10 → through the sympathetic trunk and lesser/greater Splanchnic nerves, postganglionic fibers are from the *superior mesenteric ganglia*
- **Parasympathetic:** Vagal trunks (Myenteric & Submucosal plexuses)



## ***The Large Intestine***

The site of reabsorption of undigested food, water and electrolytes and production of feces  
1.5 – 1.8 meter long, composed of:

- Cecum
- Appendix
- Colon (3 parts: Ascending, Transverse, Descending)
- Sigmoid
- Rectum
- Anal canal

The *cecum colon and sigma* can be distinguished from the small intestines by unique features:

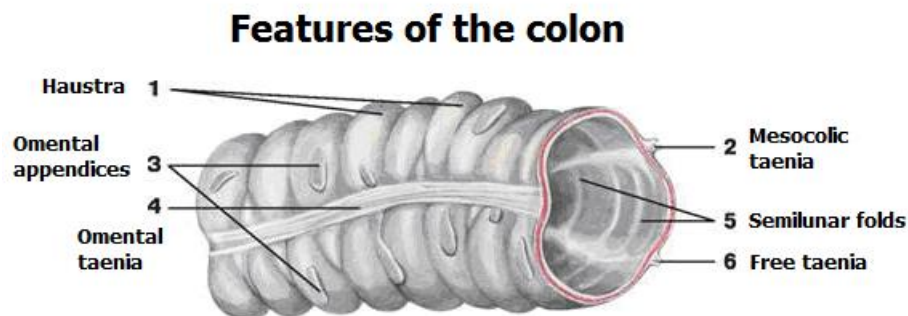
- a) ***Omental appendices***: small, sub-peritoneal fatty appendages
- b) ***Teniae coli***: three bands in which the longitudinal muscular fibers of the large intestine, except the rectum & anal canal, are collected
  - a. *Mesocolic*: to which transverse and sigmoidal mesocolons attach (mesentery)
  - b. *Omental*: site of adhesion of the greater omentum to the transverse colon
  - c. *Libera (free)*: opposite to the Mesocolic teniae, no attachment

This nomenclature is used for the teniae of the *transverse colon*, in the case of the ***ascending & descending colons***, the teniae change position due to the right and left colic flexure:

- Mesocolic tenia *becomes the Posterio-medial tenia*
- Free tenia *becomes the Anterior tenia*
- Omental tenia *becomes the Posterolateral tenia*

\*\* The omental tenia (the posterolateral tenia) is thin and almost not visible on the ascending & descending colons, because there is no greater omentum adhered there. On the sigmoid colon it totally disappears, therefore we can say that the sigmoid colon has 2 teniae only.

- c) ***Haustra***: sacculations of the wall of the colon between the teniae, in the interior aspect of the wall, between the sacculations, the *semilunar folds* of the colon are produced.
- d) ***Greater caliber***: not totally correct and reliable, because if the small intestines are filled with food or blocked due to some sort of Ileus, they can have almost the same caliber as the large colon



## ***Cecum & Appendix:***

Cecum is the 1<sup>st</sup> part of the Large intestine. It's a blind intestinal pouch, 7.5cm in length and width, located in the Right lower quadrant, in the iliac fossa. It has no mesentery and considered as a *Semi-intraperitoneal* organ that is bound to the lateral abdominal walls by cecal folds.

### ***- Ileocecal junction:***

Terminal ileum enters the cecum and partly invaginate in it, producing Ileocolic lips (superior & Inferior) at the ileac orifice, which forms the *Ileac papilla*

The folds meet laterally to form Frenula (ridges) (only seen in cadavers) & ***Ileocecal valve of Bauhin***

→ Around the *Ileocecal junction* we have to distinguish 2 folds:

- ***Vascular fold*** of the cecum: a peritoneal fold that arches over a branch of the Ileocolic artery and bounds in front a narrow recess, the superior Ileocecal (or Ileocolic) recess
- ***Bloodless peritoneal fold of Treves or the Ileocecal fold***: bounding the inferior *Ileocecal* or *ileoappendicular fossa*

\*\* The ***Retrocecal recess*** is found behind the right side of cecum

The ***Vermiform Appendix*** is a blind intestinal diverticulum (5-9cm) that contains masses of lymphoid tissue. It has a variable position, but usually arises from *posteriomedial* aspect of the cecum inferiorly to the Ileocecal junction (***Retrocecal position***)

It has a short triangular mesentery called the *Mesoappendix* which is attached to the cecum and proximal part of the appendix

Clinically the appendix can be found by using the ***McBurney point***, which a point located onto the lateral 1/3 of the line connecting the anterior superior iliac spine with the umbilicus (***Monro-Richter line***)

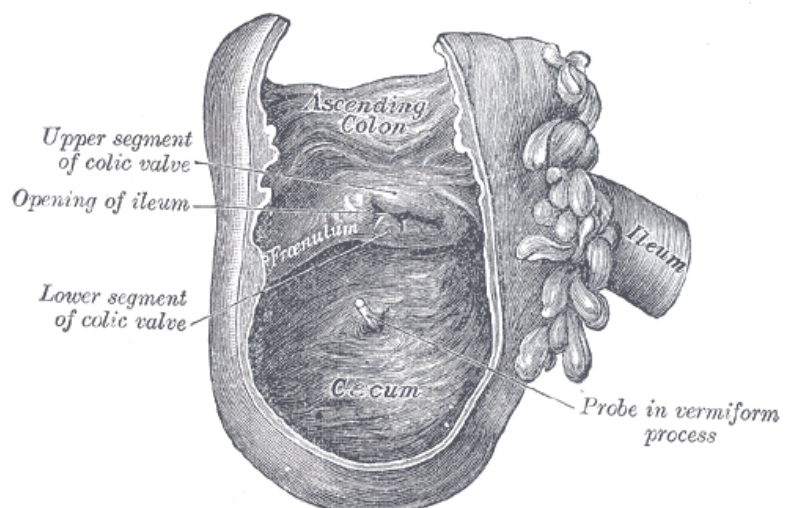
### ***→ The arterial supply of the cecum and appendix:***

By the ***Ileocolic artery***, a terminal branch of the Superior mesenteric artery that has 7 branches:

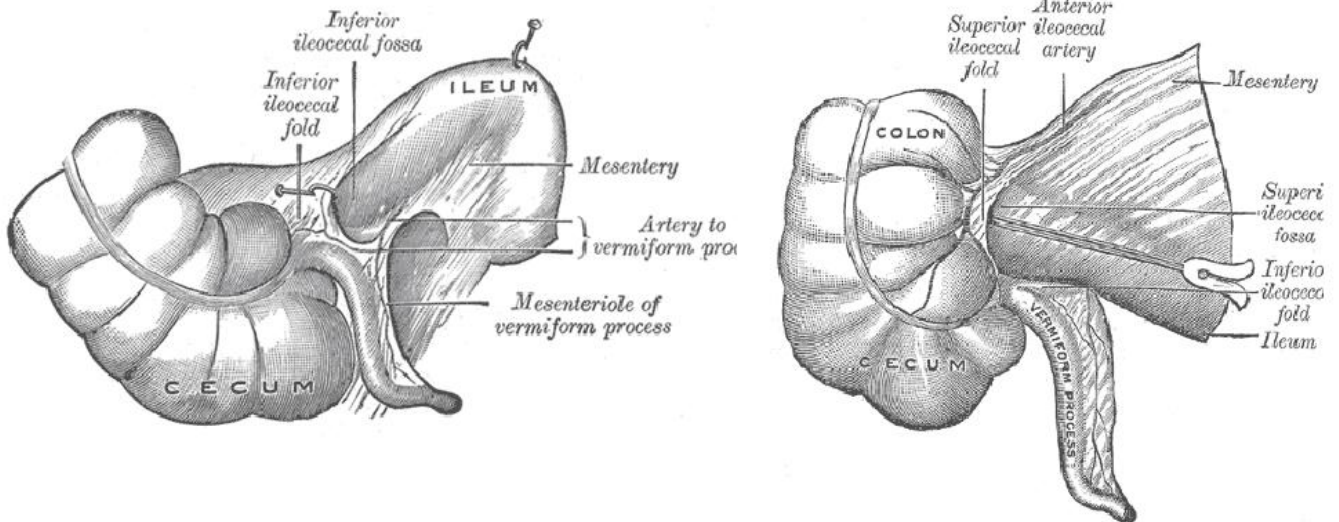
- 3 branches to the terminal ileum
- 2 branches to the cecum (anterior & posterior cecal)
- 1 branches to the appendix (appendicular artery) that descends in the mesoappendix
- 1 branch to the ascending colon (Ascending Colic)

→ ***Venous drainage*** is done by the ***Ileocecal vein*** that drains into SMV

→ ***Ileocolic lymph nodes***: into the SMLN



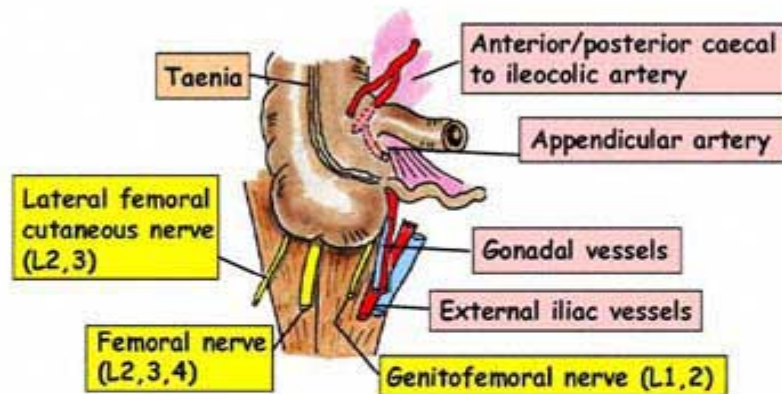




## CAECUM AND APPENDIX

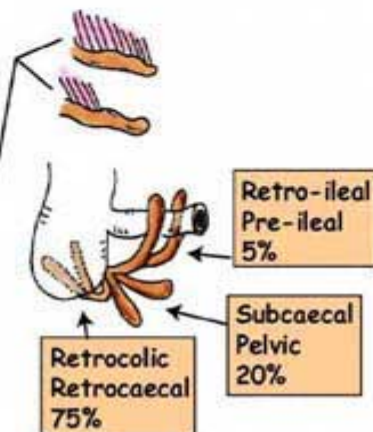
### CAECUM

- On mesentery
- Below ileocaecal valve
- Retrocaecal fossa behind it
- 3 taenia meet at base of appendix in child
- Ileocaecal valve is a double fold of mucosa & circular muscle of ileum which acts as an anti-reflux mechanism



### APPENDIX

- At McBurney's point
- 1/2"-9" (2-25cm) average 7-8cm
- Fully coated diverticulum
- Variable mesentery
- Appendicular artery usually from posterior caecal artery. It is an end artery hence appendix can easily become gangrenous
- Appendix moves posterior and medial with caecal expansion



## The Colon

A hollow tube that is 1.5 meter part of the Large intestines that has 4 major parts:

### 1) *Ascending colon:*

2<sup>nd</sup> part of the large intestines, it passes superiorly on right side of abdominal cavity from cecum until right lobe of liver, where it turns to the left forming the **Right Colic (hepatic) flexure**.

Its partly (1/2) *Intraperitoneal* and (1/2) *retroperitoneal* (secondary) so it's considered as **Semi-intraperitoneal**

Between the lateral aspect of ascending colon and adjacent abdominal wall, lies deep vertical groove lined with parietal peritoneum → **Right Paracolic gutter**, this recess is continuous upward with the Subhepatic recess and Morrison's pouch, and downward with the right side of pelvic cavity (iliac fossa), so it can play a role in infection spread

It has the so called **Jackson's membrane**, a thin vascular membrane or veil like adhesion, covering the anterior surface of the ascending colon from the cecum to the right flexure; it may cause obstruction by kinking of the bowel.

→ **Arterial supply:** *Ileocolic (Ascending colic), Right Colic + middle colic arteries* of SMA (right colic artery anastomose with right branch of middle colic artery and ascending colic)

→ **Venous drainage:** *Ileocolic and right colic veins* → SMV

→ **Lymphatic drainage:** *Epicolic + Paracolic LNs + Right colic LNs* → SMLN

### 2) *Transverse colon:*

45 cm long, most mobile part of large intestine. Crosses the abdomen as an arch from *right colic (hepatic) flexure* to *left colic (splenic) flexure* where it bends inferiorly.

The Splenic flexure lies anteriorly to the inferior part of left kidney; connected to diaphragm by **Phrenicocolic ligament** (holds the spleen in place as the *Sustentaculum Lienis* forming the splenic nest)

The **mesentery** of **transverse colon** loops downward as the **transverse mesocolon**. The root of transverse mesocolon lies along inferior border of pancreas and continues with parietal peritoneum posteriorly. It forms the inferior border of the *omental bursa* and fuses with the greater omentum forming the 6 layered *Gastrocolic ligament*

Being freely movable → the transverse colon varies in position, but we can say it's around *L3 vertebral level*

→ **Arterial supply:** mainly *Middle colic artery* (right 2/3) (branch of SMA) and *Left colic artery* (left 1/3) (branch of IMA)

\*\* The *left colic artery* (of IMA) anastomosis with *middle colic artery* (of SMA) to form the **2<sup>nd</sup> Riolan anastomosis or (Major/Greater) arc**, around the left colic (splenic) flexure, which marks the border between midgut and hindgut.

→ **Venous drainage:** *middle colic vein* → SMV

→ **Lymphatic drainage:** *Middle colic lymph nodes* into SMLN

### 3) **Descending colon:**

*Thinner* than the ascending colon, partly intra, Semi-intraperitoneal, as it descends, passes anterior to lateral border of Left kidney, and has left Paracolic gutter, which is closed upward by the Phrenicocolic ligament.

→ **Arterial supply:** - *Left colic artery* (branch of IMA)

**\*\* The marginal artery of Drummond:** an artery formed by anastomosis between the right and left colic arteries; it passes downward from the left colic flexure to the aboral (away from the mouth) end of the pelvic colon. The 2<sup>nd</sup> Riolan arc is a specific part of this artery

→ **Venous drainage:** L. colic veins → IMV

→ **Lymphatic drainage:** L. Colic lymph nodes – IMLN

### 4) **Sigmoid colon:**

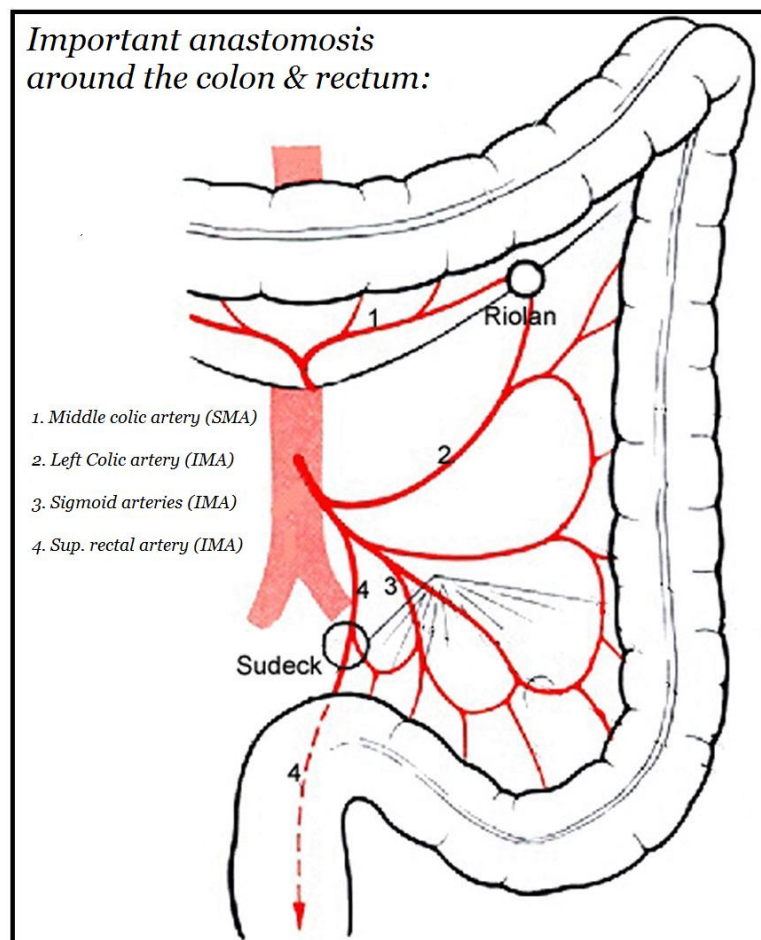
S-shaped loop (40 cm) extends from iliac fossa to S3 vertebra

The termination of *teniae* (15 cm from anus) indicates the Rectosigmoidal junction. It has a long **Mesosigmoidium** → freely moving. The root of Mesosigmoidium is an inverted-V shaped, from the Medio-superior to the anterior aspect of the sacrum.

**\*\* The Left Ureter + bifurcation of external iliac** → behind apex of the Mesosigmoidium

→ **Arterial supply:** *Sigmoidal artery* (IMA) with ascending/ descending branches

\* The **Sudeck** point (discussed in the Rectum) is an anastomosis between the sigmoidal artery and the Superior rectal artery.



→ *Venous drainage*: Sigmoidal vein into IMV

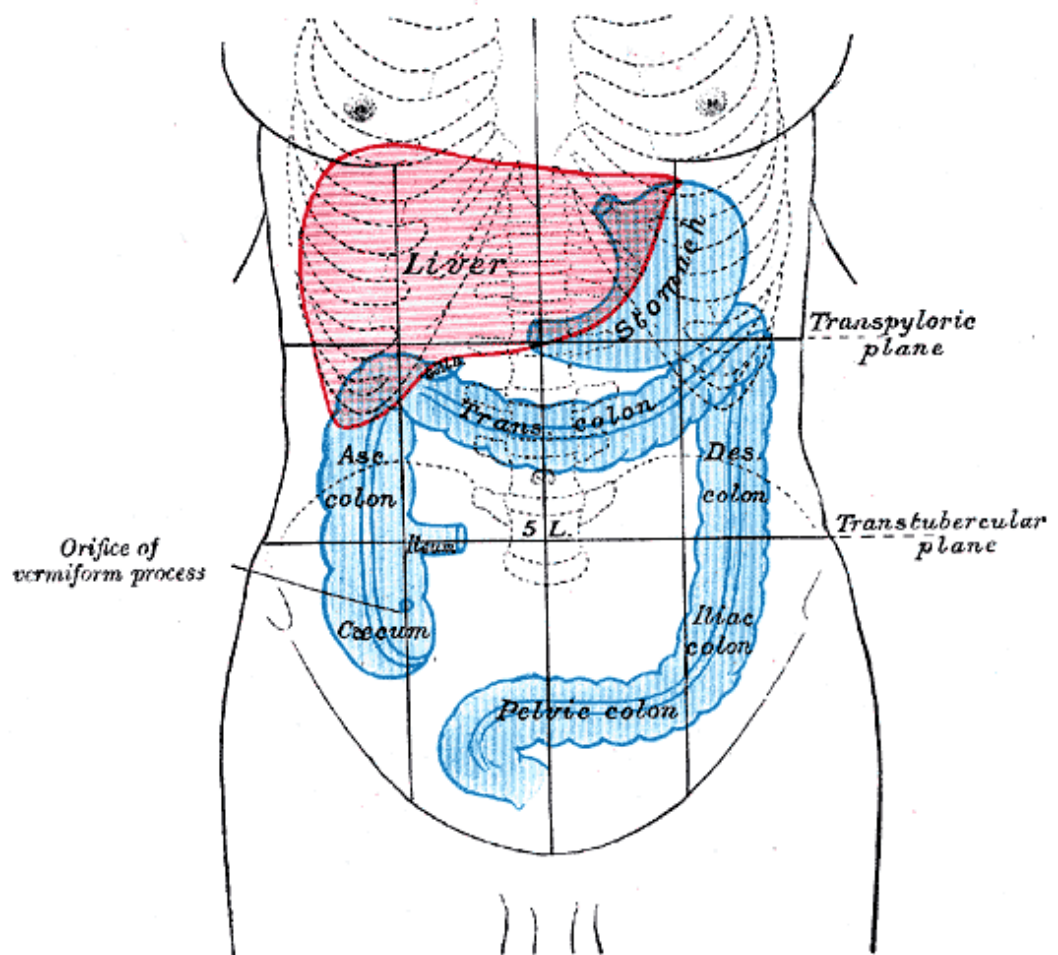
→ *Innervation of the Colon*:

- *Parasympathetic*:

- Vagus nerves (until the 2/3 of the transverse colon)
- Sacral Plexus (1/3 of the transverse colon until anus)
  - The point where the innervation changes from the Vagus to the sacral plexus, located between the right 2/3 and left 1/3 of transverse colon, is called **Cannon-Böhm point** (also marks the site of the 2<sup>nd</sup> Riolan arch)

- *Sympathetic*:

- Superior hypogastric plexus





## ***The Rectum***

The rectum is a 15-20 cm long S-shaped hollow organ located after the sigmoid colon in the pelvis. ***The Rectosigmoidal junction*** is located anteriorly to the ***S3*** vertebra.

At this point, the teniae of Sigmoid colon spread out to form outer *continuous smooth muscle layer* and the omental (*Epiploic*) appendices & the haustrations disappear.

The rectum follows the shape of sacrum and coccyx → ***Sacral flexure of rectum*** at first, then it turns backward to the anterior inferior tip of coccyx, forming a sharp turn → ***Perineal (Anorectal) flexure (80°)***, that occurs when the rectum perforates the pelvic diaphragm posterior-inferiorly, and its very important in the defecation mechanism.

From the outside, in the frontal plane, we can observe ***3 sharp lateral flexures*** of the rectum (*Superior, Intermediate and Inferior*) which are formed in relation to internal folding → ***The transverse rectal folds (Houston valves)***:

- 2 on left side (Superior and inferior)
- A prominent 1 on the right side (middle one), ***Kohlrausch's fold*** (5-6 cm from the anus)

→ ***The Ampulla of rectum***: dilated terminal part of rectum below the Kohlrausch's fold, lying directly superiorly to and supported by Pelvic diaphragm (Levator Ani muscle) and the Anococcygeal ligament. Receives and holds accumulated feces, and when it full, the sensation of defecation occurs

→ The ***Superior 1/3 of rectum semi-intraperitoneal, Middle 1/3 is Retroperitoneal (Due to its S shape) and the inferior 1/3 is Infraperitoneal***

- In ***males***, the peritoneum reflects on the *Urinary Bladder* forming → ***Rectovesical pouch*** and the ***Rectovesical septum*** that is found between fundus of UB and rectum. It is possible to feel the prostate by digital examination in front and below the Kohlrausch's fold
- In ***females***, the peritoneum reflects on posterior part of the ***vaginal fornix*** forming → the ***Rectouterine pouch of Douglas***. (also below the Kohlrausch's fold, used to reach the posterior vaginal fornix for gynecological purposes)
- In both sexes, peritoneum reflects around ***Pararectal fossa***, enabling the distention of rectum.

### ***→ Relations of the Rectum:***

Lies posteriorly against inferior 3 sacral vertebrae and coccyx, Anococcygeal ligament, median sacral vessels and inferior ends of the sympathetic trunk + sacral plexus

- ***In males*** → related anteriorly to the fundus of Urinary bladder, terminal ureters, ductus deferens, seminal glands and prostate.
- ***In females*** → related anteriorly to the vagina.

### ***→ Arterial supply:***

- ***Superior rectal artery***: branch of ***IMA***, for proximal part of rectum (upper 1/3), forming the ***Corpus Cavernosum Recti***, or the ***Anal cushions***, which is the internal hemorrhoidal plexus of rectum. The "functional unit" of the corpus cavernosum recti is a narrow, partly tortuous, thick-walled artery which terminates, without interposed capillaries, directly into the hollow spaces which are separated from each other by tough connective tissue septa. The blood remains arterial and acts as "filler," playing no metabolic role. For this

reason, bleeding from hemorrhoids is always bright red. It also plays an important role normally in maintaining anal continence.

- **Middle rectal artery:** branch of **internal iliac artery (hypogastric)**, for middle/inferior parts (middle 1/3).
- **Inferior rectal artery:** branch of **internal Pudendal** artery, for the Anorectal junction (lower 1/3).

\*\* The **Sudeck critical point/ anastomosis:** between *superior rectal artery of IMA* and *Sigmoidal artery (IMA)*.

Very important in rectal tumors surgery, when the middle portion of the rectum must be removed with the preservation of the superior portion and the anal canal (together with the sphincter), we cut the superior rectal artery *above* the anastomosis and then the superior part of the rectum that was preserved can be pulled down towards the anal sphincter (and it will still receive blood through the anastomosis)

→ **Venous drainage:**

- *Through the superior, middle, inferior rectal veins.*

An anastomosis occurs between portal and systemic veins (IVC) in the wall of anal canal, the superior rectal vein drains into the **portal venous system (IMV)** and the middle and inferior rectal veins drain into the **systemic system (Internal Iliac vein)** → part of the Porto-caval anastomosis.

- *The superior rectal v. drains into the IMV, then to the portal vein and into the liver*
- *The middle & inferior rectal veins drain into the internal iliac vein and then into the IVC*

\*\* This is of great clinical significance in the case of the *hematogenous metastasis* of tumors of the rectum, if the tumor is located in the upper part, its likely to metastasize via the portal system to the liver mainly, and if its in the lower part of the rectum, its more likely to metastasize into the lungs.

- The **Submucosal rectal plexus (venous)**, communicates with:
  - Vesicle v. plexus in males
  - Uterovaginal v. plexus in females

Has 2 parts:

- *Internal venous rectal plexus* (deep to Anorectal junction mucosa) also part of the corpus cavernosa recti, forming vascular prominences formed by clusters of normally sacculated veins of the superior rectal venous plexus which are fed by arteriovenous anastomosis that cause their engorgement, and which are usually found to lie laterally on the left and antero- and posterolaterally on the right side of the anal canal.
- Subcutaneous external rectal venous plexus (external to the muscular wall of rectum)

→ **Innervation of rectum:**

- **Sympathetic:** lumbar part of sympathetic trunk + superior hypogastric plexus through IMA.
- **Parasympathetic:** pelvic splanchnic nerves.

## RECTUM - RELATIONS

### ANTERIOR

#### Female

- Recto-uterine pouch (of Douglas)
- Small bowel
- Vagina (posterior fornix)
- Uterus & bladder



#### Male

- Rectovesical pouch
- Small bowel
- Denonvillier's fascia
- Bladder
- Vas, seminal vesicle
- Prostate

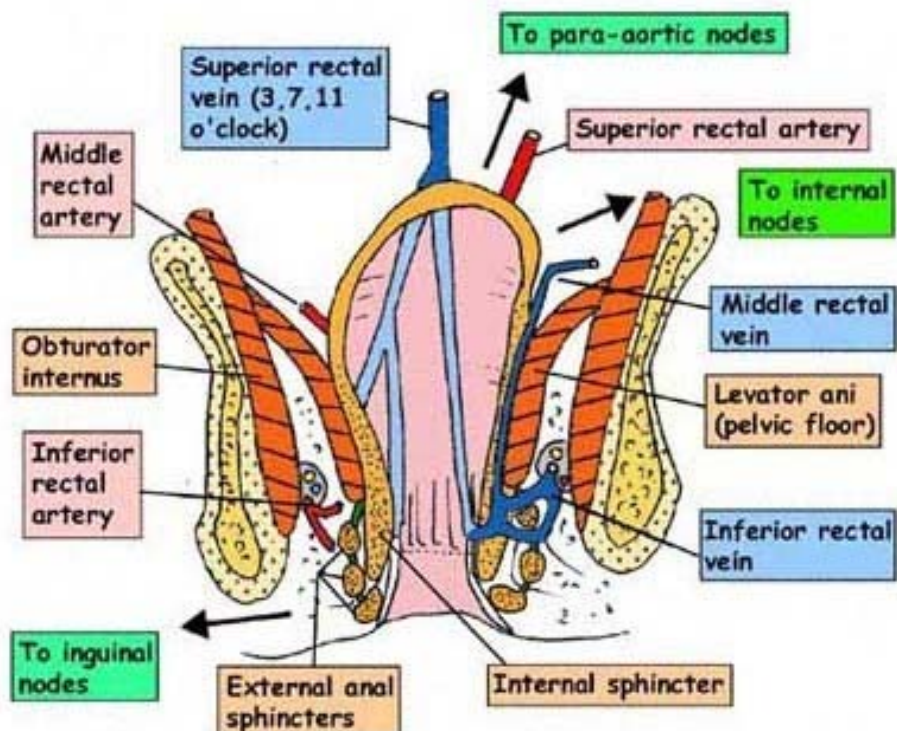


### POSTERIOR

Fascia, median sacral & rectal vessels, sympathetic trunk, pelvic splanchnic nerves, piriformis, sacral & coccygeal roots, sacrum, coccyx, anococcygeal body

### LATERAL

Peritoneum, fat, nodes, obturator internus & its fascia, Alcock's canal & contents, levator ani & coccygeus, ischio-anal fossa, lateral (fascial) ligaments of rectum



## ***The Anal Canal***

The 4 cm anal canal is the terminal part of the large intestine that extends from the upper aspect of the pelvic diaphragm to the *anus*. The anal canal begins where the rectal Ampulla narrows at the level of the U shaped sling formed by the *puborectalis* muscle, and ends at the anus, the external outlet of the GI tract. The anal canal, surrounded by the internal and external anal sphincters, descends *posteroinferiorly* between the *Anococcygeal ligament* and the perineal body. The canal is collapsed except during passage of feces. Both sphincters must relax for defecation to occur.

→ ***The internal anal sphincter:*** is an involuntary sphincter surrounding the superior two thirds of the anal canal. It's a thickening of the circular smooth muscle layer and is innervated by parasympathetic fibers that pass through the pelvic splanchnic nerves. The sphincter is tonically contracted most of the time to prevent leakage of fluid; however it relaxes in response to the pressure of feces or gas distending the rectal Ampulla, requiring voluntary contraction of the *puborectalis* and external anal sphincters if defecation is not to occur.

→ ***The external anal sphincter:*** is a large voluntary striated muscle sphincter that forms a broad band on each side of inferior two thirds of the anal canal. It has 3 components:

- Subcutaneous part
- Superficial part
- Deep part

The sphincter blends superiorly with the *puborectalis* muscle and is supplied mainly by S<sub>4</sub> through the inferior rectal nerve.

\*\* the 2 sphincters are separated by a thin *layer of longitudinal smooth muscle cells*, a contribution of the longitudinal layer of the intestinal wall and fans out as the ***Corrugator cutis muscle of anus*** into the perianal skin

In the interior of the anal canal, the superior half of the mucous membrane is characterized by a series of ***longitudinal ridges, anal columns of Morgagni***. These columns contain the terminal branches of the *superior rectal artery and vein*. The ***Anorectal junction***, indicated by the superior ends of the anal columns, is where the rectum joins the anal canal. At this point the wide rectal Ampulla narrows as it traverses the pelvic diaphragm. The inferior ends of the anal columns are joined by delicate crescent-shaped mucosal folds which called the ***anal valves of Morgagni***. Superior to the valves are small pocket like recesses, ***anal sinuses***. When compressed by feces, the anal sinuses release mucus that aids in evacuation of feces from the anal canal.

***The pectinate line (dentate line, pain line)***, an inferior comb shaped limit of the anal valves that indicates the junction of the superior part of the anal canal (*derived from hindgut*) and the inferior part (*derived from proctodeum*). The anal canal above the pectinate line differs from the part below the line in its arterial supply, innervation, and it's venous and lymphatic drainage. The mucosa of the anal canal has 3 zones:

- 1) ***Zona Columnaris:*** from the *Anorectal junction* until the *pectinate line*, on the columns, composed of Simple columnar epithelium
- 2) ***Zona Hemorrhagica:*** from the pectinate line until the ***Hilton white*** or ***Anocutaneous*** line, made of *Stratified squamous non-keratinized epithelium*
- 3) ***Zona Cutanea:*** *Stratified squamous keratinized non-hairy epithelium*, continues below the anocutaneous line with the normal hairy perianal skin. This sharp turn



where the squamous epithelium lining the lower anal canal becomes continuous with the skin of the perineum is called the **Anal verge**

\*\* The **Hilton White line** is found at the level of the interval between the subcutaneous part of the external sphincter and the lower border of the internal sphincter, that's why it's called also the **intersphincteric groove**

\*\* Zona hemorrhagica & Cutanea are called together **Anoderm** which is the lining of the anal canal immediately inferior to the pectinate line and extending for about 1.5 cm until the anal verge; it is devoid of hair and sebaceous and sweat glands, and so is not true skin, although it is squamous epithelium; it is pale, smooth, thin, and delicate, and shiny when stretched; it is especially vulnerable to abrasion (as from rough toilet paper), chemical irritants (soaps), and is well provided with tactile and nociceptive (pain, itch)

#### → Arterial supply of the anal canal:

1. **Superior rectal artery:** supplies canal above the pectinate line.
2. **Inferior rectal arteries:** supply the inferior part of the canal, as well as the surrounding muscles and perineal skin.
3. **Middle rectal arteries:** assist with the blood supply to the anal canal by forming anastomosis with the superior and inferior rectal arteries.

#### → Venous drainage of anal canal:

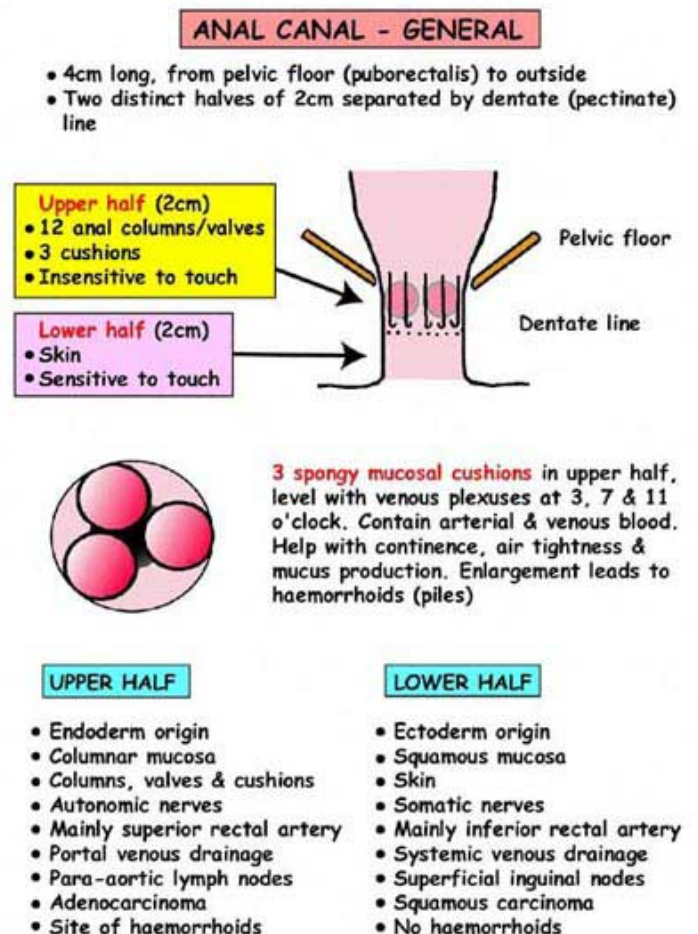
The internal rectal venous plexus drains in both directions from the level of the **pectinate line**.

1. Superior to the line, the internal rectal plexus drains chiefly into the **superior rectal vein**
2. Inferior to the pectinate line, the internal rectal plexus drains into the **inferior rectal veins**
3. The **middle rectal veins** mainly drain into the muscularis externa of the Ampulla and form **anastomosis** with the superior and inferior rectal veins.

#### → Innervation of the anal canal:

**1. Superior to the pectinate line:**  
inferior hypogastric plexus (SYM + PSY). Sensitive to stretching but not pain and temperature (that's why deep hemorrhoids are not painful)

**2. Inferior to the pectinate line:**  
inferior anal (rectal) nerves, branches of the pudendal nerve. Somatic innervation sensitive to pain, touch and temp.



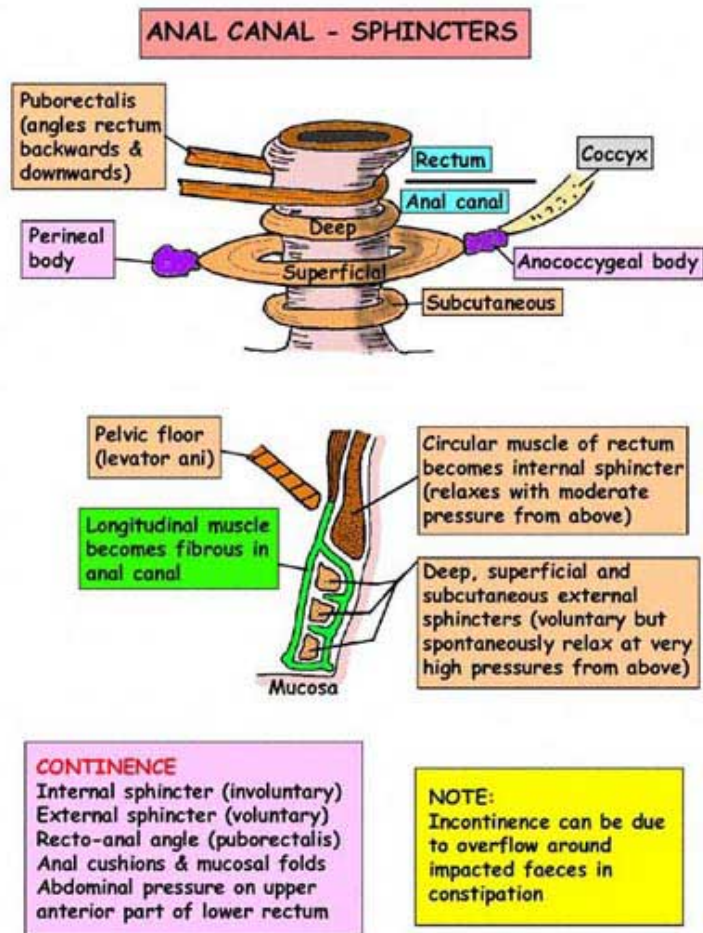
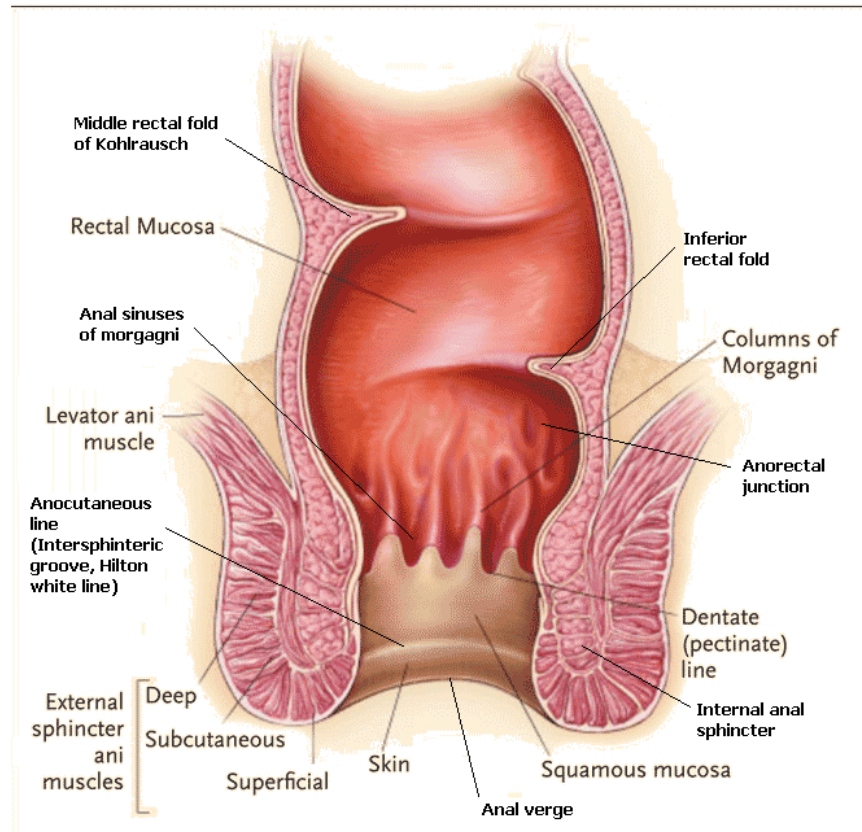


Figure 4. Anatomy of the Anus



From Ryan D, Compton C, and Mayer R. Medical progress: carcinoma of the anal canal. *N Engl J Med.* 2000;342(11):792-800. Reprinted with permission from the publisher. Copyright © 2000 Massachusetts Medical Society. All rights reserved.

## The Liver

The largest gland and abdominal organ in the body, 1500g in weight, (about 2.5% of TBW)  
Occupies most of the right Hypochondrium and upper epigastrium  
It lies deep to the right 7<sup>th</sup>-11<sup>th</sup> ribs, position varies a bit during respiration, but if its palpable under the right rib cage margin, then it's pathologically enlarged.

Has a lot of metabolic functions, detoxification, glycogen storage and bile production.

### → Surfaces of the liver:

1. **Diaphragmatic:** Convex (Ant., Sup., and some Post.). Smooth and dome shaped.
  - **Subphrenic recess:** between diaphragm and Anterio-Superior aspect of diaphragmatic surface of liver. Separated into right and left by the falciform ligament.
  - **Falciform ligament:** extends between the liver and umbilicus.
  - **Subhepatic recess:** the part of the peritoneal cavity between the visceral surface of the liver and the transverse colon, it is subdivided right and left Subhepatic recesses
    - **Hepatorenal recess (Right Subhepatic recess or Morrison's pouch):** a posterior-superior extension of the *Subhepatic recess*, between the *right part of visceral surface and the Right kidney*. Fluid draining from the omental bursa flows into this recess, as it's the deepest point of the abdomen. It also communicates down with the *right Paracolic gutter*, reaching the cecal area, which plays an important role in spreading infections in the peritoneal cavity

\*\* This surface is covered by peritoneum except at the **Bare area of liver (nude area)** where it lies in direct contact with the diaphragm. The bare area is demarcated by peritoneal reflection from the diaphragm to it as the Anterior (Sup.) and Posterior (Inf.) layers of the **right coronary ligament**. These layers meet at:

- Right: forms the *Right triangular ligament*, and extending over the right kidney as the *Hepatorenal ligament*
- Left: forms the *Left triangular ligament* (near the apex of liver)

- The **IVC** leaves a groove on the liver, in the bare area.

\*\* Directly on the surface of liver, below the peritoneum, we can find the **fibrous capsule of liver of Glisson**, a layer of connective tissue ensheathing the outer surface of the liver and also the hepatic artery, portal vein, and bile ducts as these ramify within the liver

2. **Visceral:** Covered with peritoneum except the **Gall bladder fossa** and **Porta Hepatica** (a transverse fissure where the Portal vein, Proper hepatic artery, Bile ducts + Lymphatics & nerve plexuses, enters and exists the liver)

This surface has multiple fissures which form a "**H**" shape, composed of:

1. **Right Saggital fissure:** a continuous groove formed **anteriorly** by the *Gallbladder fossa* and **posteriorly** by the *IVC groove*. It is located between the Quadrate, Caudate and right lobes

2. **Left Saggital fissure:** a continuous groove formed **anteriorly** by the *fissure of Ligamentum Teres hepatis (round ligament of liver, the remnant of left umbilical vein)* and **posteriorly** by the *Fissure of ligamentum Venosum (remnant of the Ductus Venosus Arantii)*

\*\* *The omental tuberosity of liver* is an eminence on the visceral surface of the left hepatic lobe to the left of the fossa for the ductus venosus.

→ **The Lesser Omentum** encloses the **Portal Triad** (PV, proper HA, BD) and passes from the liver to the less curvature of the stomach, and to the first 2 cm of the Superior part of the duodenum. Composed of:

1. **Hepatoduodenal Ligament:** encloses structures that passes through the *Porta hepatica* as a tent shaped covering and reaches the 1<sup>st</sup> 2cm of the superior part of duodenum
2. **Hepatogastric ligament:** between the fissure for *ligamentum Venosum* of liver and the lesser curvature of stomach. (Contains the right & gastric arteries anastomosing on the lesser curvature + coronary vein of stomach)

→ **The visceral surface has impressions made by the nearby structures contacting the liver:**

- \* *Duodenal impression:* Sup. Part of Duodenum, near gallbladder
- \* *Gastric impression:* by the stomach, on the left lobe
- \* *Gallbladder impression:* fossa at the Right Anterior groove, by the gallbladder
- \* *Colic impression:* by the Right colic flexure & transverse colon at anterior aspect of the right lobe
- \* *Renal & adrenal impressions:* by the Right kidney and suprarenal gland (Right side to IVC)

→ **In lying position, fluid will accumulate at 2 recesses in the body:**

- **In the abdomen:** Hepatorenal recess (Morisson's pouch)
- **In the pelvis:** Rectovesical recess (in males) or Rectouterine pouch of Douglas (in females)

→ **Anatomical lobes:**

*Anatomically:* 2 topographic lobes + 2 accessory lobes by the peritoneal reflections, the fissures between them and the vessels supplying the liver and the Gallbladder

- *On midline plane:* attachment of the falciform ligament + L. saggital fissure separates the large right lobe from the left lobe.

- *On visceral surface:* R. /L. saggital fissures + Porta hepatica forms the morphologic 2 accessory lobes (which are parts of the anatomic right lobe):

- **Quadrate lobe:** Anterior and inferior.
- **Caudate lobe:** posterior and superior. It has the *caudate process*, which is a narrow band of hepatic tissue connecting the caudate and right lobes of the liver posterior to the Porta hepatis. On the left inferior angle of the caudate lobe, opposite to the caudate process there is the **papillary process** of caudate lobe)

→ **Functional Sub-division of the liver:**

Functionally independent right and left livers (parts or portal lobes), each receives its own branch of the Proper hepatic Artery and drains its own Hepatic Duct.

Caudate lobe may be considered as a 3<sup>rd</sup> liver.

The liver may be divided into 4 divisions and 8 surgically resectable segments:

- I – Post. Caudate segment
- II – Left Post. Lateral Segment

- III – Left Anterior lateral segment
- IV – Left Medial segment
- V – Right Anterior Medial segment
- VI – Right Anterior Lateral segment
- VII – Right Posterior Lateral segment
- VIII – Right Posterior Medial segment

→ **Blood supply to liver:**

Like the lungs, the liver has dual blood supply (afferent vessels). A dominant venous source, and a lesser arterial one.

- **Portal Vein:** 70-80% of blood to liver (40% more O<sub>2</sub> than IVC)
- **Proper hepatic Artery:** 20-25% of blood to liver.

**The Portal vein** is formed behind the *neck* of pancreas by joining of the superior (and sometimes inferior) mesenteric vein and the splenic vein. It ascends anteriorly to the IVC as part of the **portal triad**, in a *posterior* position.

**The Proper hepatic artery** a branch of the celiac trunk (from the common hepatic artery) lies *medially and anteriorly in the portal triad*.

→ At the Porta hepatica, the proper HA and PV terminate by dividing into R., L. and middle branches, Primary branches which supply the right and left livers. (Also the caudate lobe) Within each part, secondary branches supply the medial and lateral divisions of the right and left livers. Tertiary divisions further supply 7 out of the 8 segments.

→ The liver is drained by **collecting veins**, which drain onto the central veins inside the liver parenchyma. The **central veins** unite to form the *sublobar veins* which drain into the left, right and middle **hepatic veins**, which drain into the **IVC** just under the diaphragm.

→ **Lymphatics of liver:**

In the *Sub-peritoneal* fibrous capsule of **Glisson**: *superficial lymphatics*

In the connective tissue of parenchyma: *deep lymphatics*

Most of the lymph is formed inside **Disse's space** (peri-sinusoidal space) and drains into the **interlobular portal triad**, within the **periportal Space of Mall**, a tissue space between the limiting lamina and the portal canal in the liver.

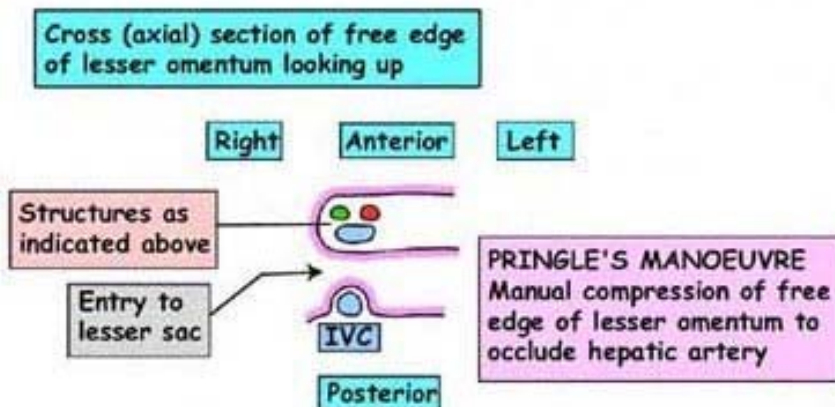
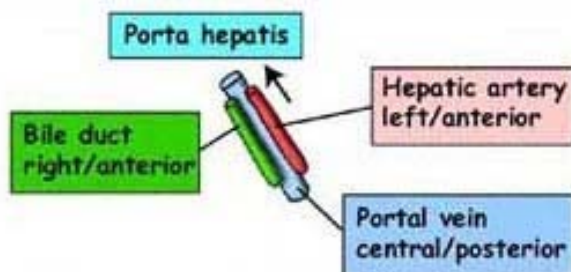
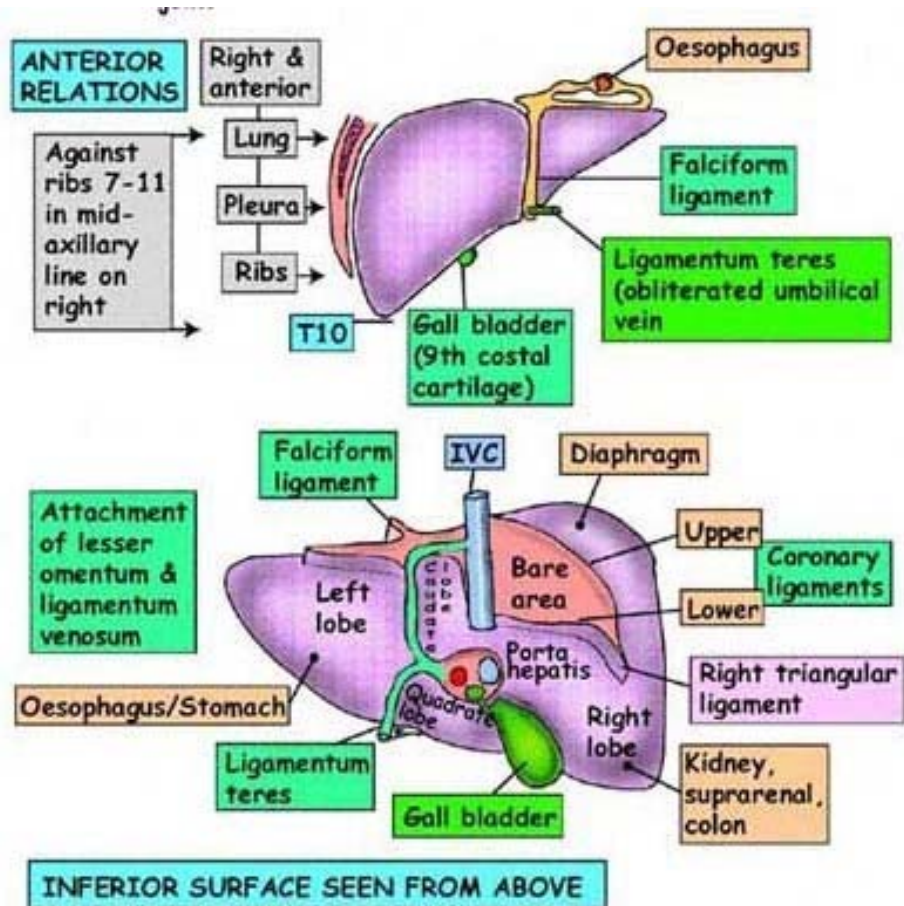
*Hepatic lymph nodes* → *celiac lymph* → **Chyle cistern** → **thoracic duct**

→ **Innervation:**

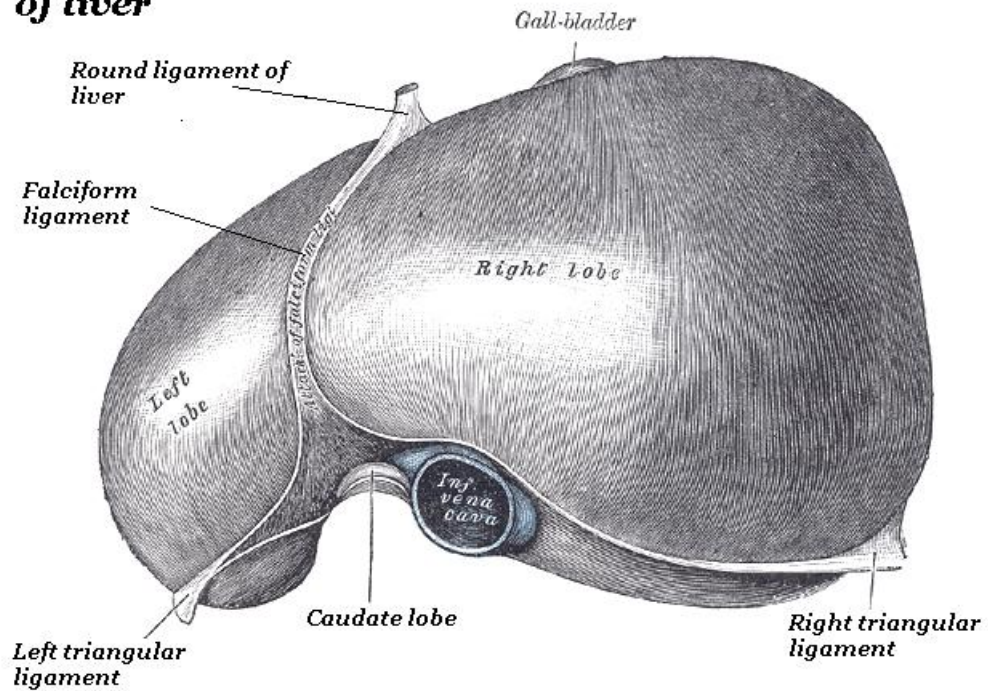
**Hepatic plexus** – largest derivative of the celiac plexus.

- **Sympathetic** from the celiac plexus
- **Parasympathetic** from the Vagus nerve.

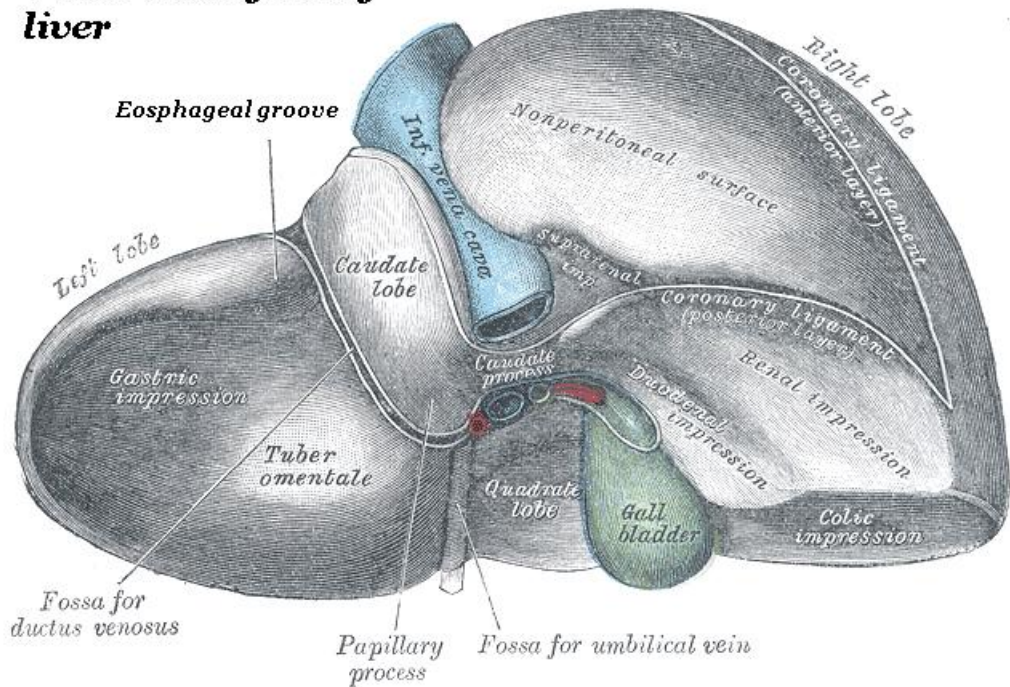




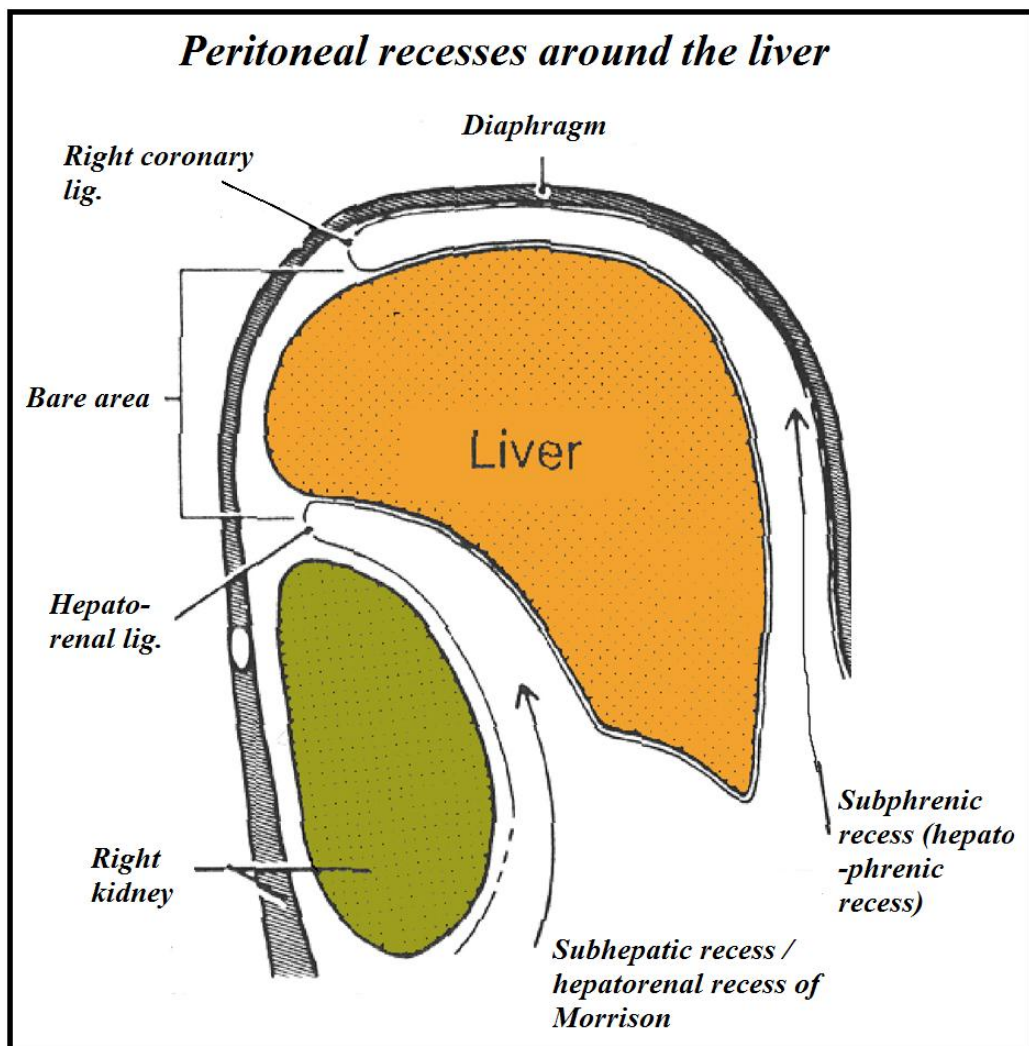
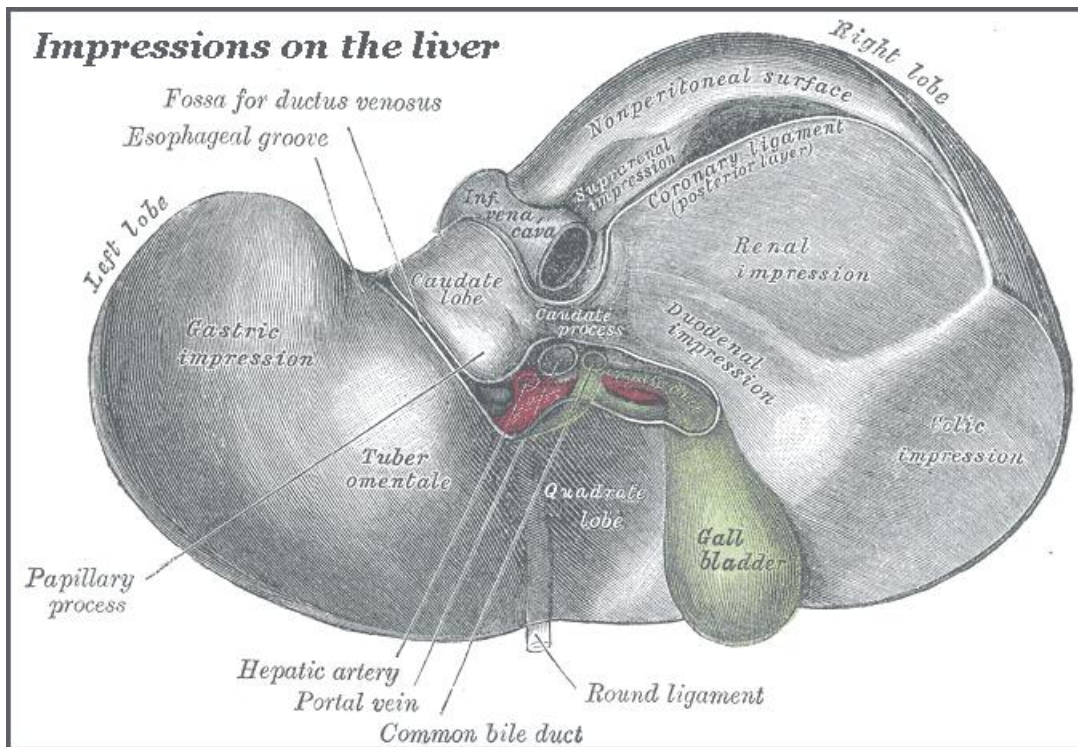
## Diaphragmatic surface of liver



## Visceral surface of liver







## ***Biliary ducts and Gallbladder***

Hepatocytes → Bile → Bile canaliculi → Herring's ducts → Interlobular Biliary ducts → collecting Bile Ducts of interhepatic portal triads → R. and L. hepatic ducts (also intermediate) → Common Hepatic duct (joins cystic ducts) → Common Bile Duct (part of portal triad)

### **→ Common Bile duct:**

5-15 cm, formed by union of the *cystic* and *common hepatic* ducts

Descends behind the superior part of duodenum, and has a groove on the posterior surface of the pancreas. Joins the pancreatic main duct of **Wirsung** and together form the **Hepatopancreatic Ampulla** (protrudes in the 2<sup>nd</sup> descending part of duodenum as **Vater's** papilla) → guarded by the sphincter of **Oddi**

**\*\* Sphincter of Bayden:** around the common bile duct. When it contracts, bile backs up into the cystic duct for storage and concentration in the gallbladder.

#### **- Arterial supply:**

- **Cystic artery:** from the right branch of proper hepatic artery, proximal part.
- **Right hepatic artery:** middle part.
- **Posterior superior Pancreaticoduodenal artery + Gastroduodenal artery:** retroduodenal part.

#### **- Venous drainage:**

- **Posterior superior Pancreaticoduodenal vein:** distal part.
- **Other parts directly from liver!**

### **→ Gallbladder:**

A 7-10 cm, pear-shaped organ that serves as a storage reservoir for bile (~50mL volume)  
Lies in fossa for gallbladder on the visceral surface of the liver, at the junction of the left and right livers

It can be palpable at the junction of the **right 9<sup>th</sup> Intercostal space & the midclavicular** line near the lateral border of the Rectus Abdominis.

Has 3 parts:

- **Fundus:** wide end, projects from the inf. Surface of the liver. Located at the tip of the 9<sup>th</sup> costal cartilage.
- **Body:** contacts the visceral surface of the liver, transverse colon and 1<sup>st</sup> part of duodenum.
- **Neck:** narrow, directed towards the Porta hepatica makes S shaped bend and joins the cystic duct.

- **Neck and cystic duct** are directly above the duodenum.

- **Fundus** is completely surrounded by peritoneum that binds it to the liver.

Hepatic surface of the gall bladder attaches to the liver by the connective tissue of the liver's fibrous capsule.

- **Cystic duct:** (around 4cm) connects the neck of gallbladder to the common hepatic duct.  
The mucosa of the neck has **Spiral folds** (*Spiral valves – Heister's valves*)

- Helps keep cystic duct open
- Offers additional resistance to sudden dumping of bile (e.g. coughing)

- **Cystic artery:** supplies the *gallbladder* and *cystic duct*, arises from the *right branch of the proper hepatic artery* in the angle between the common hepatic and cystic ducts. Variation may occur.

- **Cystic veins:** drain the neck and cystic duct to the PV.  
The fundus and body drain directly to the hepatic sinusoids.

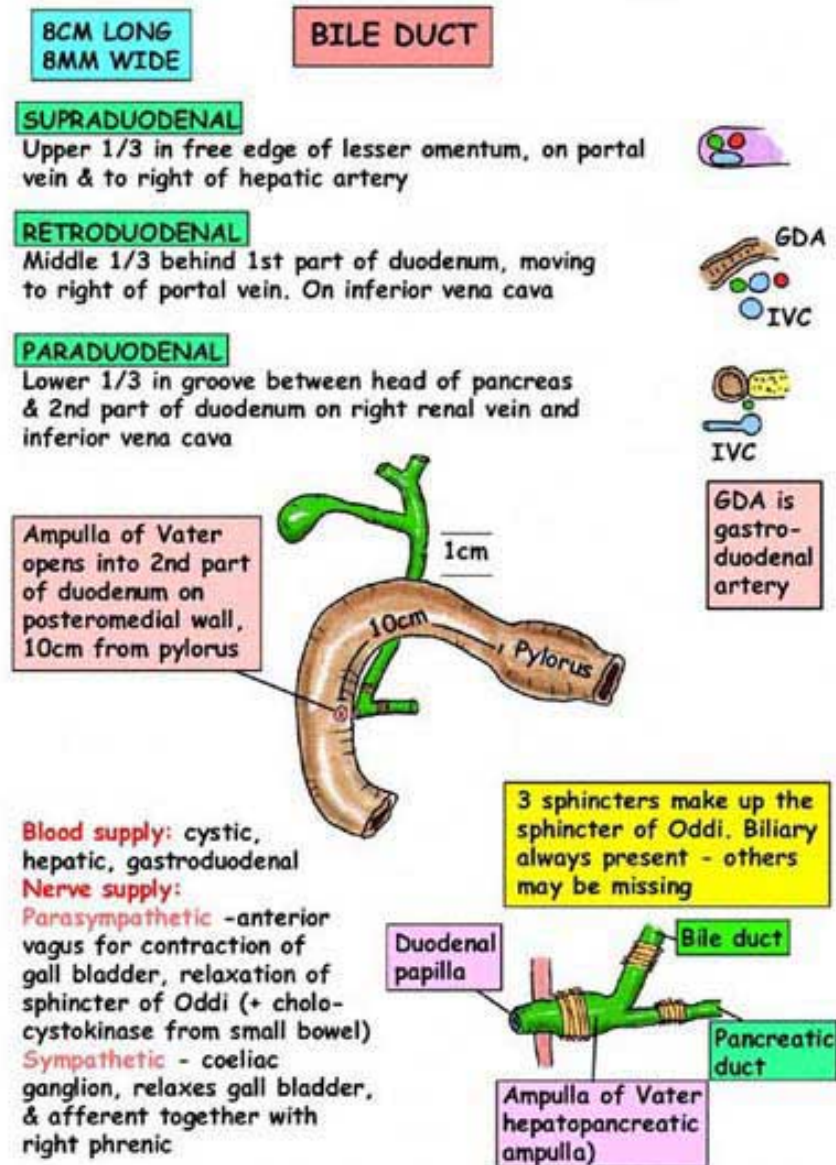
- **Lymphatics:** *Hepatic lymph nodes* → *cystic lymph nodes*

- **Innervation:**

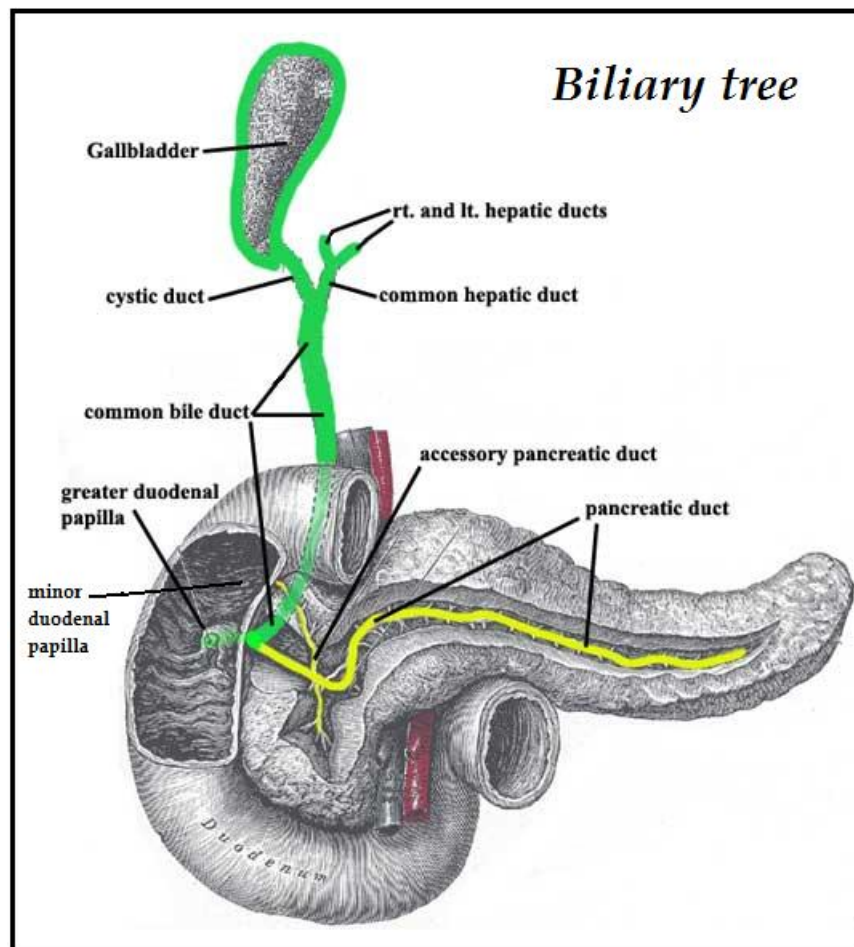
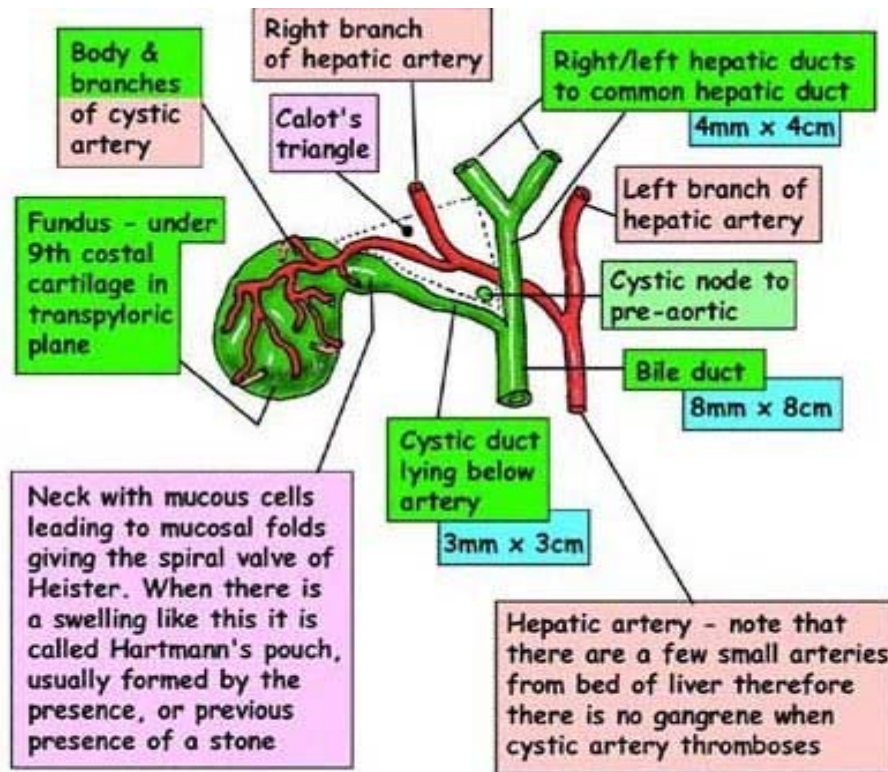
- **Sympathetic:** Celiac nerve plexus

- **Parasympathetic:** Vagus, also right Phrenic nerve (somatic) – contraction of the GB and relaxation of the sphincters, though it is usually done by **Cholecystikin** (CCK)

\*\*At diseased states, a dilation/ pouch appears at the junction of the neck of GB and the cystic duct – **Infundibulum of GB (Hartmann's pouch)**







## ***Portal system of veins, Communication between portal and systemic veins (Porto-caval or porto-systemic anastomosis)***

Main vein of portal system: **hepatic portal vein**.

Blood from GI tract, gallbladder, pancreas, spleen go through portal vein to liver. Terminal branch in liver: sinusoids → drain into hepatic veins → inferior vena cava.

→ ***Hepatic portal vein formed behind the neck of pancreas from:***

- *Splenic vein*
- *Inferior mesenteric vein* (usually join to splenic vein)
- *Superior mesenteric vein*

It then ascend to the liver posteriorly to bile duct and proper hepatic artery in the Porta hepatis and branches into right and left branches that terminate at the sinusoids.

The blood carried by the hepatic portal .v contains the products of digestion (carbohydrates, fats, proteins, product of RBC destruction from spleen etc').

→ ***Communication between portal and systemic veins: porto-caval anastomosis:***

In which portal venous system anastomose with systemic venous system (IVC).

When portal circulation through liver is diminished/ obstructed due to liver disease (e.g. Cirrhosis) or physical pressure from tumor, blood from the alimentary tract can still reach the heart via IVC or SVC via these collateral routes.

These routes are available, because of an important fact regarding the Portal vein as a vein: It has no valves in the lumen! So a retrograde flow of blood inside it is possible!

***Main portal-systemic anastomosis:***

**1) *Gastro-esophageal region:***

- Between the Submucosa of Inferior esophagus, at the cardia, where ***Esophageal veins*** are drained into SVC (or azygos) (systemic side) and between
- ***Left Gastric (coronary vein of stomach)*** (Portal side)

\*\* In the lumen of the esophagus, ***Varicosities*** (enlargement) of the Esophageal veins can be seen, but the esophageal veins are thin and small and can easily rupture due to the high pressure → stop the bleeding by inflating balloon in esophagus.

**2) *Paraumbilical region:***

- Around umbilicus, between ***Paraumbilical veins*** (*portal side*) and
- ***Subcutaneous veins*** in the *anterior abdominal wall* (***superficial epigastric .v*** and ***lateral thoracic v.***)

\*\* Dilation of epigastric veins around the umbilicus → ***Medusa Head (caput)***

### 3) Anorectal region:

- *Submucosa of Anal canal*, between **Superior Rectal veins** that drain into the IMV (Portal side) and the
- **Middle & Inferior Rectal veins** which drain into Internal Iliac veins (systemic side), which drain into the IVC

\*\* Bleeding in rectum can be seen in the pathological case

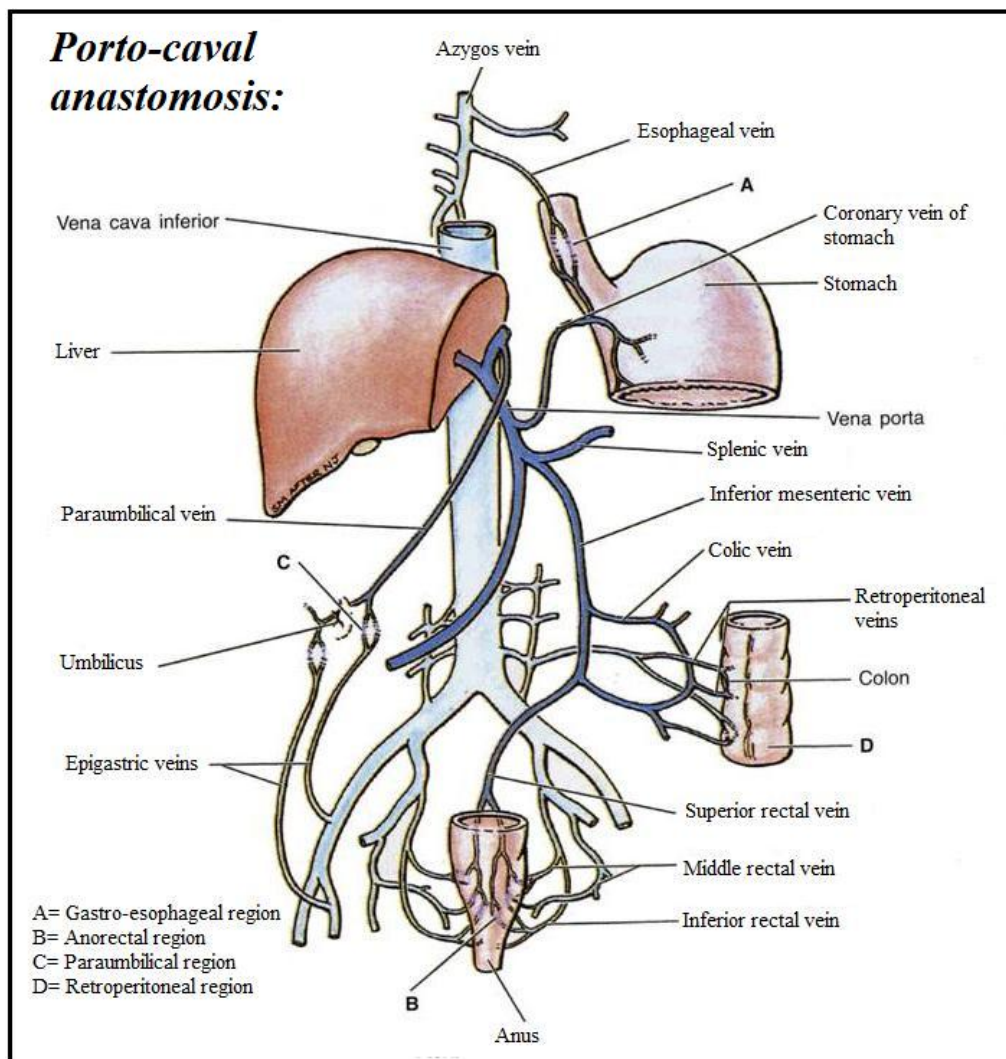
### 4) Retroperitoneal region:

Porto-caval anastomosis formed from veins in the walls of retroperitoneal viscera, such as the ascending and descending colon, or in bare areas of secondary retro-peritoneal viscera or liver, around kidney and pancreas passing to the tributaries of the inferior vena cava in the posterior body wall instead of those of the portal vein

E.g. pancreatic veins → PV (portal side) and the renal veins → IVC (systemic side)

OR

Colic veins, splenic veins, PV itself (portal side) and the retroperitoneal veins (systematic) → IVC



## ***The Pancreas:***

An elongated accessory digestive gland that lays retroperitoneally, Posterior to the stomach, between the duodenum loop to the right and spleen to the left, 2 basic functions in:

1. ***Exocrine secretion:*** pancreatic juice (Digestion)
2. ***Endocrine secretion:*** Insulin/Glucagon/ Somatostatin (Regulation of blood sugar)

### **→ Has 4 parts:**

1. ***Head:*** expanded part held in the C-shaped curve of duodenum to the right of the Superior Mesenteric Art. /V. Has an ***Uncinate process*** that projects from the inferior part of the pancreatic head. Rests posteriorly on *the IVC, right renal artery lumen and left renal vein*, the ***Common Bile duct*** is on the posterior superior surface or embedded inside.

2. ***Neck:*** (1-2cm) overlies the superior mesenteric Art. /V., which form a groove on its posterior aspect next to the stomach pylorus. Behind it the superior mesenteric vein joins the splenic vein to form the portal vein.

3. ***Body:*** continues from neck, moves horizontally, and passes over L2 vertebra, posterior to omental bursa. Posteriorly it's in contact with *aorta, SMA, Left suprarenal gland, left kidney and renal vessels*.

\*\* Between the body (neck) and head there is a notch – ***Pancreatic notch*** (under it goes the SMA, SMV above the uncinate process) and an *angle* between them which forms an elevation – the ***Omental tuber***, which is a well marked prominence that can be seen bulging behind the lesser omentum

4. ***Tail:*** lies anteriorly to *left kidney*, closely related to the *splenic hilum* and *left colic flexure*. Relatively mobile and inside the ***Splenorenal ligament*** with splenic vessels (~ L-1)

### **→ Ducts:**

- ***Main pancreatic duct of Wirsung:*** begins at the tail and goes all along the pancreas, through parenchyma until the head. Most of the time it joins the common bile duct to form the ***Hepatopancreatic Ampulla of Vater's*** which opens into the major duodenal papilla of Vater, controlled by *Oddi's sphincter* (smooth muscle).
- ***Accessory pancreatic duct of Santorini:*** opens into duodenum at a higher position (2cm) above the greater papilla, begins at head and opens into ***Santorini's papilla*** (minor duodenal papilla)

→ ***Blood supply:*** very dense and rich arterial supply, making the pancreatic surgeries one of the most difficult ones also due to the deep retroperitoneal position of the pancreas.

1. ***Superior Pancreaticoduodenal artery (anterior/posterior)*** (branch of the *Gastroduodenal artery*): superior part of the *head* of pancreas
2. ***Inferior Pancreaticoduodenal (anterior/posterior)*** (branch of SMA): Inferior part of the *head* of pancreas.



\*\* The Superior & Inferior Pancreaticoduodenal arteries form the 1<sup>st</sup> *Riolan anastomosis* around the head of pancreas.

The importance of those arteries surgically, is that tumors of the head of pancreas require total surgical removal together with their arterial supply & the *duodenum as well*, due to the common arterial supply. This process is known as ***pancreatoduodenectomy*** or ***Whipple's operation***

3. ***Dorsal pancreatic artery*** (branch of the *Splenic Art.*): to the head and body of pancreas
4. ***Pancreatic branches*** (from *Splenic Art.*): Around 10 branches for *body* and *tail*
5. ***Great pancreatic artery*** (from *Splenic Art.*): for the *body* and *tail*
6. ***Artery of tail of pancreas*** (branch of distal part of *Splenic Art.*): for the *tail*

→ ***Venous drainage:***

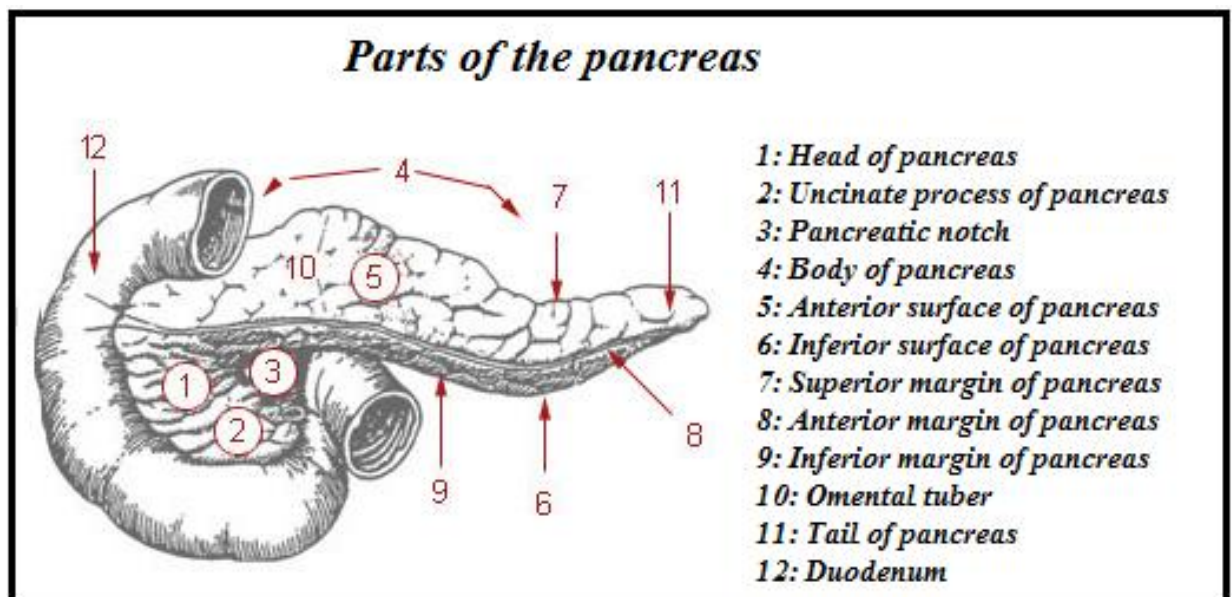
1. ***Pancreatic veins:*** to splenic veins
  2. ***Pancreaticoduodenal vein:*** to SMV
- (Both drain to PV)

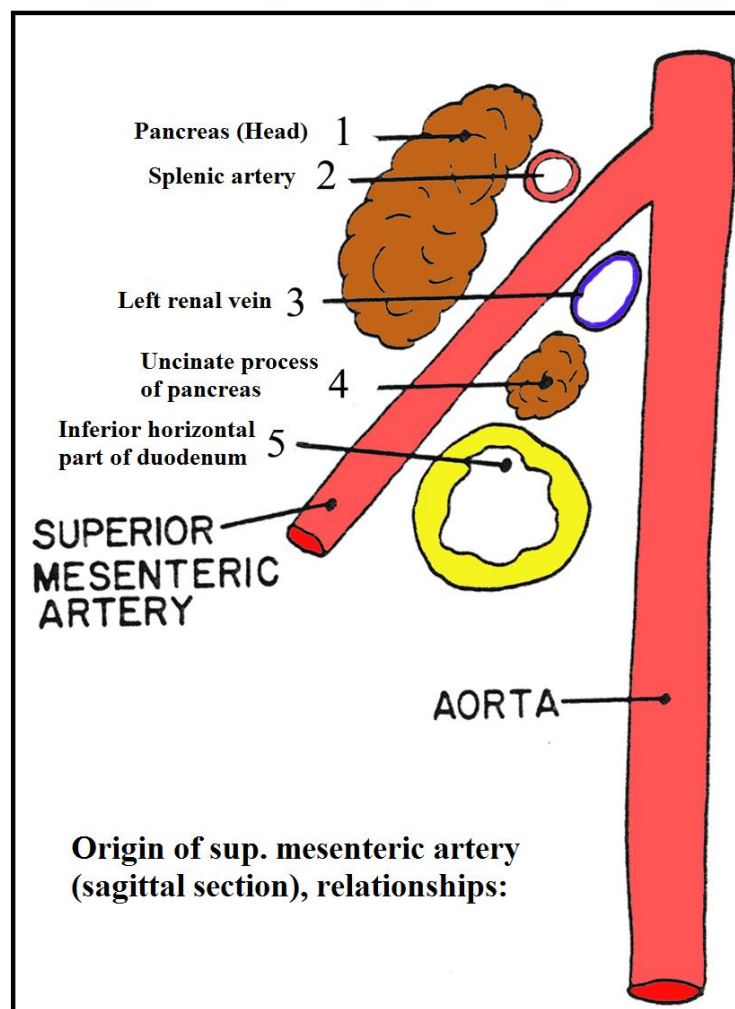
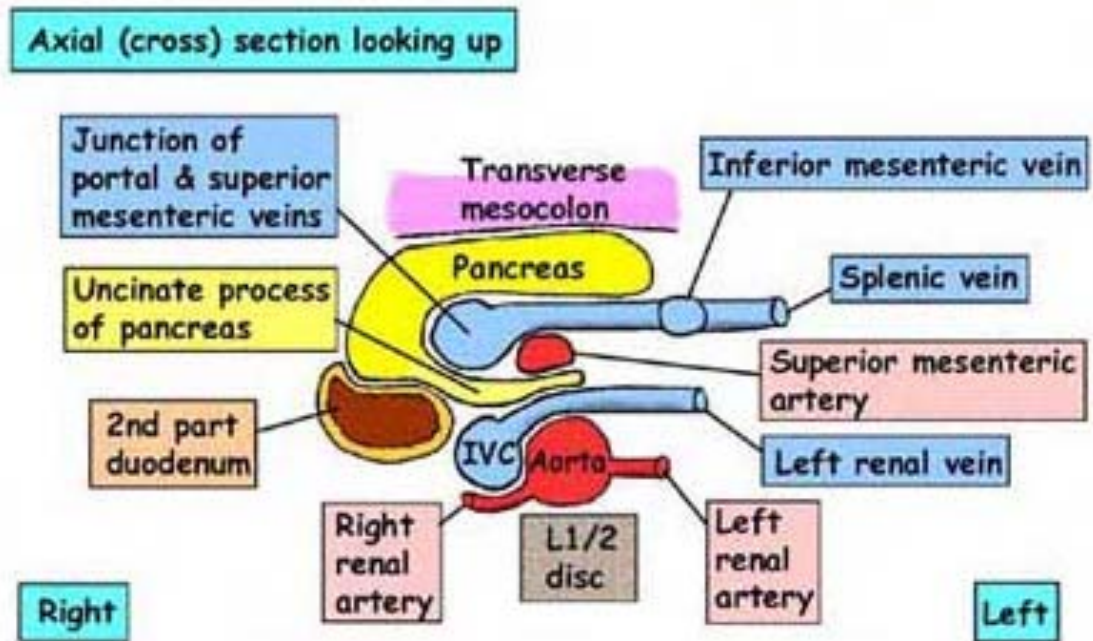
→ ***Lymphatic drainage:***

*Pancreatico-splenic lymph nodes* – to SMLN

→ ***Innervation:***

- ***Sympathetic:*** abdominopelvic splanchnic nerves
  - ***Parasympathetic:*** Vagus
- Both to the superior mesenteric plexus





## ***The Spleen***

An ovoid, purplish, pulpy mass, delicate and most vulnerable internal organ, located in left upper abdominal quadrant or Hypochondrium between the **9<sup>th</sup> and 11<sup>th</sup> rib in the midaxillary line, with its longitudinal axis parallel to 10<sup>th</sup> rib**

**\*\*Clinically**, any fracture of the ribs 9, 10 & 11, due to trauma (Boxing, ski, car accident etc.) will require an immediate investigation of the spleen for suspicion of rupture and fatal hemorrhage, due to its position and delicate structure

It is the largest lymphatic organ, participates in the immune defense, hematopoietic functions and filtering blood but still, its **not** a vital organ for living.

Usually it is 12 cm long, 7 cm wide, weights about 150 gram

It has a delicate **fibro-elastic capsule**, (containing collage, elastic fibers & smooth muscle) surrounded entirely by peritoneum (**Intraperitoneal organ**) except at the splenic hilum. It Rests on **Left colic (Splenic) flexure**, related to 9<sup>th</sup>-11<sup>th</sup> ribs and separated from them by the **Diaphragm** and **Costodiaphragmatic** recess.

### **→ Relations of the Spleen:**

- *Anteriorly* → Stomach
- *Posteriorly* → Left part of diaphragm separates it from the pleura, lung and ribs (9<sup>th</sup>-11<sup>th</sup>)
- *Inferiorly* → Left colic flexure (produces the *colic impression* on spleen).
- *Superiomedially* → Left kidney (produces the *renal impression*)

### **→ Surfaces of the Spleen:**

- *Diaphragmatic* surface: convex, related to the diaphragm upward.
- *Visceral* surface, concave facing downward, has 3 parts:
  - Anterio-Superior → Gastric surface (related to the Stomach)
  - Postero-Superior → Renal surface (related to the L. Kidney)
  - Inferior → Colic surface (related to the left colic flexure)

→ The 3 visceral surfaces meet each other at a sharp margin forming the **Hilum of Spleen**, which is the point of entry & exit of vessels & nerves at the **Hilar groove**.

### **→ Ligaments of the Spleen:**

- **Gastrosplenic ligament** : to the greater curvature of stomach, contains:
  - Left Gastroepiploic vessels (proximal parts)
  - Short gastric vessels
- **Splenorenal (Lienorenal) ligament**: to left kidney and the posterior wall of trunk. The Splenic recess of the omental bursa extends to this ligament. It contains:
  - Splenic artery /vein
  - Tail of pancreas
- **Phrenico-splenic ligament**: developing from posterior mesogastrium (Phrenicogastric ligament), attaching the spleen to the diaphragm. it is a portion of the greater omentum, extending also to the posterior wall as the *phrenico-pancreatic ligament*
- **Phrenicocolic ligament**: very important supportive ligament holding the spleen as the *Sustentaculum Lienis* and forming the most important component of the Splenic Nest (*Nidus Linealis*) on the which the spleen rests. It arises from the diaphragm and reaches the splenic flexure, prevents spleen from descending and holds it firmly in position.

**\*\* Splenic nest (Nidus linealis):** the space / fossa in which the spleen is laying and held firmly in its position by the nearby splenic ligaments and organs (L. kidney, stomach, L. colic flexure). The most important component keeping the spleen fixed in position is the Phrenicocolic ligament, which supports the spleen from below as a shelf, known as the *Sustentaculum lienis*

→ **Arterial supply of the Spleen:**

- **Splenic artery** (branch of celiac trunk, largest). Has a tortuous course posterior to omental bursa, anterior to left kidney and runs at the superior border of pancreas. At hilum divides to 5-6 branches, but since no anastomosis between them → 3 vascular segments +2 avascular ones.

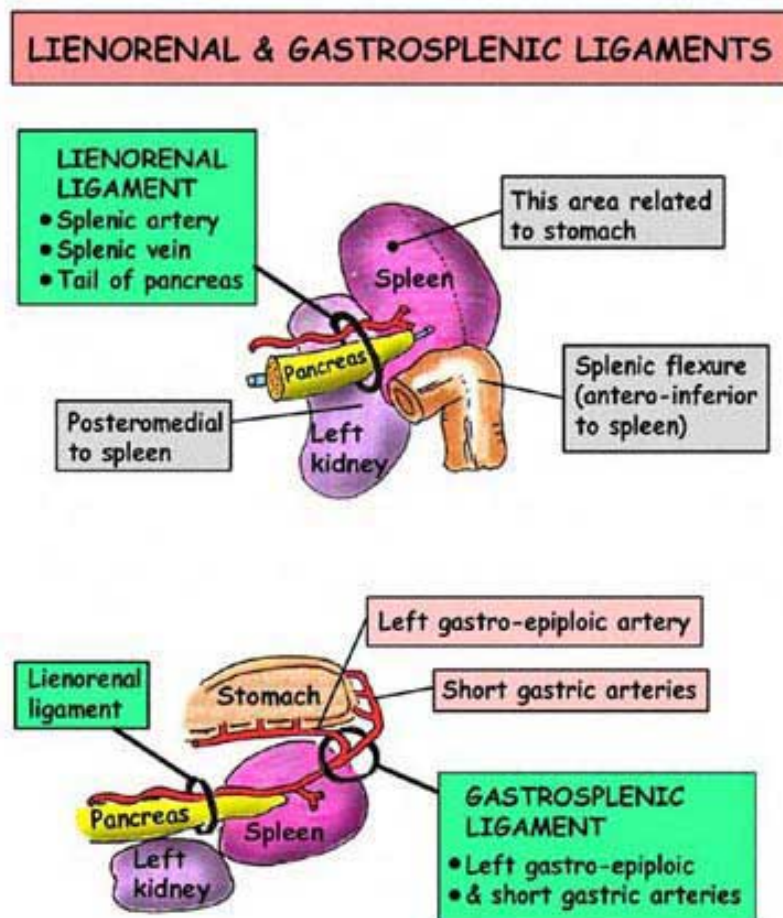
→ **Veins of the Spleen:**

- **Splenic vein**, formed by:
  - Short gastric vein
  - Pancreatic veins
  - Left Gastroepiploic vein

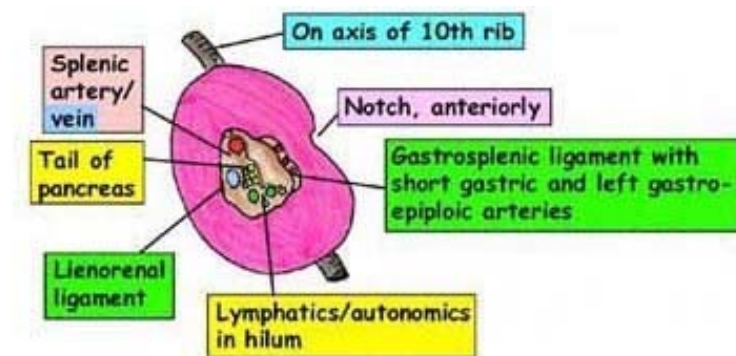
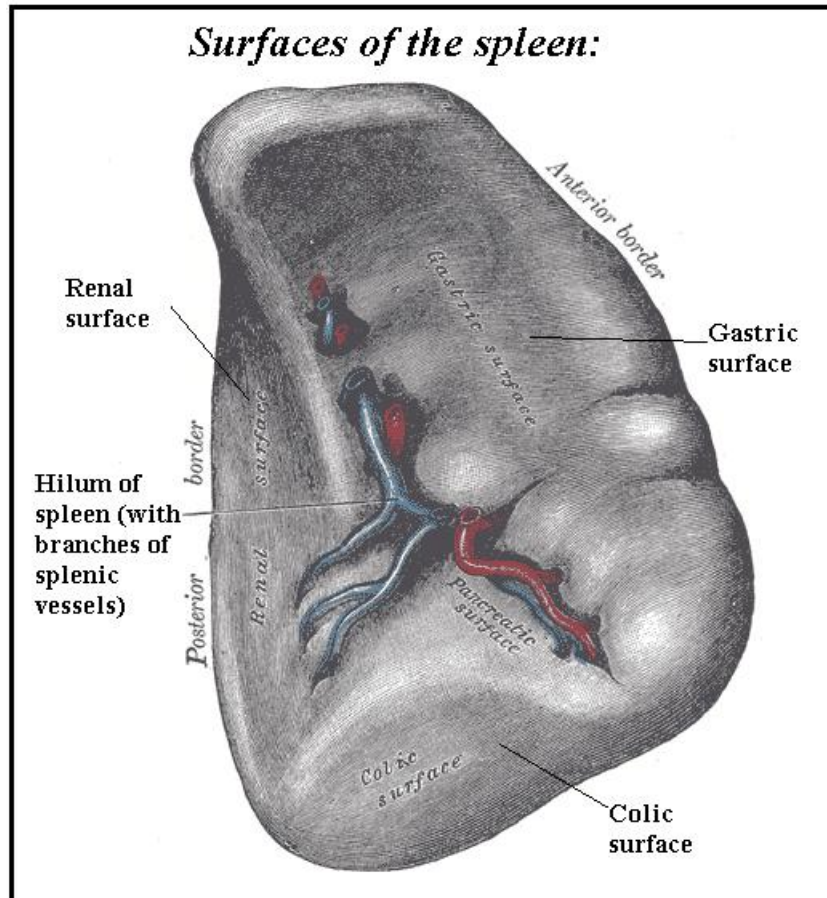
→ It joins the *Inferior Mesenteric vein* behind *body of pancreas* and at the *Neck* they both join the *Superior Mesenteric vein* to form the *Portal vein*.

→ **Lymphatic drainage:** Pancreatico-splenic and celiac lymph nodes

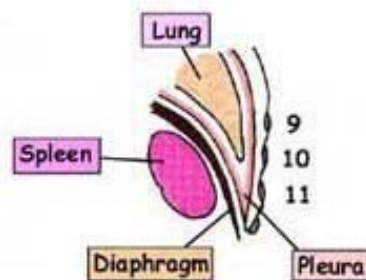
→ **Nerves:** Celiac nerve plexus





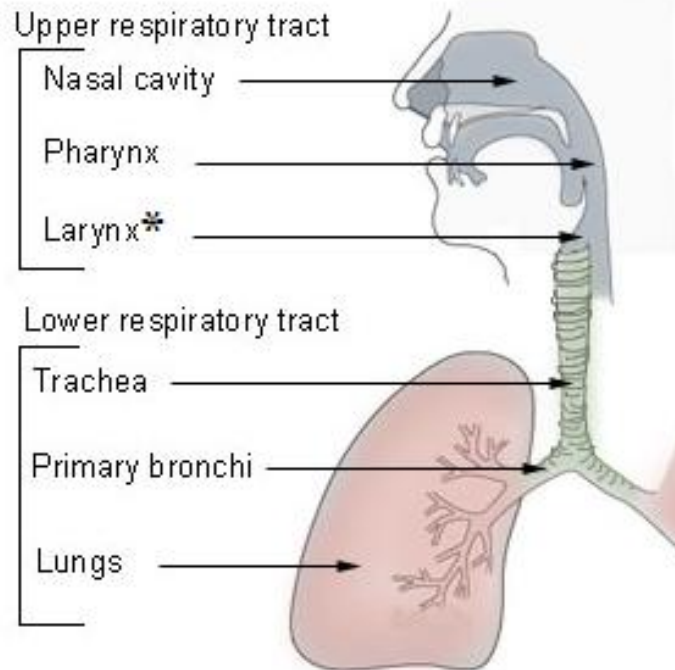


Note: lower pole is normally no further antero-medial than mid axillary line



# *The Respiratory Tract*

## **Conducting Passages**



\* *The larynx can be considered part of the upper respiratory tract or the lower respiratory tract depending on the source. Some specify that the glottis (vocal cords) is the defining line between the upper and lower respiratory tracts; others make the line at the cricoid cartilage.*

## Airways

- **Conducting Portions (Airways):** External nose, Nasal cavity, Nasopharynx, Oropharynx, Laryngopharynx, larynx, trachea, bronchi, bronchioles (terminal bronchiole, no cartilage)
- **Respiratory Portions:** Respiratory bronchioles, Alveolar ducts, Atria, Alveolar sacs

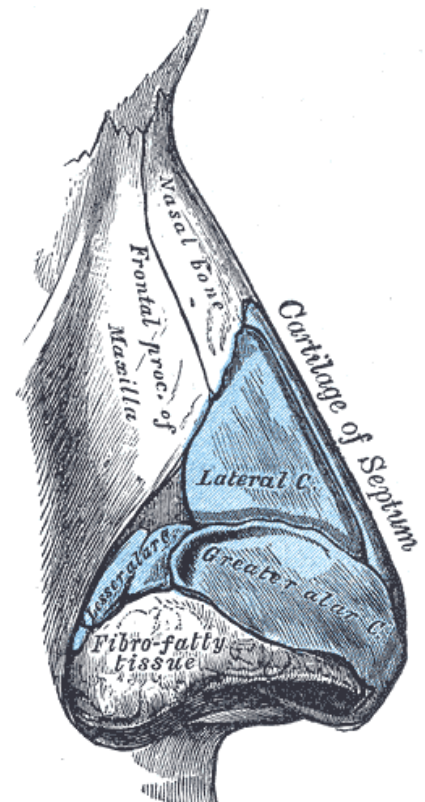
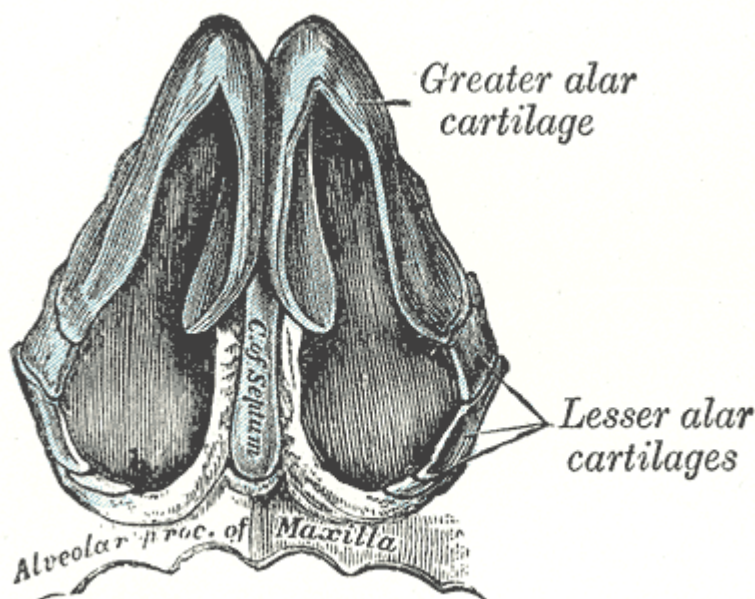
## The Nose

That portion of the respiratory pathway above the hard palate; includes both the **external nose** and the **nasal cavity**

### → External nose:

That part of the nose that protrudes from the face and made up of an osseocartilaginous framework consists of:

- **Root:** the upper least protruding portion of the external nose situated between the two orbits made of,
  - 2 *nasal bones*
  - 2 *frontal processes of maxilla* (framing the piriform aperture)
- **Dorsum:** the *external ridge* of the nose, looking forward and upward, a continuation of the 2 nasal bones joined with the triangular cartilaginous plate of the lateral process of septal nasal cartilage
- **Apex (tip):** anterior most pointed *end* of external nose, has a groove formed in the middle between the 2 major Alar cartilages
- **Ala of nose:** supporting framework, formed on each side by,
  - Large curved *major alar cartilages*, surrounding the nostrils with the lateral and medial cruses
  - 3 or 4 small *minor alar cartilages* (sesamoid)
  - *Fibroareolar* or *fibro-fatty* tissue



**The nostrils (nares)** are elliptical, forming the entrance to the right and left nasal cavity, in front of which lies the **Nasal vestibule** (the anterior part of the nasal cavity, especially that enclosed by cartilage)

The lumen of the nasal vestibule is lined with skin and short **brush-like hairs (Vibrissae)** of vestibule of nose. The skin over the external nose is rich with sebaceous glands.

Around the nose there are many subcutaneous mimetic muscles, whose fibers mainly insert on the skin of the ala of nose and **Nasolabial groove** (a furrow between the wing of the nose and the lip), these muscles also contribute to the dilation and constriction of the nostrils, and they are:

- *Procerus*: from membrane covering bridge of nose, assists the Frontalis
- *Nasalis*:
  - o Transverse part: forming an aponeurosis across the bridge of the nose; compresses the nostrils
  - o alar part: attaching to the wing of the nose; dilates the nostril
- *Levator labii superioris alaeque nasii*: elevates upper lip and wing of nose

**Arterial supply of the external nose:**

- *Angular artery*: the terminal branch of the facial artery, for muscles and skin of side of nose by lateral nasal branches
- *Dorsal nasal artery*: terminal branch of the ophthalmic artery, for the skin of side of root of nose
- *External nasal branches*: terminal branch of anterior ethmoidal artery, runs on the dorsum of nose
- *Infraorbital artery*: branch of 3<sup>rd</sup> part of maxillary artery, for the upper lip and tip of the nose

**Venous drainage of the external nose:**

- *Facial vein*
- *Superior ophthalmic vein*
  - o There is an **anastomosis** between these 2 veins, between the medial angle of eye and root of the nose. So in inflammation involving the lateral part of face and external nose, bacteria can reach the deep venous sinuses of the cranial cavity causing venous sinus thrombosis and meningitis

**Innervation of the external nose:**

- *Ophthalmic nerve, sensory* via Nasociliary (anterior ethmoidal → External nasal branches) and Infra and supratrochlear branches (Root and dorsum)
- *Maxillary nerve, sensory* via infraorbital branch (wings of nose)
- *Facial nerve, motor*, for mimetic muscles



### → **Nasal Cavity:**

The cavity divided by the nasal septum into right and left halves, lined with ciliated respiratory mucosa, extending from the external nostrils anteroinferiorly to the choana posteriorly to continue as the nasopharynx. Communicates with the paranasal sinuses through their orifices in the lateral wall

Each half of the nasal cavity has 4 walls:

**A. Superior wall:** the *roof* of nasal cavity, made by:

- *Nasal bone*, most anterior portion
- *Nasal part of frontal bone*
- *Cribriform plate of ethmoid bone*, central superior part, a horizontal lamina from which are suspended the labyrinth, on both sides, and the perpendicular plate in the center; it fits into the ethmoidal notch of the frontal bone and supports the olfactory lobes of the cerebrum, being pierced with numerous openings for the passage of the olfactory nerves.
- *Body of sphenoid bone*, most posterior part

**B. Inferior wall:** the *floor* of nasal cavity is the hard palate (See palate section in the digestive tract), made by:

- *Palatine process of maxilla* anteriorly, containing:
  - Median and transverse palatine suture
  - Nasal crest, on which the vomer inserts
  - Incisive canal and foramen, a Y-shaped passage which connects the two halves of the nasal cavity with the bony palate.
- *Horizontal plate of palatine bone*, containing:
  - Greater palatine foramen and canal
  - Lesser palatine foramina and canals
  - Posterior nasal spine
  - Median and transverse palatine suture
- *The Premaxilla or incisive bone:*
  - the anterior and inner portion of the maxilla, which in the fetus and sometimes in the adult is a separate bone; the incisive suture runs from the incisive canal between the lateral incisor and the canine tooth

**C. Lateral wall:** bony wall, formed by

- *Maxilla anteriorly* (frontal process)
- *Lacrimal bone*
- Parts of Ethmoid bone:
  - *Superior Nasal Concha* (+ *ethmoid air cells*)
  - *Middle Nasal Concha*
- *Inferior Nasal Concha* (separate bone)
- *Perpendicular process of Palatine bone* posteriorly (+ orbital & sphenoidal processes)

Each concha projects over a nasal meatus of the same name, into which paranasal sinuses and nasolacrimal ducts open:

- **Superior nasal meatus:** under the small, superior nasal concha, into which the *Posterior ethmoidal air cells* open

**\*\* *Sphenoethmoidal recess:*** a small cleft like pocket of the nasal cavity situated between the superior concha, nasal septum and body of sphenoid bone into which the sphenoid sinuses drain

Immediately below the sphenoethmoidal recess, there is the ***Sphenopalatine foramen*** bordered by the body of sphenoid bone superiorly and the sphenopalatine notch inferiorly (composed of 2 processes: Orbital and Sphenoidal)

- **Middle Nasal meatus:** below the large middle nasal concha, into which the frontal, maxillary and Anterior Ethmoidal air cells open  
The inferior portion of the ethmoid bone forms the ***uncinate process***, which projects into the middle nasal meatus and covers the orifice of the maxillary sinus. The middle Ethmoidal air cells form the ***Ethmoidal bulla*** that bulges over the uncinate process  
Between the Ethmoidal bulla and uncinate process we can find the ***semilunar hiatus***  
The middle nasal meatus has an anterior expanded portion, just above the nasal vestibule called: the *atrium of middle nasal meatus*.

There is a groove leading from atrium of middle nasal meatus to the olfactory area of nasal cavity called the *olfactory sulcus of nasal cavity*

**\*\* *Agger nasi (nasal ridge or Nasoturbinal concha):*** is an elevation on the lateral wall of the nasal cavity lying between the atrium of the middle meatus and the olfactory sulcus; it is formed by the mucous membrane covering the base of the *ethmoidal crest of the maxilla* (a ridge on the upper part of the nasal surface of the frontal process of the maxilla that gives attachment to the anterior portion of the middle nasal concha)

- **Inferior nasal meatus:** under the inferior nasal concha, into which the nasolacrimal duct open

**\*\*** The 3 nasal meatus open medially into the ***Common nasal meatus*** (space between the conchae and nasal septum)

***The nasal mucosa in the lateral wall has 3 parts:***

- 1) ***Nasal vestibule:*** anteriorly, forming the entrance to the nasal cavity proper. Lies within the nostrils and lined with skin (*stratified squamous keratinized epithelium*)
- 2) ***Respiratory region:*** reflects the bony relief pattern of the lateral nasal wall, the mucosa is covered by pseudostratified ciliated columnar epithelium (respiratory epithelium) and contains many nasal glands

**\*\*** The *border* between the nasal vestibule and respiratory region is a curved ridge called: ***Limen Nasi***

- 3) ***Olfactory region:*** circumscribed area on the lateral nasal wall above the superior nasal concha (upper 1/3)

**Arterial supply of the lateral wall of nasal cavity:**

- *Anterior Ethmoidal artery*: branch of ophthalmic artery, supplies the anterior Ethmoidal air cells, anterior upper part of nasal mucous membrane, and skin of dorsum of nose.
- *Posterior Ethmoidal artery*: branch of ophthalmic artery, supplies the posterior ethmoidal cells and upper posterior part of lateral wall of nasal cavity.
- *Sphenopalatine artery*: branch of 3<sup>rd</sup> part of maxillary artery, supplies and inferior posterior portion of lateral nasal wall and septum (lateral posterior superior branches)
- *Lateral nasal branches of facial artery*: supply the vestibule of nose
- *Alar branches of superior labial artery*
- Perforating branches of Greater palatine and anterior superior alveolar

**Venous drainage of the lateral wall of nasal cavity**, has same course and name as arteries, drain into:

- Ophthalmic vein via Ethmoidal foramina
- Pterygoid venous plexus via sphenopalatine foramen
- Facial vein via nasal vestibule

**Innervation of the lateral wall of nasal cavity:**

- 1) *Ophthalmic nerve*: anterior and superior portion
  - *Anterior Ethmoidal nerve* (giving the internal nasal branches)
  - *Posterior Ethmoidal nerve*
- 2) *Maxillary nerve*: Inferior and posterior portion
  - *Posterior Superior lateral nasal branches*: branches of pterygopalatine ganglion to upper posterior part of lateral wall of nasal cavity, including superior and middle nasal concha/meatuses, and posterior Ethmoidal sinuses.
  - *Posterior inferior nasal branches*: branches of greater palatine nerve to posterior inferior lateral wall of nasal cavity, including posterior aspect of mucosa over posterior portion of inferior nasal concha and meatus
  - *Infraorbital nerve*: gives the internal nasal branches for the most anterior and inferior portion of the nasal septum and lateral wall
  - *Anterior superior alveolar branches*: of infraorbital nerve, supply the anterior inferior portion of the lateral wall

**D. Medial wall:** the *nasal septum*, the wall dividing the nasal cavity into halves; it is composed of a central supporting skeleton covered on each side by a mucous membrane. Composed of 2 parts:

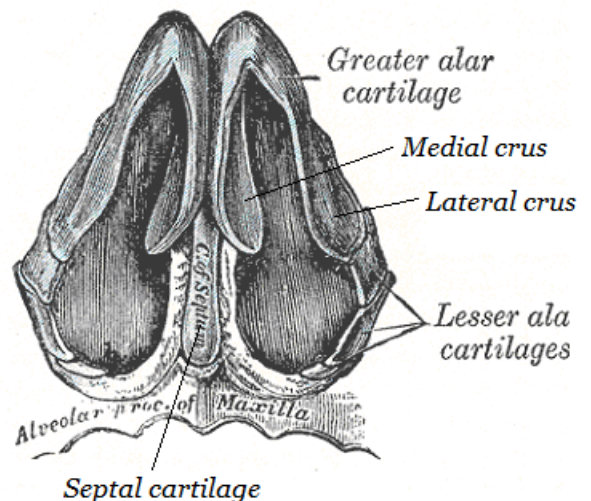
- **Bony part:** its posterior and inferior portion, sub composed of:
  - *Perpendicular plate of ethmoid bone*, inserting on the roof of nasal cavity sagittally, forming the upper part of the bony septum
  - *Vomer*, trapezoidal shaped bone articulating with the anteroinferior part of the perpendicular plate of ethmoid, and body of sphenoid bone and inserting caudally on the nasal crest of the nasal cavity formed by the maxillae, and two palatine bones  
The free posterior margin of the vomer forms the medial boundary of the Choana
  - *Nasal crest of maxilla* + (anterior nasal spine)
  - *Nasal crest of palatine bone* + (posterior nasal spine)

- **Cartilaginous parts:** a thin cartilaginous plate located between vomer, perpendicular plate of the ethmoid, and nasal bones, and completing the nasal septum anteriorly. It has a long posterior process that is inserted in the gap between the 2 thin bony plates (vomer and perpendicular plate of ethmoid).

At the dorsum of nose, the cartilaginous septum contributes to the formation of the external nose with a T-shaped lateral process, and joined inferiorly by the medial crus of the Major alar cartilage

**Vomeronasal cartilage:** a narrow strip of cartilage located between the lower edge of the cartilage of the nasal septum and the vomer.

**\*\* Vomeronasal organ of Jacobson:** a fine vestigial horizontal canal, ending in a blind pouch, in the mucous membrane of the nasal septum, beginning just behind and above the incisive duct; a structure that usually regresses after the 6th month of gestation. In many lower animals, it functions as an accessory olfactory organ.



#### **The mucosa of the nasal septum:**

- 1) **Respiratory region:** the mucosa is covered by pseudostratified ciliated columnar epithelium (respiratory epithelium) and contains many nasal glands
- 2) **Olfactory region:** in the upper part of the nasal septum where it meets the Cribriform plate

#### **Arterial supply of the nasal septum:**

- **Anterior Ethmoidal artery:** branch of ophthalmic artery, supplies the anterior Ethmoidal air cells, anterior upper part of nasal mucous membrane, and skin of dorsum of nose.
- **Posterior Ethmoidal artery:** branch of ophthalmic artery, supplies the posterior ethmoidal cells and upper posterior part of lateral wall of nasal cavity.
- **Sphenopalatine artery:** branch of 3<sup>rd</sup> part of maxillary artery, supplies and inferior posterior portion of lateral nasal wall and septum. Travels on the septum until reaching the incisive canal, where it anastomosis with the ascending branch of greater palatine artery
- **Superior Labial artery:** branch of facial, gives the nasal septal branch for the anteroinferior aspect of the septum

**\*\* Kiesselbach (Little's) area:** a well developed cavernous area on the anterior portion of the nasal septum rich in capillaries (Kiesselbach plexus) and often the site of epistaxis

**Venous drainage of the nasal septum,** has same course and name as arteries, drain into:

- Ophthalmic vein via Ethmoidal foramina
- Pterygoid venous plexus via sphenopalatine foramen
- Facial vein via nasal vestibule

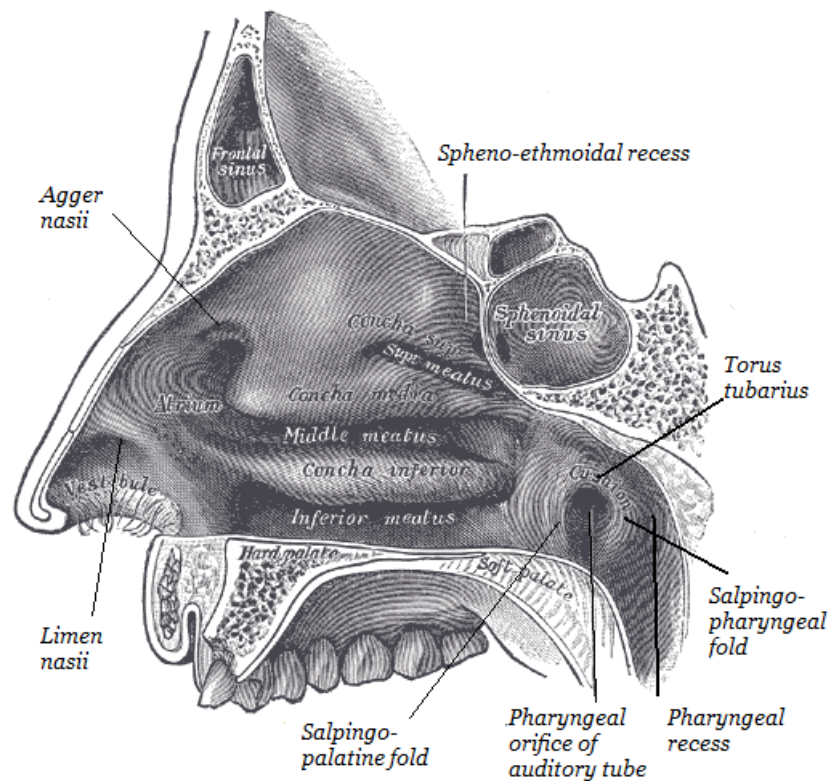


***Innervation of the nasal septum:***

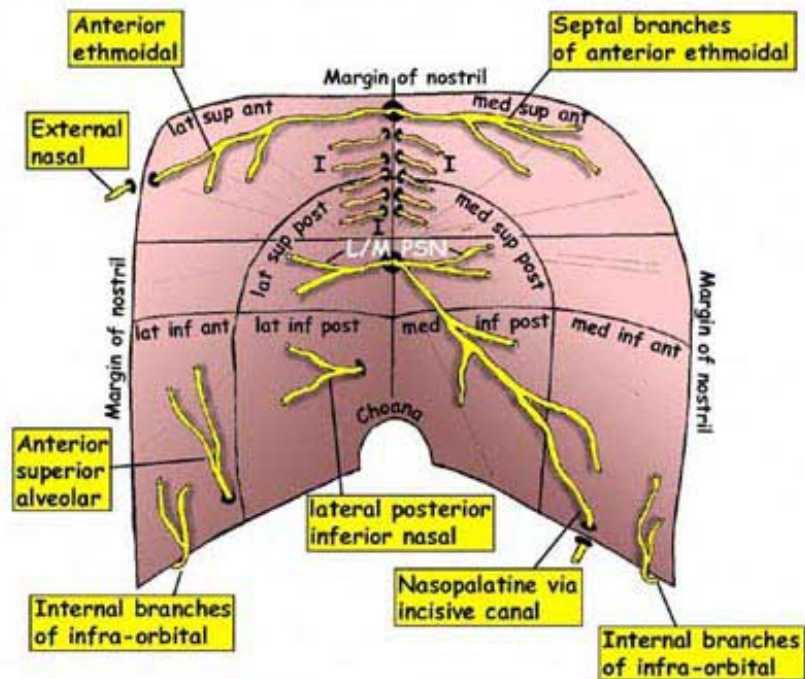
- 1) *Ophthalmic nerve*: anterior and superior portion
  - Anterior Ethmoidal nerve (giving the internal nasal and septal branches)
  - Posterior Ethmoidal nerve
- 2) *Maxillary nerve*: Inferior and posterior portion
  - ***Nasopalatine nerve of Scarpa***: a branch from the pterygopalatine ganglion, passing through the sphenopalatine foramen, crossing to and then down the nasal septum, reaching through the incisive foramen to anastomose with the greater palatine nerve and supply the nasal septum and mucosa of the hard palate
  - *Posterior Superior medial nasal branches*: branches of maxillary nerve or nasopalatine nerve to upper posterior part of lateral wall of nasal cavity, including superior and middle nasal concha/meatuses, and posterior Ethmoidal sinuses
  - *Infraorbital nerve*: Give the internal nasal branches for the most anterior and lower part of septum and lateral wall
- 3) *Olfactory nerves*: concerned with smell, arising from cells in the olfactory epithelium in the superior part of the lateral and septal walls of the nasal cavity. The central processes of these cells (forming the olfactory nerve CN-I) pass through the numerous holes in the cribriform plate and end the olfactory bulb, the rostral expansion of the olfactory tract

***Lymphatic drainage of the nose:***

- *Anterior part of nose*: drains into the Submandibular and Superficial lymph nodes of the anterior neck
- *Posterior part of nose*: drains into the retropharyngeal and deep cervical lymph nodes

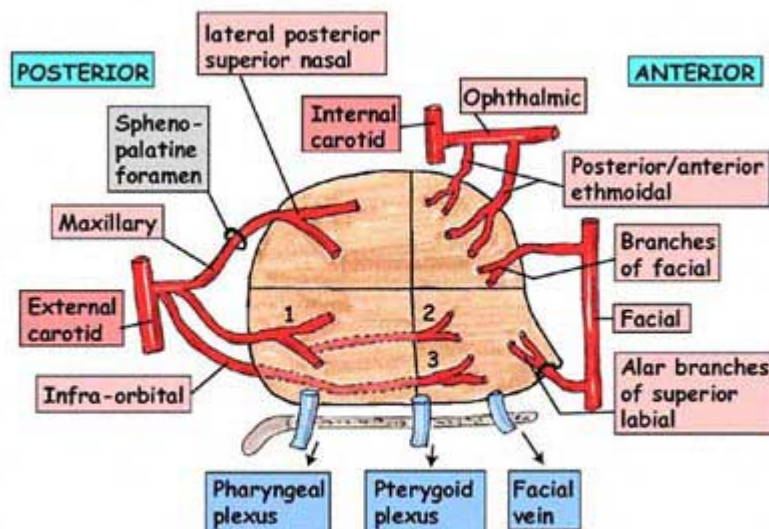


### NASAL CAVITY - NERVE SUPPLY VIEW UP RIGHT NOSTRIL



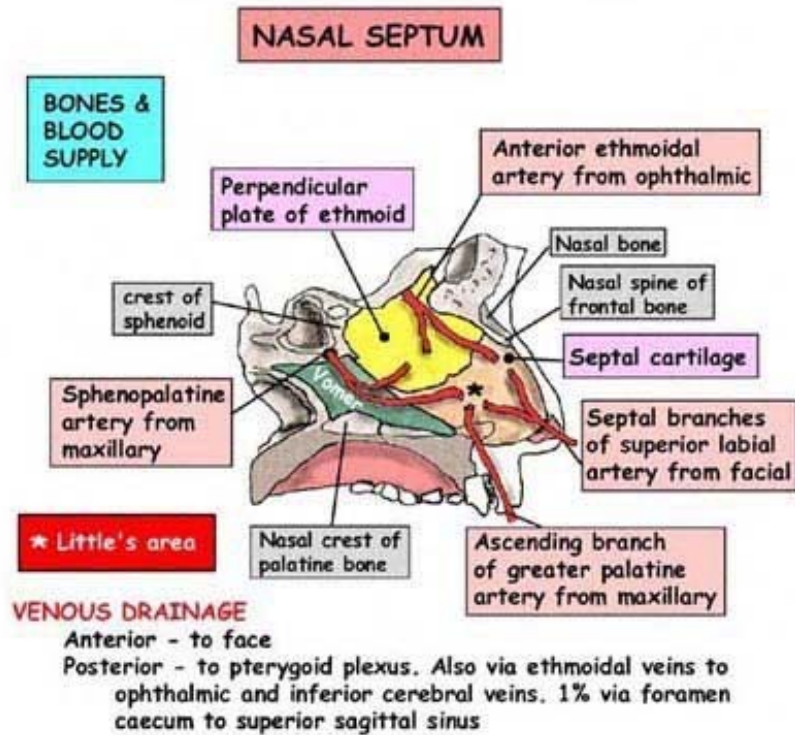
L/M PSN = Lateral & medial posterior superior nasal, from nasopalatine, from Vb, from pterygopalatine ganglion  
 Anterior ethmoidal from nasociliary, from Va  
 Lateral posterior inferior nasal, from greater palatine, from Vb, from pterygopalatine ganglion  
 Nasopalatine from Vb, from pterygopalatine ganglion  
 Anterior superior alveolar, from infra-orbital, from Vb  
 Infra-orbital, from Vb

### BLOOD SUPPLY OF LATERAL WALL OF NOSE

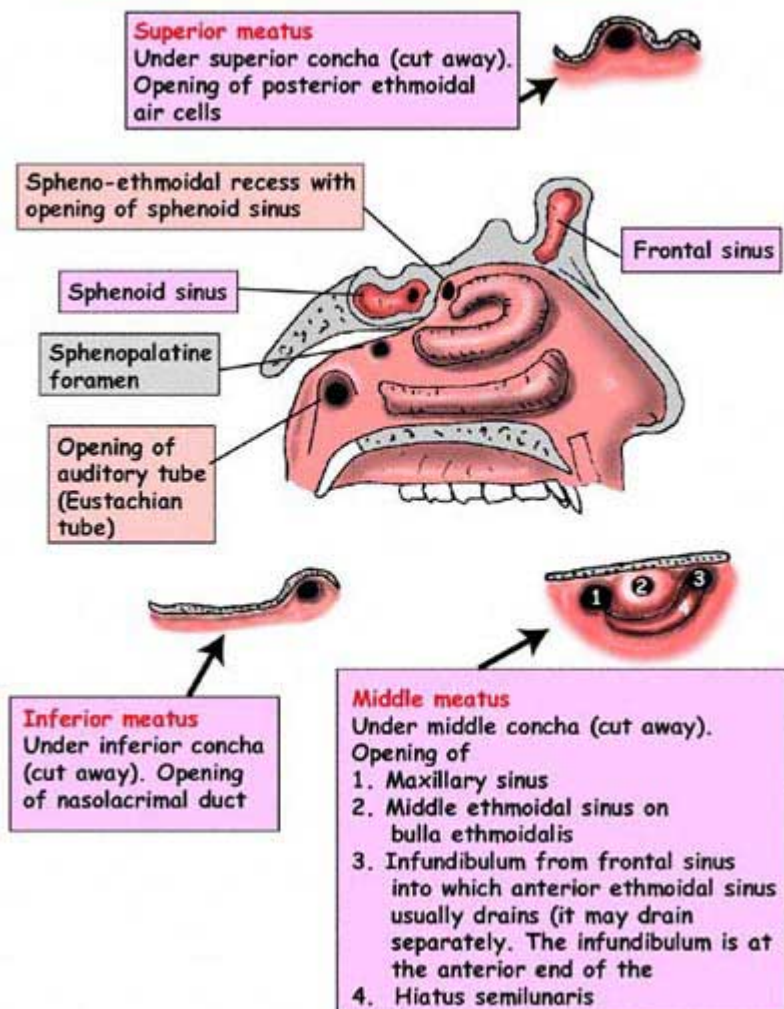


1. Branch of greater palatine
2. Perforating branches of greater palatine
3. Anterior superior alveolar from infra-orbital





**LATERAL WALL OF LEFT NASAL CAVITY**



## ***Paranasal sinuses***

These sinuses are air filled extensions of the respiratory part of the nasal cavity. They are named according to the bones in which they are located and lined by mucous membrane continuous with that of the nasal cavity, there are 4 sinuses:

### **1. Frontal sinus:**

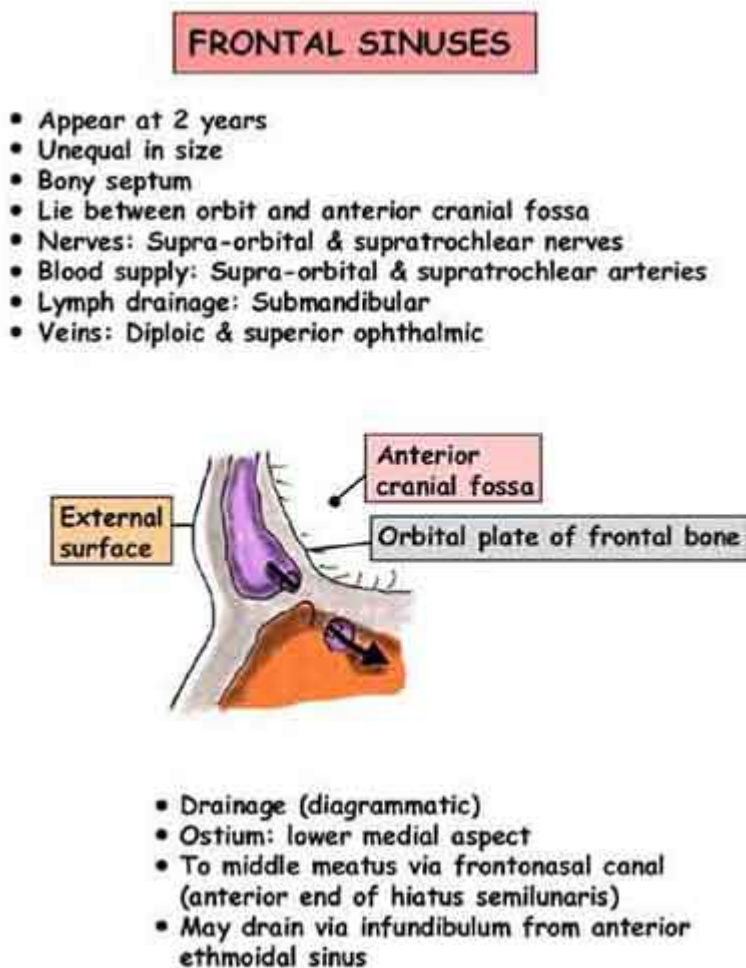
Located between the outer and inner tables of the frontal bone, posterior to the *superciliary arches*, the sinuses can be seen by the age of 7. The right and left sinuses are rarely equal in size in the same person, and the *septum* between them usually is not found in the median plane.

The sinus often has 2 parts:

- *Vertical*: found in the squamous part of the frontal bone
- *Horizontal*: found in the orbital part of the frontal bone

When the supraorbital part is large, its roof forms the floor of the anterior cranial fossa and its floor forms the roof of the orbit.

The frontal sinuses vary in size from as small as 5 millimeters to large spaces extending laterally to the greater wing of the sphenoid bone. The sinus may have separate **frontonasal ducts** that drain inferiorly to the *infundibulum* of the *middle nasal meatus*, which drains to the *semilunar hiatus*.

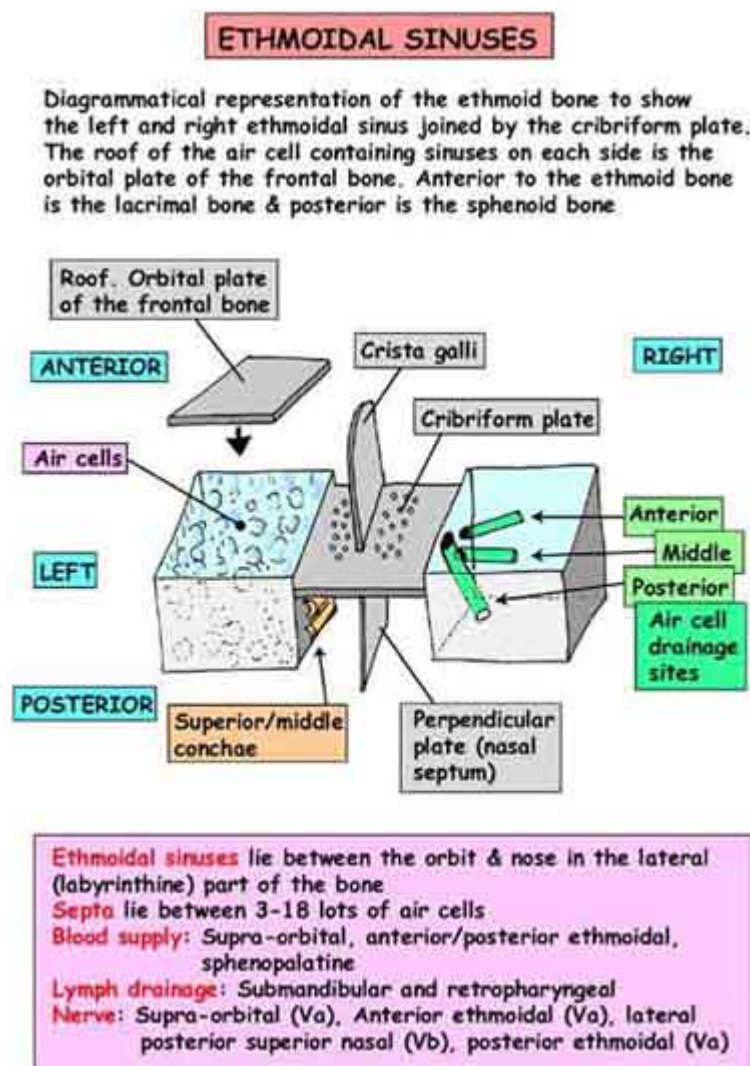


## 2. **Ethmoidal sinuses:**

These sinuses comprise several small cavities called Ethmoidal cells that are found within the *ethmoidal labyrinth* of the lateral mass of the ethmoid bone. The cells form the ethmoid labyrinth which is located between the nasal cavity and the orbit. The number of cells varies between 3-18. The cells are larger when the number is small. Medially they are separated by the orbit by a thin paper plate of the ethmoid bone, *Lamina Papyracea*

For purposes of description the cells are divided into *anterior, middle and posterior groups*. The middle cells, or "*bullar cells*", because they form the **Ethmoidal bulla**, a swelling on the superior border of the semilunar hiatus.

According to their variable location, the cells can open into the *superior nasal meatus (posterior group) or the middle nasal meatus (anterior group)*



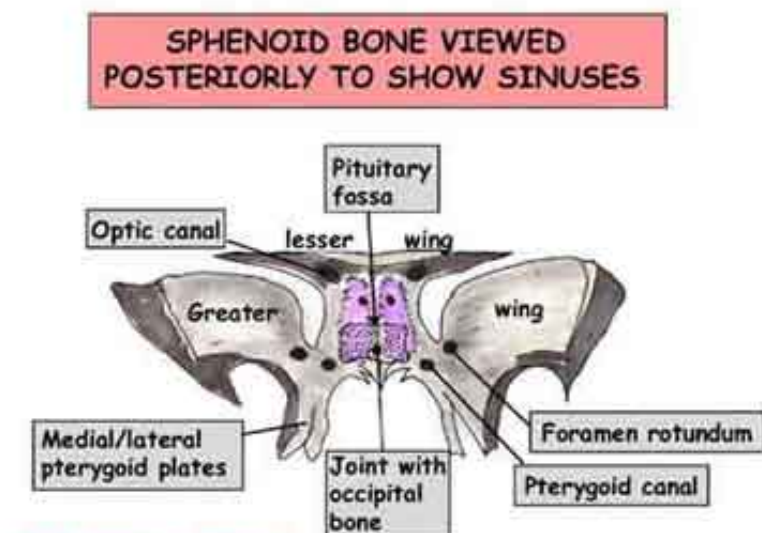


### 3. *Sphenoidal sinuses:*

Found *in the body* of the sphenoid bone. They occupy a variable amount of this bone and may extend to the *wings*. The Sphenoidal sinuses are located posterior to the superior part of the nasal cavity, bordered anteriorly by the *ethmoidal air cells*. The 2 sinuses are separated by a bony *septum* that is usually not found in the median plane. Because of these sinuses the body of the sphenoid bone is fragile hollow structure. Only thin plates of bone separates the sinuses from several important structures:

- Optic nerve in the canal (*anteriorlysuperiorly*)
- Optic chiasm (*superiorly*)
- Hypophysial fossa with pituitary gland (*posteriorly*)
- Internal carotid arteries in the carotid sulcus and cavernous and intercavernous sinuses and their contents (*laterally*)
- Roof of nasopharynx, Pterygoid canal, Palatovaginal canal (*inferiorly*)

The sphenoid sinus drains into the *sphenoethmoidal recess*



#### SPHENOIDAL SINUSES

Paired in body of sphenoid

**Septum:** Asymmetrical

**If small:** Then anterior to pituitary fossa

**If large:** Then beneath pituitary fossa, extending posteriorly to basi-occiput and laterally into greater wing

**Ostium:** In anterior wall opening into spheno-ethmoidal recess

**Laterally:** Cavernous sinuses, internal carotid artery and maxillary nerve

**Posteriorly:** Posterior cranial fossa and pons

**Inferiorly:** Roof of nasopharynx, nerve of pterygoid canal and palatovaginal canal (containing pharyngeal branch of Vb)

**Walls:** Indented by pterygoid & palatovaginal canals, internal carotid artery and maxillary nerve (Vb)

**Nerve supply:** Posterior ethmoidal (Va) & branches of pterygopalatine ganglion

**Blood supply:** Posterior ethmoidal & sphenopalatine branches of maxillary artery

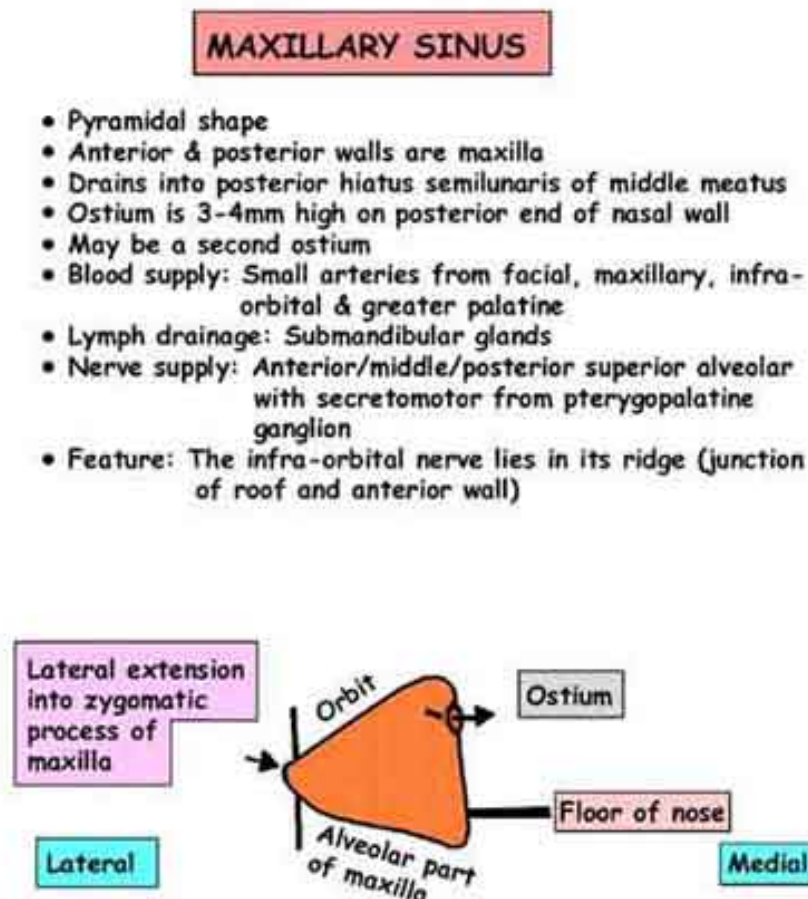
**Lymph drainage:** Retropharyngeal

#### 4. Maxillary sinuses (*Antrum of Highmore*):

This is the *largest* pair of paranasal sinuses. They are *pyramidal-shaped* cavities that may occupy the entire bodies of the maxillae. The **apex** of the sinus may extend as far as the *zygomatic* bone; the **base** forms the inferior part of the *lateral wall of the nasal cavity*. The *roof* is formed by the *floor of the orbit*, and its narrow *floor* is formed by the *alveolar process of the maxilla* (dental arch). *Anterolaterally* it's bounded by the *Facial surface* of maxilla, and *posteriorly* by the *maxillary tuberosity*

The roots of the maxillary teeth, especially the first 2 molars often produce conical elevations in the floor of the maxillary sinus. The lowest point of the sinuses is between the *1<sup>st</sup> premolar tooth and molars*.

The maxillary sinus drains into the *middle nasal meatus* through the hiatus semilunar by an aperture in the superior part of its base. Because of the location of this opening, it's impossible for fluid in the sinus to drain when the head is erect until the sinus is nearly full.



### → *Posterior nasal apertures (Choanae)*

Both nasal cavities open into the nasopharynx through posterior nasal aperture each, or the *choana*. The *margins* or *borders* of the choana are:

- **Superior border:** formed by the body of sphenoid bone, continuous superolaterally with the medial plate of the Pterygoid process (penetrated by the Pterygoid canal)
- **Medial border:** formed by the vomer, which has an Ala inserted superiorly on the roof of the choana, and inferiorly with the posterior nasal spine of palatine bone
- **Inferior border:** formed by the horizontal plate of the palatine bone
- **Lateral border:** perpendicular plate of palatine bone

From a posterior view of the choana, we can *visualize* the *inferior and middle conchae*, as well as the *Ethmoidal bulla and Uncinate process*

\*\* (The following brief description of pharynx is taken from the pharynx section of the GI tract, more details; you should read that section above about the pharynx)

**The pharynx** is the continuation of the GI tract after the oral cavity. It's a *fibromuscular* tube that is used as a pathway for both food and air.

The Pharynx is divided to 3 parts:

- An upper **Nasopharynx** that receives the posterior openings of the nasal cavity, the 2 Choana.
- Middle **Oropharynx**, behind the oral cavity (after the isthmus of fauces)
- A lower **Laryngopharynx**, behind the Larynx.

In the respiratory tract, the air will usually go from the choana into the nasopharynx into the Oropharynx and then into the larynx

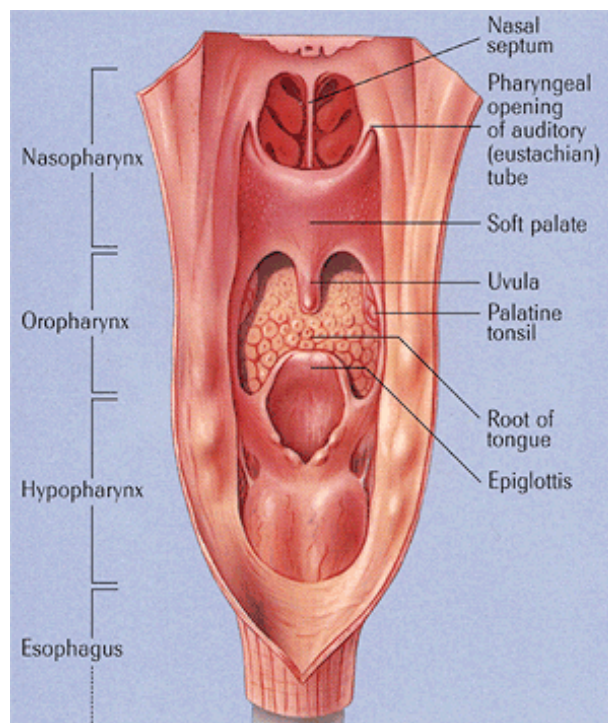


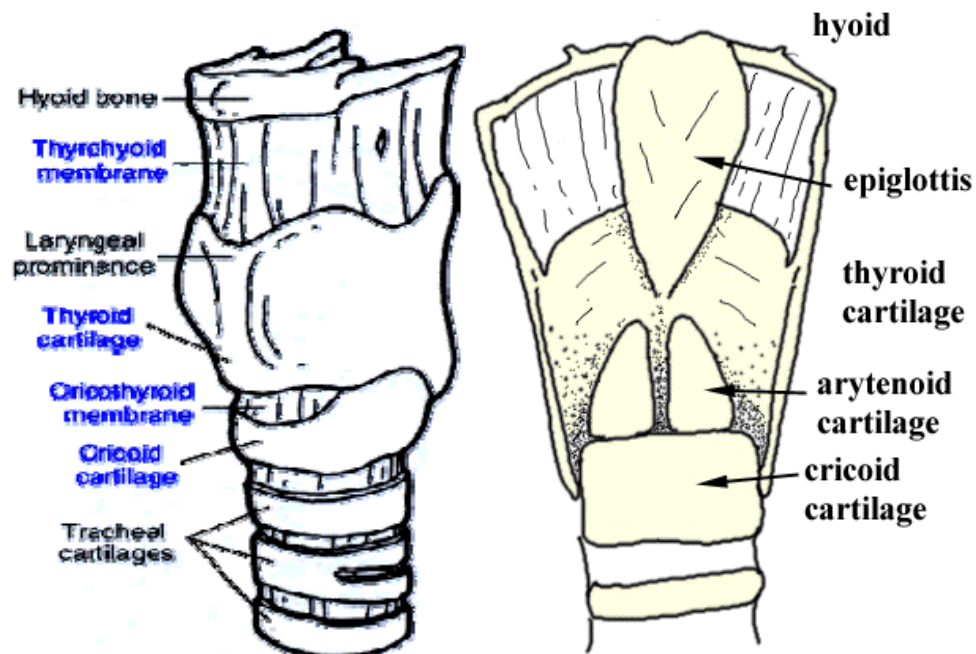
Image made available by a generous grant from Bristol-Myers Squibb

## The Larynx

The organ of voice production; the part of the respiratory tract between the pharynx and the trachea; it consists of a framework of cartilages and elastic membranes housing the vocal folds and the muscles which control the position and tension of these elements.

### → Cartilages, Skeleton of the larynx:

1. **Thyroid cartilage:** Hyaline type of cartilage that has 2 lamina, forming anteriorly the *superior thyroid notch* (Adam's apple) has an oblique line that divides the lamina into anterior and posterior facet. Has a *superior & inferior horn*. The *inferior* one articulates with the cricoid cartilage as a *Cricothyroid joint* (*Hinge type*, 1 transverse axis). Superior horn articulates with the hyoid bone.
2. **Cricoid cartilage:** ring shaped, with the lamina laying posteriorly, on the upper edge of lamina 2 articular surfaces for *Arytenoid* cartilage. On the lateral surface articular surface for the inferior horn of thyroid cartilage. Hyaline type of cartilage.
3. **Arytenoid cartilage:** sits on the upper lamina of cricoid cartilage, and articulates with the cricoid cartilage via the *Cricoarytenoid joint* (Pivot type with 1 vertical axis). It is an Elastic type of cartilage. Has 3 process:
  - *Anterior process (vocal process):* from this process arise the vocal cords
  - *Lateral process (muscular process):* Where Posterior and lateral cricoarytenoid muscles are inserted.
  - *Posterior process:* Is the apex of the pyramidal shaped cartilage. It carries the *corniculate cartilage* (of Santorini).
4. **Epiglottis cartilage:** lies against the internal surface of thyroid cartilage, elastic type of cartilage. Has the Stem (*petiolus*) that forms the *Epiglottic tubercle* on the mucosa and an *oval lamina* which is concave posteriorly and perforated by the glands.





→ **Ligaments of the larynx:**

**A) Extrinsic (Outer) ligaments:** fix the larynx between the hyoid bone and trachea.

**1. Thyrohyoid membrane:** attached from the superior border and the superior horns of the thyroid cartilage to the posterior surface of the body and greater horn of the hyoid bone. The outer surface of this membrane is covered by the *Infrahyoid* muscles. The middle part of the membrane is thicker and forms the **median thyrohyoid ligament**, which is pierced on each side by the *superior laryngeal artery and vein* and the *internal laryngeal nerve*. The tips of the superior horns of the thyroid cartilage are connected to the greater horns of the hyoid bone by the **lateral thyrohyoid ligaments**, which are also thickenings of the thyrohyoid membrane. A small cartilage named **triticeal** cartilage lies in this ligament.

**2. Thyroepiglottic ligament:** Attaches the anterior lower part of the epiglottis to the thyroid cartilage.

**3. Hyoepiglottic ligament:** Attaches the anterior upper part of the epiglottis to the hyoid bone.

**4. Cricotracheal ligament:** The inferior border of the cricoid cartilage is joined to the first ring of the trachea by this ligament

**B) Intrinsic (Inner) Ligaments:** fibroelastic membranes that joins the different parts of laryngeal skeleton.

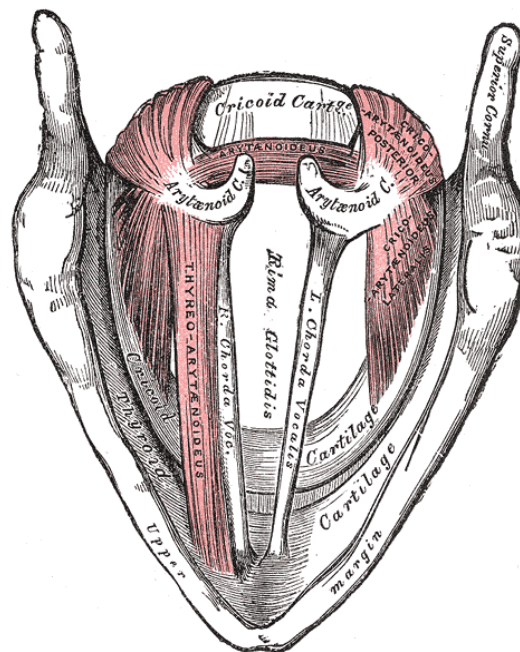
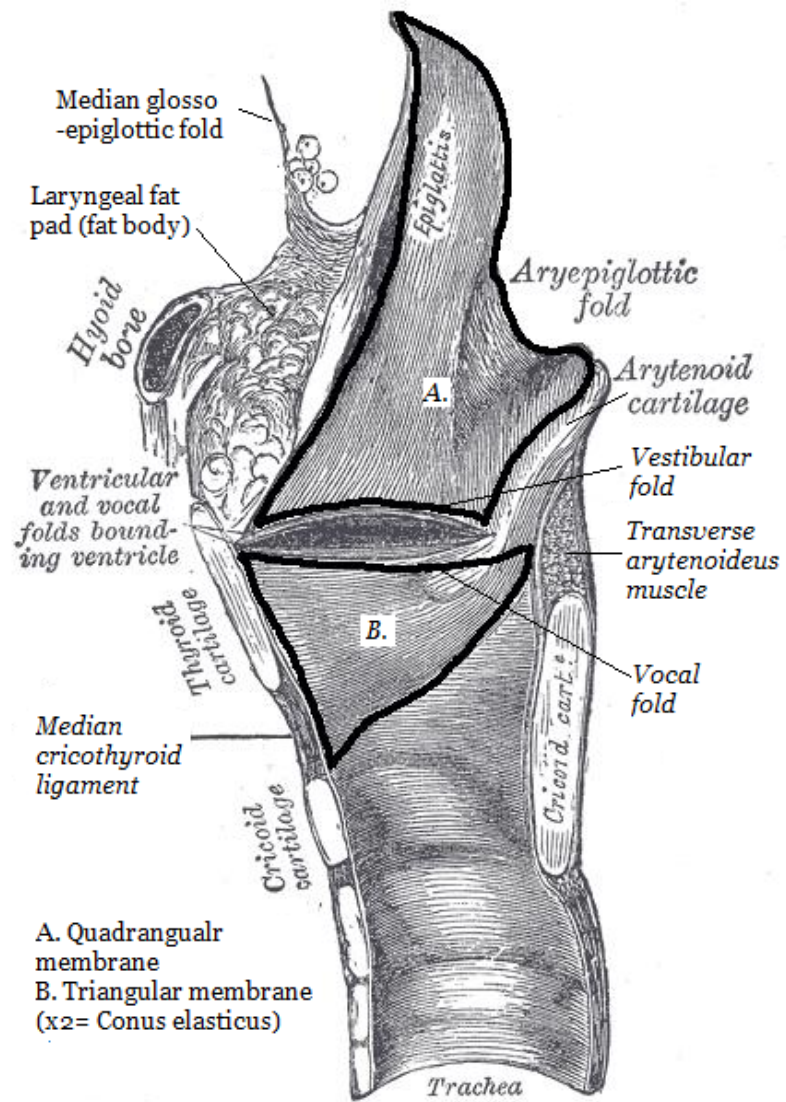
**1. Quadrangular membrane:** It extends from the lateral margins of the epiglottis within the *aryepiglottic fold* and attaches to the *Arytenoid* and *corniculate* cartilages. The *inferior* free edge is thickened to form the **vestibular ligament (false vocal cord)**. The *superior* edge is also **free** and it is covered with *aryepiglottic fold of mucosa*. The *anterior* fixed border is the lateral edge of **epiglottis**, and the *posterior* fixed border is the *Arytenoid* cartilages

**2. Triangular membrane:** This membrane arises from the inner surface of the cricoid arch and it consists of:

- The *anterior* or *superficial* part or border, also called the **median Cricothyroid ligament (Coniccate ligament)**, extends from the upper border of the anterior part of cricoid cartilage to the lower border of the thyroid cartilage
- The *lateral* part of this membrane extends from the superior inner border of the *cricoid cartilage* to the inner surface of the thyroid angle and posteriorly to the tip of the vocal process of the *Arytenoid* cartilage.
- The *superior edge* is free and thickened between its two attachments, the angle of the thyroid lamina and vocal process of the *Arytenoid* cartilage, and forms the **vocal ligament (true vocal cord)**.

→ The 2 triangular membranes on both sides form together the **Conus elasticus (cricovocal membrane)**, or the **Cricothyroid membrane** (extending between the arch of the cricoid cartilage and the inferior edge of the thyroid lamina on each side of the midline, which is occupied by the thicker median Cricothyroid ligament)

→ The Conus elasticus and the 2 vocal cords form the elastic wall of the tone producing mechanism, Lip Pipe



→ **Muscles of the Larynx:**

1. **Cricothyroideus:** originates from the anterior surface of the arch of the cricoid, inserted either on the wing of the thyroid on inferior horn of the thyroid.

*Nerve supply:* superior laryngeal nerve (the only laryngeal muscle innervated by it).

*Action:* tense the vocal folds by tilting the thyroid cartilage via the Cricothyroid joint.

Controls the rough tension of the voice

2. **Lateral cricoarytenoid:** originates from the upper margin of the arch of the cricoid cartilage, inserted to the muscular process of the Arytenoid cartilage.

*Nerve supply:* recurrent laryngeal nerve.

*Action:* narrows the Rima glottides

3. **Posterior cricoarytenoid:** originates from both sides of the posterior surface of cricoid, inserted on the muscular process of the Arytenoid.

*Nerve supply:* recurrent laryngeal nerve.

*Action:* widens the Rima glottides. It is the antagonist for lateral cricoarytenoid, and the **only** muscle that opens (widens the Rima glottides)

4. **Vocalis:** originates from posterior surface of thyroid cartilage, runs within the vocal fold and inserted into the vocal process of the Arytenoid.

*Nerve supply:* recurrent laryngeal nerve.

*Action:* shortens and relaxes the vocal cords. (Control the soft tension of voice).

5. **Thyroarytenoid:** originates from the inner surface of the thyroid cartilage (together with Vocalis muscles). Inserted into the muscular process and the outer surface of the Arytenoid

*Nerve supply:* recurrent laryngeal nerve.

*Action:* constrict the Rima glottides.

6. **Thyroepiglottic:** originates from the inner surface of the thyroid cartilage (in common with the Thyroarytenoid muscle), inserted into the epiglottis.

*Nerve supply:* recurrent laryngeal nerve.

*Action:* depresses the base of the epiglottis, constricting the entrance into larynx.

7. **Oblique and transverse Arytenoid (Inter-Arytenoid):** the fibers crossing each other between the posterior surfaces of the 2 Arytenoid cartilages.

*Nerve supply:* recurrent laryngeal nerve.

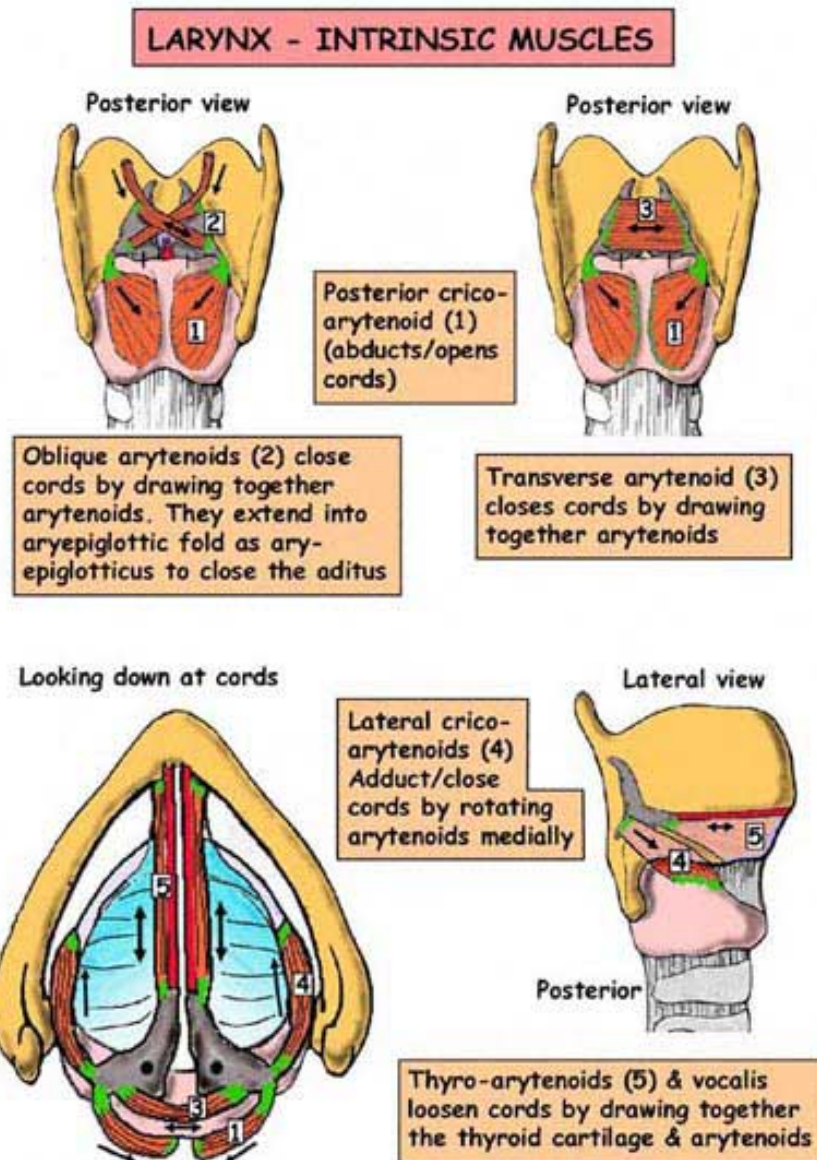
*Action:* narrows the Rima glottides.

8. **Aryepiglottic:** A band of fibers of the oblique Arytenoid muscle that extend to the side of the epiglottis.

*Nerve supply:* recurrent laryngeal nerve

*Action:* lowering the epiglottis therefore assists in narrowing the entrance to larynx.

→ The aryepiglottic fold envelops the **corniculate cartilage (of Santorini)** and the inconstant **Cuneiform cartilage (of Wrisberg)**. Both are visible through the mucosa and produces *tubercles* with similar names



→ **Mucous membranes & spaces of larynx:**

Most parts of larynx are covered by mucous membranes, mucosal folds that can be seen:

- **Aryepiglottic folds:** runs from both sides of epiglottis to Arytenoid cartilage.
- **Piriform recess:** between aryepiglottic folds to thyroid cartilage. It diverts food away from the entrance to larynx.
- **Plica Nervi laryngei (Laryngeal nerve folds):** runs across the piriform recess, seen in the mucosa. Formed by the underlying *superior laryngeal vessels* and nerve.
- **Vestibular folds (ventricular folds), or false vocal cord:** formed by some muscle fibrils and the tip of *quadrangular membrane* (lower margin).
- **True vocal folds or cords:** formed by Vocalis muscle and vocal ligament.



- **Laryngeal Aditus or inlet:** the aperture between the pharynx and larynx, bounded by the superior edges of the epiglottis (anteriorly), the aryepiglottic folds (laterally), and the mucosa between the arytenoids (posteriorly)
- **Vestibule of larynx:** the space from laryngeal Aditus until the vestibular folds or rima vestibuli. Its anterior wall is formed by epiglottis; the lateral wall is the quadrangular membranes and posteriorly the mucosa over the arytenoids.
- **Rima vestibuli:** the interval between the false vocal cords or vestibular folds
- **Rima glottides:** the interval between the true vocal folds, subdivided into:
  - *Intermembranous part (glottis vocalis):* the portion of the opening anterior to the vocal processes of the arytenoid cartilages bounded by the vocal ligaments; this portion is closed by contraction of the lateral cricoarytenoid (muscle) only during whispering
  - *Intercartilaginous part (glottis respiratoria):* the opening between the vocal processes of the arytenoid cartilages; this part is open during whispering and is closed during phonation and the Valsalva maneuver.
- **Laryngeal ventricle:** the space between the vestibular folds to true vocal folds. Has lateral extension called *laryngeal saccule*
- The space below vocal cords is called **infraglottic cavity**

#### → Epithelium

*Epiglottis:* is covered by stratified squamous epithelium on its anterior portion (continuous with posterior surface of tongue). The posterior surface has a *ciliated, pseudostratified columnar epithelium* containing *seromucous* glands. This epith' continuous over vestibule of larynx, false vocal cords and ventricle

**True vocal cords:** covered by *stratified squamous non keratinized epith'*.

#### → Innervation:

**1) Superior laryngeal nerve:** (is a direct branch of Vagus) It divides into internal and external branches. The internal branch pierces the thyrohyoid membrane to supply innervation to the mucosa above the vocal folds; the external branch travels with the superior thyroid artery and supplies the *Cricothyroid muscle*.

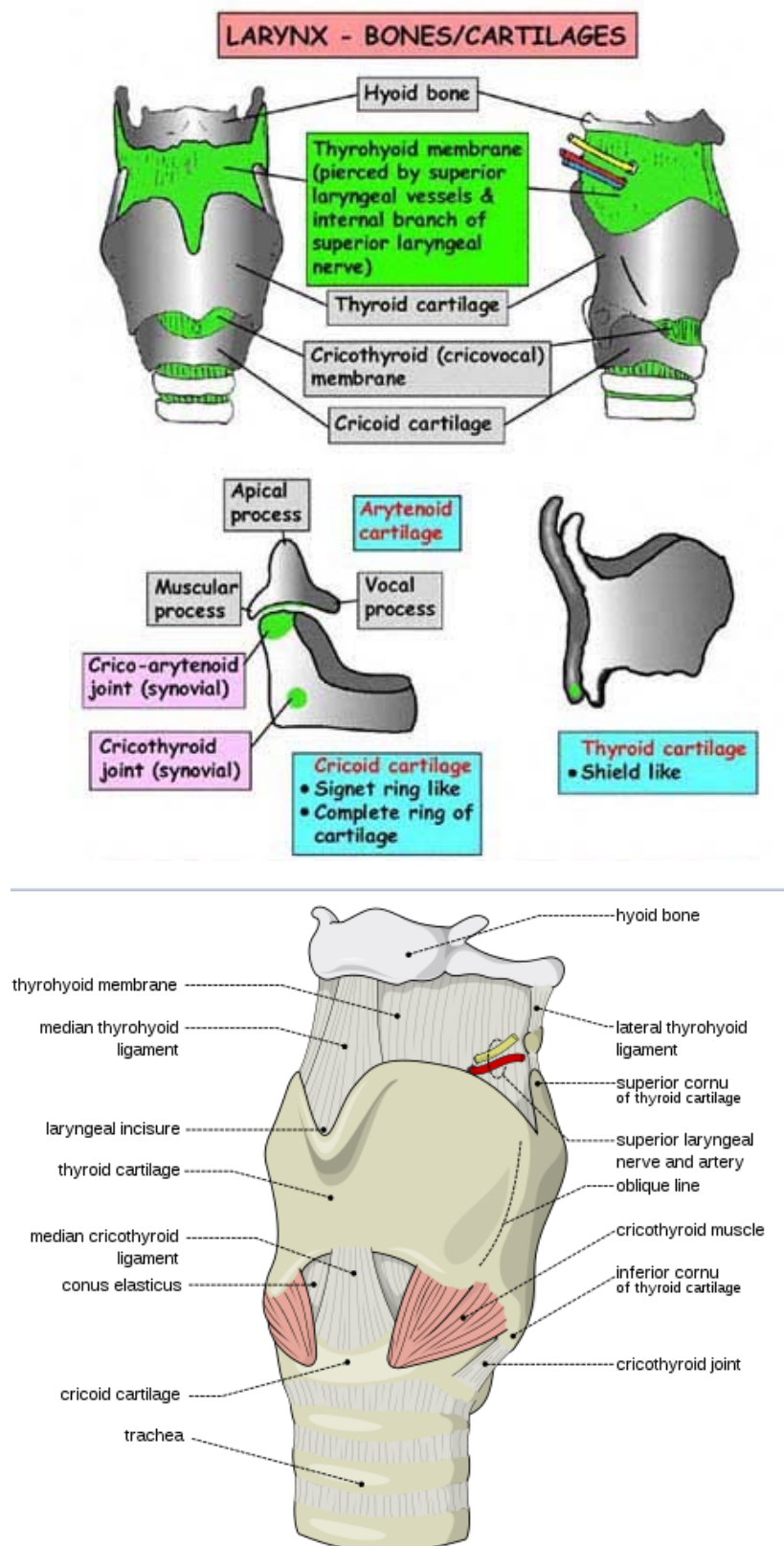
**2) Inferior laryngeal nerve:** (is a terminal branch of recurrent laryngeal) innervates the mucosa of the larynx below the vocal folds, and the upper trachea, in addition to providing motor innervation to all of the intrinsic muscles of the larynx *except Cricothyroid*.

#### → Blood supply:

The arterial blood supply of the larynx is derived from the laryngeal branches of superior and inferior thyroid arteries. Most of the arteries anastomose freely with each other.

**1) Superior laryngeal artery:** It is a branch of the superior thyroid artery, which comes off of the external carotid artery. It enters the larynx through the thyrohyoid membrane, along with the internal branch of the superior laryngeal nerve, and supplies the muscles and mucous membranes in the superior portion of the larynx.

**2) Inferior laryngeal artery:** It is a branch of the inferior thyroid artery, which comes off of the thyrocervical trunk of the subclavian artery. It ascends on the trachea, together with the recurrent laryngeal nerve, and enters the larynx beneath the lower border of the inferior constrictor muscle. It supplies the muscles and mucous membranes in the lower part of the larynx.

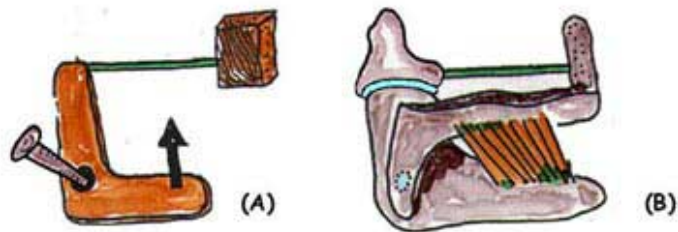


## LARYNX - CRICOTHYROID MUSCLE

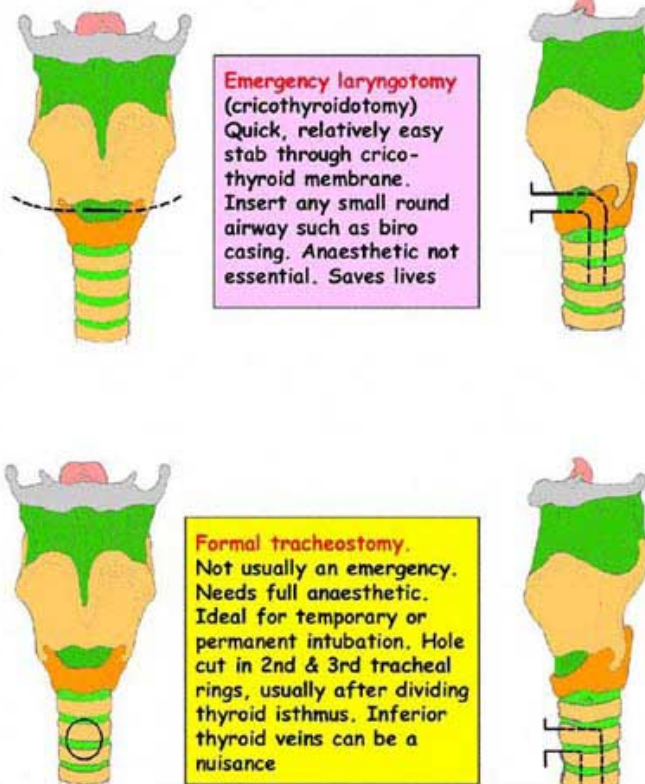
**CRICOTHYROID** has 3 special features that makes it different from other laryngeal muscles. These are:

- It is the only muscle that tightens the cords
- It is supplied the external branch of the superior laryngeal nerve and not the recurrent laryngeal
- It is the only intrinsic muscle of the larynx which on the outside of larynx

It is not initially obvious how this muscle tightens the cords but the illustration below helps with the understanding. If you can imagine a block of wood attached to the wall (A) with a strong piece of very slightly elastic string joining the block to the top of an angle-iron. The angle-iron can rotate on a nail in such a way that lifting the other end of it will tighten the string. The equivalent situation in the larynx (B) it that the cricothyroid muscle does the lifting of the angle-iron (cricoid) to tighten the cords. The thyroid cartilage is not fixed as is the block of wood in (A) so that both cartilages are tilted when the cords are tightened. Note that the cords are attached to the back of the thyroid cartilage and the vocal processes of the arytenoids



## LARYNX - SURGICAL ACCESS



## Trachea

- Located in superior mediastinum, begins at the larynx at the level of cricoid cartilage (C6), and bifurcates into 2 main bronchi (Right & Left) at the level of *transverse thoracic plane* (or *sternal angle*, Th4/Th5). The angle of bifurcation is  $50^{\circ}$ - $100^{\circ}$ , with an average of  $75^{\circ}$  with  $35^{\circ}$  on the right side (right bronchus is more vertical) and  $40^{\circ}$  on the left
- 9-15 cm length, has horse shoes *C-shaped* supporting cartilage rings (16-20)
- Has *carina* → downward/backward projection of last tracheal cartilage forms keel like ridge separating openings of R/L bronchi

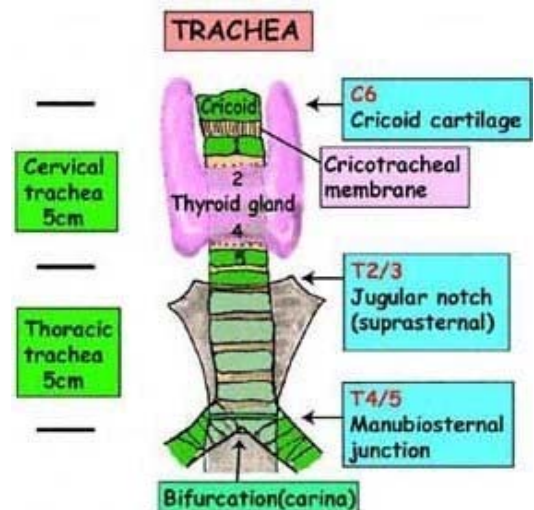
### → R. Bronchus (primary/main/principle) ~ 2 cm

- Shorter, wider, more vertical than L. bronchus. So the tendency for foreign bodies to go in it.
- Runs *under* arch of *azygos* vein and divides into 3 lobar bronchi. The *R. sup. Lobar bronchus* → *Epiarterial position* (above the right pulmonary artery), all rest → *hypoarterial* (below the artery)

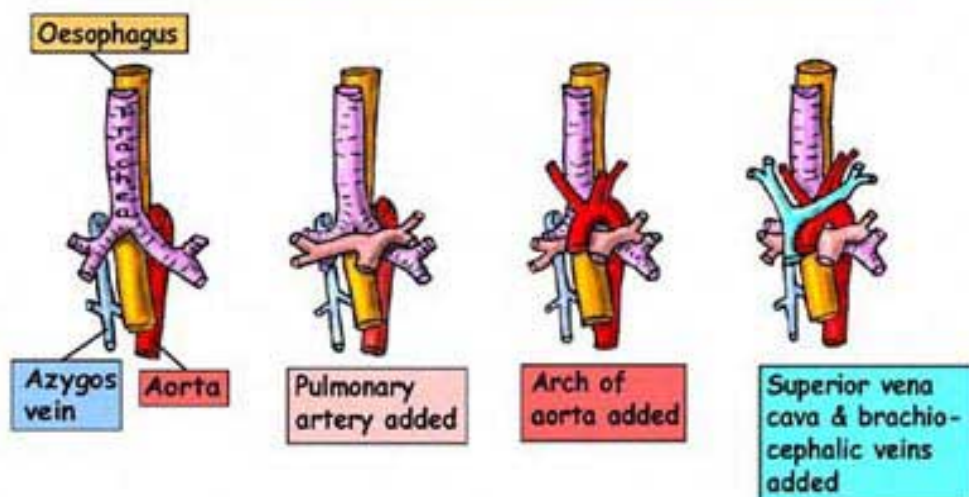
### → L. Bronchus (primary/main/principle) ~ 5 cm

- Runs Inferolaterally to the arch of aorta, crosses anteriorly to esophagus and thoracic aorta, and divides in 2 lobar bronchi
- Also crossed by arch of aorta superiorly and proximally, and by left pulmonary artery distally.

→ **Important Note:** the bronchitic tree always divide in the so called **Dichotomic** pattern of division, which means every branch divides into 2 only all the time!!! (Not more!!)(Like cells splitting each into 2)



## KEY RELATIONS IN SUPERIOR MEDIASTINUM IN PROGRESSIVE LAYERS





## ***The Pleurae (Membrana Succingens)***

Each lung is invested and enclosed by a serous pleural sac that consists of 2 continuous membranes:

- ***Visceral pleura:*** invests all surfaces of the lung and gives the shiny smooth appearance
- ***Parietal pleura:*** lines the different parts of the wall of the pleural cavity; called costal, diaphragmatic, and mediastinal, according to the parts invested

→ ***Pleural cavity:*** is a potential space between the 2 layers that contains capillary layer of *serous pleural fluid* for lubrication of plural surfaces. The tension (Pressure) in this cavity is more negative than the lung itself, which good for the Inspiration (enables the inflation of the lung), but in the case of perforation of the thoracic wall, this tension becomes normal and same as the lung so this leads to the collapse of the lung. Clinically, air in the pleural cavity is known as ***Pneumothorax***

### → ***Visceral Pleura (Pulmonary pleura):***

Covers the lungs and adheres to all of its surfaces, including surfaces within fissures.

Provides the lung with a slippery surface, continues with the Parietal pleura at the hilum of the lung → around the root of the lung → ***pulmonary ligament***

### → ***Parietal Pleura:*** covers the pulmonary cavities, has 4 parts:

- 1) ***Costal part:*** covers the surfaces of the thoracic wall, separated from the internal surface of the thoracic wall by the Endothoracic fascia, good for extra-pleural intrathoracic surgical access
- 2) ***Mediastinal part:*** in the lateral parts of the mediastinum and at the hilum of the lung it reflects laterally onto structures making up the root of lung (pulmonary sleeve). Inferior to the root of lung → the reflection or continuity between the parietal and visceral pleura is an empty double layer of pleura, the ***Pulmonary ligament*** (immediately anterior to the esophagus)
- 3) ***Diaphragmatic part:*** covers the thoracic superior surface of the diaphragm, except along its costal attachments and the fusion with the pericardium. At this elastic layer of Endothoracic fascia, the *Phrenicopleural fascia*, connects the diaphragmatic pleura with the muscular fibers of the diaphragm

***\*\* Cervical pleura:*** (special part) or the ***Pleural Cupula*** extends through the *superior thoracic aperture 2-3 cm above the midclavicular line into the root of neck*, forming a cup shaped pleural dome over the apex of lung. It's reinforced by a fibrous extension of the Endothoracic fascia or the ***Supra-pleural membrane of Sibson***, which attaches to the inner border of the first rib and to the transverse process of the seventh cervical vertebra. The whole structure is covered by the *Scalenus tent*, made by the 3 scalene muscles

### → ***Lines of Pleural reflections:***

- 1) ***Sternal line:*** sharp, between costal and mediastinal pleura
- 2) ***Costal line:*** sharp/ oblique, between diaphragmatic and costal pleura
- 3) ***Vertebral line:*** rounder reflection, between costal and posterior mediastinal part

→ **Borders of the Pleura:**

We start 2-3 cm above the midclavicular line (Cupula), then we descend in an oblique line into the 2<sup>nd</sup> Intercostal space, near the sternum after that we go down parallel to sternum into:

- 6<sup>th</sup> rib: Sternal line
- 7<sup>th</sup> rib: Midclavicular line
- 8<sup>th</sup> rib: Anterior Axillary line
- 9<sup>th</sup> rib: Midaxillary line
- 10<sup>th</sup> rib: Posterior Axillary line
- 11<sup>th</sup> rib: Scapular line
- 12<sup>th</sup> rib / Spinous process of Th-11: Vertebral line,
- then go up vertically to Th-1

\*\* on the left side, due to the presence of the heart, we descend from the 2<sup>nd</sup> IC space up to the 4<sup>th</sup> rib, and then we go laterally in an arch shaped line and then go to the 6<sup>th</sup> rib. This notch created by this arched line is shallower than the normal Cardiac notch of lungs and allows a part of the pericardium to be in direct contact with the anterior chest wall, which is very important in **Pericardiocentesis** (Needle or catheter drainage of the pericardium)

→ **Recesses:**

Lungs do not occupy the whole pulmonary cavity during expiration, and because of this, you can find recesses at the junction (or reflection) of the different parts of the **parietal pleura**. This is different from the recesses in the pericardium because in that case they are made by **both** parietal and visceral layers.

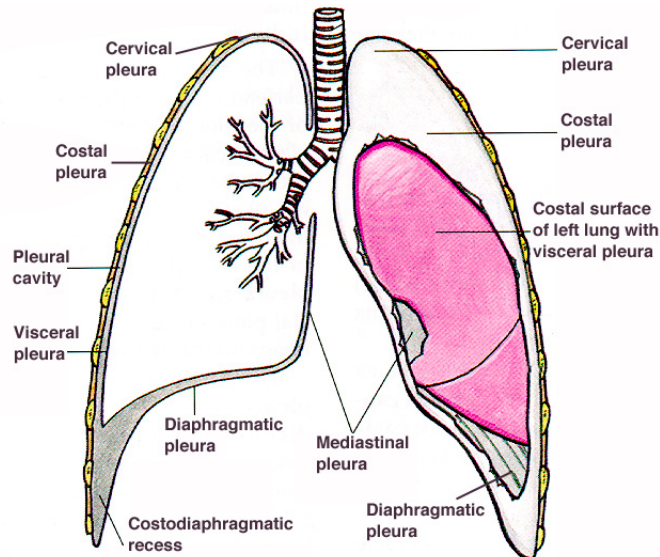
- **Costodiaphragmatic (phrenicocostal) recess:** between the diaphragm and the costal layer of the parietal pleura. It is the lowermost point of the Pleural cavity, and fluid is accumulated in it, which gives the ability to drain it by a needle via the 9<sup>th</sup> IC space. During inspiration, the inferior border/ margin of the lung descends to it and fills it
- **Phrenicomedial recess**
- **Costomedial recess**

→ **Blood supply of the parietal pleura:**

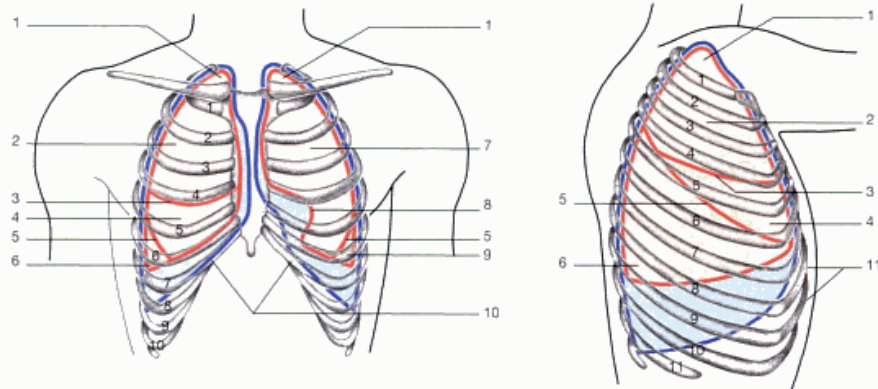
- **Arterial supply:** from the arteries that supply the thoracic wall → *internal thoracic, intercostals and Musculophrenic arteries*.
- **Venous drainage:** into systemic circulation by adjacent parts of thoracic wall.

→ **Blood supply of visceral pleura:**

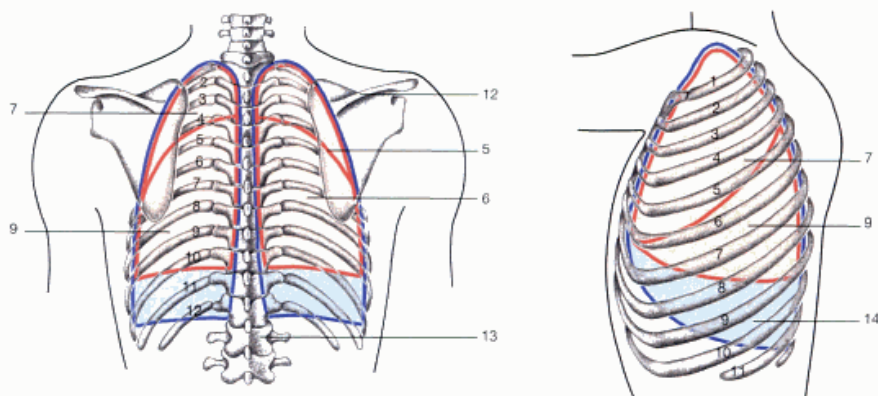
- **Arterial supply:** *bronchial arteries*
- **Venous drainage:** into (via bronchial veins) pulmonary veins.



### Surface projections of lungs & pleura



Surface projections of lungs and pleura on the thoracic wall. Left: anterior aspect; right: right-lateral aspect. Red = margins of the lung; blue = margins of pleura. The numbers indicate ribs.



Surface projections of lungs and pleura on thoracic wall. Left: posterior aspect; right: left-lateral aspect. Red = margins of lung; blue = margins of pleura. The numbers indicate ribs.

- |                                     |                               |                                                                    |
|-------------------------------------|-------------------------------|--------------------------------------------------------------------|
| 1. Apex of lungs                    | 6. Lower lobe of right lung   | 11. Costal margin                                                  |
| 2. Upper lobe of right lung         | 7. Upper lobe of left lung    | 12. Spine of scapula                                               |
| 3. Horizontal fissure of right lung | 8. Cardiac notch of left lung | 13. First lumbar vertebra                                          |
| 4. Middle lobe of right lung        | 9. Lower lobe of left lung    | 14. Space border between lung & pleura (costodiaphragmatic recess) |
| 5. Oblique fissures of lungs        | 10. Infrasternal angle        |                                                                    |

## ***The Lungs***

A pair of viscera occupying the pulmonary cavities of the thorax, the organs of respiration in which aeration of the blood takes place

→ The ***root of the lungs*** is all the structures entering or leaving the lung at the *hilum*, forming a pedicle invested with the pleura, it contains:

- *Bronchi (Major, Principal)*
- *Pulmonary arteries*
- *Pulmonary Veins (superior, inferior)*
- *Pulmonary plexus & Nerves (Parasympathetic, Sympathetic, VSA)*
- *Bronchial Arteries & Veins*
- *Lymphatics*

→ Medial to the hilum, the root is enclosed within the area of continuity between the parietal and visceral layers of pleura as → ***Pleural Sleeve or mesopneumonium***, which continues downward as ***The Pulmonary ligament***

→ Each lung has:

- ***Apex***: blunt, rounded superior end of lung ascending above the level of 1st rib into the root of neck that is covered by cervical pleura (***cupula***)(see the discussion of the pleura)
- ***3 Surfaces***:
  - ***Costal***: large, smooth, and convex. Related to costal pleura, separated from ribs/cartilage and intercostal muscles. Posterior part of it is related to vertebrae.
  - ***Mediastinal***: concave, includes hilum and root of lungs. It has grooves and impressions. (See later)
  - ***Diaphragmatic***: concave, forms base of lungs rests on dome of diaphragm  
Lateral & posterior → thin, sharp margin → projects into ***costodiaphragmatic recess***
- ***3 Borders (Margins)***: where 2 of the surfaces meet, forming a sharp end
  - ***Anterior***: between costal and mediastinal surfaces meet anteriorly and overlap the heart
  - ***Inferior***: circumscribes the diaphragmatic surface of lung and separates it from costal and mediastinal surfaces
  - ***Posterior***: where costal and mediastinal surfaces meet posteriorly



→ **The Right Lung:**

- Larger and heavier than the left lung, but shorter and wider because of high R. dome of diaphragm and the inclination of the heart on the left
  - Has **3 lobes**
    - Upper
    - Middle
    - Lower

} Separated by **oblique and horizontal fissures**
  - **Oblique fissure:** separates the upper and middle lobes from the lower lobe, begins posteriorly above the head of 4<sup>th</sup> rib in the paravertebral line and reaches the **6th sternocostal joint** anteriorly after crossing the 5<sup>th</sup> rib
- ⇕
- Both fissures are important in percussing *dullness*
- **Horizontal fissure:** deep fissure that separates the upper and middle lobes of the right lung. runs from oblique fissure in the midaxillary line following the **4th rib** and cartilage obliquely (so its not really horizontal)
  - Has **3 lobar (secondary)** bronchi and **10 segmental (tertiary)** bronchi (with artery)
  - Has **Impressions** for:
    - *Azygos Vein* : above hilum (arch like) (sometimes produces the *Azygos lobe* if it passes through the tissue of the lung to reach the SVC)
    - *SVC* (after root)
    - *Esophagus* (behind root)
    - *Cardiac impression* (by right atrium)
    - *Costal* impressions made by the ribs on the costal surface
  - **Position of structures at the right hilum**
    - Superior & inferior anterior – 2 pulmonary veins
    - Superior posterior – pulmonary arteries
    - Posterior – 2 R. Pulmonary bronchi (**Bronchus** is in **Epiarterial** position or at the same level at the hilum)

→ **The Left Lung**

- Has **2 lobes**
  - upper
  - Lower

} divided by the **oblique fissure**
- **Oblique fissure** follows **6th rib** line
- Has the **Lingula** – a tongue shaped inferomedial projection from the anterior aspect of the upper lobe of the left lung which bounds the cardiac notch inferiorly and corresponds the middle lobe of the right lung
- Has **impressions**
  - *Cardiac* impression by *L.ventricle*
    - Cardiac notch (deep indentation of superior lobe)
  - *Left subclavian artery groove*
  - *Aortic impression* (arch, descending)
- Has **2 lobar (secondary)** bronchi and **8 segmental** bronchi (see the table of segments later)
- **Position of structures at the left hilum**
  - Anterior inferior - 2 pulmonary veins
  - Superior posterior - 1 pulmonary artery (**Artery is in Epibronchial position**)
  - Posterior Inferior - Left principle bronchus

→ **Borders of Lungs**

1. *Apex*: 2 fingers above clavicle in mid-clavicular line
2. *Anterior*: from apex go down (parallel to the sternum) to → *6th rib*
3. *Lower Border*
  - a. *Sternal line* – 6th rib
  - b. *Midclavicular line* – 6th rib
  - c. *Anterior axillary line* – 7th rib
  - d. *Midaxillary line* – 8th rib
  - e. *Posterior Axillary line* – 9th rib
  - f. *Scapular line* – 10th rib
  - g. *Paravertebral line* - 11th rib
4. in the case of right lung, borders are the same, but when we descend from the apex, we reach the 4<sup>th</sup> rib (or costosternal junction) then we make a little curve around the cardiac notch after that we continue normally from the midclavicular line

(the Lungs border is 1 rib level higher than the pleural border)

→ **Bronchopulmonary segments:**

**Segments of Lungs:** morphological, functional, pathological and surgical units of the lung. The largest subdivision of lobes, *pyramidal* shaped, has *apex* and *base*.

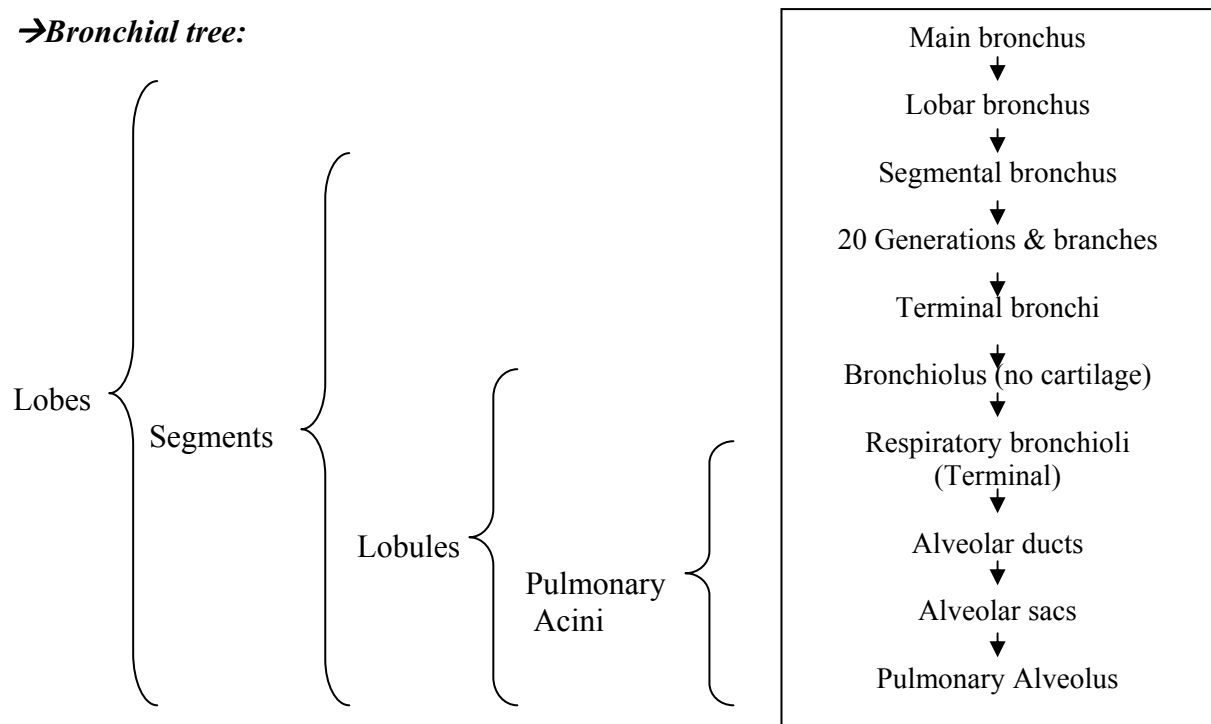
- *Apex facing lung hilum and base facing pleural surface*
- Separated from adjacent segments by connective tissue septa
- Segmental bronchus (+ tertiary branch of pulmonary art) run in *center* of *segment*
- Inside CT septa separating segments we find *Pulmonary veins* and *lymph vessels* running towards hilus (intersegmental parts of veins)
- Usually 18 in number (R. lung has 10, L. lung has 8 *usually because 1+2 → fused together & cannot be separated surgically, and the medial basal segment is absent*)
- Segments are surgically resectable / removable
- Segment is composed of smaller units → *lobules*
- *Lobules* also *pyramidal* shaped, inside them → *Bronchiole*
- One lobule belongs to 1 terminal bronchus which is branching inside lobule forming bronchioli (*no cartilage, no glands but smooth muscle*)

|                      | <b>Right Lung</b>         |                          | <b>Left Lung</b>                                                        |
|----------------------|---------------------------|--------------------------|-------------------------------------------------------------------------|
| <b>Superior Lobe</b> | 1) Apical segment         | <b>Superior Lobe</b>     | 1+2) Apico-posterior segment (Apical & posterior <b>fused</b> together) |
|                      | 2) Posterior segment      |                          | 3) Anterior segment                                                     |
|                      | 3) Anterior segment       |                          | 4) Superior Lingular seg.                                               |
| <b>Middle Lobe</b>   | 4) Lateral segment        | <b>Inferior division</b> | 5) Inferior Lingular seg.                                               |
|                      | 5) Medial segment         |                          | 6) Superior (apical) seg.                                               |
| <b>Inferior Lobe</b> | 6) Superior (apical) seg. | <b>Inferior Lobe</b>     | 7) <b>Absent</b>                                                        |
|                      | 7) Medial basal seg.      |                          | 8) Anterior basal segment                                               |
|                      | 8) Anterior basal seg.    |                          | 9) Lateral basal segment                                                |
|                      | 9) Lateral Basal seg.     |                          | 10) Posterior basal seg.                                                |
|                      | 10) Posterior basal seg.  |                          |                                                                         |

\* On the surface of the lung → 1-2 cm areas bordered by black color (surrounded). These areas are lobules (base). *Black area* → pollution inside CT → phagocytic work (Anthraxis)

\* **Bronchopericardial membrane:** central CT membrane that firmly attach root of lung and tracheal bifurcation to posterior wall of pericardium, into the tendinous center of diaphragm. Forms partition between middle and posterior Mediastinum

→ **Bronchial tree:**



\*the lobules are subunits of the bronchopulmonary segments

→ **Vasculature of lungs: Double circulation (functional + nutritive)**

**A) Pulmonary trunk:**

- a. Extends upwards from the conus arteriosus of RV of heart, and carries poorly oxygenated blood to the lungs for oxygenation
- b. Passes superior and posterior from the front of the *ascending aorta* to its left side about 5cm and bifurcates into *R. & L. Pulmonary arteries at sternal angle of Louis*
  1. **RPA:** Deoxygenated blood to right lung runs horizontally towards the hilus of R. Lung under aortic arch, behind ascending aorta and SVC. Anterior & under the right bronchus
  2. **LPA:** shorter and narrower than RPA, arches over the *L. pulmonary bronchus* Connected to aortic arch **by ligamentum arteriosum** (remnant of Ductus arteriosus of *Botalo*)

**B) Pulmonary Veins:**

- 2 on each side, oxygenated blood into the Left Atrium
- Are intersegmental (between segments, in the CT space) in drainage not parallel with art. / bronchi)
- Carry *oxygenated* blood from respiratory part of bronchioles (alveoli) and *deoxygenated* blood from visceral pleura

**C) Bronchial arteries:**

- Supply oxygenated blood to the *non respiratory* conducting tissues of the lung (root, hilum) and *visceral pleura*
- Anastomose with pulmonary artery

- 2 *L. bronchial arteries* + 1 *R. bronchial artery* Usually arise directly from aorta or maybe from upper posterior intercostal arteries

**D) Bronchial Veins:**

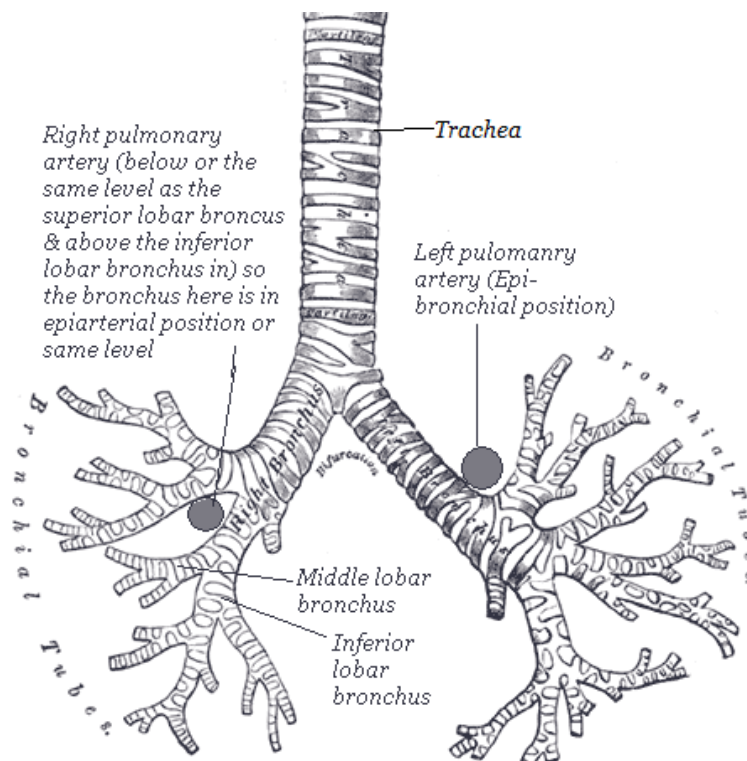
- Drain part of lungs supplied by bronchial arteries
- *R. bronchial vein drains into Azygos vein*
- *L. bronchial vein drains into Accessory hemiazygos or sup. Intercostals vein on L. side*
- They also receive blood from esophageal veins

**→ Lymphatic drainage of lungs:**

- Drain bronchial tree, pulmonary vessels and CT septa
- **Superficial (sub-pleural) Lymph plexus** → into bronchopulmonary nodes (Hilar lymph nodes)
- **Deep lymphatic plexus (Submucosa of bronchi) and CT** → into pulmonary lymph nodes (along lobar branches)
- Superficial + deep → into **tracheobronchial** nodes (sup. / inf.) around bifurcation of trachea into **Bronchomediastinal** lymph nodes/trunks. Right → *R. lymphatic duct*, Left → *Thoracic duct*, finally into **Venous angle**

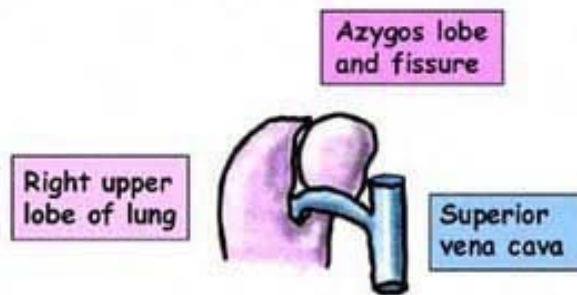
**E) Nerve supply of lungs:**

- Pulmonary plexus: Aff. + eff. (Parasympathetic) from CN-X + branches of sympathetic trunk and cardiac plexus accompany BV and bronchi into the lung
- Effects:
  1. **SYM** → dilute lumen of bronchi + constrict vessels
  2. **PARA** → dilated vessels + constrict bronchi + secretion
- *Phrenic nerve*: mediastinal pleura
- *Intercostals nerves*: costal + diaphragmatic pleura



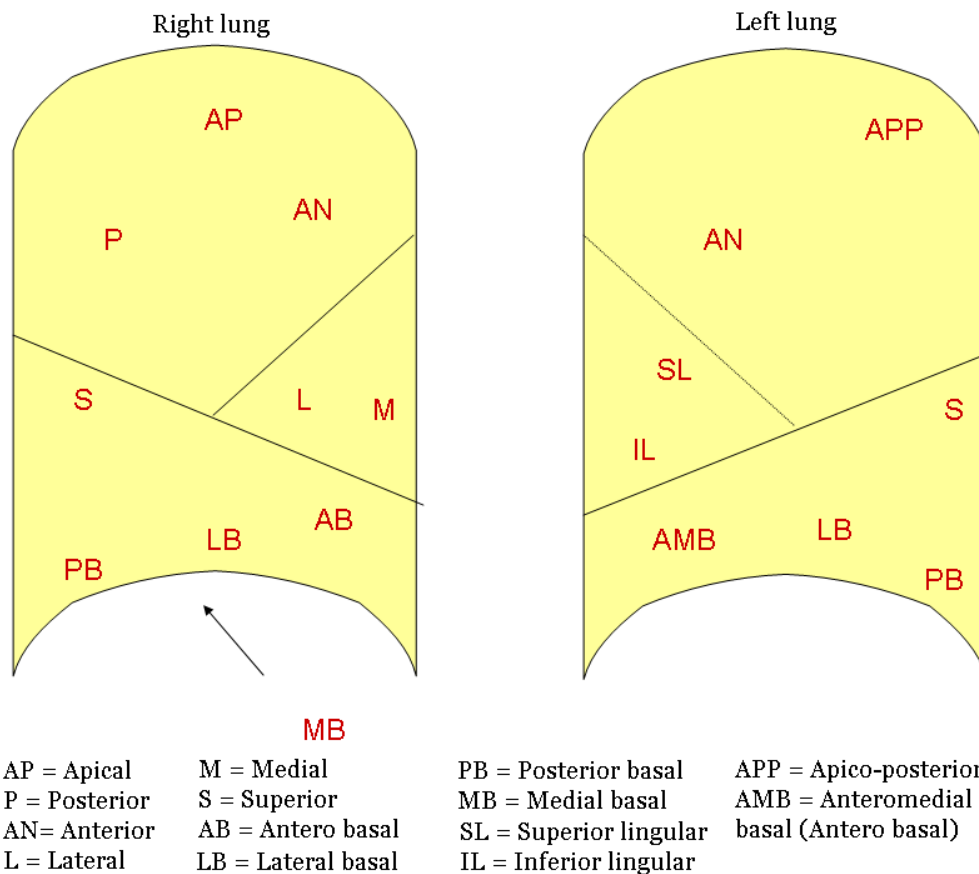


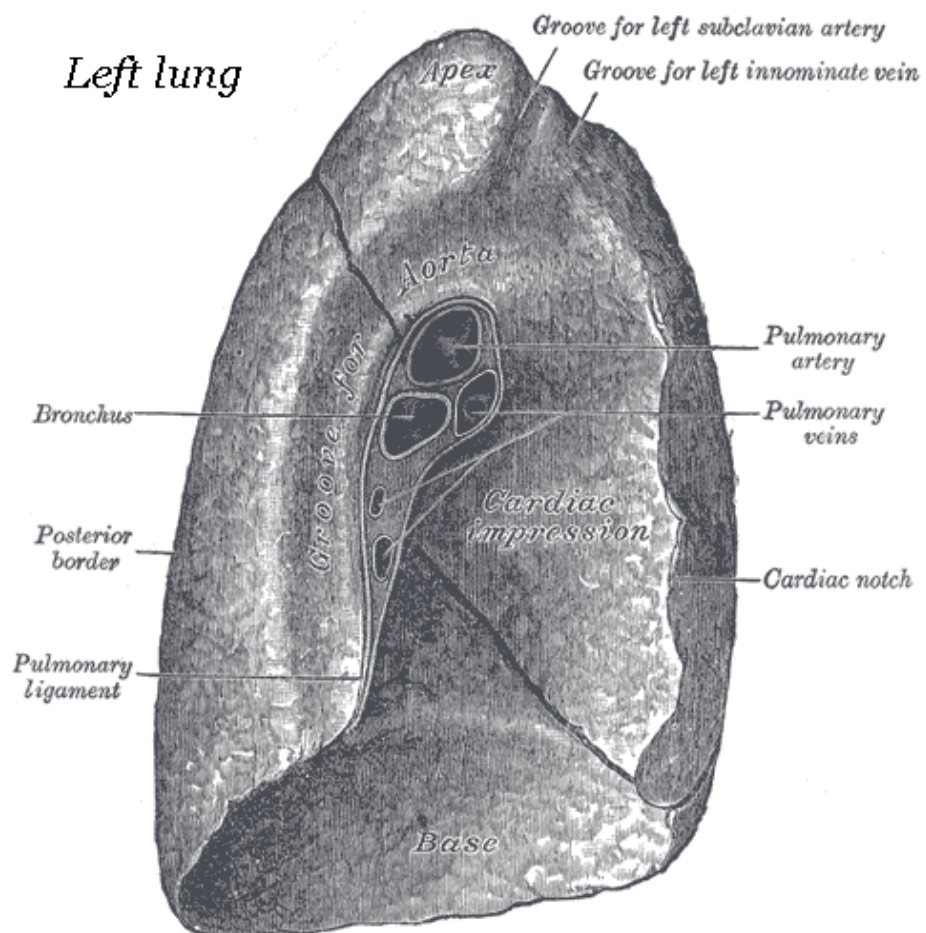
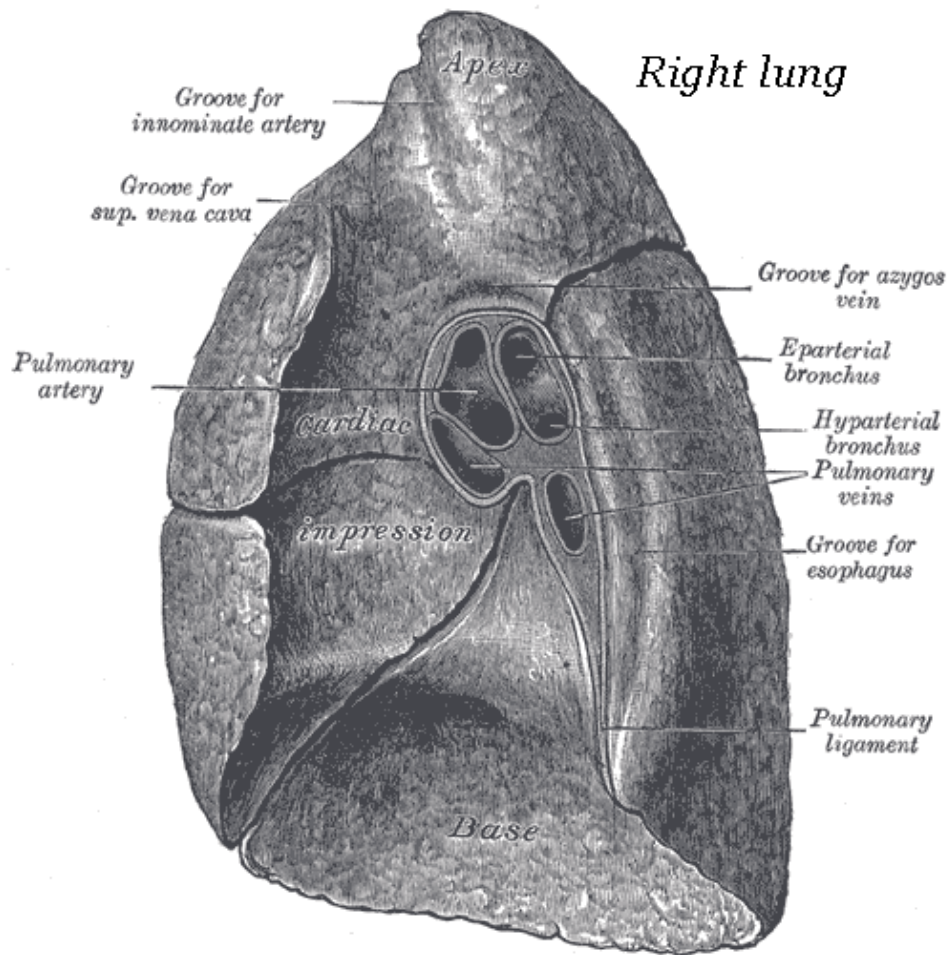
## AZYGOS LOBE OF RIGHT LUNG



Occasionally the azygos vein reaches the superior vena cava by passing through the substance of the right lobe trapping a segment of upper right lobe and creating an azygos fissure

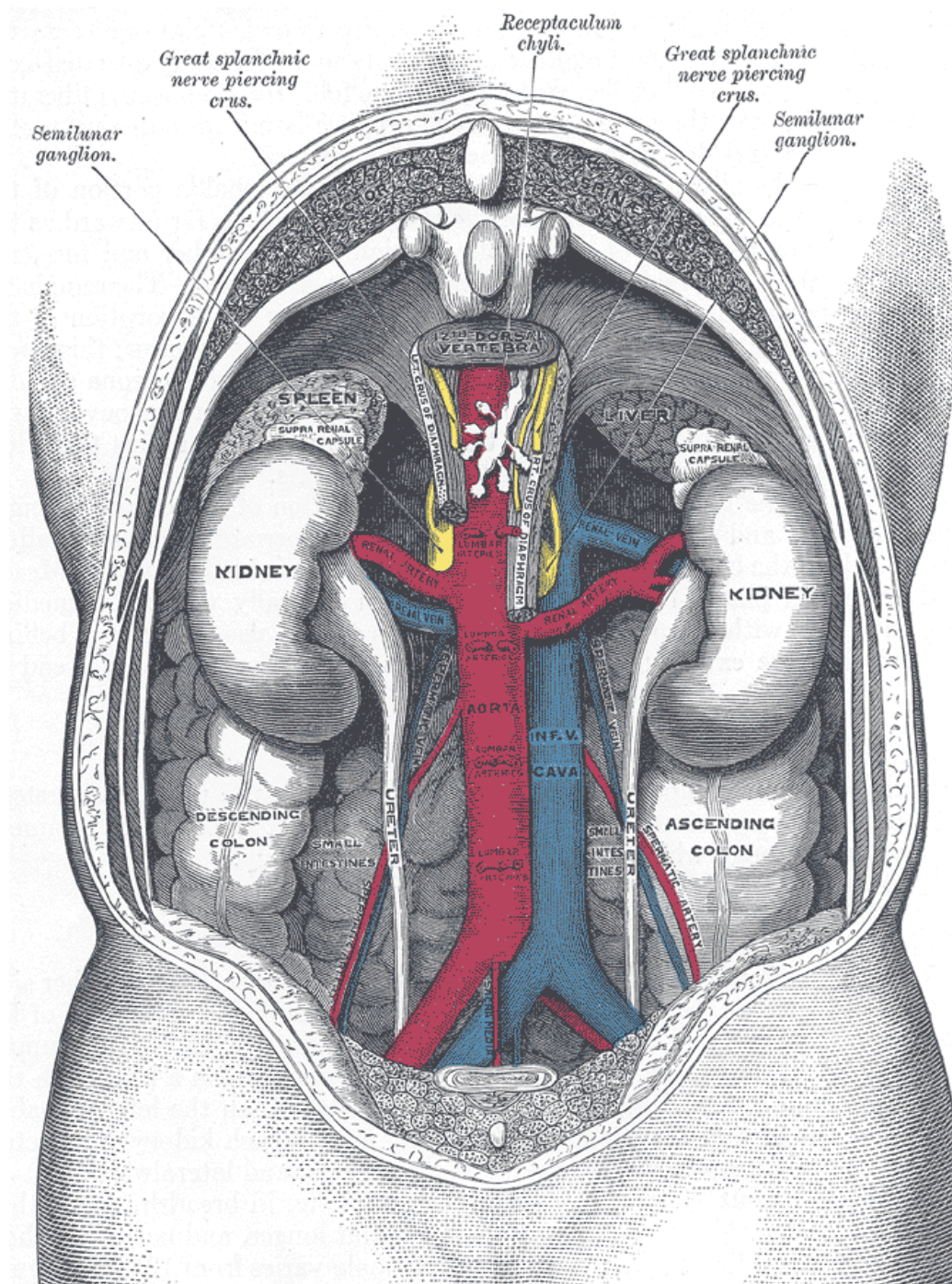
### Bronchopulmonary segments







# *Gross anatomy of the retro-peritoneal organs*



## ***The Kidneys***

The kidneys are of the paired organs that excrete urine by removing excess water, salt and wastes of protein metabolism from the blood while returning nutrients and chemicals back. They are *bean-shaped* reddish brown organs (about 11 cm long, 5 cm wide, and 3 cm thick) lying on either side of the vertebral column, *retroperitoneally*.

### **→ Position and orientation of kidneys:**

- The **right kidney** is a little lower than the left because the liver presses it down. Positioned at the level of **Th12-L3**. The superior pole of the right kidney is crossed from behind by the 12<sup>th</sup> rib.
- The **left kidney** is positioned at the level of **Th11-L2**. The 12<sup>th</sup> rib divides it into two parts: a superior, smaller part (1/3), and an inferior, larger part (2/3).

They have *anterior* and *posterior surfaces*. The anterior surface is a little lateral; the posterior surface is a little medial. So, they're not exactly in the frontal plane (obliquely placed to the protrusion of lumbar vertebrae).

They have *superior* and *inferior poles*. Above the superior pole, sits the *suprarenal glands*. On the *medial border* which is *concave*, we have the *renal hilum*. The *lateral border* is *convex*.

### **→ Relationships of the kidneys:**

- *Inferioposteriorly*: Quadratus lumborum muscle
- *Medially*: Psoas major muscle
- *Superiorly*: lumbar part of the diaphragm, suprarenal glands
- *Anteriorly*:
  - *R. Kidney*: Liver (separated from the R. kidney by the *Hepatorenal recess* or *Morrison's pouch*), duodenum (2<sup>nd</sup> part), right colic flexure
  - *L. Kidney*: Stomach, spleen, pancreas, jejunum, left colic flexure
- *The Subcostal nerve + vessels, ilioinguinal and iliohypogastric nerves* descend diagonally behind the kidneys

### **→ The kidneys are fixed to the abdominal cavity by 3 capsules:**

1. ***Fibrous capsule***: The innermost capsule which is directly on the surface of the kidney. Between the fibrous capsule and the renal fascia, there are connective tissue fibers through the adipose tissue. So, finally, the renal fascia is connected to the fibrous capsule and the fibrous capsule to the kidneys.
2. ***Perirenal adipose capsule (Perinephric)***: adipose tissue that surrounds the kidney and vessels up to the renal sinus
3. ***Renal fascia of Gerota***: The outermost capsule which covers the anterior surface of the kidneys, continues to the posterior layer at the lateral margin, and continues above the kidneys. So, it is a closed capsule superiorly and laterally, but it is open inferiorly and medially, where the renal fascia blends with vascular sheaths of renal vessels and continues as the periureteric fascia over the ureters.  
Medially, the anterior layer passes in front of the aorta and inferior vena cava, continues to the other side (anterior surface) the posterior layer passes behind the aorta and inferior vena cava.

The anterior layer of the renal fascia is fused with the parietal peritoneum (superiorly it continues under the diaphragm as the diaphragmatic fascia that forms the primary attachment of the suprarenal glands). The posterior layer is fused with the *transversalis fascia*. This is the most important support for the kidneys. Between the two layers, the middle capsule, the *adipose capsule*, fills the space.

4. **Pararenal fat body (paranephric):** not a real capsule but a mass of Extraperitoneal fat of the lumbar region laying behind the renal fascia, contain small collagen bundles

\*\* The collagen bundles, renal fascia, perinephric and paranephric fat, the renal vessels and ureter, hold the kidneys in a relatively fixed position; however its *3 cm mobile* during respiration

→ **Renal hilum (hilus):** is the depression on the medial border of the kidney through which the segmental renal vessels and nerves pass and where the apex of the renal pelvis occurs.

Order of structures at the *hilum*:

- *Anterior:* Renal vein
- *Middle:* Renal artery
- *Posterior:* Ureter

→ **The renal sinus:** is the cavity of the kidney, filled with *adipose tissue* and surrounded by renal parenchyma and containing the calices (lesser and greater), pelvis and the segmental vessels embedded within. Due to the renal sinus the kidney appears hollow or *C-shaped* on cross section. The adipose capsule continues into the sinus.

Sinus = cavity

Hilus = entrance of this cavity

Order of structures at the *sinus*:

- *Anterior:* 3-4 primary segmental branches of renal artery
- *Middle:* Renal pelvis
- *Posterior:* Renal segmental veins, with 1 primary segmental branch

→ **The renal pelvis (Pyelos):** flattened, funnel shaped expansion of the superior end of the ureter, receives 3 major calices, which receive 3-4 minor calices each (so 9-12 total minor calices), the renal papillae (apexes of renal pyramids) open into the minor calices.

(So: 1 ureter, 1 pelvis, 3 major calices, 9-12 minor calices per kidney)

→ On a **frontal section** of the kidney through the largest plane (pole to pole), we can distinguish:

- **Outermost** layer is the *fibrous capsule* on the surface.
- The next layer is called the **cortex cortices** (right below the fibrous capsule), consisting of a narrow peripheral zone where the renal corpuscles are not visible.
- Inside the cortex, there are striations called **medullary rays** (*Stria medullaris corticis*), the center of the renal lobule, consisting of straight tubular parts; these may be either ascending or descending limbs of the nephronic loop or collecting tubules
- The cortex continues into the medulla as *cortical columns* (**Renal columns** or **Bertin's columns**), which are the prolongations of cortical substance separating the pyramids of the kidney
- The next part of the kidney is the medulla, formed by the **medullary pyramids**. The apical (**papillae**) openings are situated on the minor calyx. On the surface of the



apex, there are tiny openings for the papillary ducts. It is called *lamina cribrosa* because of these openings.

→ **Kidney lobes:** The kidney develops from lobes. One original renal lobe is composed of one pyramid and a half of the cortical column (renal column) around it. Approximately 25-30 original lobes have fused with each other and open to one minor calyx. The lobes are visible on the external surfaces of kidneys in the fetuses.

→ **The kidney segments** are regions of the kidney supplied by end arteries branching from the renal arteries; they are named anterior inferior segment, anterior superior segment, inferior segment, posterior segment, and superior segment.

→ **Vasculature of the kidney:**

- **Renal arteries:** arise as paired visceral branches from the abdominal aorta immediately below the SMA (between L1 & L2). they enter the kidney through the hilus and divide into:
  - **Segmental branches:** 4-5 branches that supply the anatomical segments of kidney, they are end arteries and give off:
    - **Interlobar arteries:** which run in the middle of the *Bertin's columns*, and give rise to:
      - **Arcuate arteries:** curved arteries at the corticomedullary border that give:
        - ❖ **Interlobular arteries** or **Cortical radiate arteries:** the branches of the arcuate arteries of the kidney radiating outward through the renal columns and cortex and supplying the glomeruli as:
          - **Afferent glomerular arteriole**

The segmental arteries are: anterior inferior segmental, anterior superior segmental, inferior segmental, superior segmental and *posterior segmental* (last one originates from the posterior continuation of the renal artery, behind the renal pelvis at sinus)

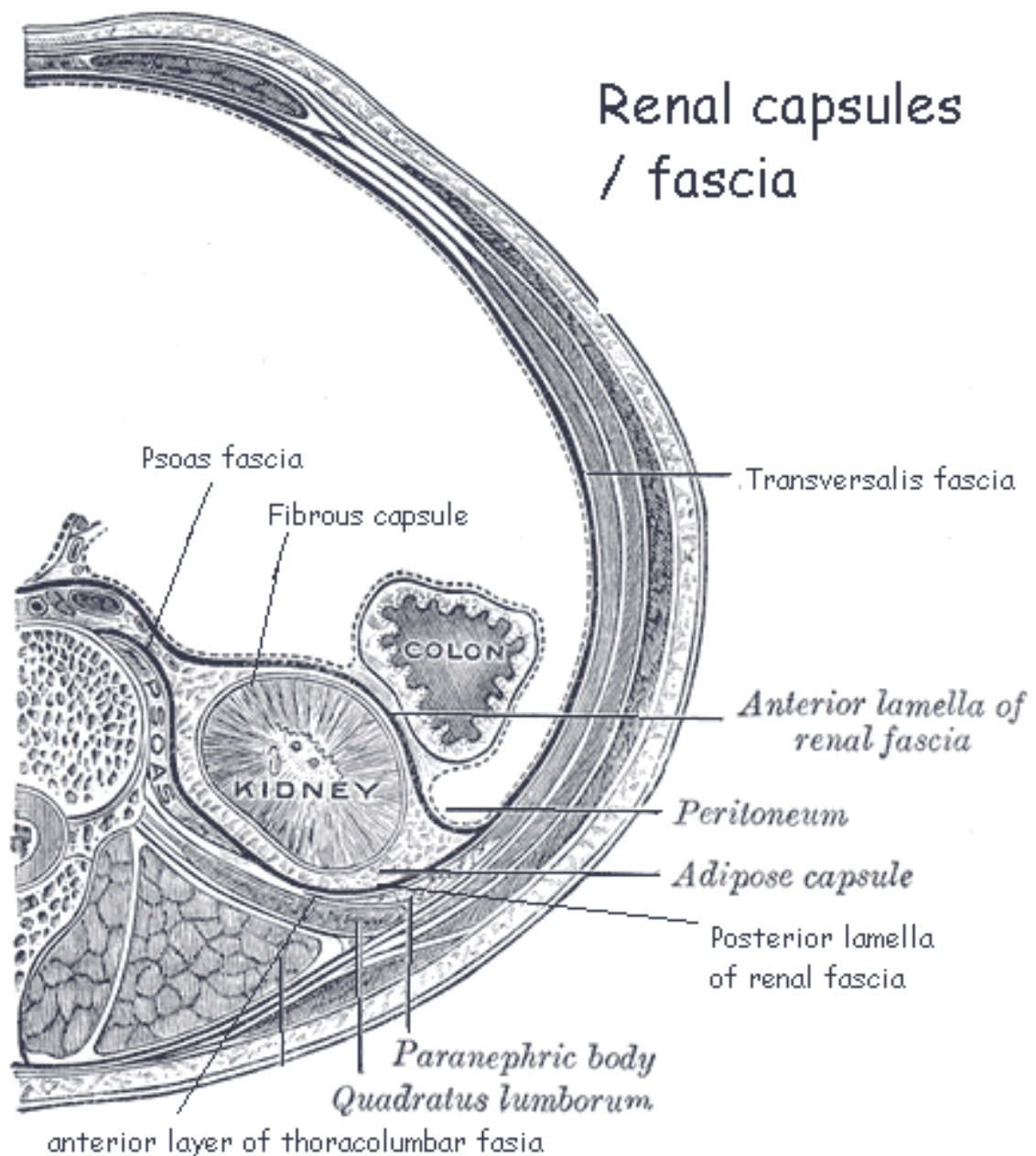
- **Uretral branches:** for the upper 1/3 of ureter
  - **Suprarenal branches** (Given before the hilus)
  - **Capsular branches**
- Usually, an **additional renal artery (accessory)** supplies the superior or inferior pole of the kidneys.
- **Renal veins:** large veins formed at the renal hilus by the merger of the segmental veins anterior to the corresponding arteries; they open at right angles into the inferior vena cava at the level of the second lumbar vertebra.  
The left renal vein passes in front of the abdominal aorta across the midline because the inferior vena cava is on the right. As a consequence of this asymmetry, the left renal vein receives the **left testicular vein** or **ovarian vein (gonadal veins)**, but the right one does not.

→ **Lymphatic drainage of kidneys:**

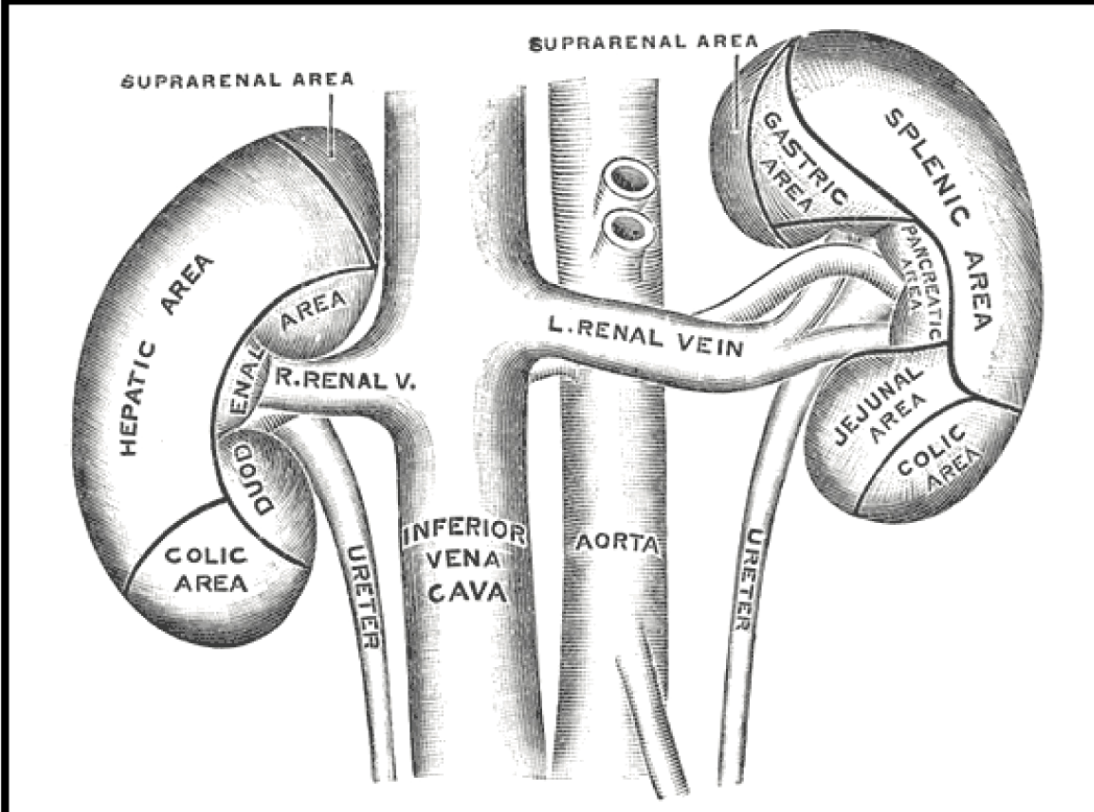
- The lymphatic vessels of the kidney form 3 *substances*: in the substance of kidney, under the fibrous capsule and in the Perirenal fat
- 4-5 lymphatic trunks leave the hilum, joined by the capsular branches, following the renal vein to the *Lumbar (Aortic and Caval) lymph nodes*

→ **Innervation of the kidney:**

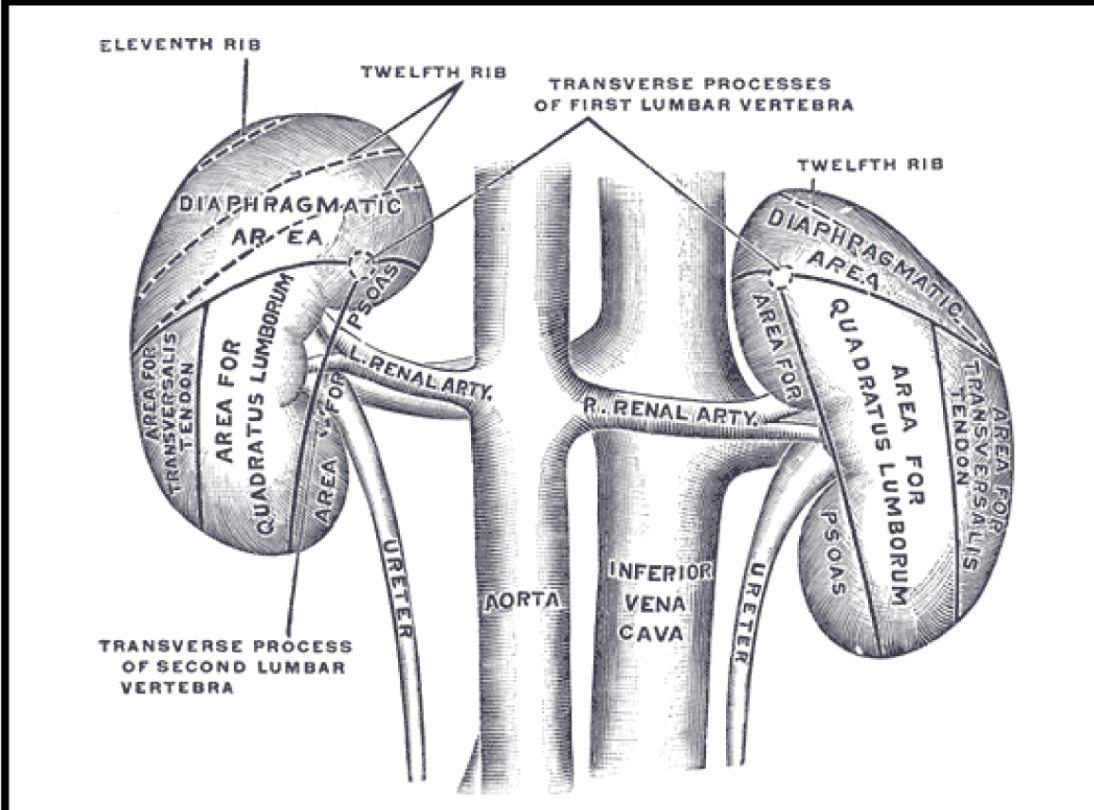
- *Renal nerve plexus*, from the celiac, aorticorenal and superior mesenteric ganglia:
  - **SYM**: supplied by fibers from the abdominopelvic Splanchnic nerves
  - **PARA**: posterior Vagal trunk



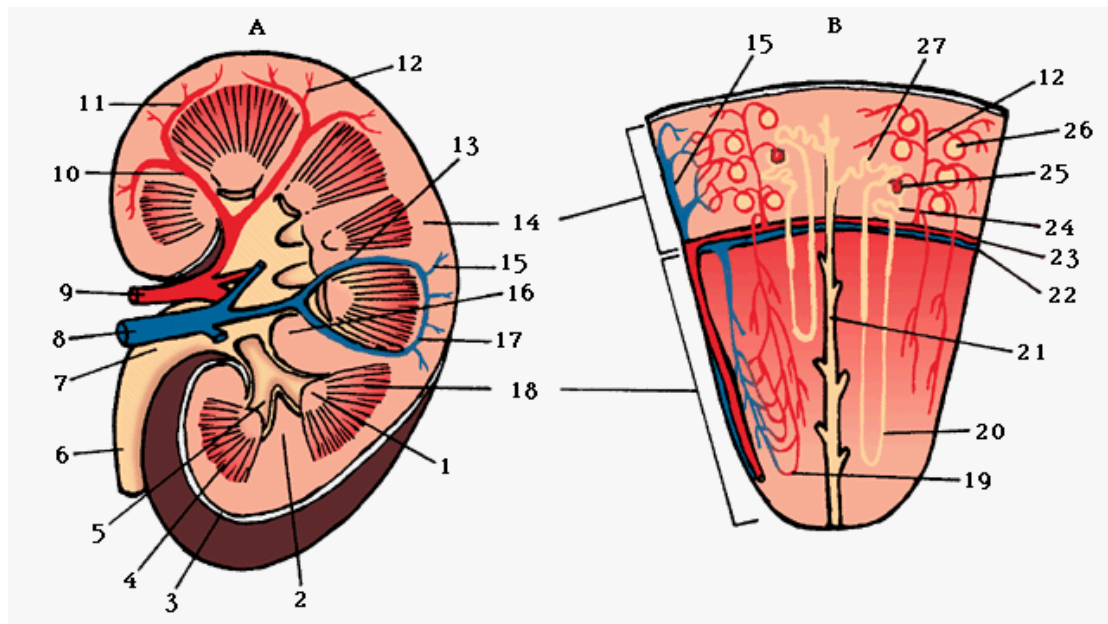
## Relationships of the kidneys, Anterior view



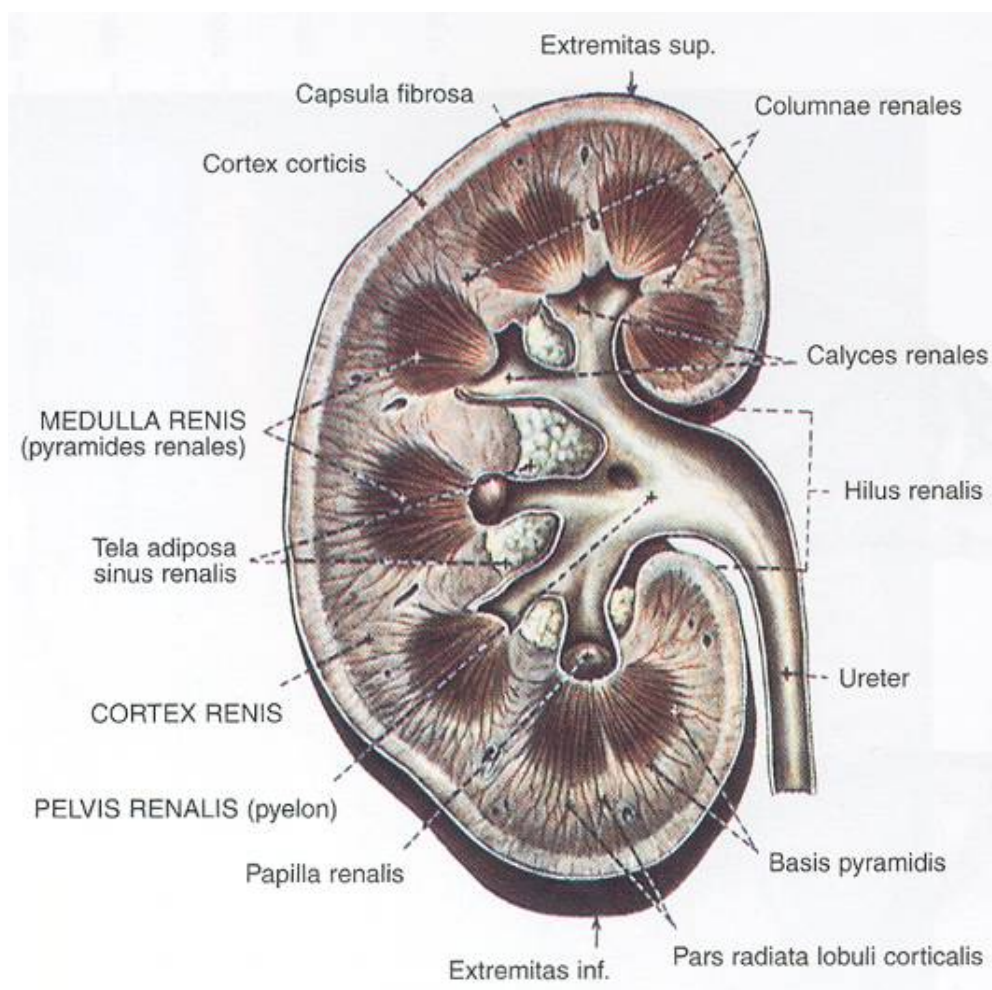
## Relationships of the kidneys, Posterior view







*Kidney 1: A longitudinal section, B nephron and adjacent blood vessels; 1 renal papilla, 2 renal column, 3 capsule, 4 renal pyramid, 5 calyx, 6 ureter, 7 renal pelvis, 8 renal vein, 9 renal artery, 10 interlobar artery, 11 arcuate artery, 12 interlobular artery, 13 interlobar vein, 14 cortex, 15 interlobular vein, 16 renal sinus, 17 arcuate vein, 18 medulla, 19 vasa recta, 20 loop of Henle, 21 collecting duct, 22 arcuate vein, 23 arcuate artery, 24 proximal convoluted tubule, 25 glomerulus, 26 Bowman's capsule, 27 distal convoluted tubule*



## ***The Ureter***

The muscular duct that conducts the urine from the renal pelvis to the bladder; it consists of an abdominal part and a pelvic part, is lined with transitional epithelium surrounded by smooth muscle, both circular and longitudinal, and is covered externally by a tunica adventitia. The ureter starts from the hilus of the kidney as the most inferoposterior structure of the hilus, and radiographically, the ureter shows **constriction** at multiple sites which are potential sites of **obstruction** by kidney stones:

1. *Junction of ureter and renal pelves*
2. The next constriction is before it enters the pelvis, at the pelvic brim, when it runs in front of the **common iliac artery** at its division into external and internal iliac arteries.
3. The last constriction is during the passage through the *wall* of the *urinary bladder*

Morphologically, the ureter is **crossed** by multiple structures which are:

1. The **Psoas major** muscle with **genitofemoral nerve** from superolateral to inferomedial (first actual crossing, the ureter is *on* the muscle and nerve)
2. **Testicular (M) or ovarian (F) artery and vein** (ureter is *behind*).
3. **Common iliac arteries**, at the pelvic brim (ureter is *in front*)
4. The last crossing is in the lesser pelvis where it crosses the **uterine artery (F) or the ductus deferens (M)**. (Ureter is *behind*)

The ureter has an **abdominal** (longer) part and a **pelvic** part. The abdominal part has the first crossing. The second crossing is exactly at the border of the abdominal and pelvic part. The third crossing is in the pelvic part.

The ureter pierces the wall of the urinary bladder from lateral to medial and posterior to anterior. So, this entrance is oblique. It forms the orifice of the ureter in the urinary bladder, and the vesicourethral valve.

→ **Arterial supply of the ureters:** comes from the neighboring blood vessels, as delicate branches:

- *Renal arteries*
- *Testicular / ovarian arteries*
- *Abdominal aorta*
- *Common iliac arteries*

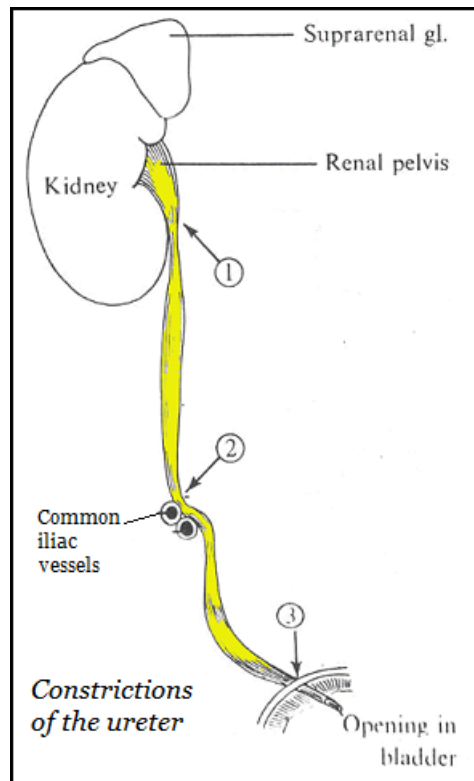
→ **Venous drainage of the ureters:**

- into the *renal and gonadal* veins

→ **Lymphatic drainage of the ureters:**

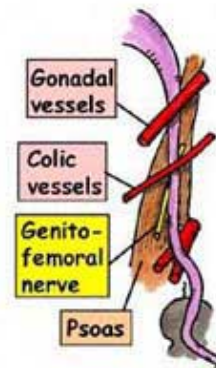
- *left or right lumbar* (caval or aortic) lymph nodes and the common iliac lymph nodes
- *internal iliac lymph* nodes in the pelvic parts of ureters





### URETER

- 25cm long
- Kidney to bladder
- Posterior: Psoas, genitofemoral nerve, sacroiliac joint, common iliac artery bifurcation
- Anterior: Right- Duodenum, right gonadal artery, right colic artery, ileal mesentery, superior mesenteric artery; Left- Left gonadal artery, left colic artery, sigmoid mesentery
- Passing under it: Vas, uterine artery
- Related to it: Lateral fornix in female
- Blood supply: Renal, gonadal, common iliac, vesical, vaginal & occasional small branches from aorta
- Nerves: Autonomic
- Points of potential hold up: Pelviureteric junction, pelvic brim, ureterovesical junction



It is recognisable as it:

- Is the most superficial structure in the pelvis
- Shows peristalsis
- Sticks to the posterior surface of the peritoneum
- Passes around the pelvic brim to 1cm short of the ischial spine the swings medially.
- Enters the bladder at the level of the pubic tubercle on a plain abdominal X-ray

## ***The Suprarenal glands (Adrenals)***

A flattened, roughly triangular body positioned at the superior pole of each kidney, enclosed by the renal fascia and attached primarily to the diaphragmatic crura; surrounded by connective tissue and considerable amount of Perinephric fat.

They are separated from the kidney by a thin septum (part of the renal fascia)

→ The **shape** of the 2 suprarenal glands is different:

- The **right** suprarenal gland is **fringian cap** shaped (**pyramidal**), contacting the IVC anteromedially, and the liver anterolaterally.
- The **left** suprarenal gland is **crescent-shaped** (**Semilunar**), related to the stomach, spleen, pancreas and left crus of diaphragm

Each gland has a **hilum** where vessel, nerves and lymphatics enter the gland

→ Each gland is composed of 2 parts:

- **Cortex**: derives from the mesoderm, and secretes corticosteroids and androgens. Has 3 histological zones: **Zona Fasciculata** (Secreting **mineralcorticoids**), **Zona Glomerulosa** (secreting **glucocorticoids**), **Zona reticularis** (Secreting **Androgens**)
- **Medulla**: mass of nervous tissue permeated with capillaries and sinusoids that derive from the **neural crest cells** associated with SYM nervous system. The **Chromaffin cells** of the medulla are related to the SYM ganglion (postsynaptic) neurons in both derivation (neural crest) and function. These cells secrete **catecholamines** (mostly Epinephrine) into the blood in response to Presynaptic signals

→ **Arterial supply of suprarenal gland:**

- **Superior suprarenal arteries**: 6-8 branches, from the **inferior phrenic artery**, the 1<sup>st</sup> paired branch of the abdominal aorta
- **Middle suprarenal arteries**: 1 branch, from the **abdominal aorta** near the level of origin of SMA
- **Inferior suprarenal arteries**: 1 branch, from the **renal arteries**

→ **Venous drainage of suprarenal gland:**

- **Right suprarenal vein**: short, drains into **IVC**
- **Left suprarenal vein**: long, joined by the inferior phrenic vein, into the **left renal vein**

→ **Lymphatic drainage of suprarenal gland:**

- lumbar lymph nodes, from the plexus deep to the capsule

→ **Nervous supply of the suprarenal gland:**

- celiac plexus, abdominopelvic (greater, lesser and least) Splanchnic nerves: myelinated Presynaptic SYM fibers (T10-L1) for the Chromaffin cells in the medulla

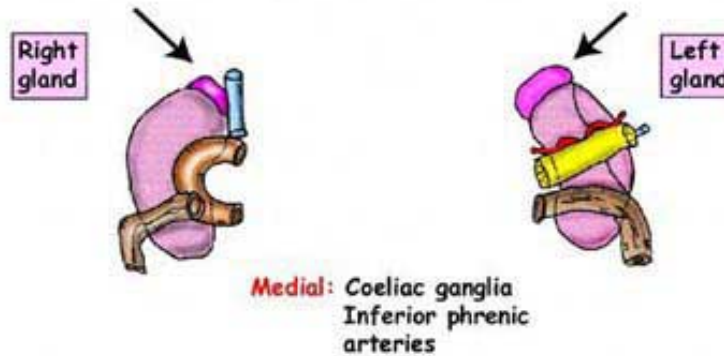
## SUPRARENAL (ADRENAL) GLANDS

- Medulla: Neural ectoderm
- Cortex: Mesoderm
- Lie: Outside Gerota's fascia
- Colour: Yellowy/brown
- Arteries: Suprarenal direct from aorta, Branches of inferior phrenic and renal
- Veins: Right short to inferior vena cava, left to renal
- Shape: Right pyramidal (hat-shaped)  
Left crescentic (cap-shaped)

### RELATIONS

**Anterior:** Right lobe of liver  
Inferior vena cava  
**Posterior:** Right crus of diaphragm

**Anterior:** Lesser sac  
Stomach  
**Posterior:** Left crus of diaphragm



**Note:** The short length of the right suprarenal vein can make it difficult to ligate it at surgery. The right gland is tucked up under the inferior vena cava

## ***Subdivision and Topography of the Aorta***

1. ***Ascending aorta:*** ascends behind the pulmonary trunk toward the right.  
*Branches:* the left and right coronary arteries
2. ***Aortic arch:*** located in the superior mediastinum (Supracardiac mediastinum). Begins at the level of sternal angle and arches posteriorly to the left → passes anterior to the trachea to reach the left side of trachea and esophagus → arches over the root of the lung → continuous as descending aorta at the level of *Th-4*.

*Related structures:* ligamentum Arteriosus passes from the root of the left pulmonary artery to the inferior concave surface of the arch.  
Left recurrent laryngeal nerve hooks around the arch before it ascends back.

*Branches* (from anteromedial to posterolateral): Brachiocephalic trunk, left common carotid, and left Subclavian arteries

3. ***Descending (thoracic) aorta:*** situated in the posterior mediastinum. It descends first on the left side and then in front of the vertebral column. This part of the aorta terminates at the level of the lower border of *Th-12*, entering the abdomen through the diaphragmatic aortic hiatus.

*Related structures:* the thoracic aorta descends through the posterior mediastinum against the left pleura, thoracic duct is running behind and right to it. It passes behind the root of the left lung and then posterior to the esophagus, close to the body of thoracic vertebrae.

*Branches:*

- Bronchial (two direct left branches and one indirect right from Intercostal)
- Esophageal
- Mediastinal (to lymph nodes)
- Posterior intercostals (nine pairs from 3<sup>rd</sup> to 11<sup>th</sup> intercostals space)
- Sub-costal
- Pericardial
- Superior Phrenic (superior aspect of diaphragm)

4. ***Abdominal aorta:*** begins at the aortic hiatus of the diaphragm at the level of *Th12* intervertebral disk and ends at *L4* by dividing into two common iliac arteries. Throughout its course, the aorta lies against the vertebral column (retroperitoneal).

*Related structures:* Anteriorly it is related to the omental bursa, pancreas, and left renal vein, posteriorly, to the vertebral column. It lays on the *right* to the thoracic duct, and the inferior vena cava

*Branches:*

- *Non-paired branches:*
  - Celiac trunk (Haller's triplet) (*Th12*)
  - Superior mesenteric (*L1*)
  - Inferior mesenteric (*L3*)

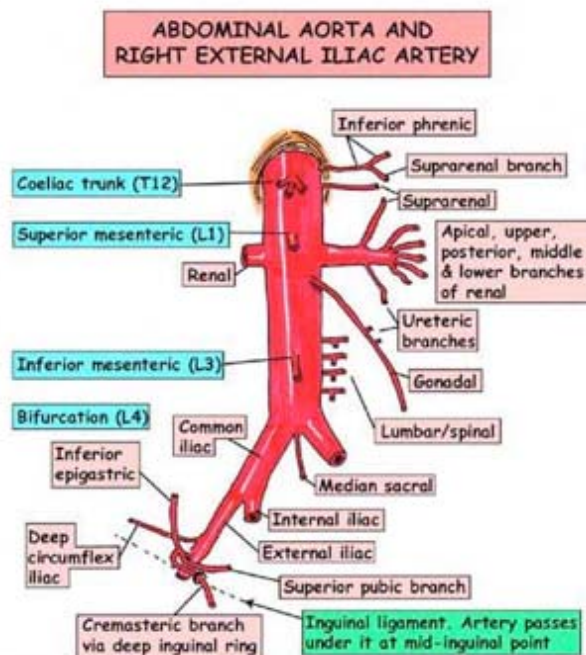
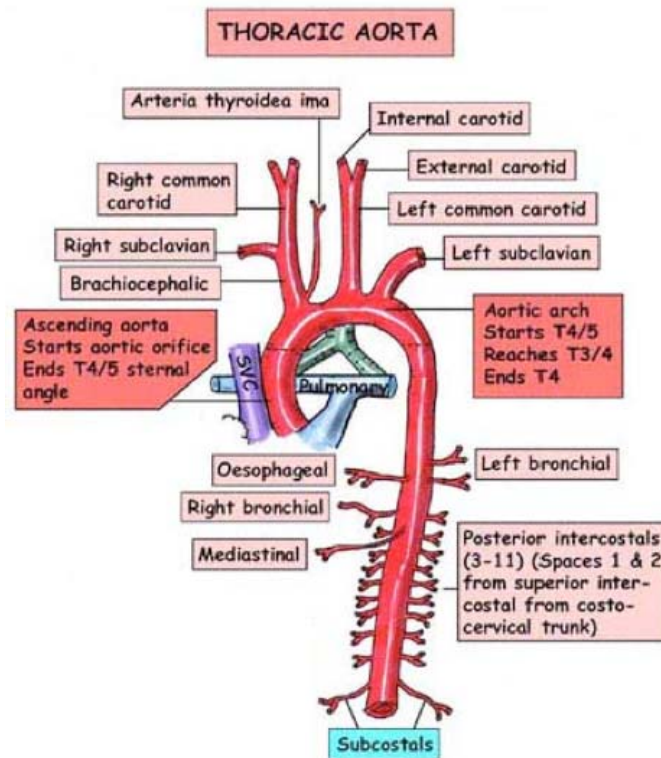
○ *Paired branches:*

▪ *Non-visceral* paired branches:

- Inferior phrenic: inferior aspect of diaphragm, gives the superior suprarenal arteries
- Lumbar (segmental branches)
- Median sacral

▪ *Visceral* paired branches:

- Middle suprarenal (L1)
- Renal
- Gonadal (L2)





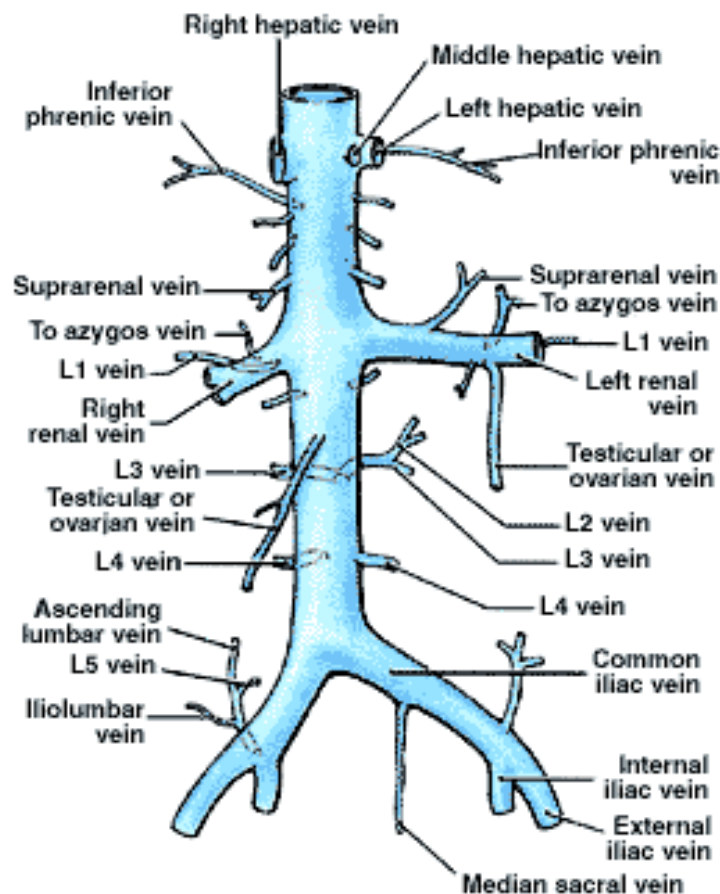
## ***The Inferior vena cava***

The IVC is formed on the *right side of L5* by the union of the two common iliac veins, below the bifurcation of the aorta.

It is longer than the abdominal aorta and ascends along the right side of the aorta, passes through the opening of the inferior vena cava in the central tendon of the diaphragm at the level of *Th8* and enters the right atrium of the heart.

→ *Receives the:*

- Right gonadal v.
- Right Suprarenal v.
- Right Inferior Phrenic v.  
(On the left side, these veins usually drain into the left renal v.)
- Left & Right renal veins
- 3<sup>rd</sup> and 4<sup>th</sup> Lumbar veins
- The three (left, middle and right) hepatic veins.  
The middle and left hepatic veins frequently unite for about 1 cm before entering the vena cava.



## ***The thoracic duct***

Consequently, essentially all the lymphatic drainage from the *lower 1/2 of the body* (deep lymphatic drainage below the diaphragm, and all superficial drainage inferior the level of umbilicus) converges in the abdomen to enter the *beginning of the thoracic duct*.

The ***common iliac lymph nodes*** receive lymph from the external & internal iliac lymph nodes, and then it passes to the ***right & left lumbar lymph nodes***

Lymph from the alimentary tract, liver, spleen and pancreas passes along the celiac and superior and inferior mesenteric arteries to the ***Preaortic lymph nodes*** (*celiac & superior & inferior mesenteric nodes*) around the origins of the arteries from the aorta. Efferents from these vessels form the ***Intestinal lymphatic trunks***, which may be single or multiple.

***The right & left lumbar (Caval and Aortic) lymph nodes*** lie on both sides of the IVC and aorta. These nodes receive lymph directly from the posterior abdominal wall, kidneys, ureters, testes / ovaries, uterus and uterine tubes. Through the inferior mesenteric and common iliac lymph nodes they also receive lymph from the descending colon, pelvis and lower limbs. Efferent lymphatic vessels from the large lumbar lymph nodes form the ***right and left lumbar lymphatic trunks***.

The thoracic duct starts *anterior to the bodies of L1-L2 vertebrae*, between the right crus of diaphragm and the aorta (in only a small proportion of people, it begins as a thin walled sac or dilation called ***Chyle cistern***, which may vary in size & shape)

The thoracic duct is formed by:

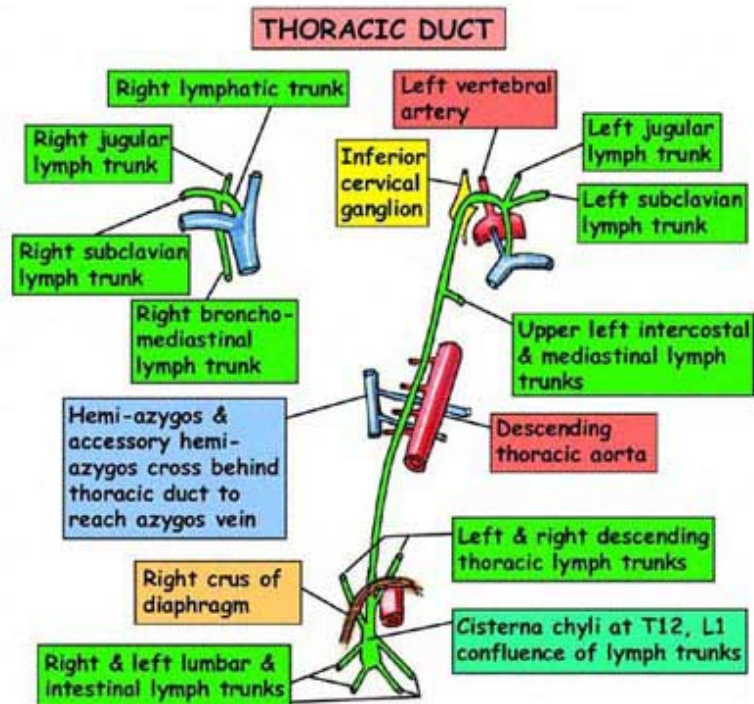
- *right & left lumbar lymphatic ducts*
- *intestinal lymph trunk(s)*
- *a pair of descending thoracic lymphatic trunks (carry lymph from lower 6 intercostal spaces)*

The thoracic duct ascends through the *aortic hiatus* in the diaphragm into the posterior mediastinum, where it lies on the anterior aspect of the bodies of the inferior 7 thoracic vertebrae among the *thoracic aorta* on its left, the *azygos* vein on its right and the *esophagus* anteriorly. At the level of *Th5-Th6* it crosses to the left, posterior to the esophagus and ascends in the superior mediastinum. It receives:

- Branches from the upper intercostal spaces of both sides through *collecting intercostal trunks*
- Branches from the *posterior mediastinal structures* (posterior mediastinal nodes, posterior intercostal nodes)
- *Jugular trunk* (draining lymph from left side of head & neck)
- *subclavian trunk* (draining lymph from the left upper limb)
- *bronchomediastinal lymphatic trunks* (draining the tracheobronchial, brachiocephalic and parasternal nodes)

The thoracic duct usually empties into the venous system, near the union of the left internal jugular and subclavian veins, the ***left venous angle*** (sometimes directly into the left subclavian or at the origin of the left brachiocephalic vein)

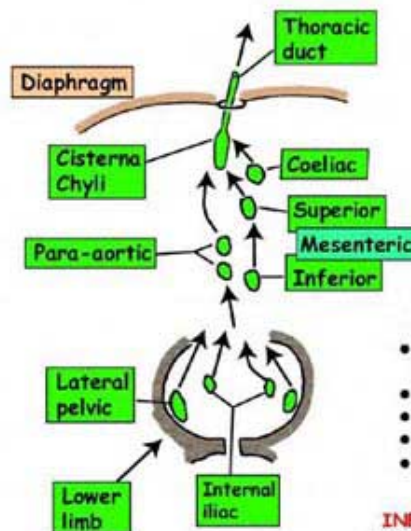
*So in summary, the thoracic duct drains all lymph from lower 1/2 of the body, left thorax, left head & neck and left upper limb*



#### THORACIC DUCT

- 45cm long (see also spinal cord, femur, vas, transverse colon, teeth/cardia)
- Several terminal branches
- Many valves in the system
- Drains all lymph below diaphragm, left head/neck & left thorax
- Commences at T12, ascends behind right crus, to right of aorta & oesophagus, then behind oesophagus, crosses midline at T5
- Lies superficial (anterior) to posterior intercostal arteries & the crossing azygos systems, over the dome of the pleura, over (anterior) to left vertebral & left subclavian arteries
- Into confluence of left subclavian and left internal jugular veins

#### ABDOMINOPELVIC LYMPHATICS



##### COELIAC

- Lower 1/3 oesophagus
- Stomach, greater omentum
- 1st/upper 2nd parts of duodenum
- Spleen
- Pancreas
- Liver
- Gall bladder

##### SUPERIOR MESENTERIC

- Lower 2nd, 3rd, 4th parts of Duodenum
- Jejunum
- Ileum
- Caecum & appendix
- Ascending/transverse colon

##### INFERIOR MESENTERIC

- Distal transverse colon
- Descending/sigmoid colon
- Upper rectum & mucosa to dentate line

##### PARA-AORTIC

- Inferior surface diaphragm
- Bare area of liver
- Suprarenal glands
- Kidneys & ureters
- Gonads (+ tube in female)
- Superior/lateral uterus
- Posterior abdominal wall

##### LATERAL PELVIC

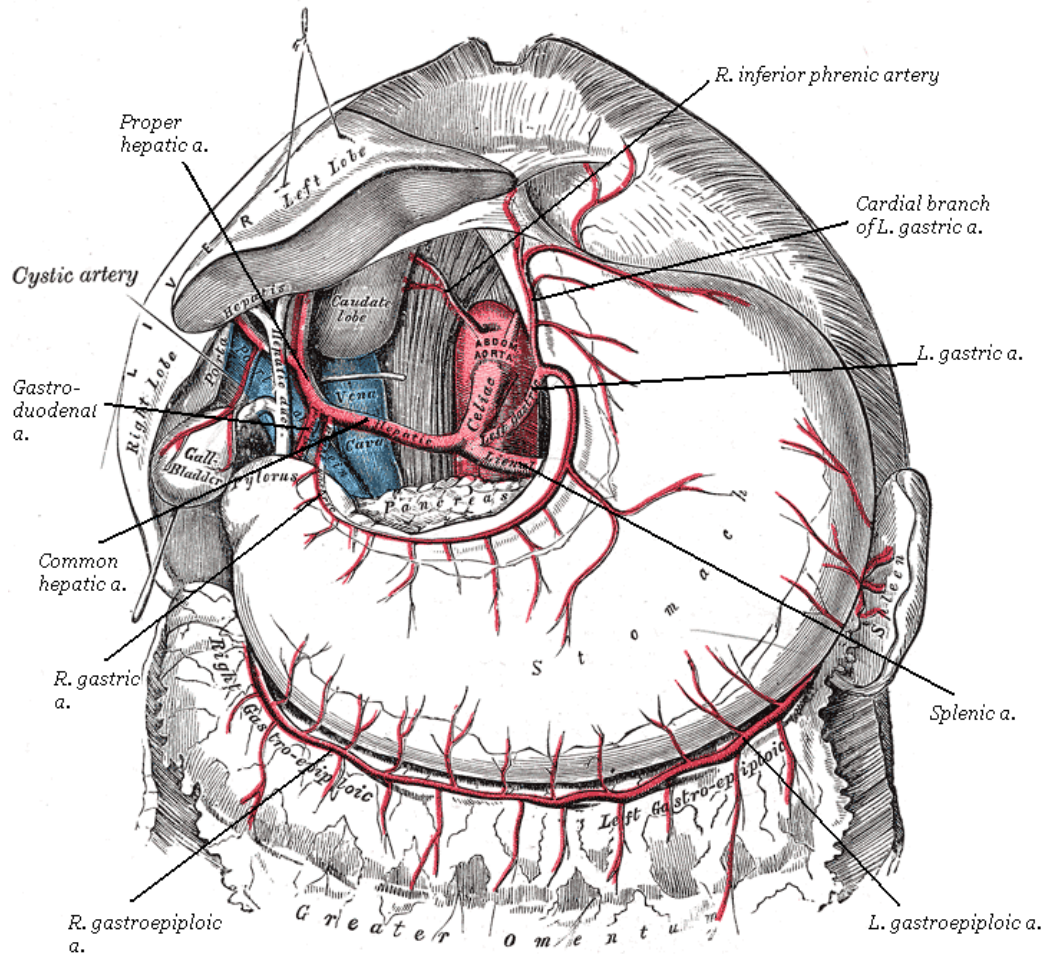
- Lower rectum & dentate line
- Bladder
- Urethra
- Lower ureter
- Uterus, cervix, upper vagina, clitoris, labia minora (female)
- Vas, seminal vesicles, prostate, bulk of penis (male)

## ***Celiac trunk (Haller tripod):***

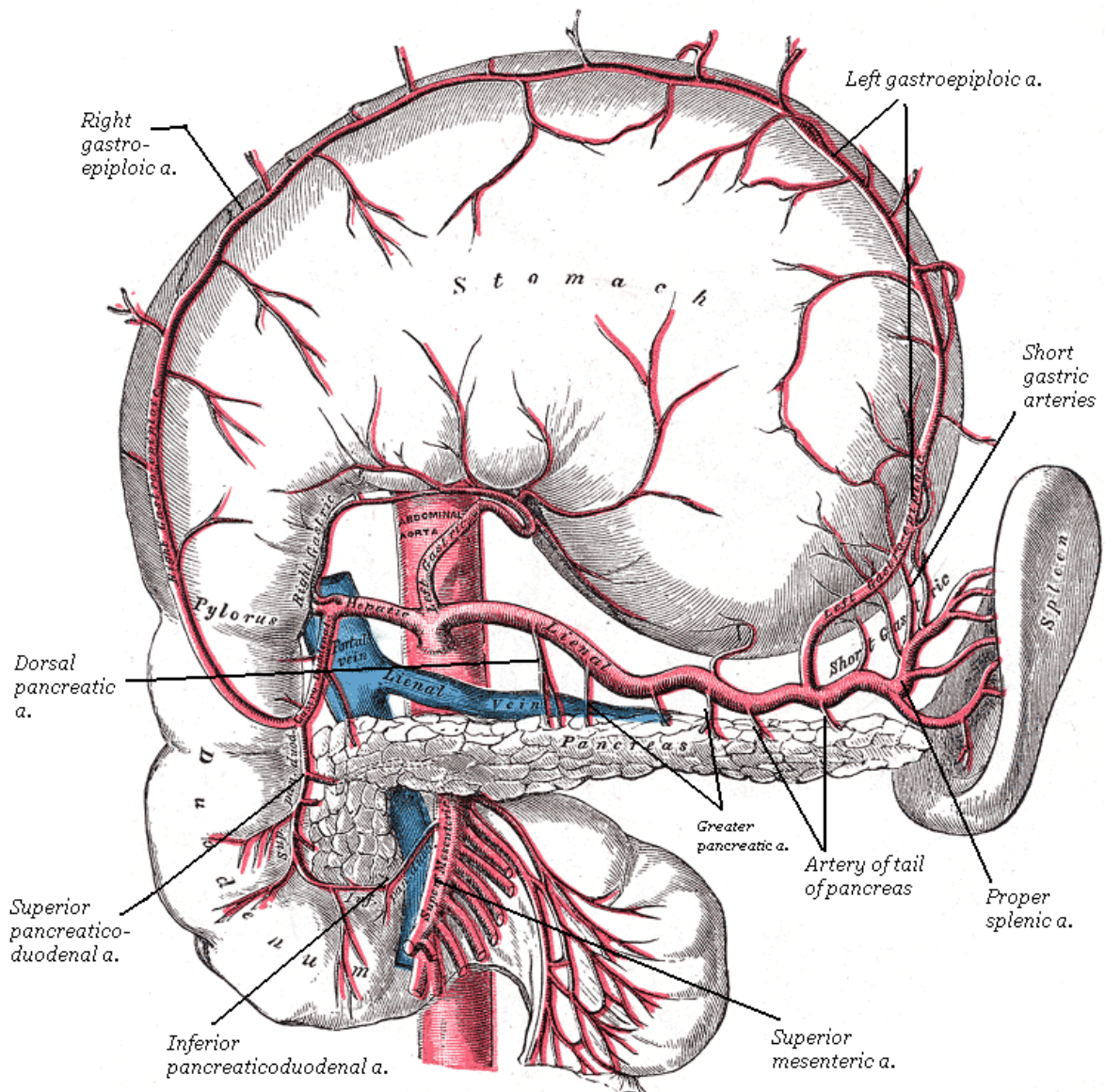
The first non-paired branch of the abdominal aorta, originating just below the diaphragm after the aorta pierces it through the aortic hiatus at the level of Th-12, it has 3 main branches:

- ***Left gastric artery:*** *ascends* to esophageal hiatus in a semi-intraperitoneal position first giving *cardial branches* to supply the cardia and abdominal portion of esophagus, forming a fold of peritoneum called *Gastropancreatic fold*, and then *descends* along lesser curvature to anastomose with *R. gastric art.*
- ***Splenic artery:*** the largest branch which has a tortuous course posterior to omental bursa, anterior to left kidney and runs at the superior border of pancreas. It reaches the hilum of spleen as the ***proper splenic artery*** through the *Splenorenal ligament* together with the tail of pancreas and there it divides to 5-6 branches, but since no anastomosis between them → 3 vascular segments +2 avascular ones. On its way it gives:
  - ***Pancreatic branches***
    - *Dorsal pancreatic artery:* for the head and body of pancreas, gives also the inferior pancreatic artery
    - *Great pancreatic artery:* for the body & tail of pancreas
    - *Artery of tail of pancreas*
  - ***Short gastric arteries:*** four or five small arteries, passing via the *gastrosplenic ligament* to the fundus of the stomach along the greater curvature
  - ***Posterior gastric artery:*** ascends retroperitoneally in posterior wall of omental bursa toward gastric fundus to reach (and supply) the gastric wall via the gastrophrenic fold
  - ***Left Gastroepiploic (gastroomental) artery:*** runs through the *gastrosplenic ligament*, to reach the greater curvatures of stomach, there, it descends along it to anastomose with the right Gastroepiploic artery, to supply the nearby region of stomach and greater omentum.
- ***Common hepatic artery:*** usually has 3 main branches,
  - ***Proper hepatic artery:*** goes through the *Hepatoduodenal ligament* in an anterior and medial position to reach the *Porta hepatis*, where there it divides into right & left branches of proper hepatic artery supplying the right & left lobes of liver respectively. The right branch will give the cystic artery to supply the gallbladder.
  - ***Right gastric artery:*** runs on the lesser curvature anastomosing with the left gastric artery to supply the pyloric portion of stomach. It can arise also as a branch of the *proper hepatic artery* quite often.
  - ***Gastroduodenal artery:*** gives 2 branches,
    - *Right Gastroepiploic artery:* runs on the right side of the greater curvature of stomach to anastomose with the left Gastroepiploic artery to supply the corresponding area on the stomach and great omentum
    - *Supraduodenal artery:* for the 1<sup>st</sup> part of duodenum

- *Superior Pancreaticoduodenal artery*: supplies the head of pancreas with the nearby duodenum with its anterior and posterior branches. This artery anastomosis with the inferior anterior & posterior Pancreaticoduodenal arteries of the superior mesenteric artery around the head of pancreas as the **1<sup>st</sup> or minor or lesser Riolan arc** or







## ***Superior mesenteric artery:***

The 2<sup>nd</sup> non-paired branch of the abdominal aorta, arising 1cm below the celiac trunk at the level of L1 vertebra, and traveling antero-inferior direction to pass below the neck of pancreas and through the pancreatic notch. Due to this course, 3 structures can be located between the SMA and abdominal aorta:

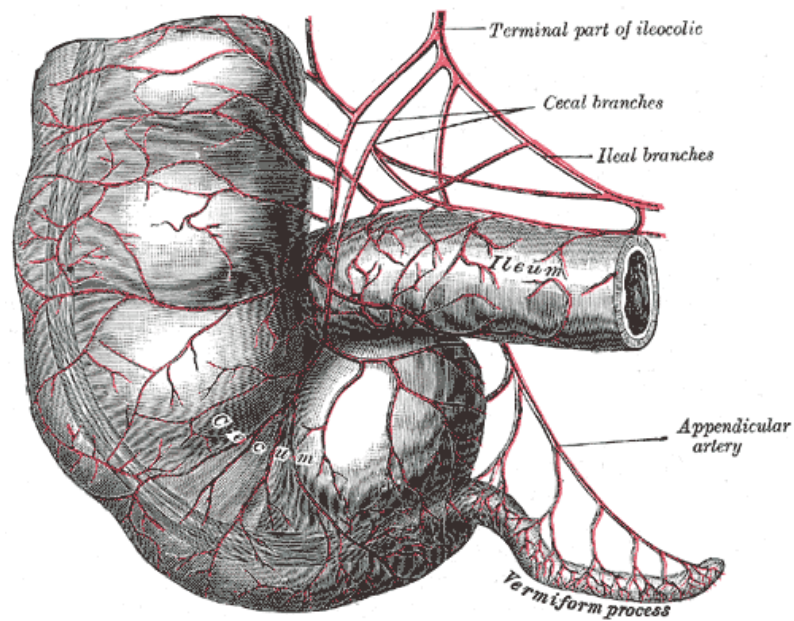
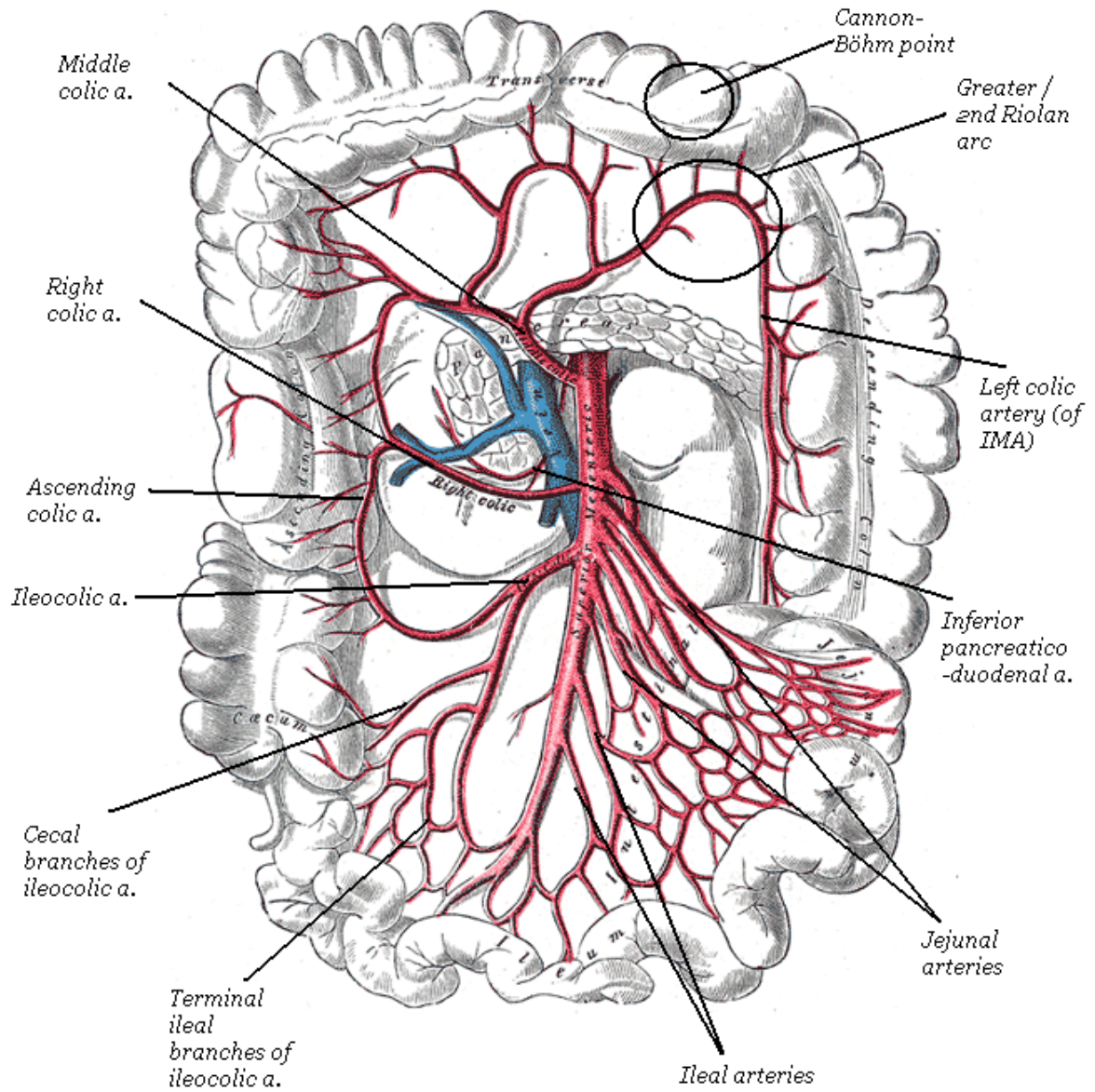
1. *Left renal vein* (can be compressed by the SMA in this region, leading to Nutcracker syndrome)
2. *Uncinate process of pancreas*, hooking around and below the SMA
3. *Inferior horizontal part of duodenum* (can be compressed here by the SMA, leading the superior mesenteric artery syndrome)

The SMA runs to the left of the SMV, and after passing through the pancreatic notch, it starts giving off branches:

1. *Inferior Pancreaticoduodenal artery*: first branch of the SMA, given on the upper border of the 3<sup>rd</sup> part of duodenum. It divides into anterior & posterior branches that supply the

head of pancreas and the surrounding duodenum, and anastomose with the superior anterior & inferior Pancreaticoduodenal arteries (branches of the gastroduodenal, of common hepatic of celiac trunk, forming the an anastomosis or arc between the celiac trunk and SMA around the head of pancreas know as the Minor or lesser Riolan arc)

2. **Intestinal arteries:** They run nearly parallel with one another between the layers of the mesentery, each vessel dividing into two branches, which unite with adjacent branches, forming a series of *arches (arterial arcades)*, the convexities of which are directed toward the intestine.  
From this first set of arches branches arise, which unite with similar branches from above and below and thus a second series of arches is formed; from the lower branches of the artery, a third, a fourth, or even a fifth series of arches may be formed, diminishing in size the nearer they approach the intestine.  
In the short, upper part of the mesentery only one set of arches exists, but as the depth of the mesentery increases, second, third, fourth, or even fifth groups are developed.  
(Arcades increase in number towards the ileum)  
From the terminal arches numerous small *straight vessels (vasa recta)* arise which encircle the intestine, upon which they are distributed, ramifying between its coats.
  - a. **Jejunal arteries:** 4-5 branches, less arcades, longer vasa recta, less fat in the mesentery
  - b. **Ileal arteries:** 11-13 branches, more arcades, shorter vasa recta, more fat in the mesentery
3. **Ileocolic artery:** lower terminal branch of the SMA, passes towards the right iliac fossa, and then branches into:
  - a. 3 branches to the *terminal ileum* (ileac branches), anastomose with Ileal arteries
  - b. 2 branches to the *cecum* (anterior & posterior cecal)
  - c. 1 branches to the *appendix* (appendicular artery) that descends in the mesoappendix
  - d. 1 branch to the *ascending colon* (Ascending Colic) and anastomose with the right colic artery
4. **Right colic artery:** It passes to the right behind the peritoneum, and in front of the testicular or ovarian vessels, the right ureter and the Psoas major, toward the middle of the ascending colon. Often, the artery arises as a common trunk with the Ileocolic artery. It gives:
  - a. *Descending branch:* anastomoses with the Ileocecal artery
  - b. *Ascending branch:* anastomoses with the middle colic artery\*\* Both branches supply the *ascending colon*
5. **Middle colic artery:** mostly supplies the transverse colon. It arises just below the pancreas, and, passing downward and forward between the layers of the transverse mesocolon, divides into two branches
  - a. *The right branch:* anastomoses with the right colic artery
  - b. *The left branch:* anastomoses with the *left colic artery*, a branch of the inferior mesenteric artery, around the left colic flexure, forming the Greater / Major Riolan anastomosis or arc.

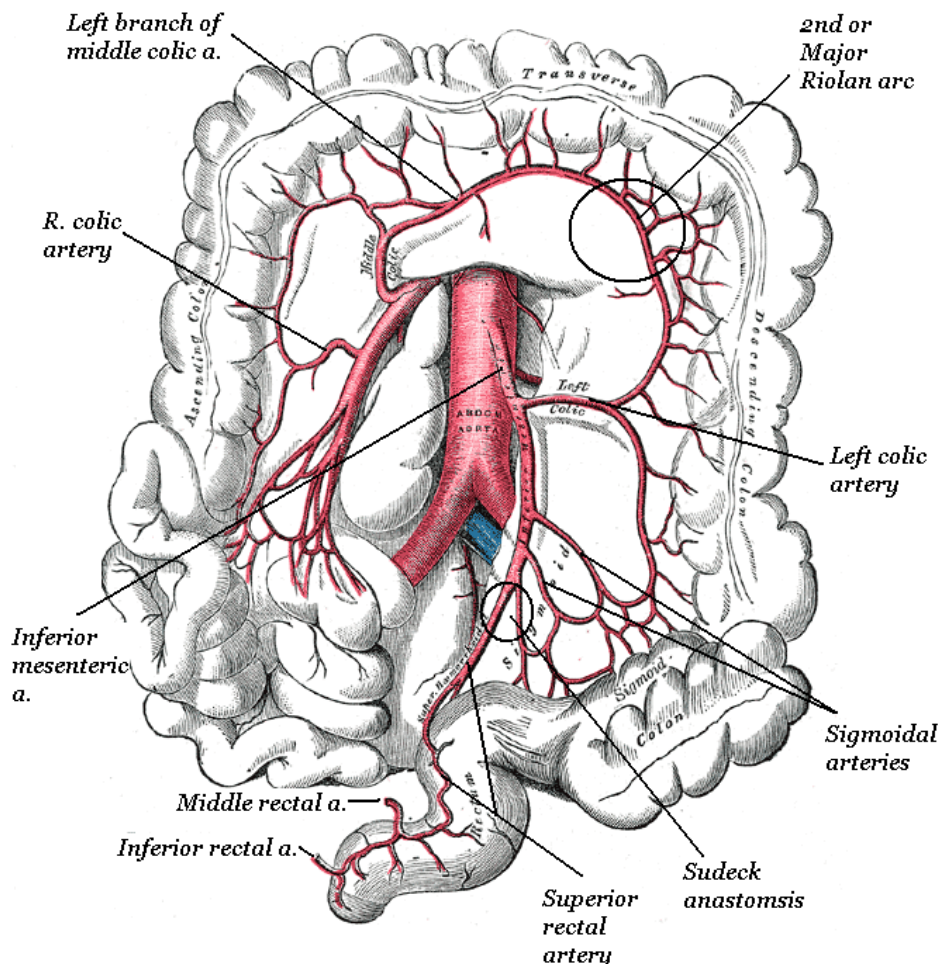




## ***Inferior mesenteric artery:***

The 3<sup>rd</sup> non paired branch of the abdominal aorta, at the level of L3, it gives:

1. ***Left colic artery:*** runs to the left behind the peritoneum and in front of the Psoas major muscle, and after a short course divide into an ascending and descending branches. The stem of the artery or its branches cross the left ureter and left gonadal vessels.
  - a. *The ascending branch* crosses in front of the left kidney and ends, forming the *paraduodenal fold*, and then passes between the two layers of the transverse mesocolon, by anastomosing with the middle colic artery (forming the 2<sup>nd</sup> Riolan arc discussed above)
  - b. *The descending branch* anastomoses with the highest sigmoid artery.
- \*\* From the arches formed by these anastomoses branches are distributed to the descending colon and the left part of the transverse colon
2. ***Sigmoidal arteries:*** usually 2-3 in number, supplying the lowest part of the descending colon (iliac colon), and the sigmoid (pelvic) colon. It anastomoses above with the left colic artery and below with the superior rectal artery as the *Sudeck critical anastomosis*
3. ***Superior rectal artery (Superior hemorrhoidal artery):*** an artery that descends to the pelvis within the Mesosigmoidium layers to supply the rectum. At the level of S3 it divides into 2 branches, which descend on both sides of the rectum supplying mainly the upper 1/3 of it. Anastomoses with the middle and inferior rectal arteries







## ***The Heart***

A hollow fibromuscular conical organ that receives the blood from the veins and propels it into the arteries, by repeated, rhythmic contractions. In mammals it is divided by a musculomembranous septum into two halves—right or venous and left or arterial—each of which consists of a receiving chamber (atrium) and an ejecting chamber (ventricle).

The heart of a vertebrate is composed of cardiac muscle, an involuntary muscle tissue which is found only within this organ. The average human heart, beating at 72 beats per minute, will beat approximately 2.5 billion times during a lifetime (about 66 years). It weighs on average 250 g to 300 g in females and 300 g to 350 g in males.

### **→ Location:**

The heart is located in the thoracic cavity, in the *inferior* part of the *anterior* mediastinum or so called *cardiac mediastinum*, below the supra-cardiac mediastinum. (See description of mediastinum at the end)

2/3 of the heart is located to the left of the midline and 1/3 to the right. The heart has an:

- **Apex:** located at the level of the 5<sup>th</sup> intercostal space, 9 cm from midsternal line
- **Base:** (Basis Cordis) located at the 3<sup>rd</sup> costal cartilage level (2cm L., 1cm R.)

The heart itself is rotated **60°** to the left in the **sagittal** axis, **30°** up in the **transverse** axis and **15°** to the **left** around the **Heart axis**.

- The *heart axis* is defined as a line passing between the middle of the spine of the right scapula and the apex of the heart.

### **→ Surfaces:**

- *Sternocostal:* Right atrium + right ventricle + small part of left ventricle
- *Diaphragmatic:* Mainly 2/5 right ventricle and 3/5 left ventricle + a small part of the right atrium around the IVC
- *Posterior (mediastinal):* Mainly the left atrium, between the 4 pulmonary veins orifices
- *Pulmonary (right):* Right atrium
- *Pulmonary (left):* Left ventricle, contacting the cardiac impression of the left lung

### **→ Anatomical Borders (Margins):**

- *Right:* Long, formed by the
  - Right atrium above, atrial portion
  - Right ventricle below, thin and sharp, named the *Acute* margin, almost horizontal
- *Left:* shorter, full and more round, formed by:
  - Left ventricle mainly, called the *Obtuse* (dull, not sharp)
  - Slightly by the left atrium above

\* Note: for the radiological borders (margins) of the heart, please check the heart X-ray section below

→ **Projections:**

- Right superior point: at the 3<sup>rd</sup> rib, right side of sternum (entrance of SVC to right atrium)
- Left superior point: at the 3<sup>rd</sup> rib, 3 cm left to the sternum (left superior end of coronary sulcus)
- Right inferior point: 1 cm right to the 6<sup>th</sup> sternocostal joint, right side of sternum (right lower end of coronary sulcus)
- Left inferior point: 5<sup>th</sup> IC space, 9 cm left of midline of sternum (apex of heart)

If we connect these 4 points with 4 arcuate lines we get: **The Margin (borderline)** of the heart, which has:

- *short axis*: between superior left and inferior right points → coronary sulcus plane (in this plane we have the Fibrous ring of the heart (Anulus fibrosus))
- *Long axis*: Oblique, from superior right (posterior) to inferior left (anterior) → means that the apex of heart is closer to the chest wall than the SVC entrance to right atrium (which means that the heart is rotated around its axis as we mentioned above)

→ **Pericardium:**

A fibrous sac that encloses the heart and the proximal parts of the great vessels and occupies the cardiac mediastinum, It is composed of 2 main layers:

- *Tough external parietal pericardium*
- *Visceral layer of serous pericardium* (Epicardium)

**1) Parietal pericardium:**

- Continues with Tunica adventitia of the great vessels and leaves the heart via the pretracheal layer of the cervical fascia, composed of an outer *fibrous* layer, and an inner *serous* layer (mesothelial cells) that reflects on the proximal parts of great vessels as the visceral layer of pericardium. It has got some connections:
  - The *Sternopericardial* ligament attaches to the posterior aspect of sternum
  - The *Pleuropericardiac* membrane attaches to the parietal pleura
  - Its connected to the central tendon of diaphragm by the *Pericardiophrenic* ligament (strong adhesion)

**2) Visceral layer of pericardium:**

- is reflected onto the heart and great vessels as the visceral layer of serous pericardium (Epicardium)
  - *Arterious reflections*: at division of pulmonary trunk and beginning of the aortic arch, so the 10-12 cm of the aorta + pulmonary trunk are inside the pericardiac cavity
  - *Venous reflections*: on the SVC at the entrance of the azygos veins into it (so 1/2 SVC in and 1/2 out). Also all the thoracic IVC which is ~1cm is totally inside the pericardium

Together with the reflection on the pulmonary veins as well, the venous reflections form the so called **Sappey's T** (T-shaped reflections of the pericardium, horizontal T)

### → *Pericardiac Sinuses:*

During embryonic development, the heart tube is initially straight down with the venous end being below and the Arterious end above.

The venous end will migrate upward and behind the Arterious end and forms a ***U-shaped*** tube which will give the venous reflection at the bending of this tube.

- *Transverse Pericardiac sinus*: found between main arteries (Aorta and Pulmonary trunk) and main veins (Pulmonary veins and SVC) reflections. Its an Open sinus Surgically important sinus because it allows the surgeon to pass his fingers in, and applies a ligature to stop the circulation for example in heart transplantation surgeries
- *Oblique Pericardiac sinus*: located at the venous reflections only (R. and L. pulmonary veins and IVC) behind the heart, after lifting the apex, between the parietal and the visceral layers. It's a *blind* recess with a dead end at the inferior part of Sappey's T. It's reached from the left side by 45°

### → *Arterial supply of the pericardium:*

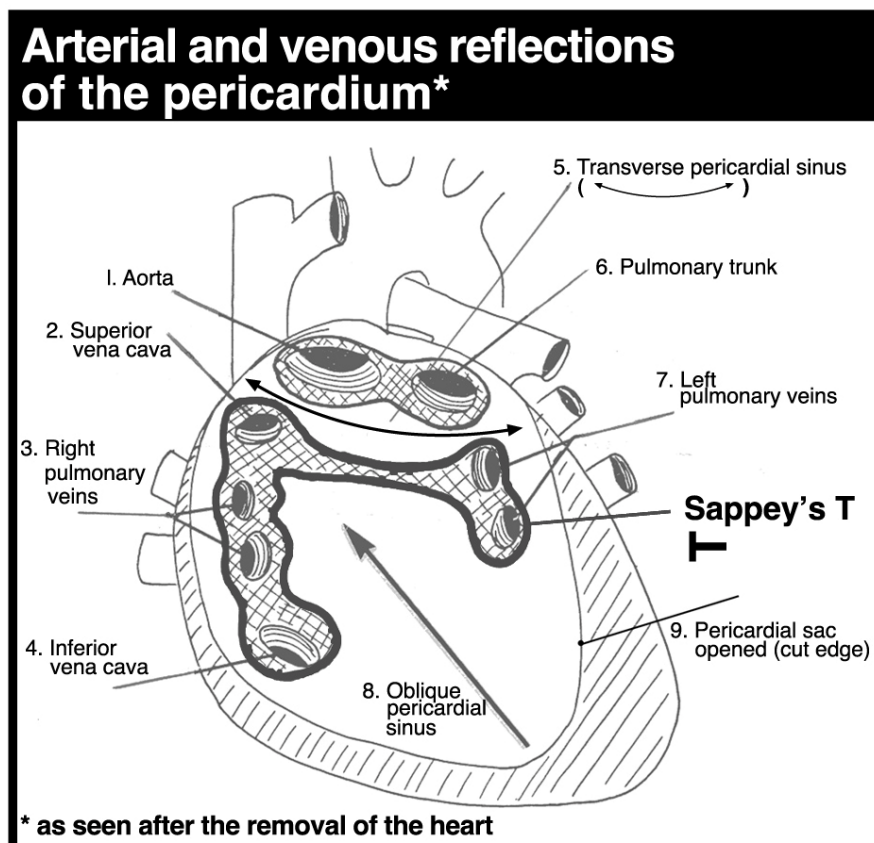
- Mainly the *Pericardiophrenic* artery (branch of the internal thoracic artery)
- Musculophrenic artery (also from the internal thoracic artery)
- Superior Phrenic arteries (branches of thoracic aorta)
- Coronary arteries (only to the visceral layer)

### → *Venous drainage:*

- Pericardiophrenic vein to the brachiocephalic veins
- Azygos venous system

### → *Nervous supply of the pericardium:*

- Phrenic nerve: Sensory innervation (reflected to ipsilateral skin dermatomes C3-C5)
- Sympathetic trunk (vasomotor)



### → Layers of the heart:

- 3 main layers: 1) **Epicardium**: Visceral layer of the pericardium  
2) **Myocardium**: muscular layer  
3) **Endocardium**: innermost endothelial layer

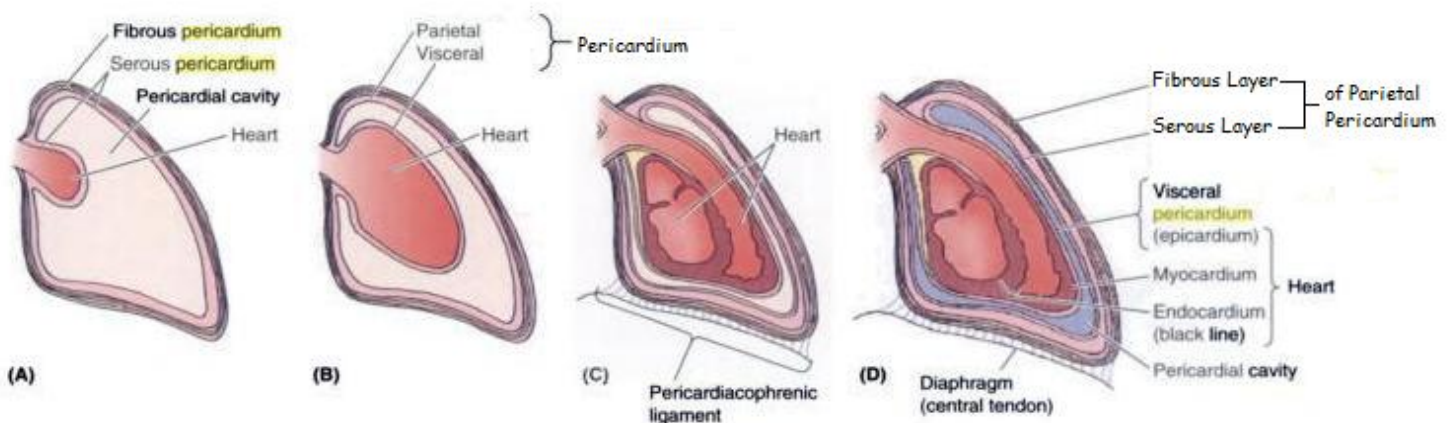
But if we start from the outermost layer of the heart including the pericardium we have:

- 1) Outer fibrous layer of parietal pericardium
- 2) Inner serous layer of parietal pericardium
- 3) **Visceral pericardium (Epicardium)**: the fat (pericardiac adipose tissue) will accumulate between this layer and the next myocardium layer. The coronary arteries run inside this fat between the 2 layers as well
- 4) **Myocardium**: originates and inserts on the fibrous skeleton of the heart with the muscular hooks, has 3 layers in the *ventricles*:
  1. **Outer (external) oblique or longitudinal layer**: subepicardiac; originates from the AV valves fibrous rings to the apex, Anticlockwise and it forms the Vortex of heart, and turns inward to form the innermost layer.
  2. **Middle circular layer**: parallel to the coronary sulcus plane, NOT found at apex (vortex)
  3. **Internal (innermost) longitudinal layer**: subendocardial, its actually the continuation of the outer (External) longitudinal at the vortex of the heart internally, tot form the Papillary muscles and the Trabeculae Carnea

The atria have only 2 layers:

1. **Outer Circular layer**: extending over both atria, thicker anteriorly
  2. **Inner longitudinal layer**: containing loops or annular fibers that pass to the respective AV orifice or surround veins openings
- 5) **Endocardium**: a thin internal layer composed of an *endothelium* + subendothelial connective tissue that form the valves (the valves are basically a duplication of the endocardium).

**\*\* The Pericardial cavity (space)** is a potential space between the parietal & visceral layers of the pericardium. Normally it contains a thin film of serous fluid that lubricates the serous surfaces and allows free movement of the hearth within the pericardium

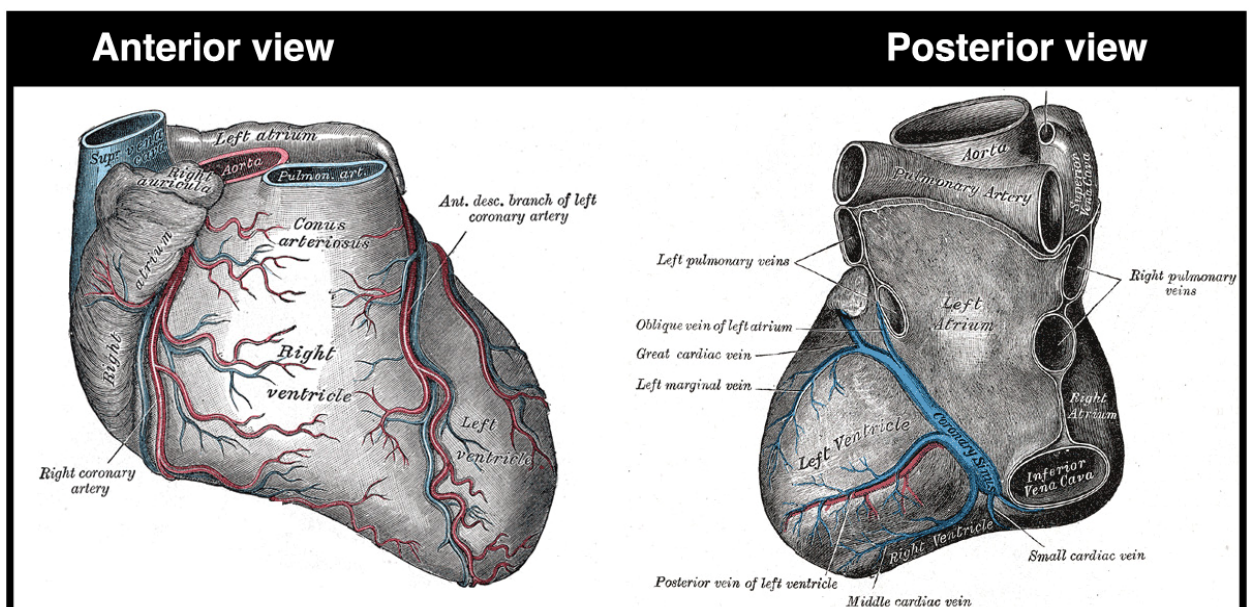




### → **External Features of the heart:**

When looking at the heart externally we can distinguish:

- Surfaces of the heart (Anterior, Inferior, Pulmonary)
  - Chambers of the heart (RA + auricle, RV, LA + auricle, LV) separated by the Coronary and Interventricular sulci, containing the cardiac veins and coronary arteries:
    - *The coronary (atrioventricular) sulcus*: a groove running around the heart, separating the atria from the ventricles
    - *The anterior interventricular sulcus*: interventricular sulcus anteriorly,
    - *The posterior interventricular sulcus*: interventricular sulcus posteriorly, in which the middle cardiac vein runs with the posterior IV branch or RCA
- \*\* The interventricular sulci join the coronary sulcus at a right angle; posteriorly it's called the *Cross of heart (Cruix Cordis)*
- \*\* *Waterston's groove*: between the right pulmonary veins and the right atrium
- Great vessels:
    - *Aorta*: arising from the LV as the ascending aorta, then the aortic arch forming (with its branches: Brachiocephalic trunk, L. common carotid and L. subclavian) and crossing over the pulmonary trunk, and after becoming the descending aorta posteriorly
    - *Pulmonary trunk*: arising from the RV, anterior to the aorta, then it winds around and goes under the aortic arch and bifurcating into R. and L. pulmonary arteries that enter the root of the R. and L. lungs respectively
    - *Superior Vena Cava (SVC)*: formed by the R. and L. brachiocephalic trunks
    - *Inferior Vena Cava (IVC)*: entering the inferior aspect of the RA
    - *Pulmonary veins*: 2 right and 2 left opening horizontally into the LA
- \*\* the **Ligamentum arteriosum**: a short band which remnant of fetal ductus Arteriosus (Botalo duct) can be found between the inferior aspect of the aortic arch and the superior aspect of pulmonary trunk, medially to it, we can find the R. recurrent laryngeal nerve arching below the Aortic arch
- \*\* **Sulcus terminalis Cordis**: a groove separating the venae cavae from the base of the right auricle



## → Cavities / Chambers of the heart:

### 1) The Right Atrium:

- Forms the right border of the heart and part of its anterior surface
- Has 2 main parts:
  1. *R. Auricle*: rough muscular unfunctional anterior wall, composed of pectinate muscles that develop from the common atrium
  2. *Atrium proper*: smooth, thin walled posterior part (sinus venarum cavarum) that develops from the Sinus venosus. Have openings for veins in it
- \*\* Both parts are separated *externally* by the **Terminal sulcus** (shallow vertical groove from the superior to the inferior vena cava) and from the *inside* by the **Crista terminalis** (well defined muscle bundle like a ridge) which develops from the septum spurium.
- **IVC orifice**: on the lateral side. Has a *Eustachian valve* (guides the blood from IVC towards the oval foramen to Left atrium in fetal life. In adults this valve has no function. This orifice is at 5<sup>th</sup> costal cartilage level
- **SVC orifice**: has no valve. Opens at the level of the 3<sup>rd</sup> costal cartilage
- **Coronary sinus orifice**: located between the IVC orifice and the oval fossa. Has a *Thebesian valve*. It collects blood from all cardiac veins except the anterior small branches.
- **Interatrial Septum**: Have the *Oval fossa + Limbus*, which is a depression in the wall covered by a valve called the valve of the oval fossa. During the embryonic period it was the Oval foramen. After birth, the valves fuse with the Limbus of oval foramen closing it. If remains open after birth it's called Patent oval foramen or Foramen apertum and plays a role in dangerous Venous-Arterious shunts that might cause paradoxal embolism formation.
- Invisible to the naked eye, inside the right atrium we have the **SA (Sinoatrial) node**, that is located at the upper end of the terminal sulcus near the SVC inlet in a *subepicardiac* position
- **Koch triangle**: a triangular area of the wall of the right atrium of the heart that marks the approximate situation of the *AV node*, it is bordered *superiorly* by the tendon of *Todaro*, *inferiorly* by the *tricuspid valve annulus* and *posteriorly* by the *Thebesian valve* of the coronary sinus
- **Sinus Septum**: a small fold forming the medial end of the valve of the inferior vena cava; it is developed from the dorsal wall of the embryonic sinus venosus
- **Todaro Tendon**: an inconstant tendinous structure that extends from the right fibrous trigone of the heart toward the valve of the inferior vena cava or Sinus septum. It runs intramyocardially to insert into the right fibrous trigone but can be easily demonstrated by superficial dissection, to demarcate the site of AV node (in the Koch triangular area)
- **The Torus aorticus**: is a normal slight bulge of the right atrial septum sited superiorly and anteriorly. It is superior to the coronary sinus and anterior to the fossa ovalis. Formed by the posterior (non-coronary) cusp and the right coronary cusp of the aortic root lean against the medial right atrium.

## 2) *The Right Ventricle:*

- Forms the largest part of the anterior surface of heart, a small part of the diaphragmatic one and the inferior border.
- The ventricles are described in terms of 3 parts :
  1. *An inlet (Inflowing part)*: containing an atrioventricular valve and its tension apparatus receiving venous blood from Right atrium
  2. *A trabecular body*
  3. *An outlet (Outflowing part)*: supporting an arterial valve, blood flows out from ventricle through the orifice of pulmonary trunk (infundibulum)
- This division is **not** meant to indicate that ventricular inlet and outlet portions lack trabeculations, although in many places they do have smooth walls. Rather it indicates that the apical trabecular zones are the most trabeculated and most distinctive atrioventricular valves is similar in each ventricle although distinctive differences exist.

\*\*So the ventricle is a V-shaped structure (a 140° angle is formed between 2 parts), the 2 parts are separated by the **Supraventricular crest** (a protrusion on the interventricular wall/septum)

\*\*The Outflowing part forms a cone on the outer surface → **Conus Arteriosus**, from where the pulmonary trunk starts (*infundibulum is from inside*) and it has a smooth surface
- The interior of the ventricle has an irregular muscular elevations → **Trabeculae Carneae** or *Rathke's bundles*
- The **Septomarginal Trabeculae** or the **Moderator band** is a unique big Trabecula crania that transverse the ventricle from side to side and has 3 functions
  - Origin of the Anterior papillary muscle
  - Transmission of the right branch of the AV bundle (R. Tawara crus) to the ventricular wall
  - Holds the walls of the ventricle, prevent over dilation
- The **Tricuspid** valve guards the right atrioventricular orifice (*AV orifice*) (surrounded by 1 coronary filum of fibrous skeleton). The base of the valve's cusps are attached to the fibrous ring, and the tip is attached like a parachute to the Cordae Tendinae, that attach to the anterior, posterior and Septal papillary muscles
- **Papillary muscles** are conical/cylindrical muscular protrusions with bases attached to the ventricular wall. 3 groups in the right ventricle:
  - Anterior papillary muscle: largest and most prominent, arises from the supramarginal Trabeculae and attaches to the Anterior + Posterior cusps
  - Posterior papillary muscle: arises from the inferior wall of the right ventricle and attaches to the Posterior + Septal cusps
  - Septal papillary muscle: usually small and infrequent, arises from the IV septum. (its almost motionless because the IV wall is not moving during contraction)

- **The interventricular septum:** composed of *muscular* part and *membranous* part. It's 2-3 times thicker than the wall of the Right ventricle, and bulges towards the right ventricle in the cross sections creating the Supraventricular crest. The membranous part located superioposteriorly in the right ventricle is a thin membrane that is part of the fibrous skeleton and attached to the Septal cusp of the tricuspid valve. (that's why this cusp is not really functional)

### 3) **Left Atrium:**

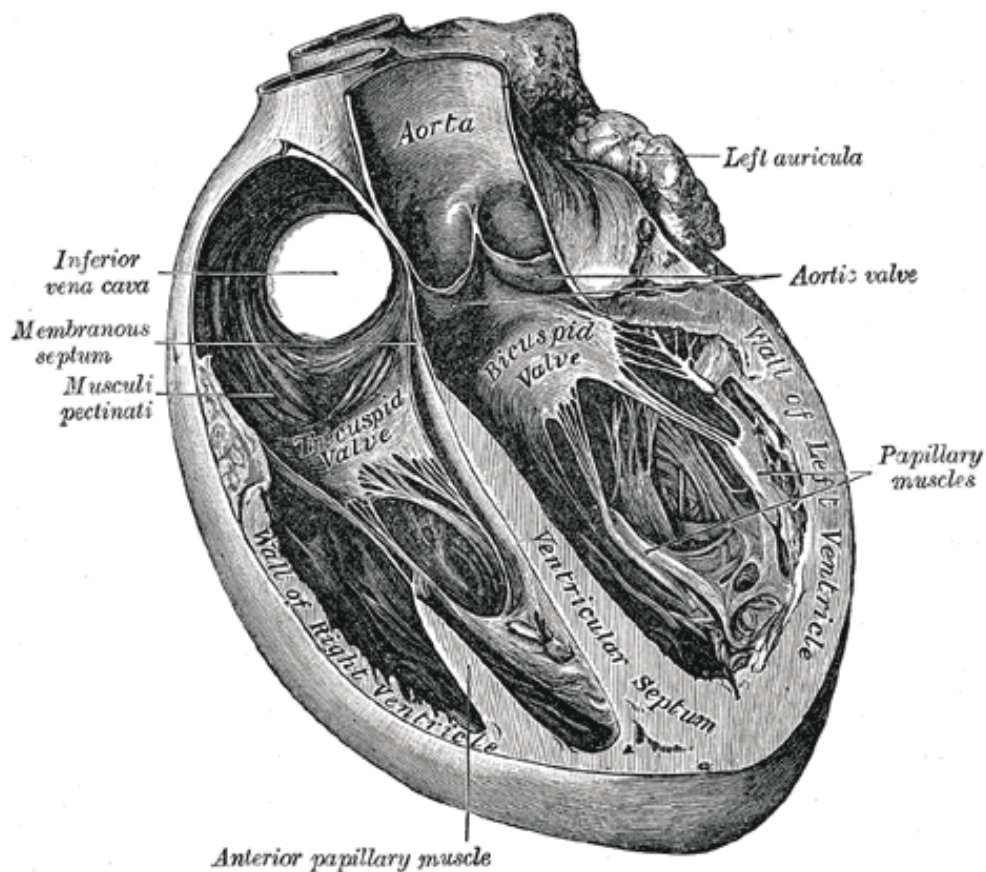
- forms most of the base of the heart (posterior surface), so it's the only chamber that cannot be seen on normal X-ray
  - Has 2 main parts:
    1. *L. Auricle:* rough muscular unfunctional anterior wall, composed of pectinate muscles. It overlaps the Pulmonary trunk (it's a remnant of the primordial atrium)
    2. *Atrium proper:* smooth, thin walled posterior part. Has openings for the valveless pairs of the L. and R. Pulmonary veins conveying oxygenated blood from the lungs
- \*\* Though there are no true valves at the junction of the pulmonary veins and the left atrium, "*sleeves*" of atrial muscle extend from the left atrial wall around the pulmonary veins for 1 or 2 cm and may exert a partial sphincter-like influence, tending to lessen reflux during atrial systole or mitral regurgitation
- Has a semilunar depression on the Interatrial wall → **Fossa ovalis**
  - Has a slightly thicker wall than the right atrium
  - The left AV orifice (mitral) leads to the left ventricle
  - Has a close relation with the esophagus which is *important clinically*:
    - contrast material can be applied in the esophagus + X-ray medially will show the enlargement of the left atrium bulging into the esophagus which is not seen in normal X-Ray
    - when using the esophagoscope → danger of injury to the esophageal wall and damaging the left atrium

### 4) **Left Ventricle:**

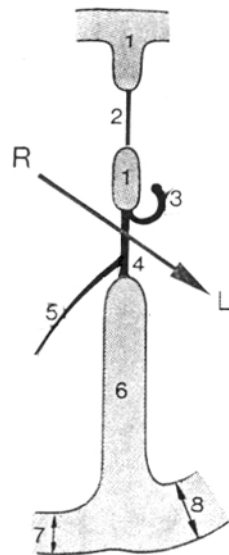
- forms the left surface (Pulmonary) + most of the diaphragmatic surface + apex of the heart
- It comprises:
  1. *An inlet portion (Inflow part):* containing the mitral valve and its tension apparatus; receiving oxygenated blood from the Left atrium
  2. *An apical trabecular zone:* characterized by fine trabeculations
  3. *An outlet zone (Outflow part):* supporting the aortic valve, which is incomplete posteriorly so that the aortic and mitral valves are in fibrous continuity



- It performs more work the right ventricle because the systemic circulation pressure is higher; therefore it has a *thicker muscular wall* (1-1.5cm) than the right ventricle (0.5cm).
- The inner surface is covered with Trabeculae Carnea that are finer and more numerous than right ventricle
- It has an *Anterio-Medial* and a *Posterio-Lateral* papillary muscles larger than the ones in the right ventricle and has a double leaflet **Mitral (bicuspid)** (with the same cusps as the papillary muscles) valve guarding the AV-orifice
- As it transverses through the LV, the blood stream undergoes 2 right angle turns → *180° direction change*. This flow direction reversal takes place around the *anterior cusp of the mitral valve*, which forms the border between the outflow and inflow portions (no Supraventricular crest here).  
The superioanterior part of the ventricle is a smooth outflow part reaching the **Aortic vestibule** and orifice that is positioned in the right posteriosuperior part and surrounded by a fibrous ring to which the Right + Posterior + Left cusps of the aortic valve are attached.

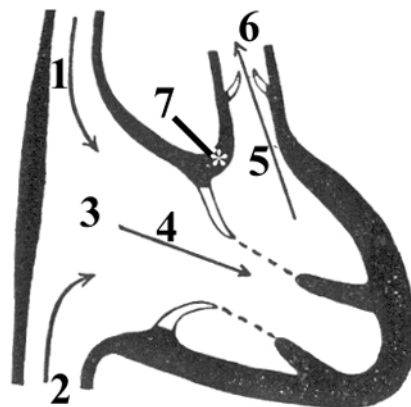


## Frontal section of the heart (illustrating the interatrial and interventricular septum)



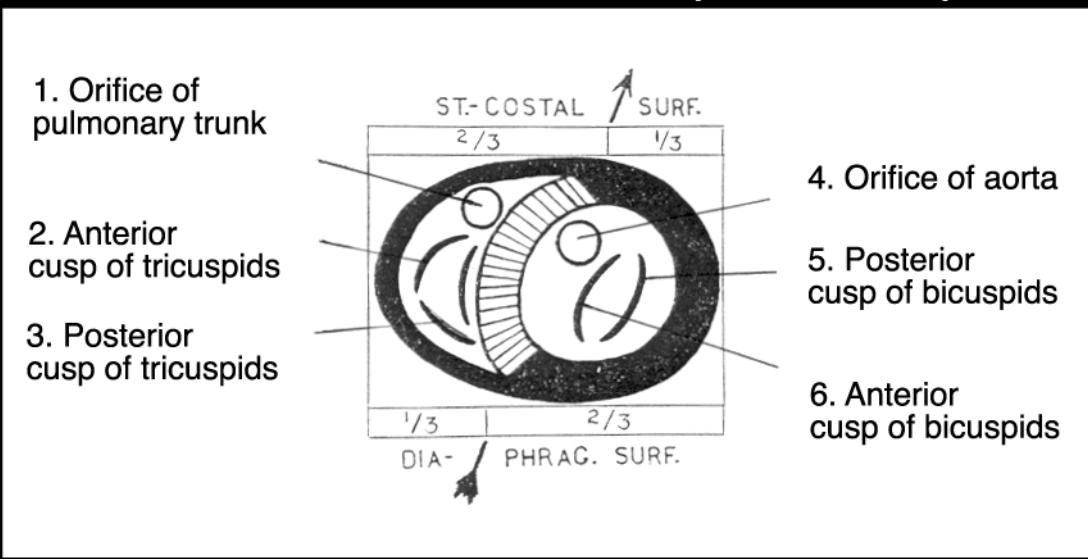
1. Interatrial septum
2. Fossa ovalis
3. Right semilunar valve of the aorta
4. Membranous part of interventricular septum
5. Septal cusp of the tricuspid
6. Muscular part of the interventricular septum
7. Wall of the right ventricle
8. Wall of the left ventricle

## Blood flow in the right atrium and ventricle



1. Sup. vena cava
2. Inf. vena cava
3. Right atrium
4. Inflow of the right ventricle
5. Outflow of the right ventricle
6. Pulmonary trunk
7. Supraventricular crest

## Transverse section of the heart (from below)



### → *Fibrous Skeleton of the heart (Annulus fibrosus Cordis):*

It's a complex network of dense collagen forming: 4 *fibrous rings* (annuli fibrosi) that surround the orifices of the valves, Right and left fibrous trigons (formed by the connections between the rings) and the membranous part of the IV septum. It is found in association with the base of the ventricles, i.e., at the level of the *coronary sulcus*;

*Functions:*

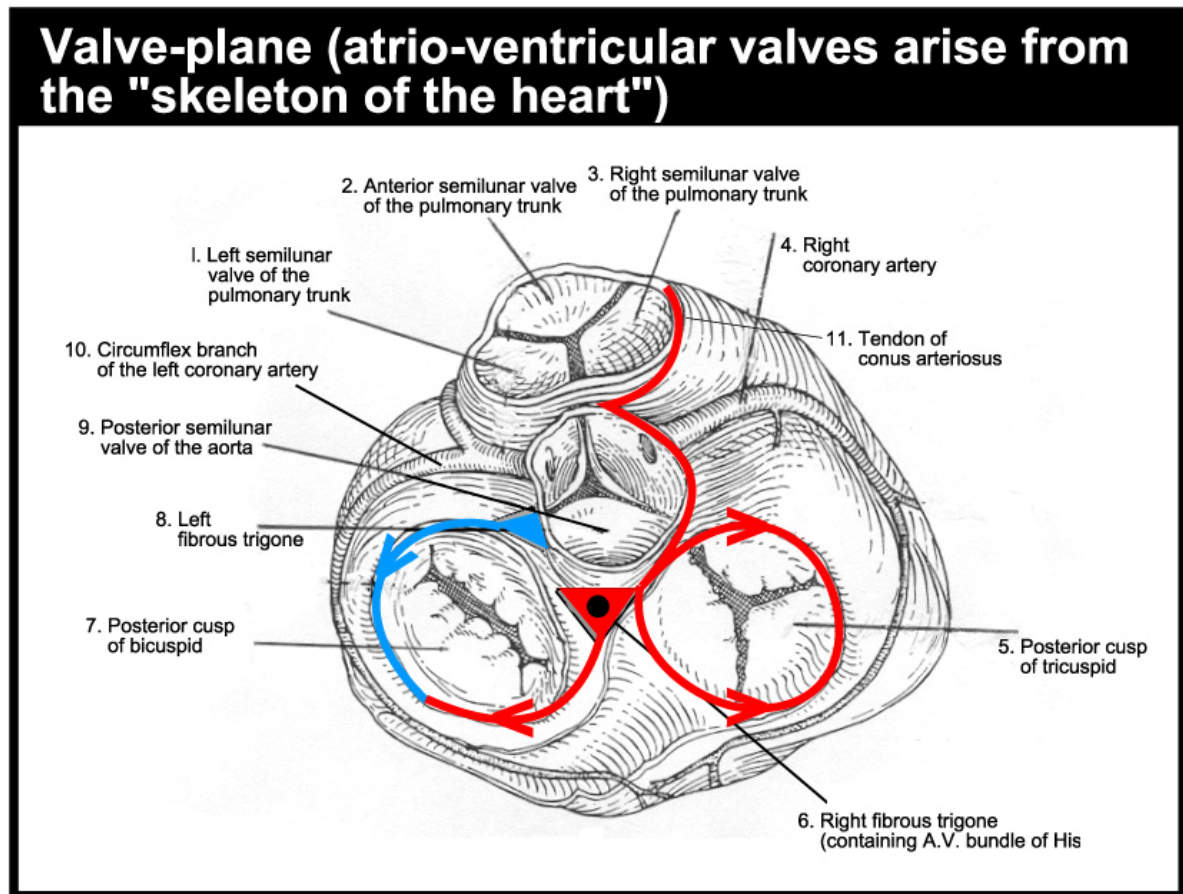
- Keeps the orifices of the AV and semilunar valves patent and prevents them from being over distended by an increased volume of blood pumped through them (reinforcement)
- Providing attachment for the leaflets and cusps of the valves
- Provides attachment for myocardium (myocardial hooks), when uncoiled → continuous ventricular myocardial band that originate primarily from fibrous ring of pulmonary valve and mainly inserts on the fibrous ring of the aortic valve
- Forms an electrical "Insulator" by separating the electrically conducted impulses of atria and ventricles so that they contract independently of the surrounding.
- Provides passage for the initial part of the AV bundle (His bundle) of the conducting system via the right fibrous trigone
- 

The area between the septal leaflet of the tricuspid valve, the anterior leaflet of the mitral valve, and the posterior or non-coronary cusp of the aortic valve forms one part of the central fibrous body (Right fibrous trigone). The remaining portion is made up of fibrous tissue connecting the left aortic cusp and the anterior leaflet of the mitral valve (Left fibrous trigone).

1. *Right trigone*: between L. and R. AV ostia and Aorta
2. *Left trigone*: between Aorta and L. AV ostium

- ***Fila coronaria***: a threadlike like structures arising from the fibrous trigons and encircling the nearby valves, from the left fibrous trigone arises 1 coronary filum, and from the right fibrous trigone 3, to form a complete fibrous rings around the AV valves.

- **Conus Arteriosus tendon:** a stripe of fibrous tissue structure that is situated between the aortic root and pulmonary trunk. Extends upward from the right atrioventricular fibrous ring and connects the posterior surface of the Conus Arteriosus to the aorta
- **Os Cordis (Cardiac bone):** represents the right fibrous trigone in big animals like ox/elephant, or the calcified tendon of Todaro



### → Valves:

The heart contains four cardiac valves: 2 semilunar and 2 atrioventricular. The two *semilunar valves*, *aortic and pulmonary*, guard the outlet orifice of their respective left and right ventricles. The two *AV valves*, *mitral and tricuspid*, guard the inlet orifice of their respective left and right ventricles.

The four cardiac valves are surrounded by fibrous tissue forming partial or complete "rings" (valve anulus). These fibrous rings join to form the fibrous skeleton of the heart, to which also is attached atrial and ventricular myocardium.

The distal two-thirds of the normal AV valve and all the semilunar valve are avascular

- **The Cusped valves** (Mitral, tricuspid) (AV) are composed of 2 (Mitral: Anteromedial, Posterolateral) or 3 (Tricuspid: Anterior, Posterior and septal) cusps that have mesothelium

on both surfaces (Duplication of the endocardium). Between the mesothelium layers we have fibrous cutaneous tissue.

The cusps are attached to Chordae Tendinae → Papillary muscles → prevent flipping back into atria.

*Chordae Tendinae*: Dense collagen tendinous strands running from the papillary muscles to the leaflets of the atrioventricular valves (mitral and tricuspid).

- **The Semilunar valves**: both aortic and pulmonary has 3 cusps
  - Aortic: Posterior / Right / Left
  - Pulmonary: Anterior / Right / Left

The small space between attachments of adjacent cusps is called a *commissure*. Each semilunar valve has three commissures. The three commissures lie equally spaced around the aorta or pulmonary trunk, and the circumference connecting these points has been termed the *sinotubular junction*, which may also be described as the portion of the great vessel separating the sinuses of Valsalva from the adjacent tubular portion of the great artery.

In the aorta a distinctive circumferential "hump" or line marks this junction, originally described by Leonardo da Vinci as the "*supraaortic ridge*"

During diastole, the elastic recoil of Pulmonary trunk/Aortic wall forces blood back towards heart, but cusps open up like pockets → close orifice → prevents regurgitation of blood.

Each semilunar valve is composed of 2 major parts:

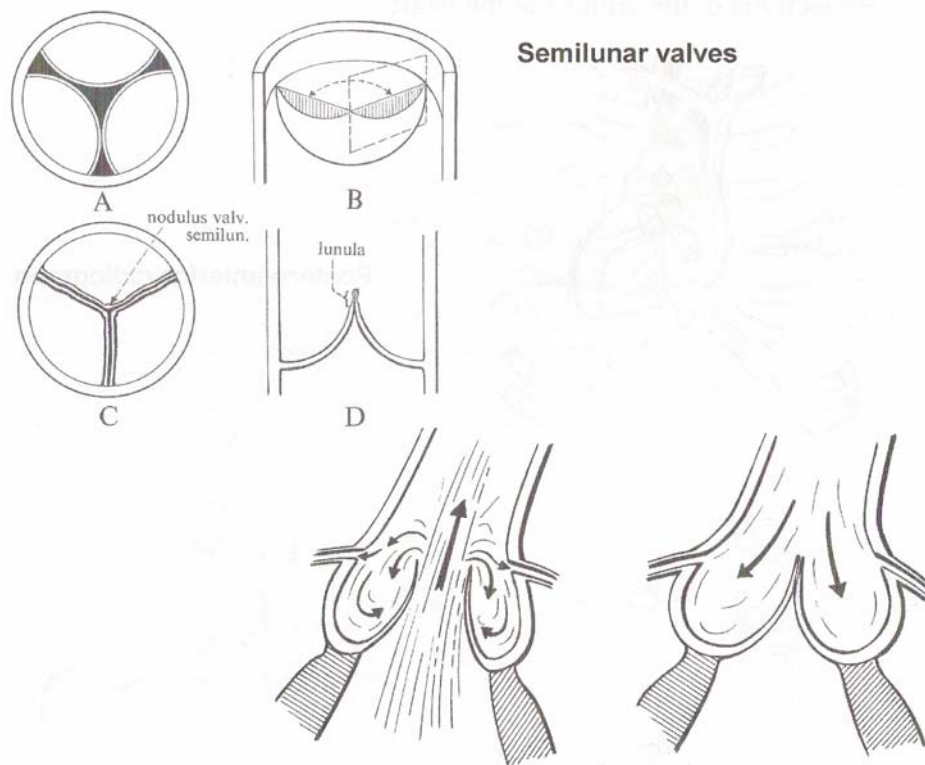
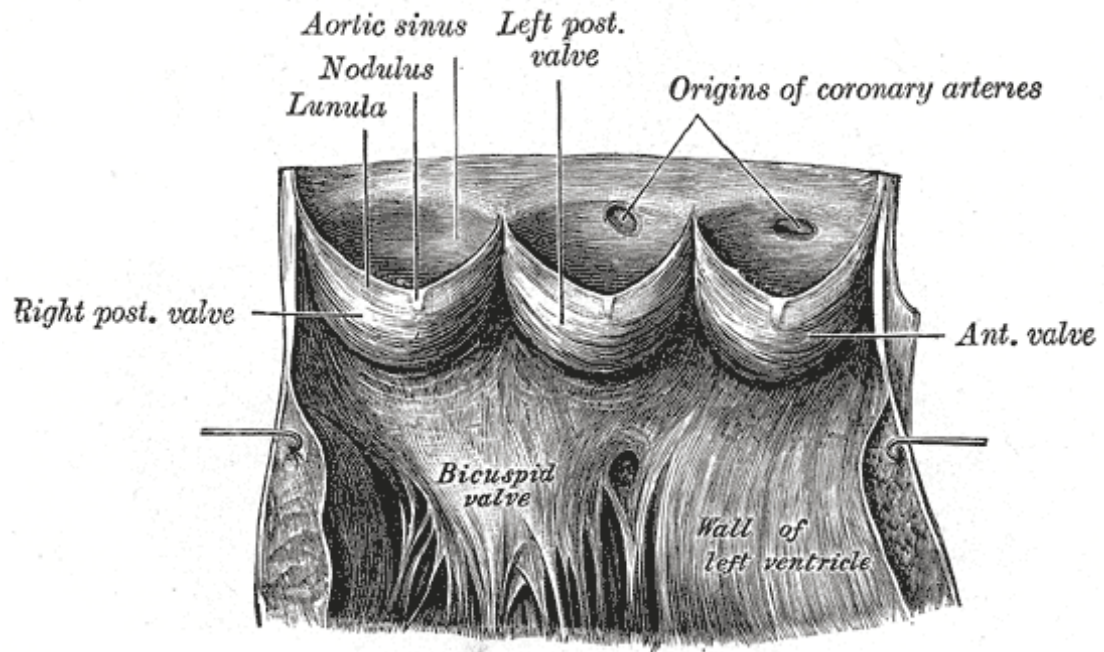
- *Pars Densa*: lower part, less flexible and mobile
- *Pars Flaccida*: upper more mobile portion of the valve, sub-composed of:
  - i. *2 Lunules*: The thickened edge of each cusp in the region of contact.
  - ii. *1 Nodule (Body of Arantius)*: the apex of the angulated free edge is thickened further.

So each semilunar valve has 6 lunules and 3 nodules. The nodules help in *tight* closure of the cusps when they approach each other

Immediately superiorly to the cusps, the walls of the origin of the aorta and the pulmonary trunk are slightly dilated → forming a Sinus → aortic/pulmonary sinuses → blood in them prevents the cups from sticking to the wall of vessel

- The mouth of R. coronary artery is at the Right aortic sinus
- The mouth of L. coronary artery is at the Left aortic sinus
- No artery at the posterior aortic sinus (non-coronary sinus)





→ **Auscultation of the heart-valves:**

- **Aortic valve:** 2<sup>nd</sup> IC space, **right** side of sternum (1-2 cm right to midsternal line)
- **Pulmonary valve:** 2<sup>nd</sup> IC space, **left** side of sternum (1-2 cm left to midsternal line)
- **Tricuspid valve:** 5<sup>th</sup> IC space, **left** side of sternum (2 cm from midsternal line)
- **Mitral valve:** 5<sup>th</sup> IC space, **left** side of sternum in the **midclavicular** line (9 cm **left** to midsternal line) (at the apex of heart)

(Remember:

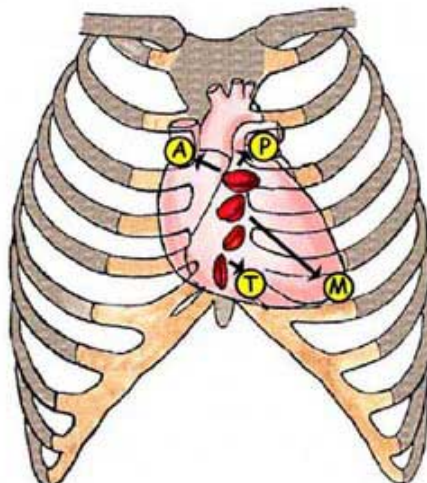
- *Sternal line:* vertical line on the edges of the sternum
- *Parasternal line:* a vertical line in the middle between the midclavicular and sternal lines)
- **Erb's point:** 3<sup>rd</sup> IC space, 3cm to midline of sternum (general auscultation point for all sounds) (Don't mix it with Erb's nervous point or Punctum nervosum)

**\*\*Note:** the auscultation point of the tricuspid valve is according to the Oxford manual of internal medicine, Oxford manual of medical examination, and Bates manual of clinical examination, although it's possible also to listen to the Tricuspid valve on the right side of the sternum according to other sources, because the location of the tricuspid valve itself which is directly behind the midline of sternum at the level of 5<sup>th</sup> IC space. So be ready to present both points of view.

**HEART - AUSCULTATION**

As the valves open and close they produce sounds that are transmitted in the direction of the flow of blood. Thus, by picturing the heart and the positioning of the four valves it is easy to work out the likely points for maximal audibility of the sounds. The position of the valves, relative to the surface is shown on this diagram and the points at which auscultation is best achieved.

- P** Pulmonary - 2nd left space, parasternally
- A** Aortic - 2nd right space, parasternally
- M** Mitral - 5th left space, mid clavicular line (apex)
- T** Tricuspid - Over lower sternum



### → *Vasculature of the heart:*

2 coronary arteries supply the heart; they arise from the aortic (Valsalva) sinuses (at the dilated portion of the ascending aorta)

They supply the: Myocardium & Epicardium

- *The Right coronary artery (RCA):*

Arises from the right anterior aortic sinus → runs in the coronary groove after passing between the right atrium and the pulmonary trunk reaching the posterior aspect of the heart to end up in the posterior interventricular groove

Branches:

- Conus Arteriosus branch: for the conus arteriosus portion of the right ventricle
- Ascending Sinoatrial (SA) branch: for SA node (in 60% of people)
- Right marginal branch: right border of heart, but *not* to the apex!
- Right atrial branch: for right atrium
- Right ventricular branch: for right ventricle, but not totally, until 1cm from anterior interventricular groove)
- Atrioventricular nodal branch: for AV-node, given at the crux of heart (Posteriorly)
- Posterior interventricular branch: descends in posterior interventricular groove, accompanying the middle cardiac vein, has perforating ventricular posterior Septal branches for interventricular septum (for posterior 1/3 of IV septum)

- *The Left coronary artery (LCA):*

Arises from the left anterior aortic sinus as the main left coronary stem → passes between left atrium (auricle) and left side of the pulmonary trunk, runs into the coronary groove and divides into 2 main branches:

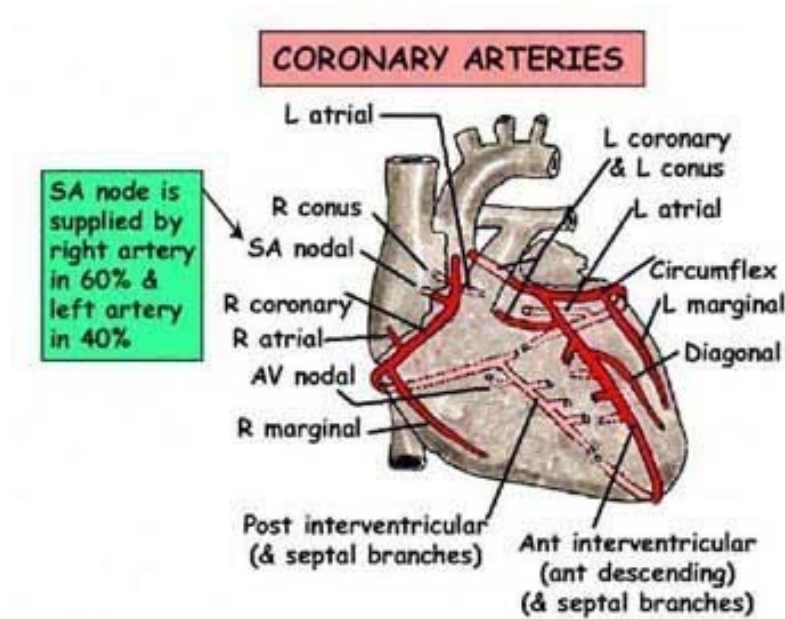
- Anterior interventricular branch (left anterior descending, LAD): passes along the anterior interventricular groove until the apex of heart together with the great cardiac vein (can anastomose with RCA), gives:
  - Anterior Septal branches: supply the anterior 2/3 of IV septum + the AV bundle system
  - Adjacent parts of Left and right ventricles (adjacent to IV groove)
  - Lateral diagonal branch: for the anterior surface of the heart
  - Ascending Sinoatrial (SA) branch: for SA node (in 40% of people)
- Circumflex branch: follows the coronary groove together with the coronary sinus until the posterior surface of the heart, gives:
  - Left marginal artery: supplies left margin of the heart
  - Left atrial artery: for left atrium and auricle
  - Left posterior lateral branch: Left ventricle (back)

### **\*\* Coronary artery dominance**

The artery that supplies the *posterior descending artery (PDA)* (a.k.a. posterior interventricular artery) and the *AV-Node* determine the coronary dominance:

- If the posterior descending artery (PDA) is supplied by the right coronary artery (RCA), then the coronary circulation can be classified as "right-dominant".
- If the posterior descending artery (PDA) is supplied by the circumflex artery (CX), a branch of the left artery, then the coronary circulation can be classified as "left-dominant".
- If the posterior descending artery (PDA) is supplied by both the right coronary artery (RCA) and the circumflex artery, then the coronary circulation can be classified as "co-dominant".

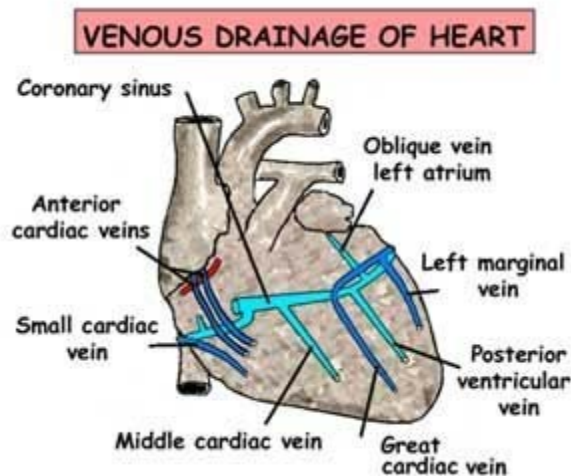
Approximately 70% of the general population are right-dominant, 20% are co-dominant, and 10% are left-dominant



### → Venous drainage of the heart:

- 1) *Great cardiac vein*: begins near the apex of heart, ascends in the anterior IV groove with the anterior IV branch of the LCA, after it goes left with the circumflex branch into the coronary sulcus and then drains into the Coronary sinus
- 2) *Middle cardiac vein*: runs in the posterior IV groove, together with posterior IV branch of RCA, drains finally in the coronary sinus
- 3) *Small cardiac vein*: runs with the marginal branch of RCA backwards until it drains into coronary sinus
- 4) *Posterior vein of the left ventricle*: found on the posterior surface of the left ventricle, drains to the coronary sinus
- 5) *Oblique vein of left atrium (Marshall's vein)*: remnant of the embryonic L. SVC, under the left auricle Posteriorly, drains into the coronary sinus
- 6) *Anterior small cardiac branches*: found on the anterior surface of the right ventricle and drain into the right atrium directly! (not into Coronary sinus)
- 7) *Least Cardiac veins (Venae Cordi Minimi)*: found on the anterior surface of the left ventricle, drain directly into the left atrium directly! (causing a small un-significant shunt)

\*\* The coronary sinus is a wide venous channel on the posterior aspect of heart in the coronary groove that collects the great, middle, small and oblique veins and drains them into the right atrium, has *Thebesian valve* guarding its exit.



### → *The Conducting system of the heart:*

Coordinates the cardiac cycle of systole/diastole, highly specialized cardiac muscle cells and conducting fibers for initiating impulses and conducting them rapidly

Starts with the pacemaker: *the SA (Sinoatrial)(Keith-Flack) node*, located *sub-epicardially* at the junction of SVC and right atrium anterolaterally, near the superior end of sulcus terminalis, it gives ~100 pulse/min

Impulses will travel in the walls of the atria by myogenic conduction through the *internodal pathways (4 bundles: anterior, middle, posterior and Bachmann's)* and will reach *the AV (atrioventricular) (Aschoff-Tawara) node*

The AV node located *sub-endocardially* at the posterior inferior region of the interatrial septum, below the fossa ovalis, demarcated by a triangular area called *Koch's triangle*. After passing through the AV node, the impulse will go through the *AV-bundle of His (fascicle)* that arises from the AV node and pierces the right fibrous trigone through the fibrous skeleton of the heart, which is the only connection between the atria and ventricles

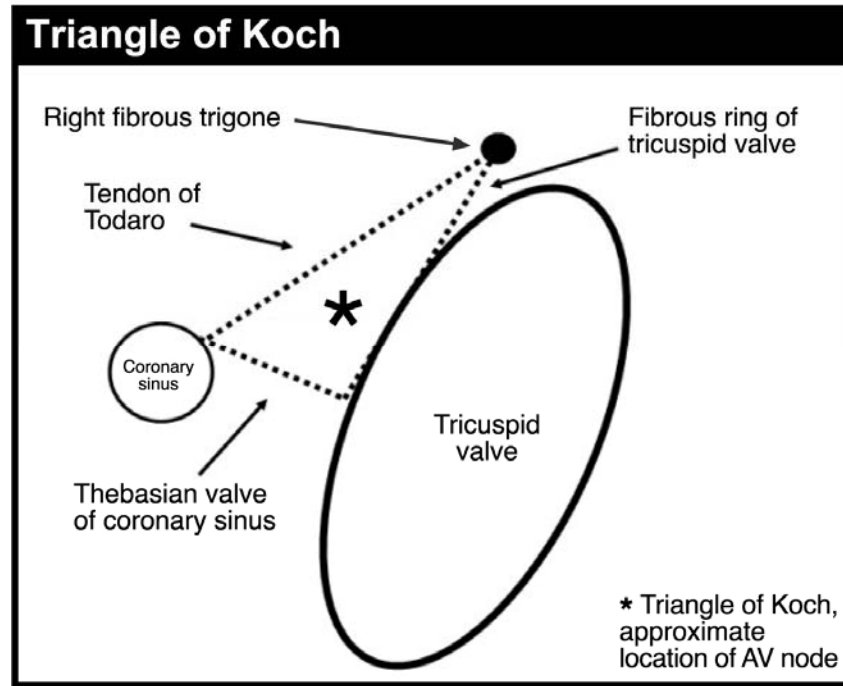
After piercing the right trigone, the impulse will travel in the *membranous* part of the IV septum, and at the junction of the membranous and muscular part, the His bundle will separate into 2 bundle branches:

- *Right bundle branch or AV crus (Right Tawara crus)*: which curves downward and enters the Septomarginal Trabecula
- *Left bundle branch or AV crus (Left Tawara crus)*: sub-divides into Anterior (which gives the septal) & Posterior fascicles

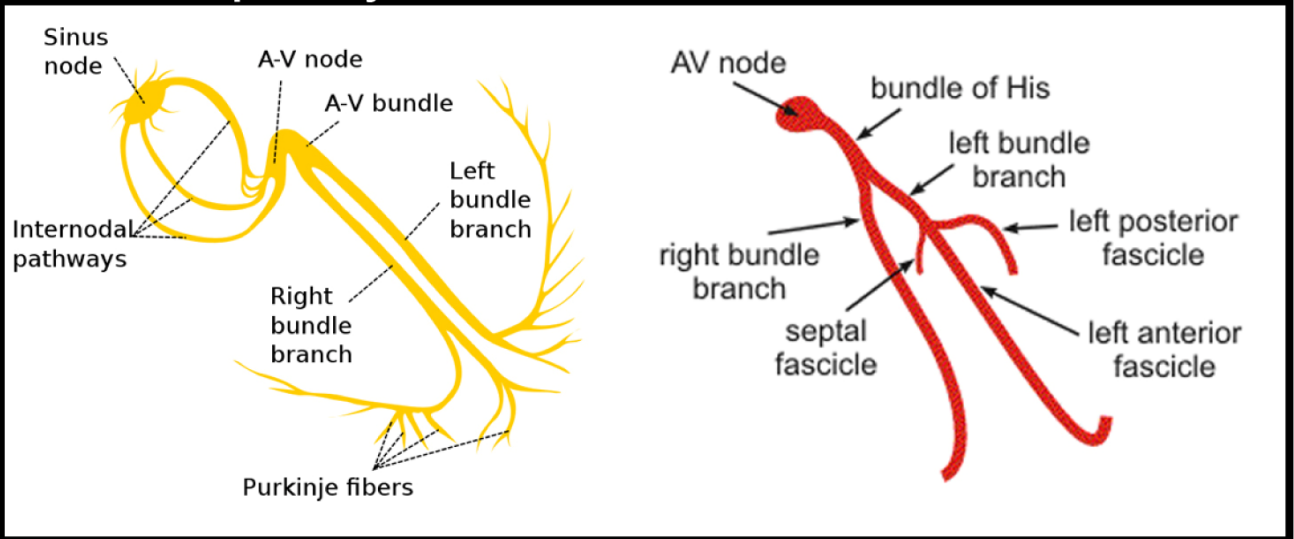
The 2 crura ride on the IV septum → then ramify into *Purkinje fibers* over the ventricle walls

So: SA node → Interatrial fibers → AV node → His bundle → Tawara crura (L. and R.) → Purkinje fibers





### Conductive pathways of the heart



### → Nervous Supply of the heart:

The heart is supplied by both Sympathetic and Parasympathetic fibers forming:

- Super
  - Middle
  - Inferior
- } cardiac plexuses

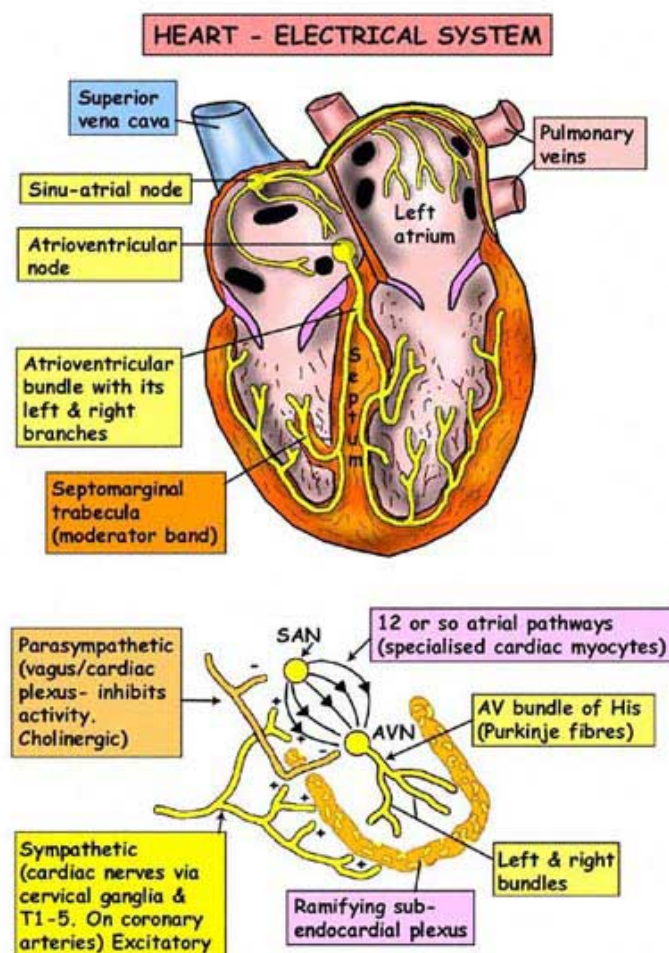
The superior Sympathetic fibers from the cervical ganglia of the Sympathetic trunk, the middle ones from the stellate ganglion and the inferior ones from the thoracic ganglion (long postganglionic fibers)

The Preganglionic parasympathetic fibers from the Vagus nerve (synapsing on intramural ganglia close to the surface of heart, in the coronary sulcus, giving rise to the short postganglionic fibers), which also carries sensory fibers

**\*\* Note:**

- The right Vagus nerve passes between the Right Subclavian vein and artery
- The left Vagus nerve passes between the aortic arch and the left Subclavian vein
- The right Vagus gives its recurrent laryngeal branch under the right Subclavian artery
- The left Vagus gives its recurrent laryngeal branch under the aortic arch

Both Vagi continue into the thorax passing behind the root of lungs, while the Phrenic nerves which arise from the cervical plexus and enter the thorax passing on the anterior scalene muscles, pass in front of the root of lungs



### → *Percussion of the heart:*

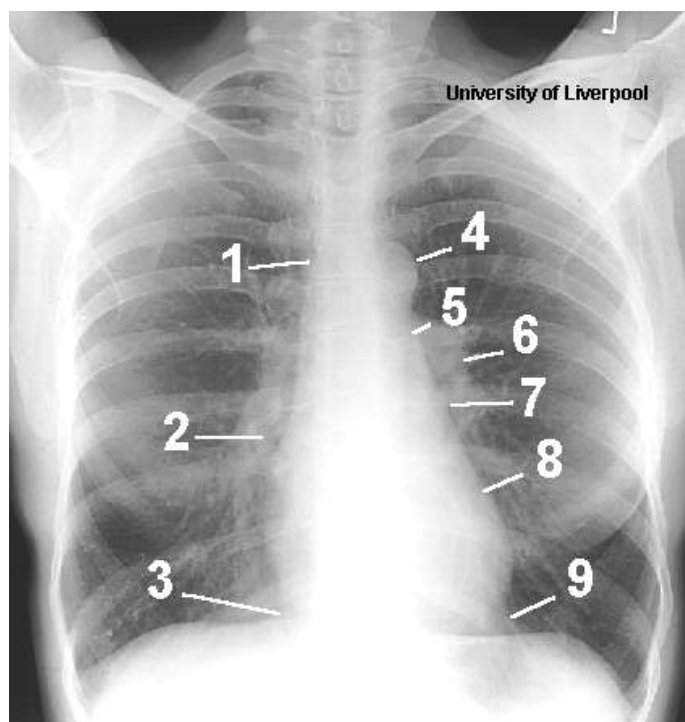
Percussion is a method of diagnosis that relies on the resonance of the air/fluid/solid filled cavities that will sound differently when tapping them with fingers

When percussing over an air filled cavity like the lung, we can hear resonance of the sound, but while percussing over a solid area like over the liver or the heart we can hear dullness and when percussing over an area of solid organ covered by an air filled organ like the lung covering the superior border of the heart we can hear a sound in between tympani and dullness so this method requires expertise and very good hearing

- *Relative Dullness of the heart:* starting within the air filled lung (1<sup>st</sup>, 2<sup>nd</sup> IC spaces) we go down to the 3<sup>rd</sup> IC space, the resonance will become dull due to the fluid filled heart being behind the heart  
Relative dullness marks the upper border of heart
- *Absolute dullness of the heart:* we keep going down to the 4<sup>th</sup>, 5<sup>th</sup> IC spaces (on left side of course) and the sound will be more dull because the lung doesn't cover the heart here, the heart is directly found behind the anterior chest wall so no resonance  
The size of the absolute dullness of the heart gives the size of the *Cardiac notch*

### → *Shadow of the heart (Silhouette / X-Ray of the heart):*

How the heart will look in an X-ray of the chest area, since its denser than the surrounding lung tissue, the X-ray will not penetrate it leaving behind a shadow corresponding with the shape, size and edges (protrusions) of the heart



On a chest radiograph, the central mediastinum area is occupied by a large radio-opaque shadow termed the **central mediastinal shadow** which is the anatomical location of the **heart** and **great vessels**. If any heart disease is suspected, it is very important to carefully examine its size, position and shape. Enlargement of the heart shadow might indicate the presence of a cardiac problem such as heart failure following a myocardial infarction. However, the heart shadow is variable in shape even in normal subjects, mainly due to the wide variation in body build of different people. Additionally, respiration will also affect the heart shadow.

Using the image above as a key, examine the **right border** of the central mediastinal shadow. In its superior parts, the right margin of the shadow represents the **superior vena cava** (1). This is a large vein which receives blood draining from the upper part of the body ie head, neck, upper limbs and part of the upper chest. Inferior to the superior vena cava, the border is formed by the **right atrium** (2) of the heart.

Inferior to the right atrium, a small part of the **inferior vena cava** can *sometimes* be identified as it enters the chest through the diaphragm, prior to returning blood to the heart (3).

Now look at the **left border**.

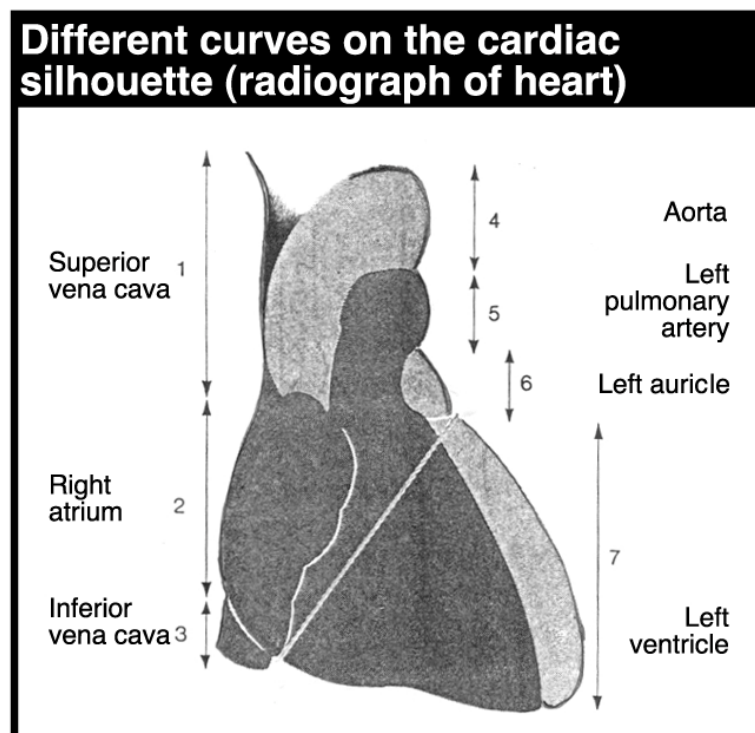
The left border of the central mediastinal shadow is formed superiorly by the prominent left projecting shadow of the **arch of the aorta** (4), also known clinically as the **aortic knuckle**. Immediately inferior to the aortic arch is a shadow formed by the **left pulmonary trunk** (5). Lateral to this shadow, a larger white shadow within the lung field is the **left pulmonary artery** (6) shadow.

In a more inferior direction, the left border of the central mediastinal shadow is composed of the left margin of the heart shadow.

In its superior region, the heart shadow is formed by the tip of the **auricle** of the left atrium (7).

Below the auricular shadow, the border is represented by the **left ventricle** (8) of the heart.

The point where the cardiac shadow meets the diaphragmatic shadow is called the **left cardiophrenic angle** (9).



## → The Mediastinum:

The middle part of the thoracic cavity, separated from the pleural cavities by the *mediastinal layer of the parietal pleura*. Anteriorly it's bordered by the *sternum*, posteriorly by the *vertebral column* and inferiorly by the *diaphragm*.

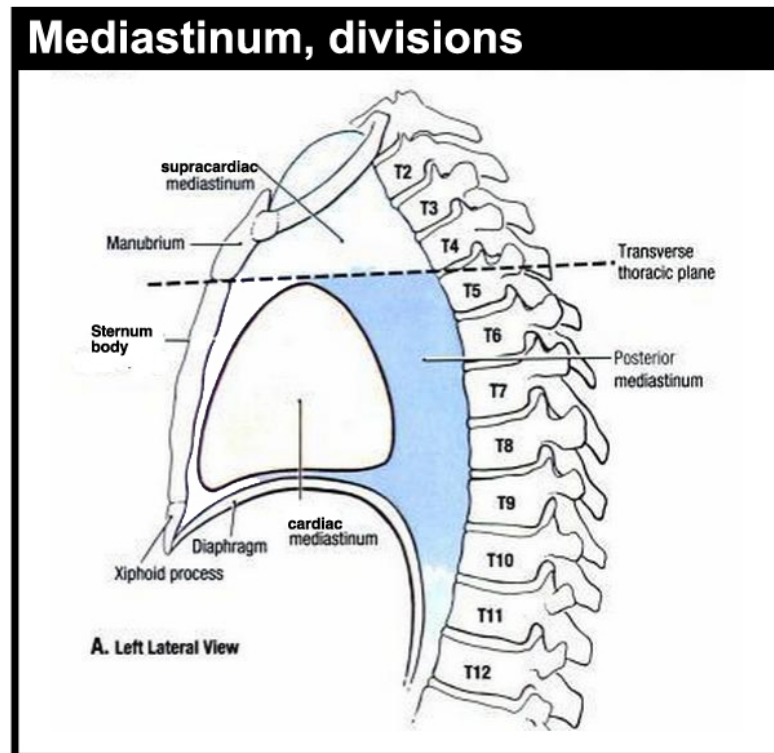
→ Separated by **the root of lung** (below the root of lung, the *pulmonary ligament* and the *Bronchopericardial membrane* continues the separation, (above it there is no boundary) into:

1) *Anterior mediastinum*, which is further divided into

- a. *Inferior*: Heart, cardiac mediastinum
- b. *Superior*: Supracardiac mediastinum

2) *Posterior mediastinum*

**\*\*Note:** This division is the one favored by our department, so please use this although many other divisions have been made which are also correct.



→ Another division by Moore suggests a *Transverse thoracic plane/ line* between Th-4 & Th-5 separating an:

1) *Superior mediastinum*

2) *Inferior mediastinum*, divided by the pericardium into:

- a. *Anterior*
- b. *Middle*, contains the pericardium and its contents
- c. *Posterior*

| Sup.       |           |           |
|------------|-----------|-----------|
| Inf. Post. | Inf. Mid. | Inf. Ant. |



→ **Supracardiac mediastinum, Layers from anterior to posterior:**

1) **Sternum bone**

2) **Thymus** (Adipose remnant in adult): primary lymphoid organ, receives Anterior Intercostal and Anterior mediastinal branches of the internal thoracic arteries

3) **Layer of Veins:**

a. **Brachiocephalic veins:** formed by behind the sternoclavicular joints by the union of the jugular and Subclavian veins

i. **Left Brachiocephalic:** oblique & long (10-12cm) passes over the branches of aorta. Into it drains the inferior thyroidal vein

ii. **Right Brachiocephalic:** straight and short (5cm)

b. **Superior Vena Cava (SVC):** on the right side of the superior mediastinum, anterolaterally to the trachea → The right Phrenic nerve lies between the SVC and the mediastinal pleura

4) **Vagus & Phrenic nerves** (Vagus continues down *behind* the root of lung, while the Phrenic *in front* of it together with the Pericardiophrenic a./v.)

5) **Layer of Aortic arch & branches:**

a. **Ascending aorta:** 2-3 cm in diameter, intrapericardial

b. **Aortic arch:** arches posteriorly and descends behind the root of L. lung on the left side of *Th-4 vertebral body*. Has 3 main branches:

i. **Brachiocephalic trunk:** at the right sternoclavicular joint level, gives the *Right common carotid & the Right Subclavian arteries*

ii. **Left common carotid artery**

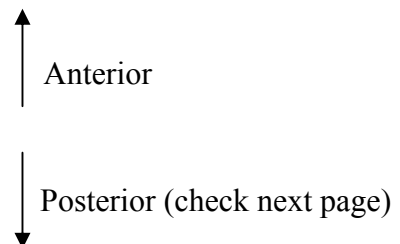
iii. **Left Subclavian artery**

6) **Trachea** (root of lung, hilum)

7) **Recurrent Laryngeal nerve**

8) **Esophagus**

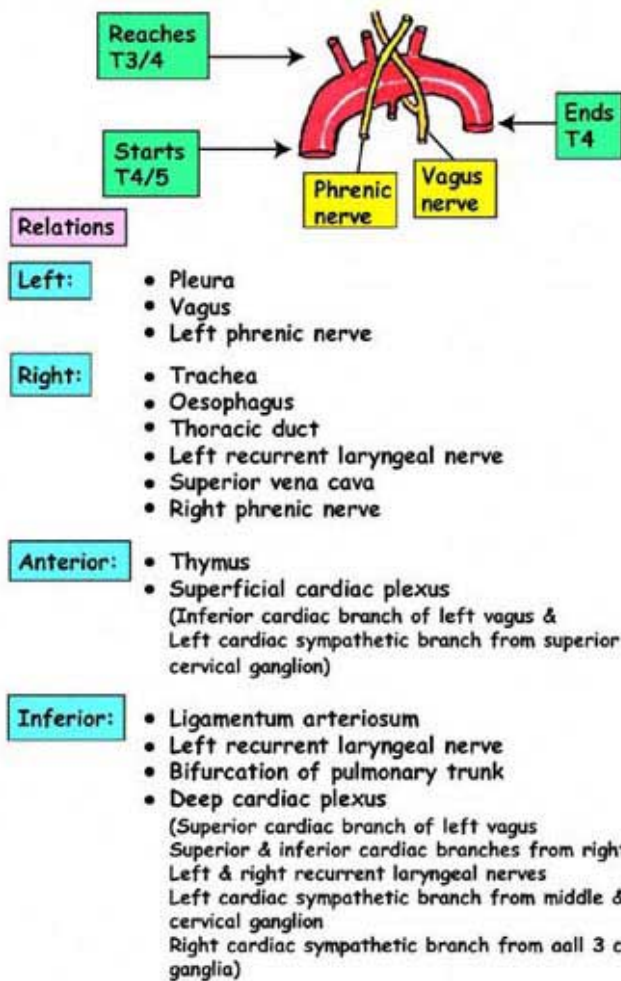
9) **Vertebral column**



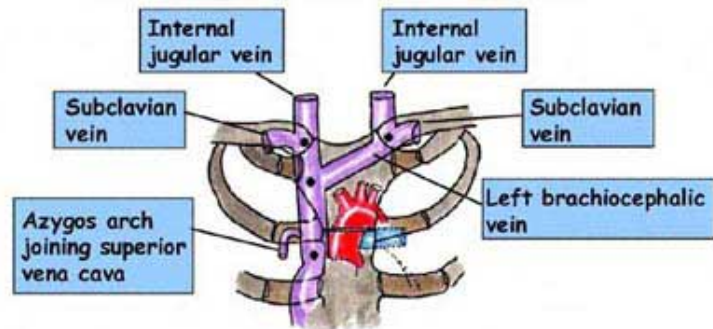
\*\* *The right Vagus* enters the thorax anteriorly to the right Subclavian artery → gives the Right recurrent laryngeal nerve, which ascends between the trachea and the esophagus to the larynx. After that the R. Vagus continues behind the root of lung and gives Pulmonary, esophageal and cardiac plexuses

\*\* *The Left Vagus* enters the thorax between the L. common carotid and the L. Subclavian artery. It gives the recurrent branch under the Aortic arch and branches for the plexuses, after it continues behind the root of lung

## AORTIC ARCH



## SUPERIOR MEDIASTINAL VEINS



### Brachiocephalic veins

Form as the junction of the internal jugular and subclavian veins on each side behind the sternoclavicular joints

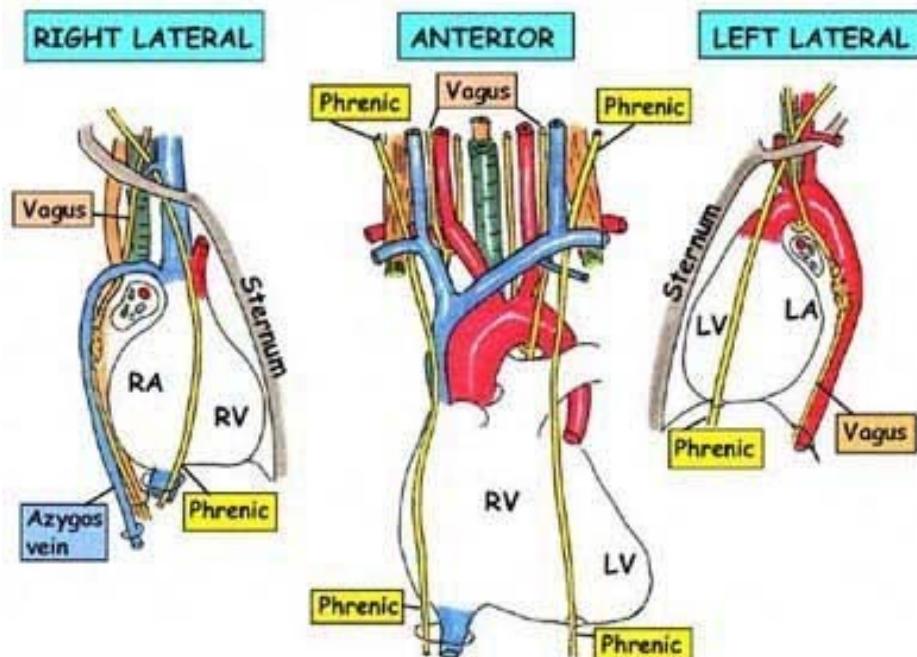
### Superior vena cava

Forms as the junction of the two brachiocephalic veins at the first right sternocostal junction (ie lower border of right 1st costal cartilage). It runs inferiorly to end at the right 2nd intercostal space parasternally

### Tributaries of left brachiocephalic vein

- Thoracic duct (into junction as it forms)
- Thymic vein
- Left vertebral vein
- Left & usually right inferior thyroid veins
- Left internal thoracic vein
- Left supreme intercostal vein
- Left superior intercostal vein

## UPPER MEDIASTINUM



→ **Posterior Mediastinum layers, contents:**

**1) Thoracic Aorta:**

- a. Begins at L. side of *Th-4 body* until *Th-12*, surrounded by thoracic aortic plexus, forms an impression on the Left lung.
- b. Gives *Esophageal arteries* (2-5 branches)
- c. Gives *Bronchial arteries*: usually from 2nd/3rd IC art. but also directly from aorta
- d. Gives *9-10 paired Posterior Intercostal arteries*
- e. Gives *Pericardial branches*

**2) Esophagus and the 2 Vagi nerves:** R. side, the Vagus behind esophagus, L. sides in front

**3) Thoracic duct and lymph trunks:**

Originates from *Chyle Cistern* in the abdomen, between *azygos vein* and *thoracic aorta/* and *esophagus* on the thoracic vertebral bodies. At the level of Th-4/Th-5 it crosses to the left and terminates into the *Left venous angle*. Collects:

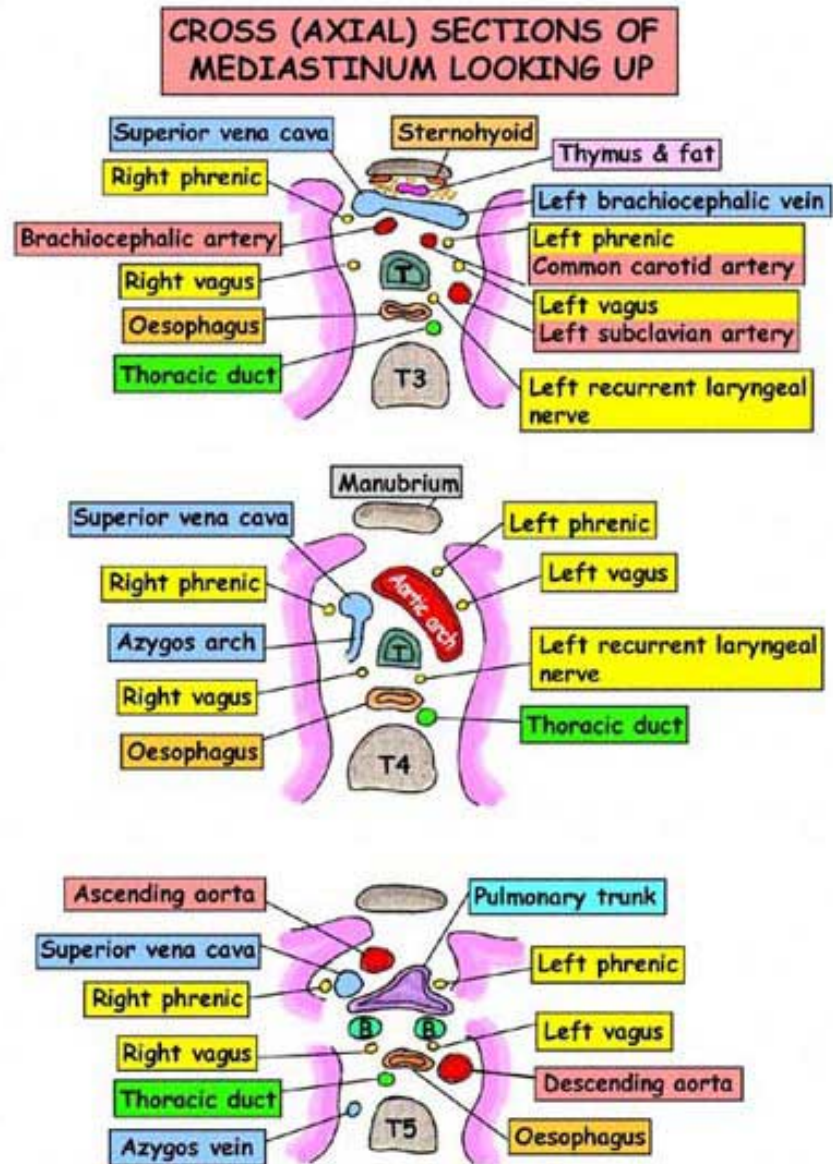
1. *Left jugular trunk*
2. *Left subclavian trunk*
3. *Left bronchomediastinal trunk*

**4) Veins/ Lymph nodes:**

- a. Post. Mediastinal lymph nodes
- b. ***Azygos vein (R. side):*** forms a collateral pathway between IVC & SVC. Drains the posterior walls of thorax/ abdominal. It arches over the superior aspect of R. root of lung to join SVC, producing an impression on the L. lung
- c. ***Hemiazygos vein:*** arises on the L. side by the union of *L. subcostal* and *ascending lumbar veins*. Ascends posteriorly To thoracic aorta until *T9 vertebra*, crosses to the right, behind the thoracic aorta, thoracic duct and esophagus and joins the azygos vein
- d. ***Accessory hemiazygos:*** begins at medial end of 4th/5th IC spaces, descends on L. side of vertebral column from Th5-Th8 and joins azygos or hemiazygos veins.

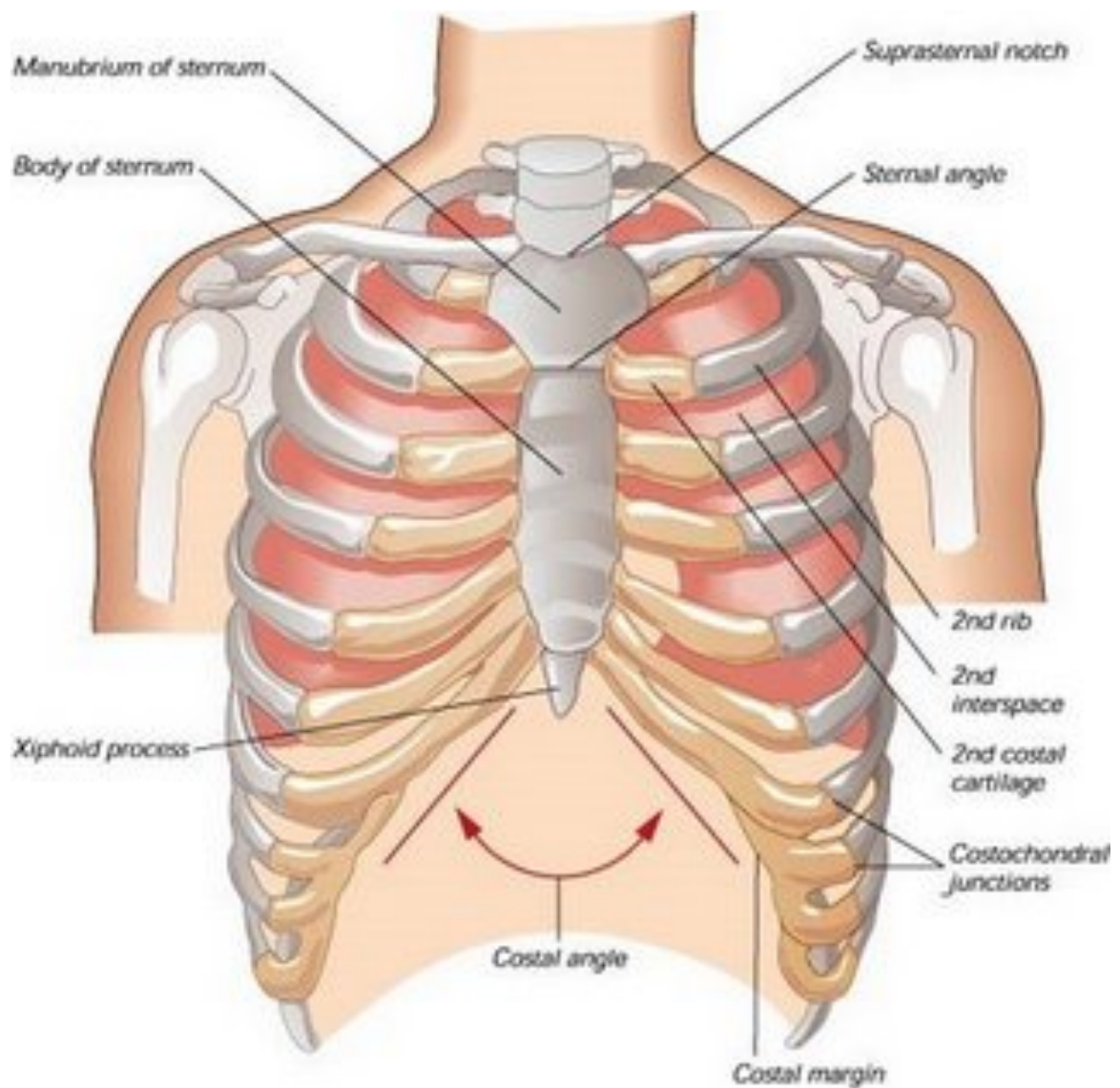
**5) Nerves:**

- a. ***2 sympathetic trunks*** (thoracic), lie parallel on the heads of ribs and vertebral bodies
- b. ***Lower Splanchnic nerves***
  1. Greater
  2. Lesser
  3. Least





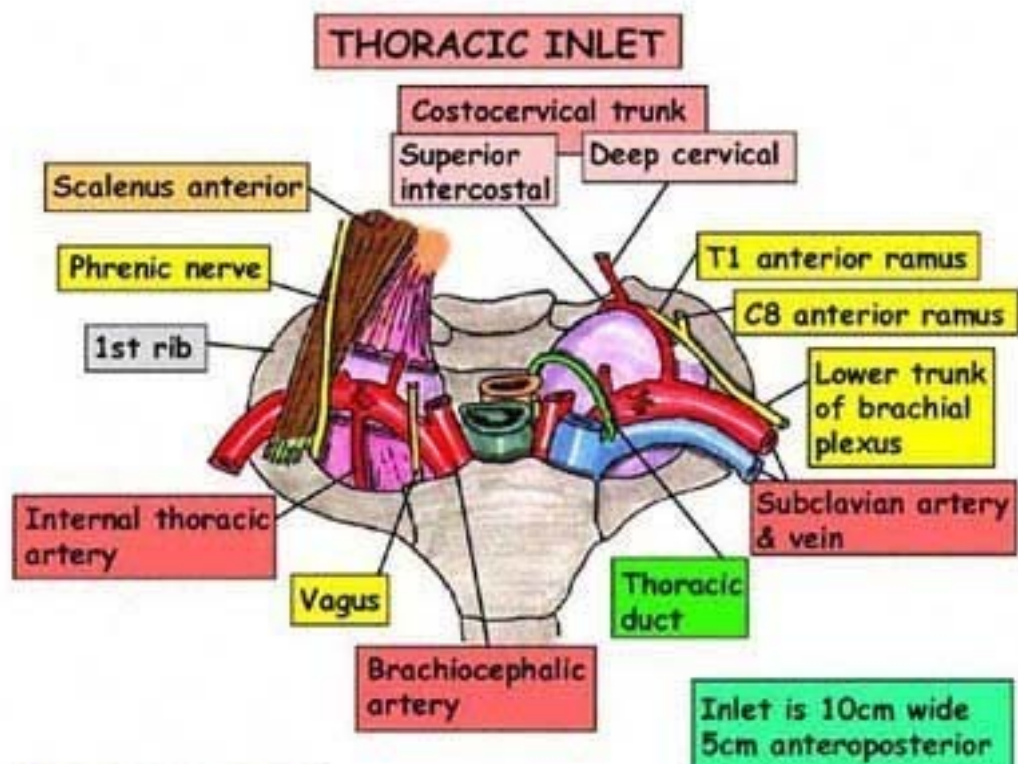
# *Topography of the thoracic cavity*



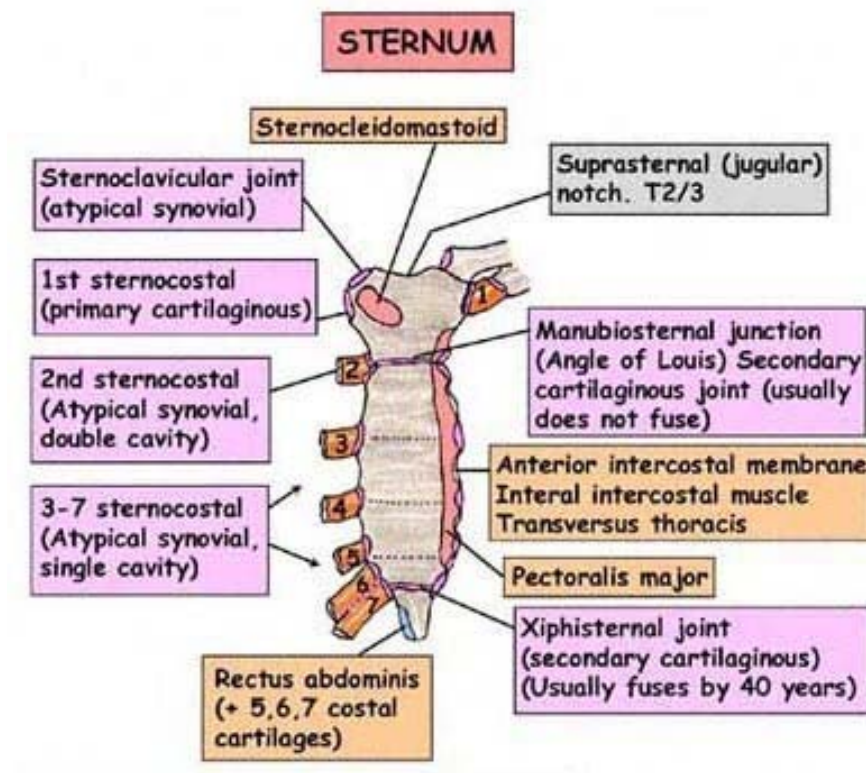


➔ **Structure of thoracic cage:**

- 12 thoracic vertebrae + Intervertebral disks
- 12 pairs of ribs (with angle) + sternum
- **Superior thoracic aperture (thoracic inlet):** narrow, borders:
  - *Posterior:* T1 vertebra
  - *Lateral:* 1<sup>st</sup> pair of ribs + cartilage
  - *Anterior:* Superior border of manubrium, Jugular notch
  - *Contents:*
    1. Esophagus
    2. Trachea
    3. Dome of Pleura (Cupula): the dome-shaped roof of the pleural cavity extending up through the superior aperture of the thorax surrounded by the 3 scalene muscles like a tent (*see the description about the lung pleura, or endothoracic fascia*)
    4. Bifurcation of the brachiocephalic trunk into right subclavian and right common carotid arteries
    5. Left common carotid and left subclavian arteries
    6. Left and right brachiocephalic veins (venous angles)
    7. Internal thoracic and superior Intercostal arteries (and veins)
    8. T1 anterior ramus (joining brachial plexus)
    9. Thoracic duct
    10. 2 sympathetic trunks
    11. 2 Vagi nerves
    12. 2 Phrenic nerves (found on the anterior scalene muscle)



- **Inferior thoracic aperture:** very wide, limited by the costal arch, borders:
  - *Posterior:* T12 vertebra
  - *Posterolateral:* 11<sup>th</sup>, 12<sup>th</sup> ribs
  - *Anterolateral:* Joined (common) costal cartilages of 7-10 ribs
  - *Anterior:* Xiphoid process (Xiphosternal joint)
- **Sternum:** A long flat bone, articulating with the cartilages of the first seven ribs and with the clavicle, forming the middle part of the anterior wall of the thorax; it consists of three portions:
  - *Corpus or body:* the middle and largest portion of the sternum, articulating with the 3<sup>rd</sup> -7<sup>th</sup> ribs
  - *Manubrium:* the upper segment of the sternum, a flattened, roughly triangular bone, occasionally fused with the body of the sternum, forming with it a slight angle, the sternal angle of Louis that marks the level of the second costal cartilage (rib) for counting ribs or Intercostal spaces. Denotes level of aortic arch, bifurcation of trachea, and T4/T5 Intervertebral disc
  - *Xiphoid process:* the cartilage at the lower end of the sternum



## ➔ *Vertebrae, IV disk, Ligaments:*

- **Vertebrae:**
  - *7 Cervical vertebrae:*
    1. C1: Atlas, C2: Axis
    2. C3-C5: typical
    3. C6: has carotid tubercle, anterior tubercle of transverse process
    4. C7: prominent v., the spinous process called Processus Prominens
    5. The transverse process has:

- Anterior root: remnant of ribs
  - Posterior root: really transverse process
- 6. Transverse foramina, Anterior/Posterior tubercles, groove for spinal nerves, origin of scalene muscles
- *12 Thoracic vertebrae*
  - 1. Costal fovea: for ribs attachment
- *5 Lumbar vertebrae:*
  - 1. Transverse process composed of:
    - Costal process
    - Accessory process
    - Mammillary process (on superior articular process)
- *Sacrum (fused 5)*
- *Coccyx (fused 4)*
- ***Intervertebral disk:***
  - a disk interposed between the bodies of adjacent vertebrae, composed of:
    - 1. *Annulus fibrosus*: outer fibrous part, a ring of fibrocartilage and fibrous tissue forming the circumference of the intervertebral disk; surrounds the nucleus pulposus, which is prone to herniation when the annulus fibrosus is compromised/cracked with injury or age
    - 2. *Nucleus pulposus*: a central gelatinous soft fibrocartilage portion of the intervertebral disk; regarded as a derivative of the notochord
- ***Ligaments:***
  - *Anterior longitudinal ligament*: wide fibrous band interconnecting the anterolateral surfaces of the vertebral bodies, blending with the outer lamellae of the intervertebral disks (loosely) as it passes between vertebra, stabilizes the vertebral column
  - *Posterior longitudinal ligament*: fibrous band interconnecting the posterior surfaces of the vertebral bodies; it narrows to pass between the pedicles and spreads out to blend with the outer lamellae of the posterior aspect of the annulus fibrosus of the intervertebral disks (strongly); forms the anterior wall of the vertebral canal, in the cervical level it becomes the Tectorial membrane, saves the IV disks in position
  - *Ligamenta Flava*: paired ligaments of yellow elastic fibrous tissue, which bind together the laminae of adjoining vertebrae, forming the dorsal wall of the vertebral canal between the vertebra or laminae (Arches); penetration of the ligamentum flavum with a trocar during epidural or spinal puncture produces a distinct feel ("pop"), letting the practitioner know that the tip of the trocar has entered the epidural space.  
Helps with the erection of vertebral column
  - *Interspinous ligaments*: bands of fibrous tissue that connect the spinous processes of adjacent vertebrae
  - *Supraspinous ligaments*: the longitudinal fibrous band attached to the tips of the spinous processes of the vertebrae; in the cervical region it is altered to form the *Nuchal ligament*

- *Intertransverse ligament*: connect the transverse processes of adjacent vertebrae

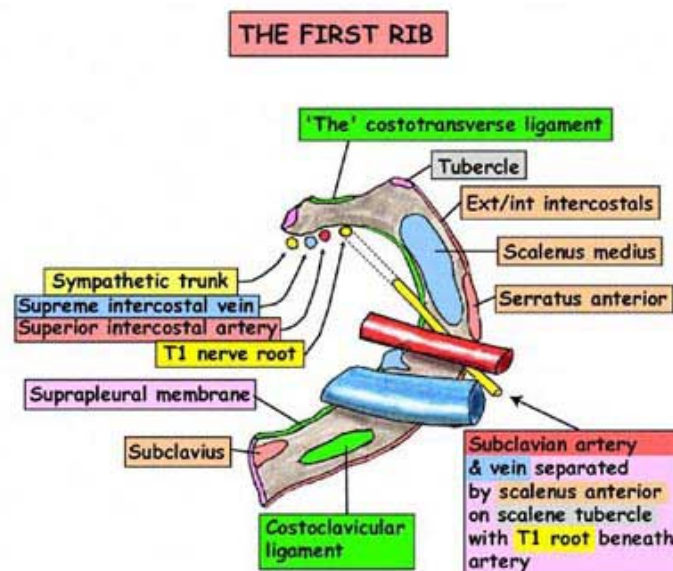
- ***Intervertebral joints, movements:***

- *Anteflexion*: mainly cervical and lumbar vertebrae
- *Retroflexion*: mainly cervical and lumbar vertebrae
- *Lateral flexion*: mainly thoracic vertebrae
- *Torsional movement*: mainly cervical and thoracic vertebrae

## ➔ ***Ribs:***

- ***12 pairs:***

- 1-7: *true ribs*, have individual cartilage connection to the sternum
- 8-10: *false ribs*, common cartilage connection
- 11, 12: *floating ribs*, no cartilage
- Each rib has: Head, Neck, Body (with inferior groove/sulcus) and a tubercle on the angle between neck and body
- Curvatures of ribs:
  1. On the edge
  2. On the surface
  3. Torsional (in plane)
- Due to the curvature of the ribs, we get a space that lies laterally on both sides of the vertebral column called the *Pulmonary groove of thorax* (paravertebral gutter)

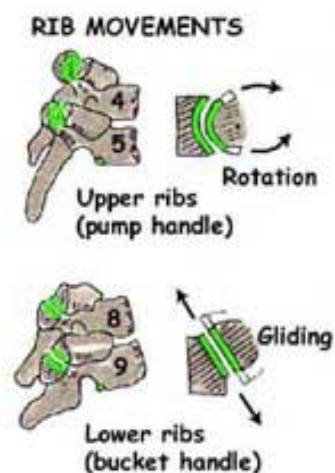
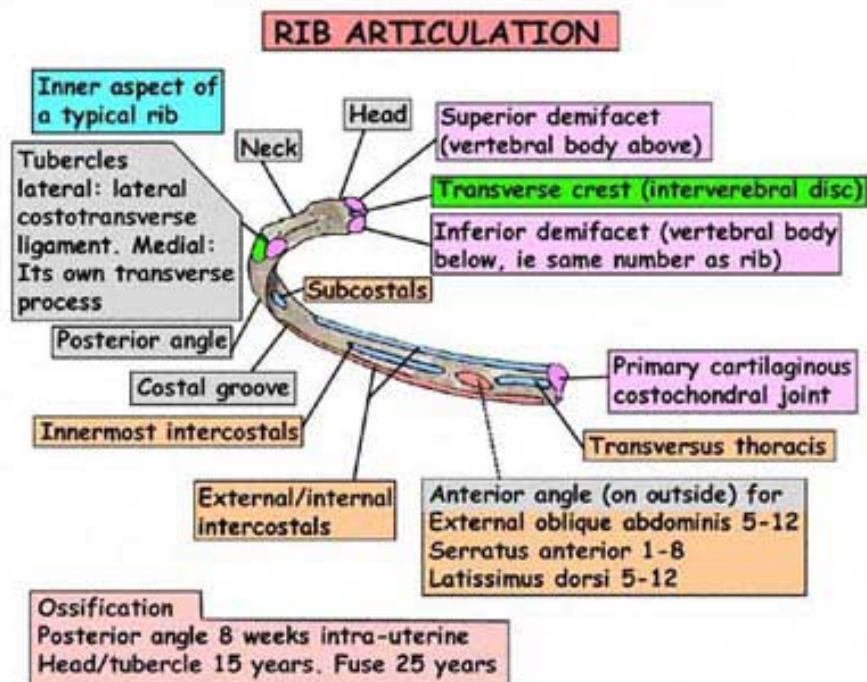


**Note:**

- The under surface of the 1st rib is smoother
- When the rib is laid on a flat surface, the head touches the flat surface when the rib is the correct way up



- **Joints of the head of ribs:**
  - Double chambered joints (except 1, 11, 12)
  - Each rib articulates with lower and upper borders of 2 adjacent vertebrae. The disk is connected by *intra-articular ligament* of head of rib
  - The *radiate*, *Stellate*, or *anterior costovertebral ligament* connecting the head of each rib to the bodies of the two vertebrae with which it articulates
- **Costotransverse joints:**
  - Between articular facet of costal tubercle and costal fovea of the transverse process
  - Has the *Costotransverse ligament* that connects the dorsal aspect of the neck of a rib to the ventral aspect of the corresponding transverse process (*lateral and superior*)
- **Sternocostal joints:**
  - 1, 6, 7: *Synchondrosis*, joined by cartilages
  - 2-5: *Synovial* joints, enforced by *radiate Sternocostal ligaments* anteriorly
  - *Interchondral joints*: the synovial joints between the contiguous surfaces of the 5-10 costal cartilages, forming the *costal arch*



- 1st rib:** Short, broad, most curved, single facet on head
- 2nd rib:** Poorly marked costal groove, rough area for serratus anterior & scalenus posterior
- 3-10 ribs:** Typical. 7th longest
- 11/12 ribs:** Floating, single facet, no tubercle, tapered end, 12th no groove
- 1-7 ribs:** articulate with sternum
- 8-10 ribs:** with each other
- Muscles attached to outer ribs:**
  - Serratus anterior 1-8
  - External oblique abdominis 5-12
  - Pectoralis minor 3,4,5
  - Latissimus dorsi 9-12
- Muscles attached to costal cartilages:**
  - Pectoralis major 1-7 (often 2-6)



## COSTOTRANSVERSE LIGAMENTS

As the name implies, the costotransverse ligament connects the rib to the transverse process but it is a complex ligament in that it has three parts to it with somewhat confusing terminology.

### THE costotransverse ligament

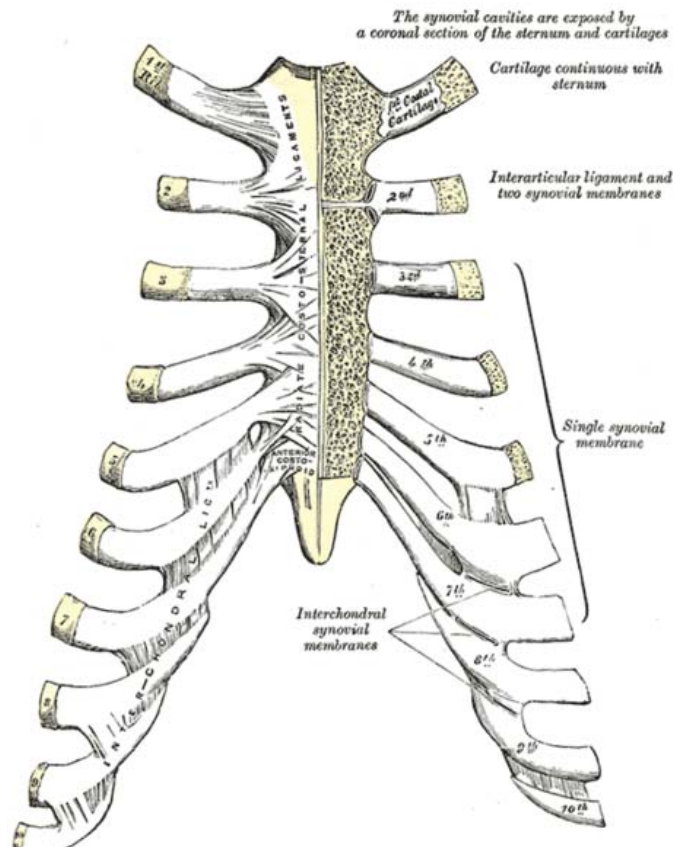
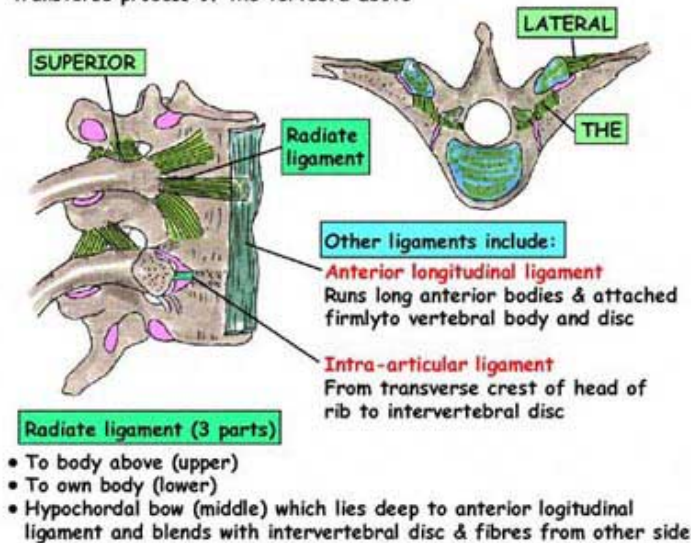
This fills the gap between the rib and its own transverse process

### LATERAL costotransverse ligament

This lies posteriorly and extends from the transverse process to its own rib, just beyond the tubercle

### SUPERIOR costotransverse ligament

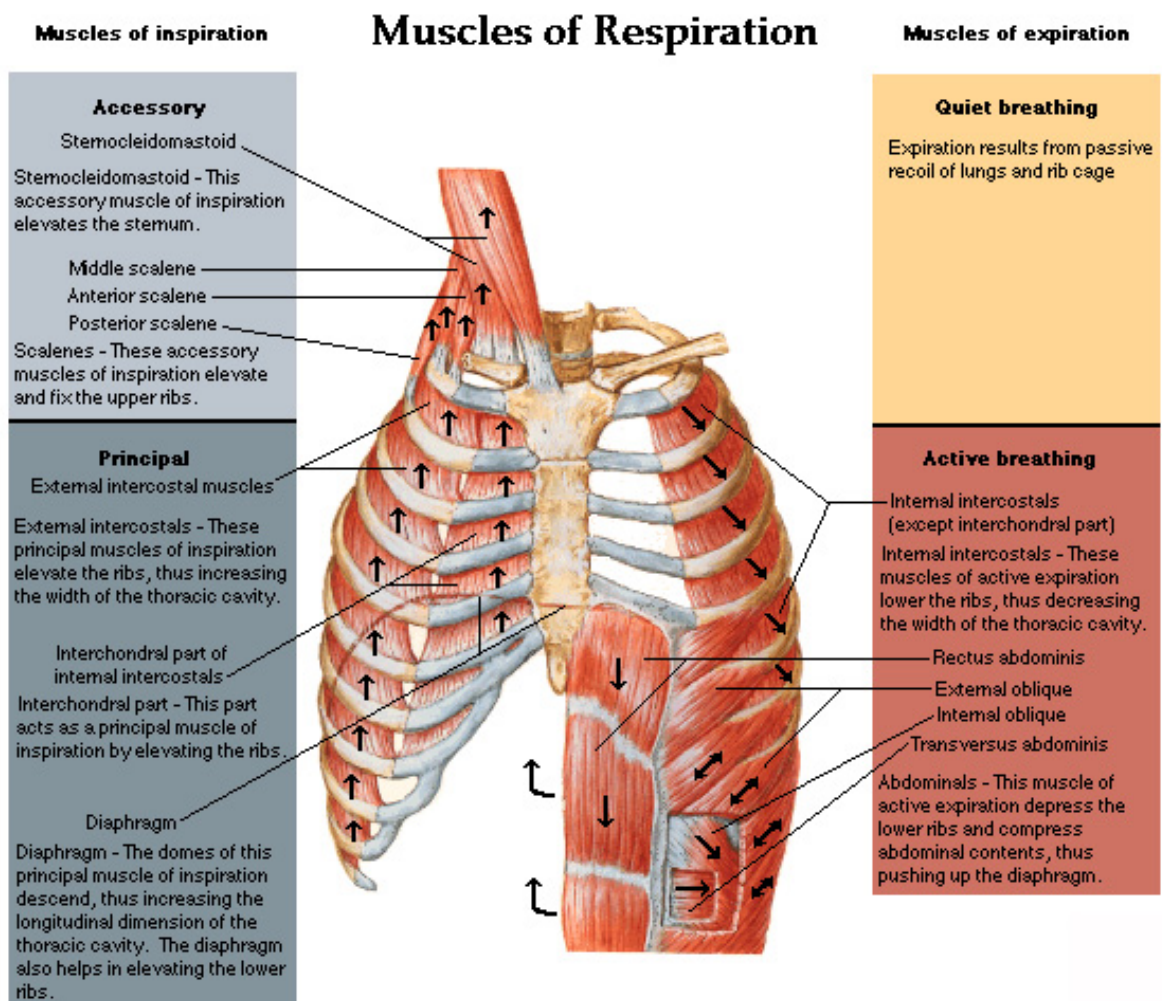
This is a two layered ligament with the fibres at right angles to each other (corresponding and continuous with the intercostal muscles) that passes from the upper border of the neck of the rib to the transverse process of the vertebra above

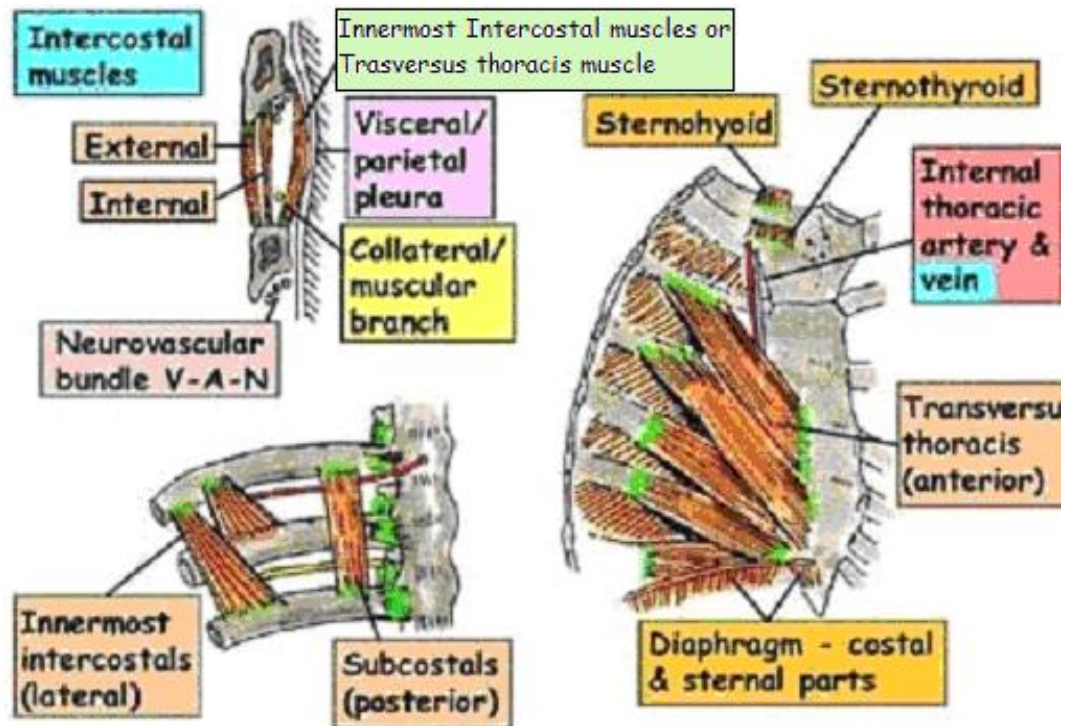


➔ **Muscles of the thoracic wall:**

- *Pectoralis major*
  - *Pectoralis minor*
  - *Subclavius*
  - *Serratus anterior*
  - *Latissimus Dorsi*
- } Thoracoappendicular muscles that attach also to the limbs and function also as accessory respiratory muscles
- 
- *Scalene Muscles* (Anterior, Middle, Posterior): important in *deep inspiration*
  - *Serratus posterior superior / inferior*: function more in stabilization and proprioception
  - *Levatorius Costorum*: Vertebral movement and proprioception
- 
- **Intercostal muscles:**
    - *Superficial layer:*
      1. External Intercostal muscles:
        - 11 pairs, occupy Intercostal spaces from the tubercles of the ribs posteriorly until the costochondral junctions anteriorly
        - Anteriorly they are replaced by the *External Intercostal membrane*
        - The fibers run from the lower sharp margin of the rib and passes obliquely inferoanteriorly to the upper border of the lower rib (like the direction of your fingers when you put your hands in your pocket)
        - *Function*: contract during *inspiration*, elevate ribs and maintain tone of the Intercostal space
    - *Inner Layer:*
      1. Internal Intercostal muscles:
        - 11 pairs, run deep at right angles with the external Intercostal muscles
        - Fibers run superioanteriorly from the inferior groove of the upper rib to the upper margin of the lower rib
        - Range from the sternum until the angle of the rib posteriorly, where they are replaced by the *internal Intercostal membranes*
        - *Function*:
          - *Interossious part*: during expiration, depresses ribs
          - *Interchondral (intercartilaginous part)*: during inspiration, elevates ribs with external Intercostals, resist mediolateral movement
      2. Innermost Intercostal muscles:
        - Similar to the internal intercostals and essentially their deeper parts, parallel to them. They are separated from the internal intercostals by the Intercostal nerves and vessels in the *intercostal space* (Neurovascular bundle, order from superior to inferior: vein, artery, nerve)
      3. Subcostal muscles (inconstant):

- In the posterior lateral thoracic wall, same direction as the internal Intercostals, but extends across more than 1 Intercostal space
- Function: *Expiration*
- *Transverse thoracic muscles*: attach on the lower part of the sternum and xiphoid process internally in the thoracic cage, and insert superolaterally on the 2<sup>nd</sup>-6<sup>th</sup> costal cartilages, function in *Expiration* (weak)
- **Diaphragm**: Main inspiration muscle





### → Fascia of the thoracic wall:

- **Pectoral fascia:** bed of breast, covering the Pectoralis major
- **Clavipectoral fascia:** stretches from the *clavicle*, *Coracoid* process and extends downward enclosing the *Subclavius* muscle then the *Pectoralis minor* (between them it's called *costocoracoid membrane or ligament*) and inferiorly it becomes continuous with the *Axillary fascia* as the *Suspensory Ligament of the Axilla*, which supports the Axillary fascia and pulls it with the skin upward, forming the *Axillary fossa*. The Clavipectoral fascia divides the Deltopectoral triangle into 2 compartments: **Superficial & Deep**.

The *Superficial Pectoral fascia* covers the Pectoralis major muscle, which continues to the Axillary region to become the *Superficial Axillary fascia* (base-floor of Axillary fossa with skin). The *Deep pectoral fascia* covers the Pectoralis major muscle from the inside.

**Note:** please check the Netter atlas, Plate 428 for this fascia and its drawing

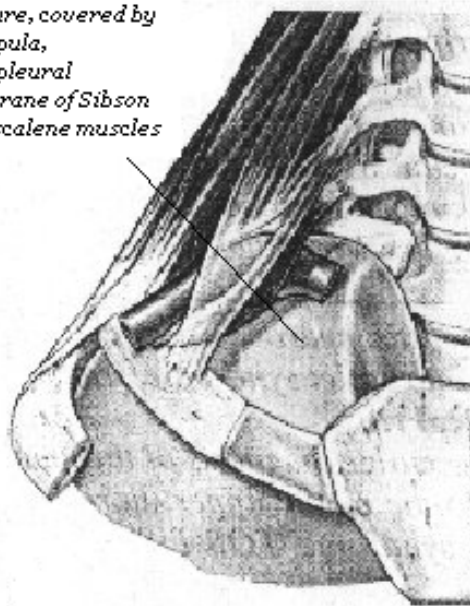
- **Endothoracic fascia:** lines the thoracic cage internally and attaches the adjacent portion of the lining of lung cavities (Costal parietal pleura) to the thoracic wall. Towards the apices of the lungs it becomes more fibrous & thickened, and reinforces the lung cupula as the *Surpaleural membrane or Sibson's fascia*, that attaches from inner border of the first rib to the transverse process of the 7<sup>th</sup> cervical vertebra (through the upper thoracic inlet)

\*\* So if we sum up the layers at the *dome of lung or cupula* will be (from ext. to int.):

- Deep cervical fascia
- Surapleural membrane (Sibson's fascia)
- Cervical pleura
- Visceral pleura
- Lung (Apex)

\*\* And all these structures are enclosed by the *3 scalene muscles as the scalene tent*, and covered by the skin and subcutaneous adipose tissue

*Apex of lung through the superior thoracic aperture, covered by the cupula, suprapleural membrane of Sibson & the scalene muscles*



### ➔ *Nerves of the thoracic wall:*

- 12 pairs of thoracic spinal nerves supply the thoracic wall, as soon as they leave the IV foramina, the mixed spinal nerve divides into:
  - *Anterior Ramus*
  - *Posterior Ramus*
- ***Posterior Rami*** of the thoracic spinal nerves pass posteriorly, immediately lateral to the articular processes of vertebrae to supply the muscles, joints and skin of dorsal thoracic region. divide into:



- *Lateral branch*
- *Medial branch*

Both of which will supply the deep (true) muscles of the back. The medial branch of the dorsal primary ramus also supplies articular branches to the Zygapophysial joints and the periosteum of the vertebral arch.

- **The anterior rami** of T1-T11 form the intercostal nerves that run along the extent of the intercostal spaces  
 \*\* Anterior ramus of T12 courses inferior to the 12<sup>th</sup> rib and known as the *Subcostal nerve*

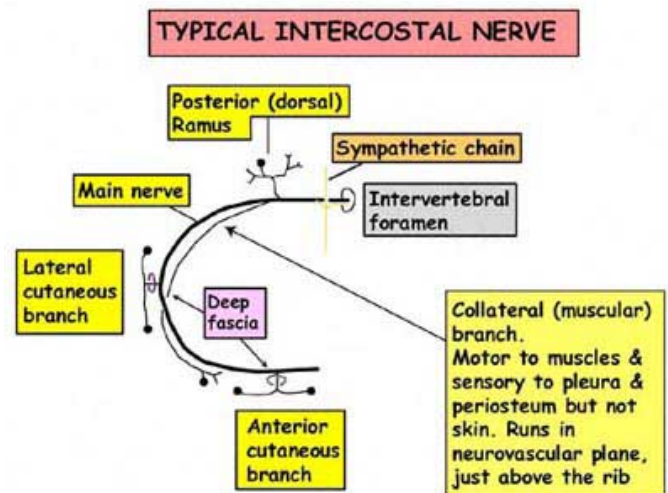
- **3<sup>rd</sup> – 6<sup>th</sup> IC nerves:** enter the most medial part of posterior IC spaces, run initially within the Endothoracic fascia between the parietal pleura and internal intercostal membrane, in the middle of IC spaces.  
 Near the angle of ribs the nerve pass *between the Internal intercostal and innermost intercostal* muscle in the costal groove (within the IC space), inferior to IC artery/vein, on the inferior margin of the rib (below it)  
 Collateral branches arise also at the angle of rib, and run on the superior border of the rib below

At the *midaxillary line (MAL)* it gives:

- *Lateral cutaneous branches*, divides into:
  1. Anterior branch
  2. Posterior branch

Near the sternum it turns anteriorly between the costal cartilages and become:

- *Anterior cutaneous branches*, that divides into:
  1. Lateral branch
  2. Medial branch

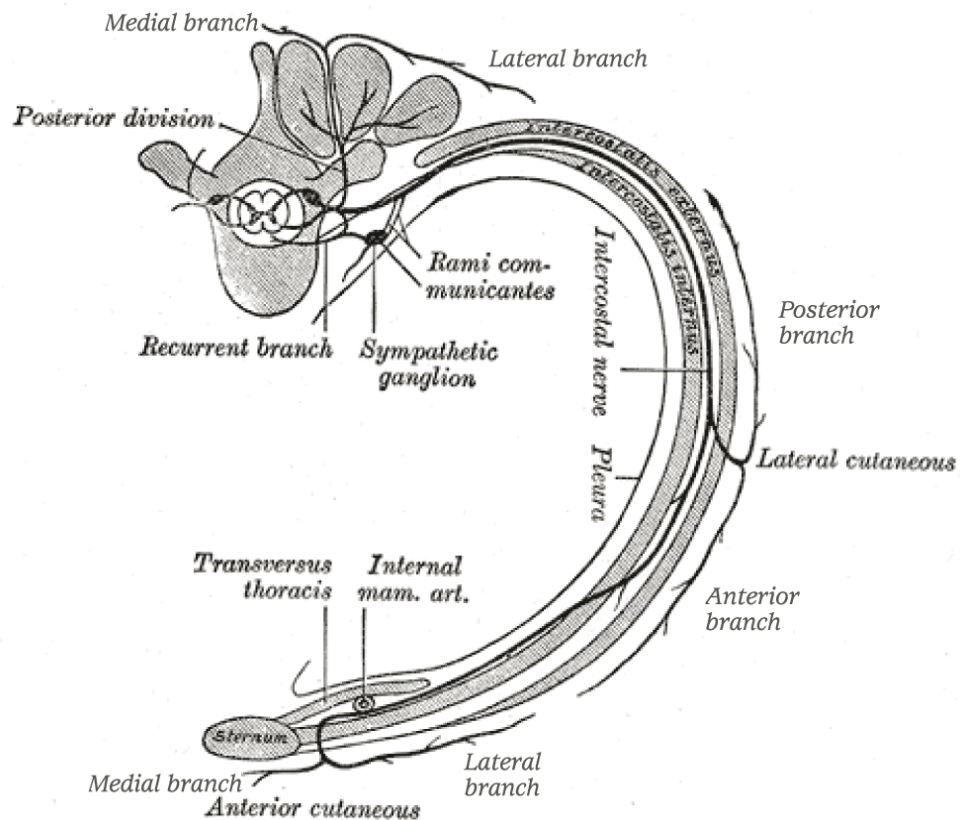


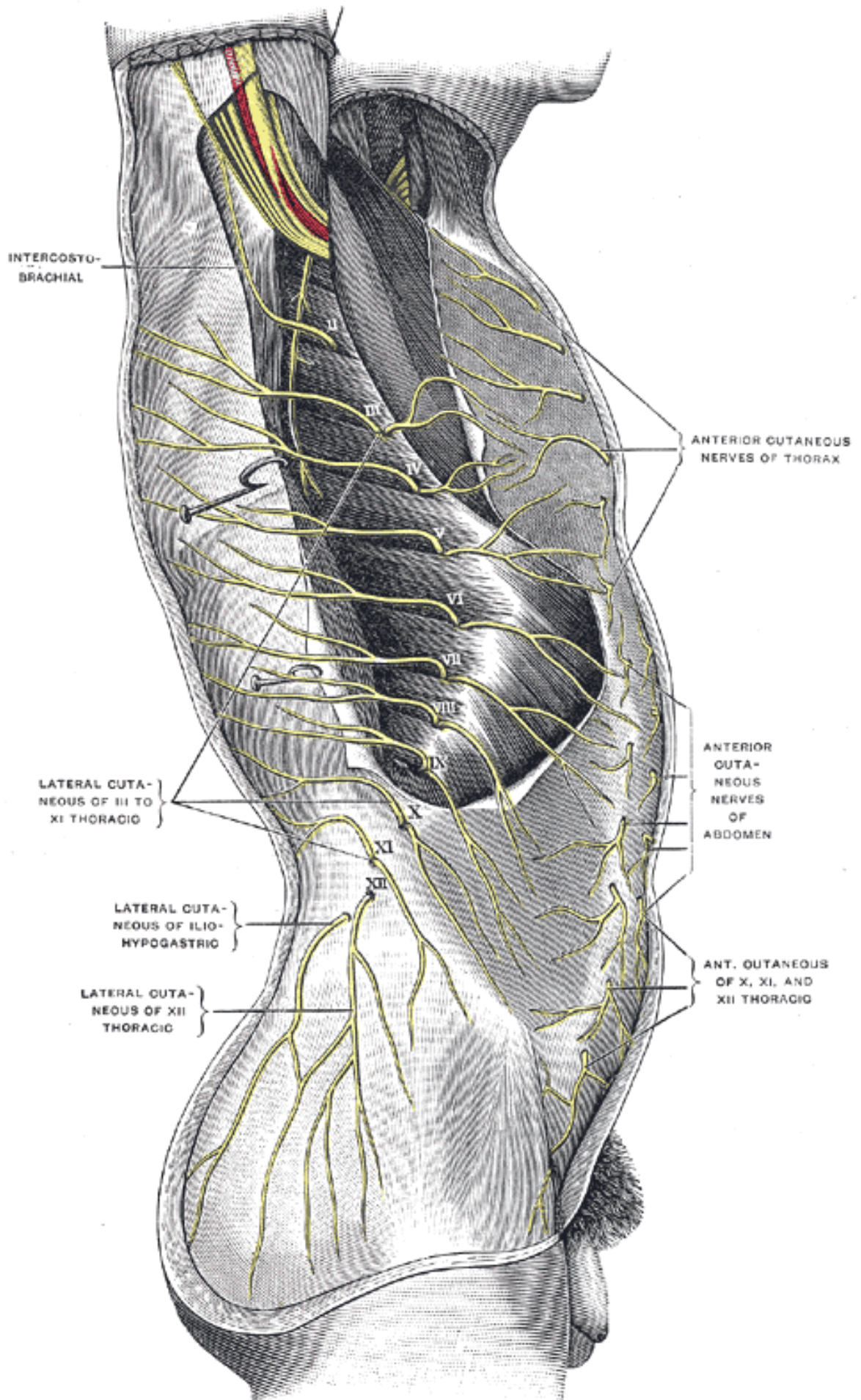
- **Atypical intercostal nerves:**

- *Anterior ramus of T1* divides into:
  1. Large superior part: joins the brachial plexus
  2. Small inferior part: 1<sup>st</sup> intercostal
 \*\* It has **no** cutaneous branches!

- **1<sup>st</sup> and 2<sup>nd</sup> intercostal nerves:** course behind the ribs, not under them in costal grooves
- **2<sup>nd</sup> intercostal nerve:** has a large lateral cutaneous branch called → *Intercostobrachial nerve* formed at the 2<sup>nd</sup> IC space, in MAL that penetrates the *Serratus anterior* muscle and enters the *Axilla* and *medial aspect of arm* to anastomose with the *Medial Brachial cutaneous* nerve as the **Intercostobrachial anastomosis** to innervate the skin of Axilla (importance in breast surgery anesthesia)

- 7<sup>th</sup>-11<sup>th</sup> IC nerves: after giving the lateral branch, they cross the costal margin posteriorly, continue to supply the abdominal skin and muscles and become the Thoracoabdominal nerves of the anterior abdominal wall
- So through the posterior ramus, the lateral and anterior cutaneous branches of its anterior ramus, each spinal nerve supplies a slip like region → **Dermatome** (The area of skin supplied by cutaneous branches from a single spinal nerve)  
**Myotome**: All muscles derived from one somite and innervated by one segmental spinal nerve





### ➔ *Vasculature of the Thoracic wall:*

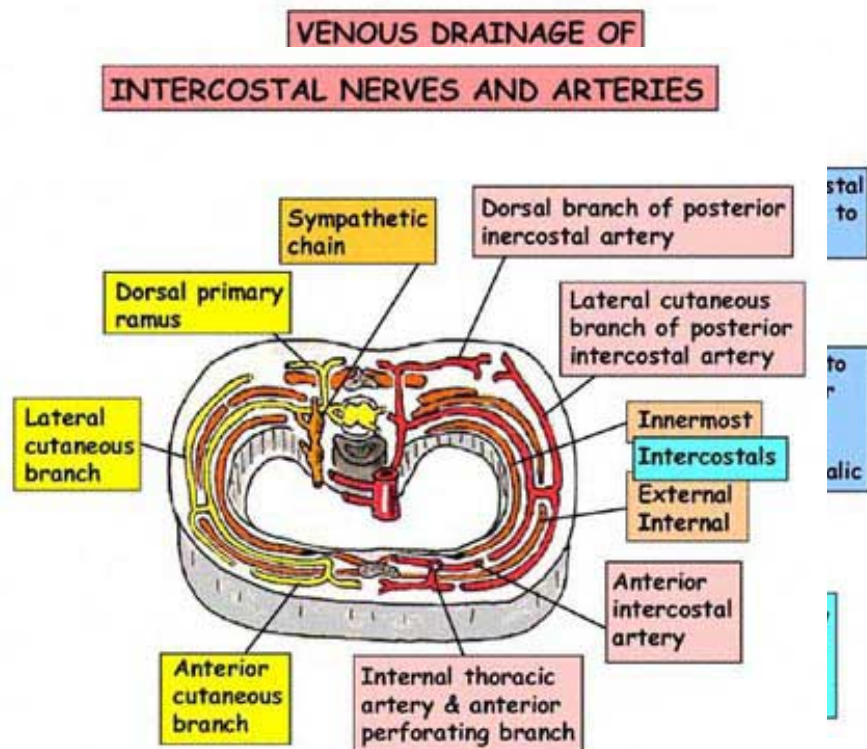
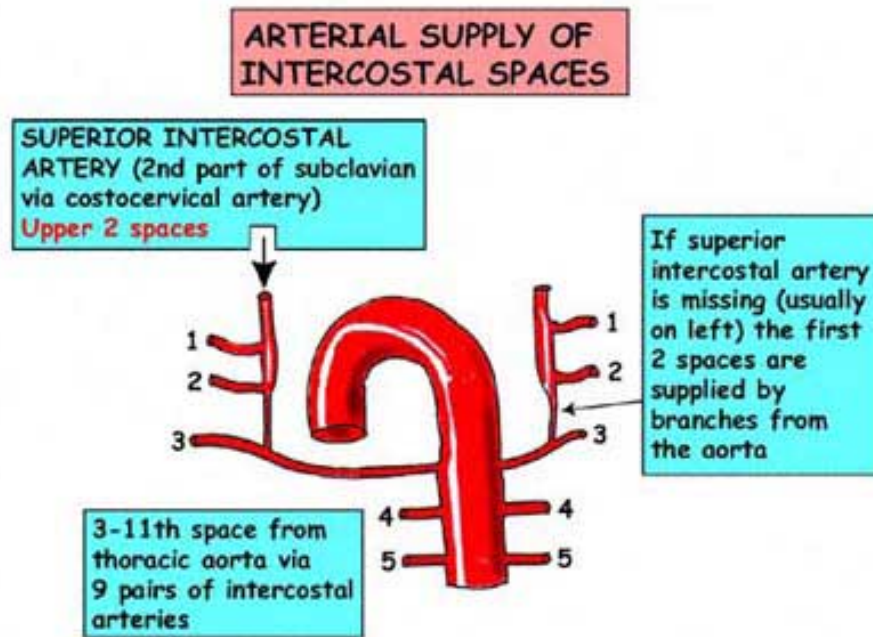
- Derived from:
  - *Thoracic aorta*: through the posterior Intercostal and Subcostal arteries
  - *Subclavian artery*: through the internal thoracic arteries and supreme intercostal artery
  - *Axillary artery*: superior and lateral thoracic arteries
- So each IC space is supplied by 3 arteries:
  - *Large posterior IC artery*
  - *A small pair of Anterior IC arteries*
- ***Posterior IC arteries:***
  - The 1<sup>st</sup> and 2<sup>nd</sup> IC arteries are given by the *Supreme (Highest) IC artery*, a branch of the *Costocervical trunk* of the Subclavian artery
  - In the 3<sup>rd</sup>-11<sup>th</sup> IC spaces + the *Subcostal artery* of the Subcostal space, are given as paired branches of the *Thoracic aorta (Aortic Intercostals)*
    - \*\* **Note:** The right side arteries cross the vertebra and pass posterior to the esophagus, thoracic duct, azygos vein, right lung and pleura
  - Each posterior IC artery has:
    1. *Posterior Ramus (branch)*: accompanies the posterior ramus of spinal nerve to supply the spinal nerve, vertebral column, back muscles and related skin
    2. *Collateral branches* as in the case with the IC nerve
    3. *Anterior Ramus (branch)*: travels along the bottom of the rib, in the IC space together with the corresponding vein and nerve, organized in VAN fashion (Vein being most superior), supplying muscles and related pleuraAt the angle of ribs it gives the Lateral cutaneous branch that divides into anterior and posterior sub-branches to supply the related dermatome skin

Terminal and collateral branches anastomose with Anterior IC branches anteriorly

- ***Internal thoracic arteries (Historically, Internal mammary arteries):***
  - Arise at the root of neck, from the inferior surface of the *Subclavian* arteries at their origin
  - Descends in the thorax behind the clavicle and 1<sup>st</sup> costal cartilage, crossed by its origin by the ipsilateral Phrenic nerve
  - Descend on the internal surface of thorax slightly both sides lateral to the sternum, posterior to the upper 6 costal cartilages, thus medial to the nipple, running between the slips of transverse thoracic muscle
  - *Branches:*
    1. Thymic branches

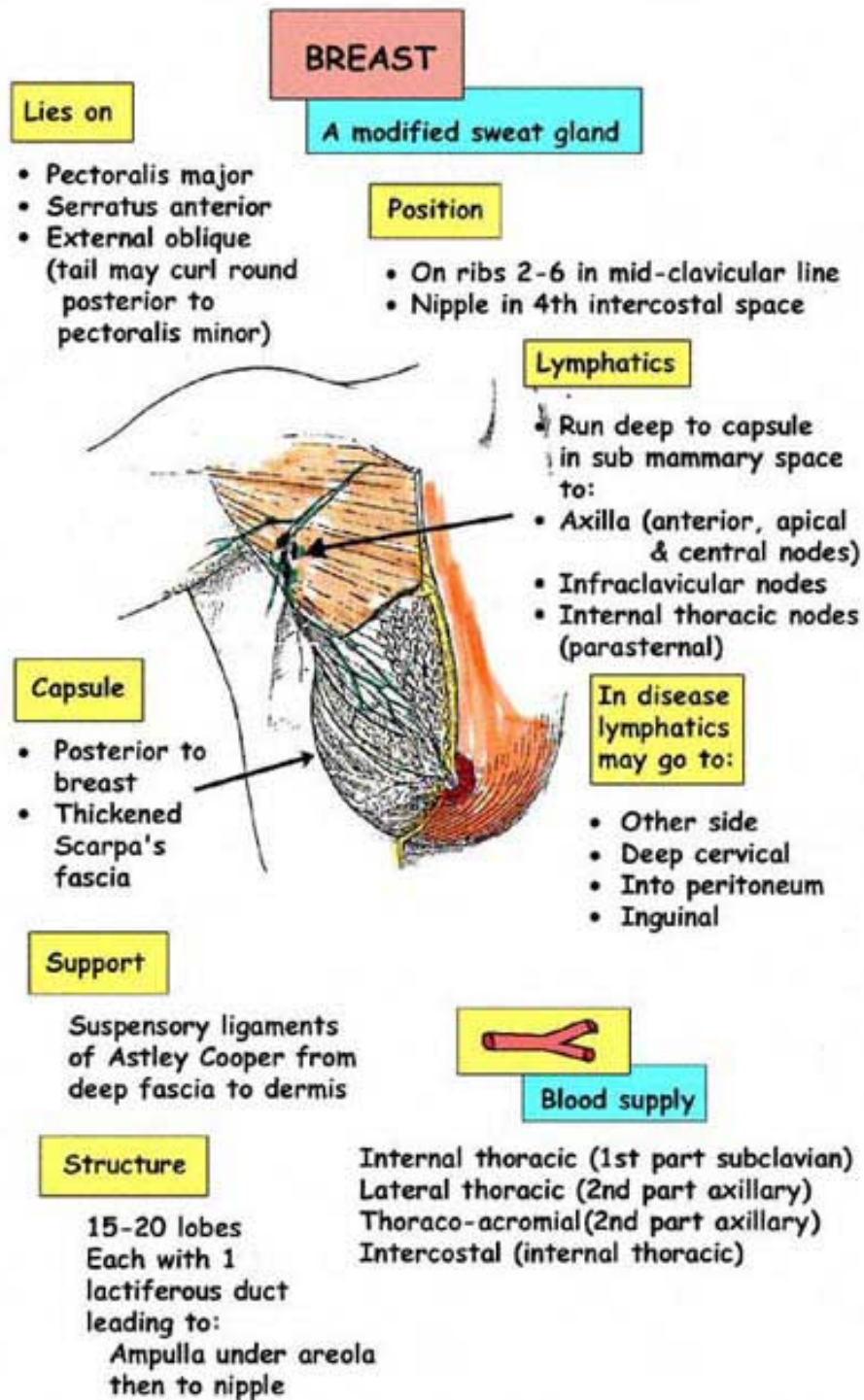
2. *Pericardicophrenic arteries*: traveling with the phrenic nerve anterior to the root of lungs, both sides of the pericardium
  3. Sternal branches
  4. Perforating branches
  5. Mediastinal branches, bronchial
  6. *Direct Anterior IC arteries*: 12 pairs, 2 in each upper 6 IC spaces, traveling in the Intercostal space laterally, until anastomose with the corresponding posterior IC arteries, supply also the pectoral muscles, skin and breast
- Terminate in the 6<sup>th</sup> IC space by dividing into
    1. *Superior Epigastric artery*: medially travels downward into the abdominal wall, anastomose with the inferior epigastric artery behind the Rectus sheath, supplying the abdominal muscles, integument and Falciform ligament
    2. *Musculophrenic artery*: laterally and roughly follows the costal margin to supply the 7<sup>th</sup>-9<sup>th</sup> IC spaces as the Indirect Anterior IC arteries to supply the spaces and diaphragm
- \*\*Note:** the inferior most 2 IC spaces don't have Anterior IC arteries; hence they are supplied by the Posterior IC branches totally
- ***Superior Thoracic artery***: branch of the 1<sup>st</sup> part of Axillary supplying muscles of the superior chest
  - ***Lateral thoracic artery***: branch of the 2<sup>nd</sup> party of Axillary, running down at the lateral border of Pectoral muscles, supplying them, mammary glands and muscles of chest
  - ***Veins of the thoracic wall***: 11 Intercostal veins, accompanying the IC artery, nerve, laying most superior in the costal groove + 1 Subcostal vein
    - The veins of the 1<sup>st</sup> IC space are called *Supreme Intercostal veins* and they drain into the nearby brachiocephalic vein
    - The most posterior IC veins drain into the Azygos/Hemiazygos venous system
    - 2<sup>nd</sup>-4<sup>th</sup> IC veins form a trunk called: *Superior IC vein*
      1. *The Right Superior IC vein* is the final tributary to the azygos vein
      2. *The left superior IC vein* drains into the L. brachiocephalic vein, passing along the L. side of the superior mediastinum and receiving the L. Bronchial and Pericardicophrenic veins





**Remember:**

- Nerve is always outside the artery
- 1st thoracic nerve has no lateral cutaneous branch
- Nerves T7 & -T11 also supply the abdomen



## THORAX - SURFACE MARKINGS CLINICAL IMPLICATIONS

### Access to thoracic aorta

Through the left chest (left thoracotomy)

### Access to oesophagus

Through right chest (right thoracotomy)

(Both of these are usually through the 5th intercostal space by stripping up the periosteum of the rib below and incising through the rib bed)

### Thoraco-abdominal incision

This is for large access to upper abdomen or chest  
Through the 9th intercostal space on either side  
9th Costal cartilage is usually excised  
The diaphragm is incised radially to avoid damaging its nerve

### Pericardial aspiration

Insert needle just to left of xiphoid

### Pericardial window

Through left thoracotomy  
5th costal cartilage excised  
Pericardial flap cut

### Heart transplantation

Via median sternotomy  
En bloc - aorta, pulmonary trunk, both atriae  
SVC, IVC, SA node are all left

### Aspiration of pleural cavity

Siting of drain in 4th space in mid axillary line

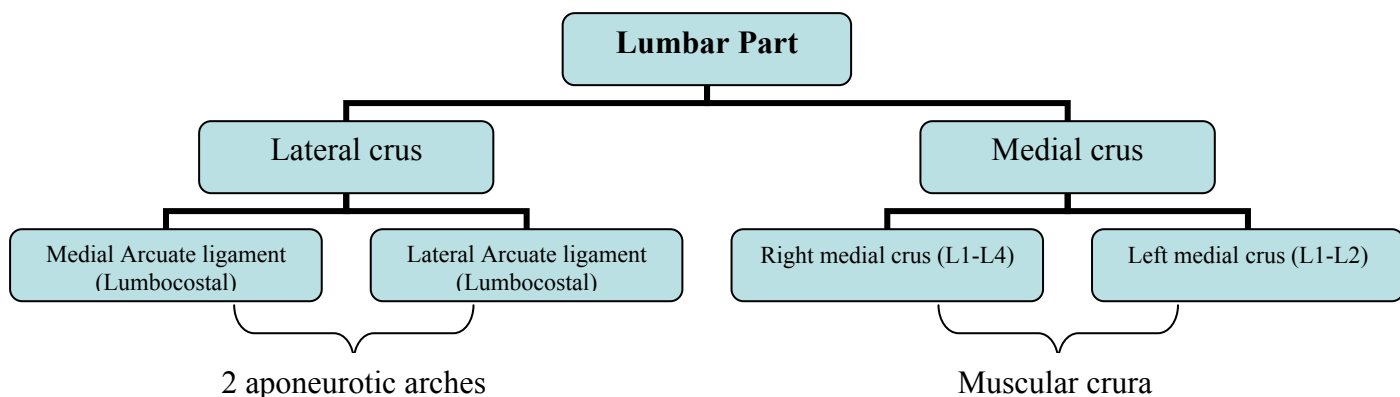
## The Diaphragm

Double domed *musculotendinous* partition between the thoracic and the abdominal cavities.

It is the chief muscle of *inspiration*, descending downward during deep breath

It has 2 main parts:

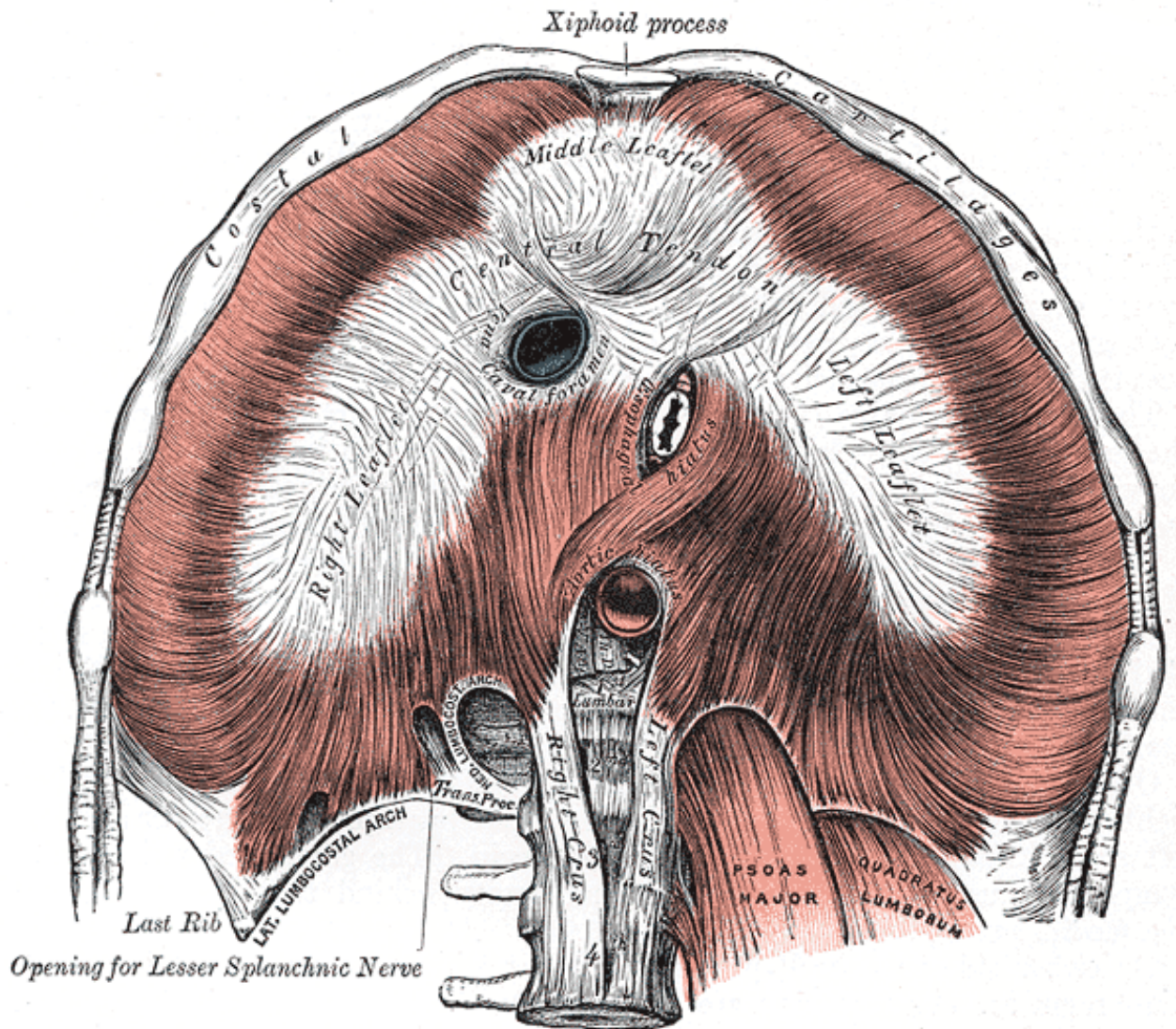
1. **Muscular part, origin:** subdivided into 3 parts,
  - a. *Sternal part:* 2 muscular slips attaching to the posterior aspect of the xiphoid process
  - b. *Costal part:* attaching to the inner surfaces of the lower 6 costal cartilages and their adjoining ribs on each side with muscular slips interdigitating with the transverse abdominis muscle origin
  - c. *Lumbar part:*



2. **Tendinous part, insertion:** trifoliate central aponeurotic part, the Central tendon of diaphragm, fused with the pericardium strongly with the *Pericardiophrenic ligament*
- *The right medial crus* is larger and longer than the left one, and although the esophagus is found on the left side of the midline in the thoracic cavity, the esophageal hiatus in the diaphragm is a formation of the **right** medial crus at the level of T10
  - Over the anterior aspect of the Aorta, the 2 medial crura (L+R) unite together to form a fibrous **Median Arcuate ligament** (T12 level)
  - *The right medial crus* consist of 3 ligaments:
    - 1 goes to the central tendon
    - 1 arises from the median Arcuate ligament, forming the right border of the esophageal hiatus
    - 1 forming the left border of the esophageal hiatus
  - The medial and lateral Arcuate ligaments are thickening of the fascia covering the Psoas major and Quadratus lumborum muscles (anterior layer of the Thoracolumbar fascia)
    - **Medial Arcuate ligament:** from the left vertebral body of L1 to the transverse process of L1, covers the *Psoas major muscle*

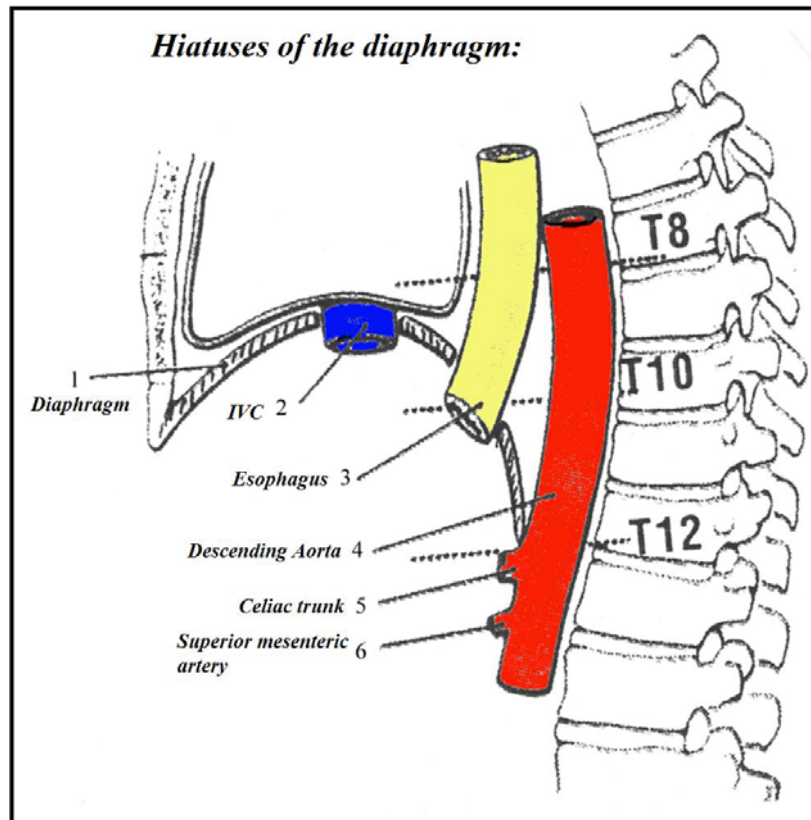


- **Lateral Arcuate ligament:** from the transverse process of L1 to the top of 12<sup>th</sup> rib, covering the *Quadratus lumborum* muscle
- **Apertures in the diaphragm:**
  - **Caval opening:** at the level of **T8/T9**, for the passage of the IVC and some branches of the right Phrenic nerve. The IVC is attached to this opening, so when the diaphragm contracts it helps the blood flow in it
  - **Esophageal hiatus:** in the right crus, **T10** level, for the passage of the *esophagus* and the *Anterior (L.) and Posterior (R.) Vagal trunks* and esophageal tributaries of the *Left gastric vein* (Coronary vein of stomach) (porto-caval anastomosis)
  - **Aortic hiatus:** at the level of **T12**, behind the *median Arcuate ligament*, for the passage of aorta (thoracic becoming abdominal) and the thoracic duct
  - **Sternocostal triangle / foramen:** between the sternal and costal attachments, transmits the superior epigastric vessel (branches of the internal thoracic arteries)
  - **Between the Lateral and Medial crura:** we have the **intermediate hiatuses**, through which Sympathetic trunk, azygos (right side), hemiazygos (left side), greater & lesser splanchnic nerves pass

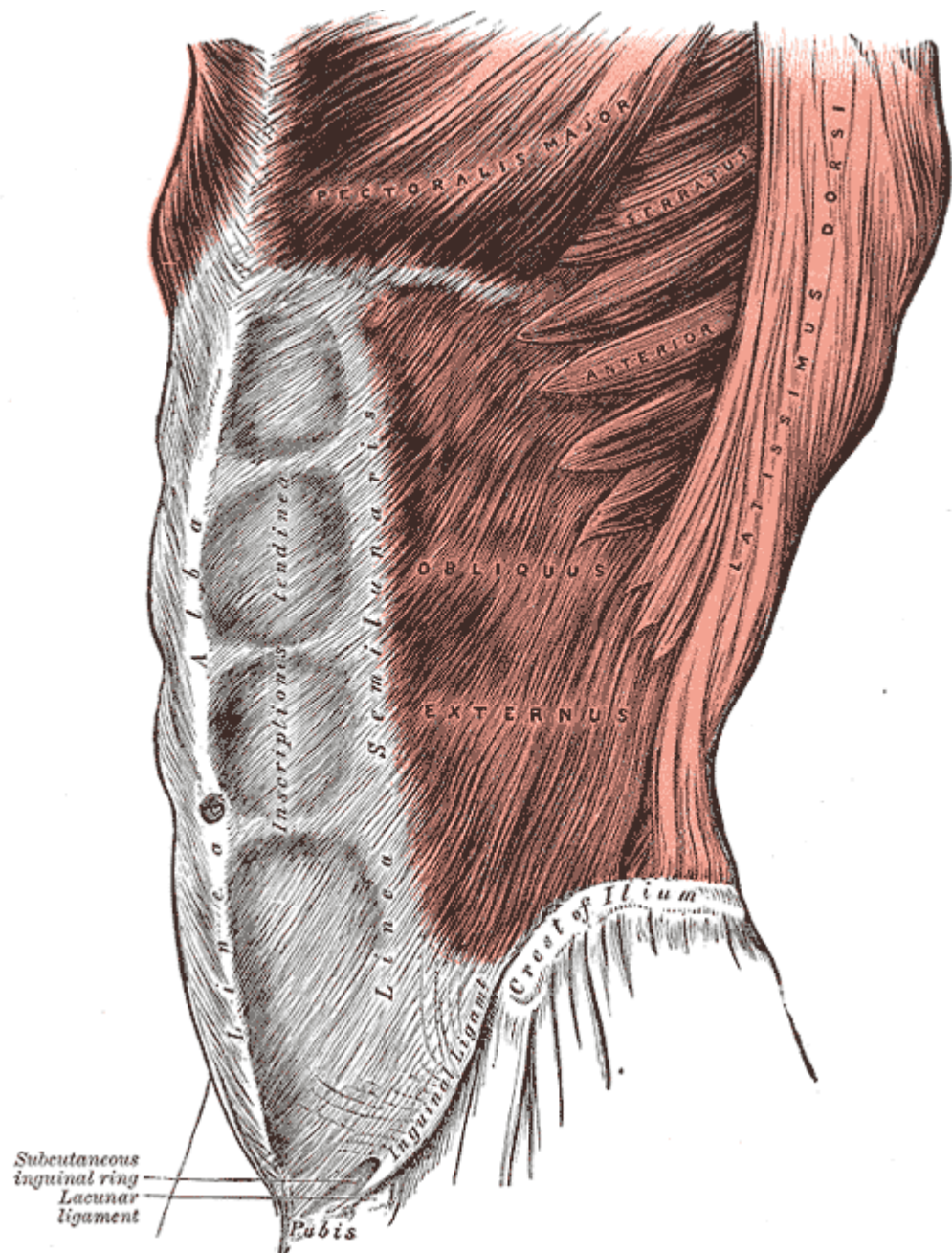




- **Arterial supply of the diaphragm:**
  - *Superior Phrenic arteries:* from the thoracic aorta
  - *Inferior Phrenic arteries:* from the abdominal aorta
  - *Musculophrenic + Pericardiophrenic arteries:* from the internal thoracic artery
- **Innervation of the diaphragm:**
  - *Phrenic nerve (C3-C5)* motor innervation, sensory also (Cut C4, breath no more!)



# *Topography of the abdominal wall & inguinal region*



➔ ***Fascia of the Anterolateral abdominal wall, layers:***

○ ***Superficial fascia:***

- *Superficial layer of the superficial fascia (Camper's fascia)*
  1. superficial portion of the subcutaneous tissue which is specialized for fat storage and thus often has an abundance of fat, especially in the overnourished individual, compared with the deeper, fibrous portion of the subcutaneous tissue; in morbid obesity, this layer forms the core of a large, sagging apron like fold
  2. Continues over the inguinal ligament to merge with the *superficial fascia of thigh*
  3. Continues over the pubic and perineum as superficial layer of *superficial Perineal fascia*
- *Deep layer of the superficial fascia (Membranous / fibrous fascia of Scarpa)*
  1. Attaches to the fascia lata just below the inguinal ligament
  2. Continues over the pubic and perineum as the membranous layer (Colle's fascia) of the superficial Perineal fascia
  3. Continues over the penis as the superficial fascia of penis and over the scrotum as the Tunica Dartos (has smooth muscle)

○ ***Deep fascia (Investing):***

- Has intermediate and deep layers
- Covers the external aspects of the 3 muscles of the anterolateral abdominal wall and their aponeurosis, cannot be easily separated or detached
- The inner aspect of the abdomen is lined by the *Trasversalis fascia* (contacting the trasversalis muscle)
- Continues over the spermatic cord at the superficial inguinal ring as the *external spermatic fascia*
- Continues over the penis as *the deep fascia of penis (Buck's fascia)* and over the pubis and perineum as the *deep Perineal fascia*

○ So if we sum up the ***layers of the abdominal wall***, from outside to inside:

- *External*
  1. Skin
  2. Subcutaneous fatty tissue
  3. Superficial fascia
    - Superficial layer of the superficial fascia (*fatty Camper's fascia*)
    - Deep layer of the superficial fascia (*Membranous fascia of Scarpa*)
- *Middle*
  1. External oblique + aponeurosis
  2. Internal oblique + aponeurosis
  3. Transversus abdominis muscle + aponeurosis (on anterior surface)
- *Internal*
  1. Trasversalis fascia
  2. Preperitoneal fatty tissue
  3. Parietal peritoneum

➔ **Abdominal muscles:**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b><i>External Oblique</i></b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Fleshy digitations from the external surface of the lower 8 ribs (5th-12th), the dorsal border is free                                                                                                                                                                                                                                                                                                                                               | <i>Origin</i>                  |
| On the anterior 2/3 of the external lip of the iliac crest, on the inguinal ligament and on the outer (anterior) layer of the rectus sheath medially (linea alba)                                                                                                                                                                                                                                                                                    | <i>Insertion</i>               |
| Intercostal nerves (T8-T11), Subcostal nerve (T12), Ilioinguinal n., iliohypogastric n.                                                                                                                                                                                                                                                                                                                                                              | <i>Innervation</i>             |
| <ul style="list-style-type: none"> <li>• Fibers direction is from Superolateral to inferomedial (Hands in pocket)</li> <li>• Muscle fibers become aponeurotic approximately at MCL and at the spinous line inferiorly</li> <li>• Maintains abdominal pressure to keep viscera in position</li> <li>• Increases abdominal pressure,</li> <li>• Both sides flex the vertebral column,</li> <li>• One side bends it laterally and rotates it</li> </ul> | <i>Notes / function</i>        |

|                                                                                                                                                                                                                                                                                                                                                                              | <b><i>Internal Oblique</i></b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| Internal lip of the iliac crest, Thoracolumbar fascia, ASIS, lateral 2/3 of the inguinal ligament                                                                                                                                                                                                                                                                            | <i>Origin</i>                  |
| Cranial part: Fleshy digitations to lower 3 ribs (10th-12th)<br>Middle part: on the linea alba, aponeurosis (anterior / posterior layers of rectus sheath)<br>Caudal part: pubic tubercle, in males: spermatic cord as cremasteric muscle and in females: reaches the round ligament of uterus                                                                               | <i>Insertion</i>               |
| Lower 6 thoracic IC nerves, first lumbar nerves (Ilioinguinal, iliohypogastric)                                                                                                                                                                                                                                                                                              | <i>Innervation</i>             |
| <ul style="list-style-type: none"> <li>• Fibers spawn Inferolaterally to Superiomedially (Hands on chest)</li> <li>• Maintains abdominal pressure to keep viscera in position</li> <li>• Increases abdominal pressure</li> <li>• One side Bends the body laterally, and rotates it ipsilaterally</li> <li>• Flexes lumbar vertebral column (bends thorax forward)</li> </ul> | <i>Notes / function</i>        |

|                                                                                                                                                                                                                  | <b><i>Transversus abdominis</i></b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Internal surface of the lower 6 ribs (7th-12th), thoracolumbar fascia, internal lip of the iliac crest, lateral 2/3 of the inguinal ligament                                                                     | <i>Origin</i>                       |
| Xiphoid cartilage, Linea alba and, through the conjoint tendon, pubic tubercle and pecten pubis. Helps forming the posterior layer of the rectus sheath                                                          | <i>Insertion</i>                    |
| Lower 6 thoracic IC nerves, first lumbar nerves (Ilioinguinal, iliohypogastric)                                                                                                                                  | <i>Innervation</i>                  |
| <ul style="list-style-type: none"> <li>• Maintains abdominal pressure to keep viscera in position,</li> <li>• Contraction and expansion of the abdominal wall</li> <li>• Rotates and flexes the trunk</li> </ul> | <i>Notes / function</i>             |

|                                                                                                                                                                                                                                                                                                                                                               | <b><i>Rectus Abdominis</i></b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 5th-7th cartilages of the ribs, Xiphoid process                                                                                                                                                                                                                                                                                                               | <i>Origin</i>                  |
| Pubic tubercle, crest and Symphysis                                                                                                                                                                                                                                                                                                                           | <i>Insertion</i>               |
| Intercostal nerves (T7-T12)                                                                                                                                                                                                                                                                                                                                   | <i>Innervation</i>             |
| <ul style="list-style-type: none"> <li>• Anchored transversely by attachment to the anterior layer of the Rectus sheath at 3 or more tendinous intersections, separating the muscle into smaller parts</li> <li>• Flexes the vertebral column</li> <li>• Pulls the sternum (thorax) toward the pubis</li> <li>• Tenses the anterior abdominal wall</li> </ul> | <i>Notes / function</i>        |

|                                                                                                                                                             | <b><i>Pyramidalis</i></b> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Pubic crest                                                                                                                                                 | <i>Origin</i>             |
| Lower Portion of linea alba                                                                                                                                 | <i>Insertion</i>          |
| Intercostal nerves                                                                                                                                          | <i>Innervation</i>        |
| <ul style="list-style-type: none"> <li>• Small triangular muscle, Tenses the linea alba, attachment is a surgical mark median abdominal incision</li> </ul> | <i>Notes / function</i>   |

- ***Superficial abdominal muscles:***
  - *Lateral group:* External oblique, internal oblique, transversus abdominis muscles
  - *Medial group:* Rectus abdominis, Pyramidalis
- ***Deep abdominal muscles***
  - Quadratus lumborum, Psoas major / minor (See the discussion of the lumbar dorsal region later)



### → **The Rectus sheath:**

It's a strong incomplete fibrous compartment of the rectus abdominis and pyramidalis muscles. In it we can find also the epigastric arteries anastomosing, lymphatics and distal portions of the Intercostal nerves (anterior rami of spinal nerves T7-T12)

The sheath is formed by the decussation of interweaving of the aponeurosis of the flat lateral abdominal muscles.

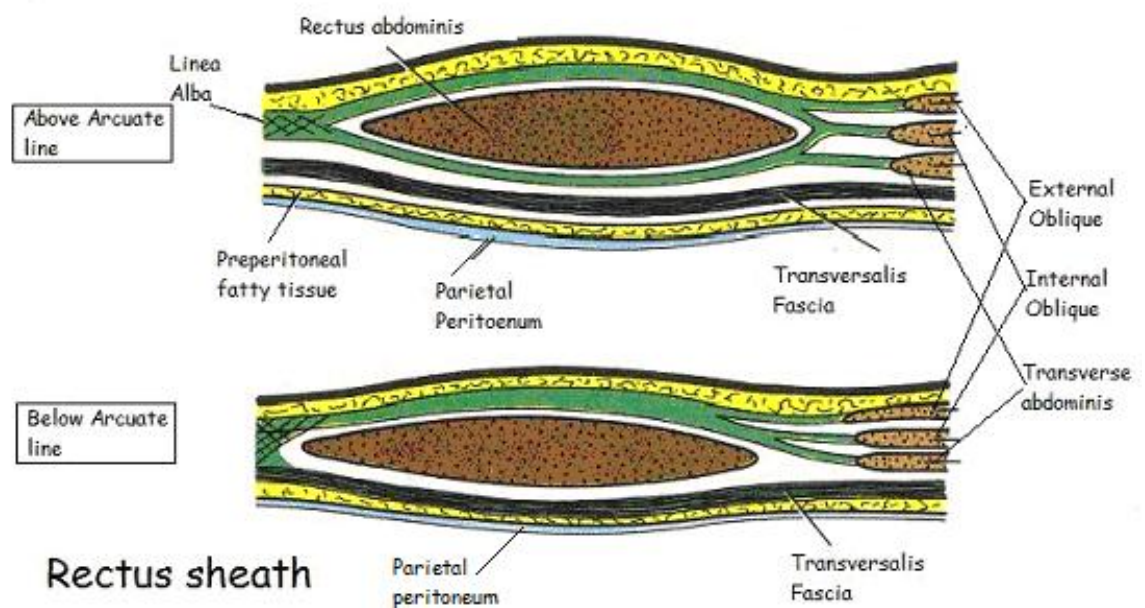
- **Anterior layer of rectus sheath:**
  - Above Arcuate line: aponeurosis of external and internal oblique muscles
  - Below Arcuate line: aponeurosis of external oblique, internal oblique and transversus abdominis
- **Posterior layer of rectus sheath:**
  - Above Arcuate line: aponeurosis of internal oblique and transversus abdominis muscles
  - Below Arcuate line: rectus abdominis is in direct contact with Transversalis fascia

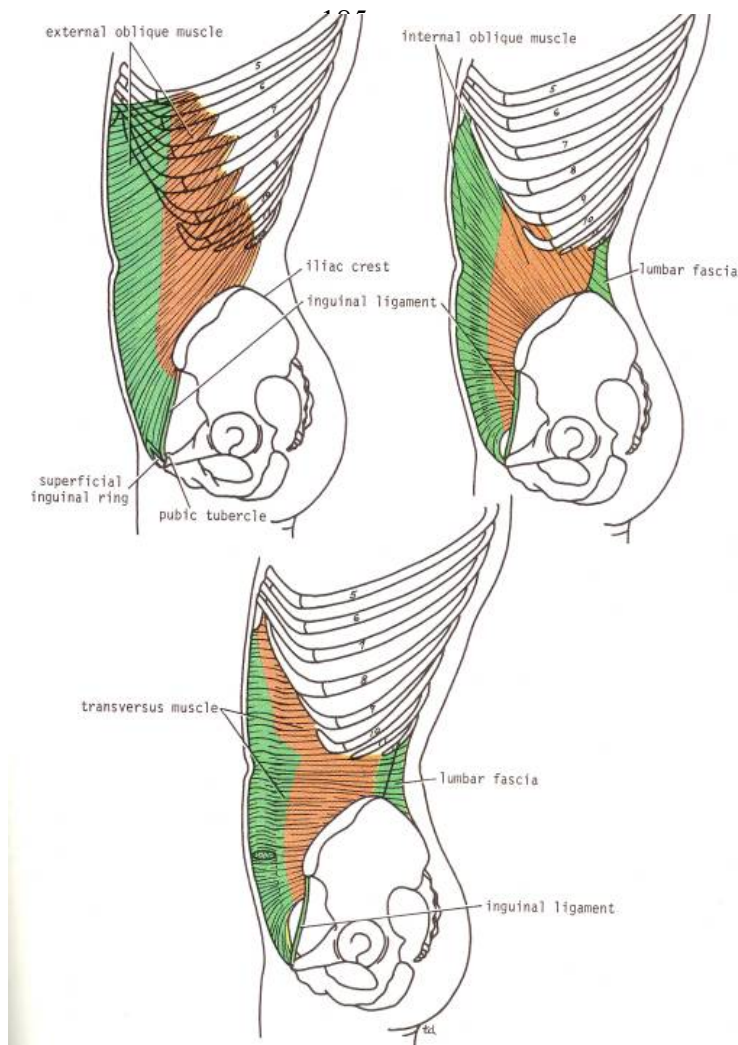
**\*\* Arcuate line of Douglas (Semicircular):** crescent shaped line marking the inferior limit of the posterior layer of the rectus sheath just below the level of iliac crest

**\*\* Semilunar line of Spigelius:** curved line along the lateral border of rectus abdominis muscle, found on the aponeurosis of transversus abdominis muscle

**\*\* Linea Alba:** a tendinous median raphe between the 2 rectus abdominis muscles, extending from the xiphoid process to the pubic symphysis. Formed by the aponeurosis of external oblique, internal oblique and transversus abdominis muscles

**\*\* Umbilical ring:** a defect in the Linea Alba through which the fetal umbilical vessels are passed to form the umbilical cord. All layers from the anterolateral wall fuse at the umbilicus, and subcutaneous fat tissue accumulates postnatally raising the skin around it (7-14 days after birth)





## POSTERIOR ABDOMINAL WALL

- 5 vertebrae
- Transverse process of L3 is largest
- Transverse process of L5 is conical

### PSOAS MAJOR

Origin: Intervertebral discs T12/L1 to L4/5  
 Bodies of L1-5, transverse processes L1-5  
 Inserts: Lesser trochanter  
 Nerve: L1,2,3  
 Action: Flexes hip

### PSOAS MINOR

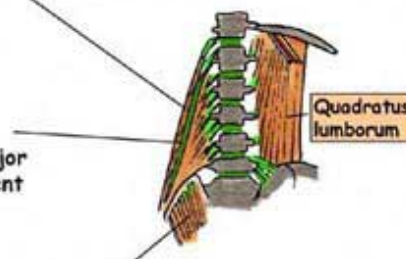
Origin: Bodies T12,L1  
 Inserts: Fascia over psoas major  
 behind inguinal ligament  
 Nerve: L1  
 Action: Weak spine flexor

### QUADRATUS LUMBORUM

Origin: Transverse process L5  
 Iliolumbar ligament &  
 posterior 1/3 iliac crest  
 Inserts: Medial 1/2 12th rib &  
 transverse process L1-4  
 Nerve: T12-L4  
 Action: Holds down 12th rib

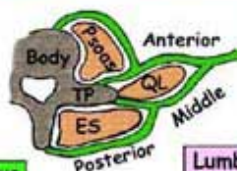
### ILIACUS

Origin: Hollow of iliac fossa  
 Inserts: Psoas tendon &  
 below lesser trochanter  
 Action: Flexes hip  
 Nerve: Femoral (L2,3,4)



## LAYERS OF THORACOLUMBAR FASCIA

TP= Transverse  
 process  
 ES= Erector spinae  
 QL= Quadratus  
 lumborum



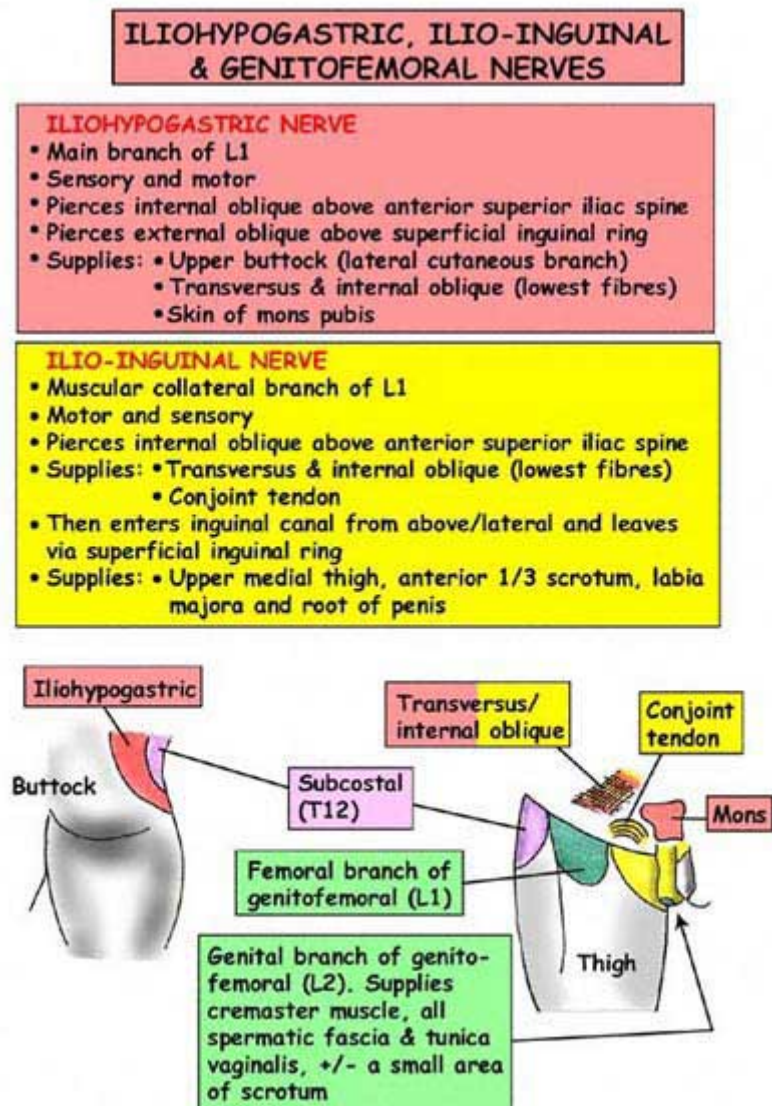
Attachment of  
 transversus &  
 internal oblique

Spine/supraspinous  
 ligaments

Lumbar regions all 3 layers  
 are present, thoracic  
 region has posterior layer only

### → Nerves of the Anterolateral abdominal wall:

- **Thoracoabdominal (Intercostal) nerves:** distal, abdominal parts of the anterior rami of the inferior 6 thoracic spinal nerves (T7-T11) (T10 is around the umbilicus)
- Lateral thoracic cutaneous branches of the thoracic spinal nerves T7-T10
- **Subcostal nerve:** larger anterior ramus of spinal nerve T12
- **Iliohypogastric nerve:** terminal branch of anterior rami of spinal nerves L1-L2
- **Ilioinguinal nerve:** terminal branch, with the iliohypogastric nerve, the first lumbar nerve, passes through the inguinal canal and superficial inguinal ring to supply the skin of the upper medial thigh, mons pubis, and scrotum or labia majora



### → Vessels of the anterolateral abdominal wall:

- **Superior epigastric vessels:** branches of the internal thoracic arteries enter the rectus sheath superiorly through the posterior layer and supply the superior part of rectus abdominis. Anastomose with the inferior epigastric vessels around the umbilical region
- **Inferior epigastric vessels:** branches of the external iliac artery/vein just superior to the inguinal ligament. Run superiorly in the transversalis fascia to enter the rectus sheath under the Arcuate line and form the Lateral Umbilical fold
- **Superficial circumflex iliac vessels:** branches of femoral artery and tributaries of greater saphenous vein, they go laterally and up above inguinal ligament to anastomose with the deep circumflex iliac vessels

- **Deep circumflex iliac vessels:** branches of the external iliac artery, have the same position as the inferior epigastric, but more laterally
- **Superficial epigastric vessels:** branches of femoral artery and tributaries of greater saphenous vein, they go medially and up above inguinal ligament to anastomose with the deep circumflex iliac vessels or external pudendal
- **Cutaneous veins:**
  - **Paraumbilical veins:** several small veins arising from cutaneous veins about the umbilicus running along the round ligament of the liver, and terminating in the portal vein. They constitute a porto-caval anastomosis and are subject to varicosity during portal hypertension; varicose Paraumbilical veins form the “caput medusae”
  - **Thoracoepigastric veins:** anastomosis between the superior epigastric vein and lateral thoracic vein, many develop as altered venous flow
  - **Inferior / Superior epigastric veins:** may shunt blood in case of IVC block

## ***Inguinal region:***

### *Borders:*

- *Superior:* horizontal line from the ASIS to the midline
- *Medial:* Linea alba
- *Inferolateral:* inguinal ligament

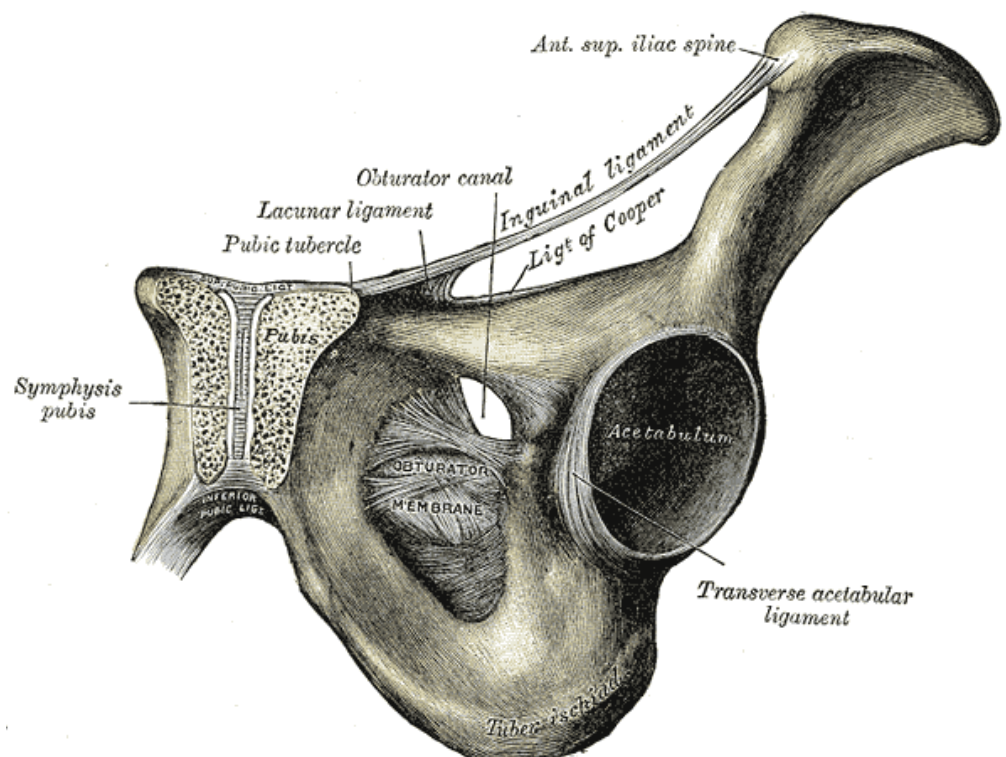
### *Structures:*

- *Internal wall:* discussed later, the 5 umbilical folds with fossa between them
- *Externally:* most important, Inguinal canal (vessels, nerves discussed previously also)
- ***Inguinal ligament of Poupart:*** a fibrous band formed by the thickened inferior border of the aponeurosis of the external oblique that extends from the ASIS to the pubic tubercle, bridging muscular and vascular lacunae; forms the floor of the inguinal canal; gives origin to the lowermost fibers of internal oblique and Transversus abdominis muscles.
- ***Iliopubic tract:*** thickened inferior margin of the transversalis fascia seen as a fibrous band running parallel and posterior (deep) to the inguinal ligament, contributing to the posterior wall of the inguinal canal as it bridges the external iliac-femoral vessels from the Iliopectineal arch to the superior pubic ramus. It marks the inferior edge of the deep inguinal ring and the medial margin of the femoral canal. Seen only when the inguinal region is viewed from its internal aspect, it is a useful landmark in laparoscopy of this region, as for repair of inguinal herniae.



Together the inguinal ligament and ilopubic tract function as the *flexor retinaculum* for the hip joint, and this retinaculum spans the *subinguinal hiatus*, allowing passage of different structures and subdividing it into compartments (see Subinguinal region for details). Some fibers arise from the inguinal ligament as:

- **Lacunar ligament of Gimbernat:** a curved triangular fibrous band that passes horizontally backward from the medial end of the inguinal ligament to the Pectineal line; it forms the medial boundary of the femoral ring (and Subinguinal hiatus)
- **Pectineal ligament of Cooper:** a thick, strong fibrous band that passes laterally from the lacunar ligament, on the Pectineus muscle, along the Pectineal line of the pubis. This fibrous tissue on the bony surface allows the purchase of sutures in various procedures to repair inguinal herniae.
- **Reflected inguinal ligament of Colles:** slightly reinforced portion of the external oblique aponeurosis, formed by fibers derived from the medial portion of the inguinal ligament of one side that run medially and slightly superiorly, defining the inferior margin of the superficial inguinal ring; the fibers then pass on the deep aspect of the ipsilateral medial crus, crossing the Linea alba and running within the contralateral aponeurosis to course parallel and superior to the contralateral inguinal ligament.
- **Iliopectineal arch / ligament:** a thickened band of fused iliac and Psoas fascia passing from the posterior aspect of the inguinal ligament anteriorly across the front of the femoral nerve to attach to the iliopectineal eminence of the hip bone posteriorly. The iliopectineal arch thus forms a septum which subdivides the space deep to the inguinal ligament into a lateral muscular lacuna and a medial vascular lacuna.
- **Inguinal Falx (conjoint tendon):** common tendon of insertion of the Transversus and internal oblique muscles into the crest and tubercle of the pubis and iliopectineal line; it is frequently largely muscular rather than aponeurotic and may be poorly developed; forms posterior wall of medial inguinal canal.





## → **Inguinal canal:**

The obliquely, inferiorly and medially directed ~4cm passage through the musculoaponeurotic layers of the lower anterior abdominal wall that transmits the spermatic cord in the male and the round ligament of uterus in the female from the pelvic cavity to the scrotum or labia majora, respectively.

We can imagine the canal as a long box with 6 sides (2 of which are openings),  
So the canal has an:

- **Entrance, the deep inguinal ring:** opening in the transversalis fascia in the lateral inguinal fossa, through which the ductus deferens and gonadal vessels (or round ligament of uterus in the female) enter the inguinal canal. Located midway between anterior superior iliac spine and pubic tubercle, it is bounded medially by the lateral umbilical fold (inferior epigastric vessels) and inferiorly by the Iliopubic tract. *Indirect inguinal hernias* exit the abdominal cavity via the deep inguinal ring.
- **Exit, the superficial inguinal ring:** slit-like opening in the aponeurosis of the external oblique muscle of the abdominal wall through which the contents of the inguinal canal emerge. Has a medial (Attached to the pubic crest) and lateral (attached to the pubic tubercle) crura bordering its sides and connected by intercrural fibers

### **Borders:**

|                                                                                                                                              |                                                                                                                                           |                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                              | <b>Superior wall (Roof):</b> <ul style="list-style-type: none"> <li>• Internal Oblique</li> <li>• Transversus Abdominis muscle</li> </ul> |                                                                                                                                                                        |
| <b>Posterior wall:</b> <ul style="list-style-type: none"> <li>• Transversalis Fascia</li> <li>• Conjoint tendon (medial 1/3 only)</li> </ul> | <b>Inguinal canal</b>                                                                                                                     | <b>Anterior wall</b> <ul style="list-style-type: none"> <li>• Aponeurosis of External oblique</li> <li>• Aponeurosis of Internal oblique (Lateral 1/3 only)</li> </ul> |
|                                                                                                                                              | <b>Inferior wall (floor):</b> <ul style="list-style-type: none"> <li>• Inguinal ligament</li> </ul>                                       |                                                                                                                                                                        |

### **Contents if the inguinal canal:**

- *In males:* the spermatic cord and its coverings (including genital branch of genitofemoral nerve) + the ilioinguinal nerve.
- *In females:* the round ligament of the uterus (including genital branch of genitofemoral nerve) + the ilioinguinal nerve.

**\*\* Note:** The ilioinguinal nerve passes between the internal oblique and external oblique, then behind the intercrural fibers of the Superficial Inguinal ring, passing through the superficial ring to meet the spermatic cord and descend into the scrotum, so technically it doesn't run through the whole canal, only the most lateral part of it.



### → **Lumbar (Dorsal) Region:**

*Borders:*

- Superior: Transverse line passing through the inferior angle of scapula
- Lateral: A vertical line along the lateral border of abdomen
- Inferior: Iliac crest
- Medial: A vertical line between the spinous processes of corresponding vertebrae

*Skin Innervation:*

- Dermatome innervation (Segmental) (Dorsal cutaneous branches of T7-L3)

Below the skin, we find the Superficial Lamina (Posterior) of the Thoracodorsal fascia covering the most important structure which is the Latissimus Dorsi muscle (the broadest muscle of the body) (*Origin:* spinous processes of thoracic T7-T12, Thoracolumbar fascia, iliac crest and inferior 3 or 4 ribs, inferior angle of scapula, *Insertion:* Crest of Lesser tubercle of Humerus with Teres Major).

Below the Latissimus Dorsi we should find the Thoracodorsal artery, vein and nerve supplying the muscle, lying on the Serratus posterior inferior muscle.

Below the Thoracolumbar fascia, we can find the Erector Spinae muscle, External oblique muscle and some fibers of the transverse abdominal muscle

The deepest structure below the Erector Spinae muscle is the Quadratus lumborum (below the deepest lamina of the thoracolumbar fascia)

### → **The Thoracolumbar (Lumbodorsal) fascia:**

A deep investing membrane which covers the deep muscles of the back of the trunk, it is made up of three layers:

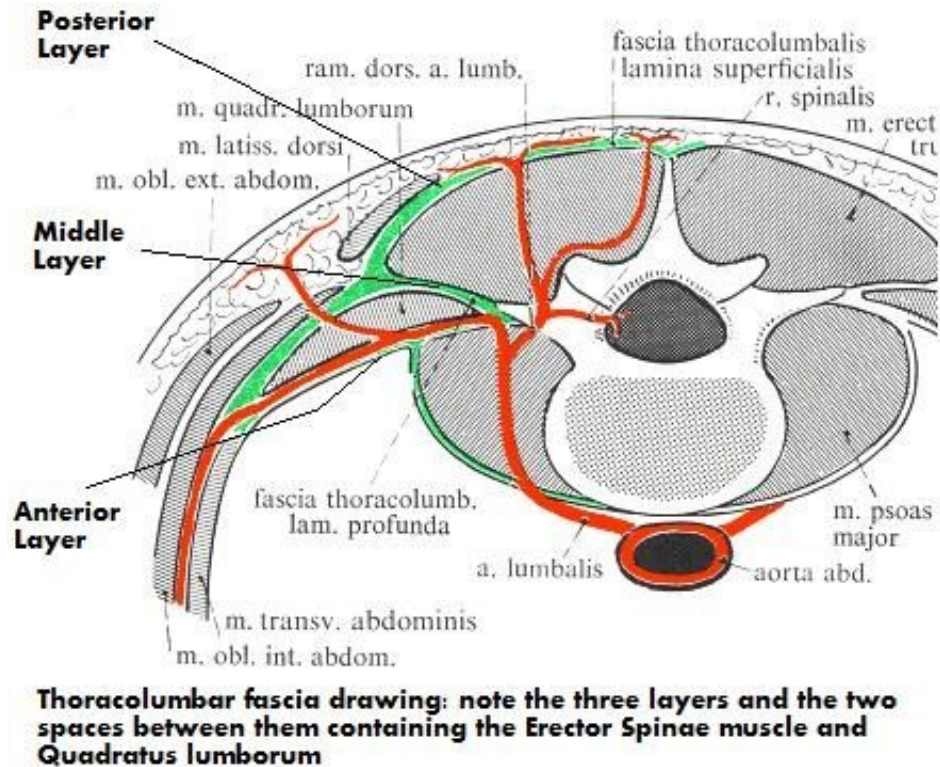
- *Anterior (thinnest)(Quadratus Lumborum fascia)*
- *Middle (Deep layer)*
- *Posterior (Thickest) (Superficial layer)*

Two spaces are formed between these three layers of the fascia:

- *Quadratus Lumborum space:* between Anterior and middle layers
- *Erector Spinae space:* between middle and posterior layers

It contains both longitudinal and transverse fibers, and is attached, medially, to the spinous processes of the thoracic vertebrae; laterally to the angles of the ribs. It gives origin to internal oblique and Transversus Abdominis muscles, and above, it passes in front of the Serratus posterior superior and is continuous with a similar investing layer on the back of the neck to become the Nuchal fascia.

In the thoracic region the Thoracodorsal fascia is a thin fibrous lamina which serves to bind down the Extensor muscles of the vertebral column and to separate them from the muscles connecting the vertebral column to the upper extremity.



\*\* Important structures to be found in this region also are 2 triangles: Petit Triangle & the Auscultation triangle:

→ **Triangle of Petit (Inferior Lumbar):** a weak point of the wall so herniation occasionally occur here, borders:

- Latissimus Dorsi muscle *superiorly*
- External oblique muscle *laterally*
- Iliac crest *inferiorly*
- Internal oblique muscle as the *floor*

→ **Auscultatory Triangle (Superior Lumbar):** the absence of musculature in this space allows respiratory sounds to be heard clearly with a stethoscope, especially If the scapula is drawn forward by folding the arms across the chest, and the trunk bent forward, borders:

- Inferior margin of trapezius *superiorly*
- Latissimus Dorsi *inferiorly*
- Vertebral margin (medial) of scapula *laterally*

## ***The Peritoneum and peritoneal coverings***

The serous sac, consisting of mesothelium and a thin layer of irregular connective tissue, that lines the abdominal cavity and covers most of the viscera contained therein; it is the largest serous membrane of the human body; develops from the lateral plates of the mesoderm.

It forms two sacs: the peritoneal (or greater) sac and the omental bursa (lesser sac) connected by the Epiploic foramen, and like any serosal covering it has 2 parts:

- **Parietal peritoneum:** lines the internal surface of abdominopelvic wall (sensitive to pain)
- **Visceral peritoneum:** Invests and covers abdominal and pelvic viscera/ organs (insensitive)

The Abdominal and pelvic organs and structures -according to their peritoneal relations- can be divided into 2 main groups:

**A) Intraperitoneal:** covered by peritoneum almost completely (like passing your fist into an inflated balloon) (There is only one „really Intraperitoneal” located organ: **the ovary** - which is situated within the peritoneal cavity)

- **Semi-Peritoneal** (used to be Intraperitoneal, then it got retracted back and remained covered only on it 1/2)

**B) Extraperitoneal** (not surrounded by the peritoneum) – within this:

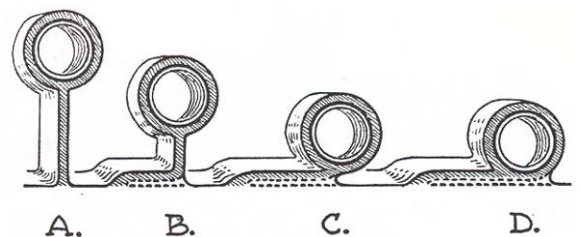
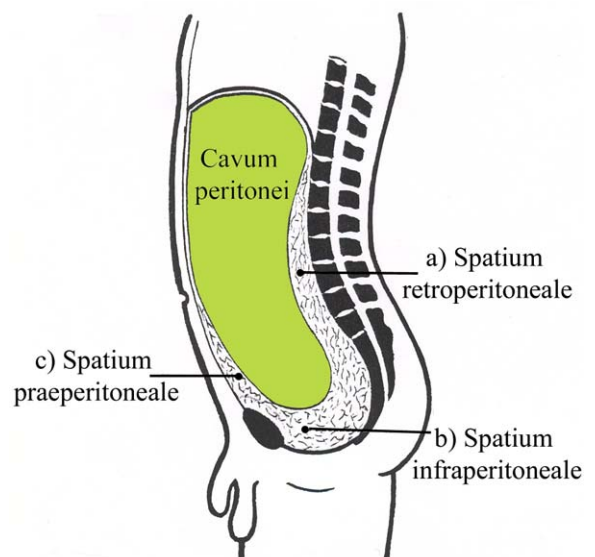
- a) **Retroperitoneal**
- b) **Infraperitoneal**
- c) **Preperitoneal structures**

### ***A) Intraperitoneal Organs:***

1. Stomach
2. Duodenum (Sup. horizontal part only!!)
3. Liver – except bare area (Nude area)
4. Spleen
5. Jejunum
6. Ileum
7. Cecum (in some cases), appendix
8. Transverse colon
9. Sigmoid colon
10. Oviduct
11. Uterus (body + fundus)

### ***→ Semi-Intraperitoneal organs:***

1. Cecum (in some cases)
2. Ascending colon (Fig. D)
3. Descending colon (Fig. D)
4. Rectum - upper 1/3 (Fig. D)
5. Urinary bladder (when filled)
6. Post. Fornix of vagina





**B) Extraperitoneal organs, within this:**

**a) Retroperitoneal**

**→ Primary retroperitoneal:**

1. Kidney
2. Adrenal gland
3. Ureter
4. Rectum – middle 1/3

**→ Secondary retroperitoneal** (originally intraperitoneal organs during their development later lose their complete peritoneal covering and become retroperitoneal):

1. *Duodenum – from superior flexure to duodenojejunal flexure*
2. *Pancreas*

**Retroperitoneal structures** also include:

1. Abdominal aorta and many of its primary and secondary branches
  - *e.g.: parietal branches*: Inf. Phrenic a., lumbar arteries, middle sacral a.
  - *Paired visceral branches*: middle suprarenal a., renal a., gonadal arteries
  - *Not primary branches*: left gastric a., sup. + Inf. Pancreaticoduodenal arteries, right colic a. (*Jackson's membrane*), left colic a.
2. Veins: IVC, common iliac veins;  
Ascending lumbar veins, azygos/hemiazygos veins (their beginning)
3. Lymphatics: lumbar trunks, the end of the intestinal trunk, Cisterna Chyli
4. Nerves: lumbar and sciatic plexuses, sympathetic trunk, Para- and prevertebral ganglia.

**b) Infraperitoneal (primary):**

1. Urinary bladder (if empty)
2. Rectum – lower 1/3
3. Prostate
4. Seminal vesicles
5. Vas deferens
6. Cervix of uterus
7. Ant. fornix of vagina
8. Urethra
9. Vagina

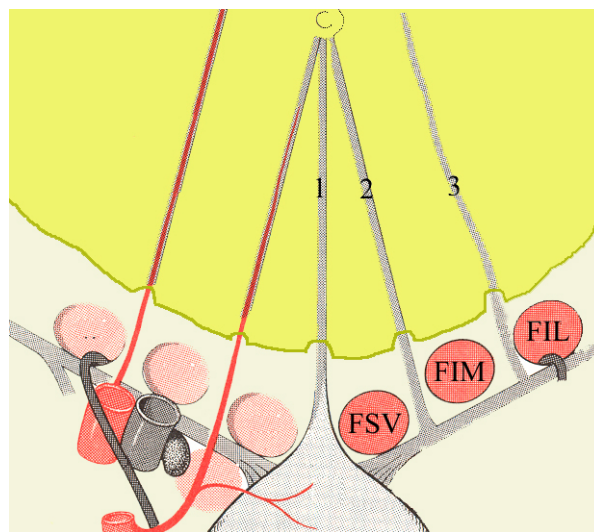
→ **Infraperitoneal structures** include also

1. Common iliac arteries, ext. iliac a., some branches of int. iliac a.
2. Pelvic venous plexuses
3. Lymphatics: beginning of lumbar trunk; pelvic lymph nodes
4. Nerves: Pudendohaemorrhoidal plexus - sacral parasympathetic. nerves, pelvic autonomic plexuses.

### C) Preperitoneal:

The Preperitoneal space is located between the anterior parietal peritoneum and the transversalis fascia. Structures situated here cause **5 folds** on the parietal peritoneum:

1. **Median umbilical fold (unpaired)**: extends from the apex of urinary bladder to the umbilicus and covers the median umbilical ligament, remnant of the urachus, which joined the apex of fetal bladder to the umbilicus
2. **Medial umbilical folds (paired)**: lateral to medial umbilical fold, cover the medial umbilical ligaments, formed by the occluded parts of *umbilical arteries* (remnants)
3. **Lateral umbilical folds (paired)**: lateral to medial umbilical folds, cover the *inferior epigastric vessels* → can cause severe bleeding in case of cut



→ FSV: *Supravesical fossa*

→ FIM: *Medial inguinal fossa* (at the level of the superficial inguinal ring)  
*Hasselbach trigone* (inguinal triangle)

→ FIL: *Lateral inguinal fossa*, lateral to lateral umbilical folds, including the deep inguinal rings, potential site for indirect hernia

Going downwards, the preperitoneal space gets wider: supravesical and prostatic spaces (and prevesical space of **Retzius**). Via the subarcuate hiatus the deep dorsal vein of penis enters the vesical venous plexus.

## Embryology of the Peritoneum

**Ventral and dorsal mesenteries** (peritoneal duplications in front and behind) surround the developing gut.

### → Ventral Mesentery:

The anterior peritoneal duplication exists only above the umbilical vein. Two parts can be distinguished: **ventral mesogastrium** and **ventral mesoduodenum**.

#### A) ventral mesogastrium

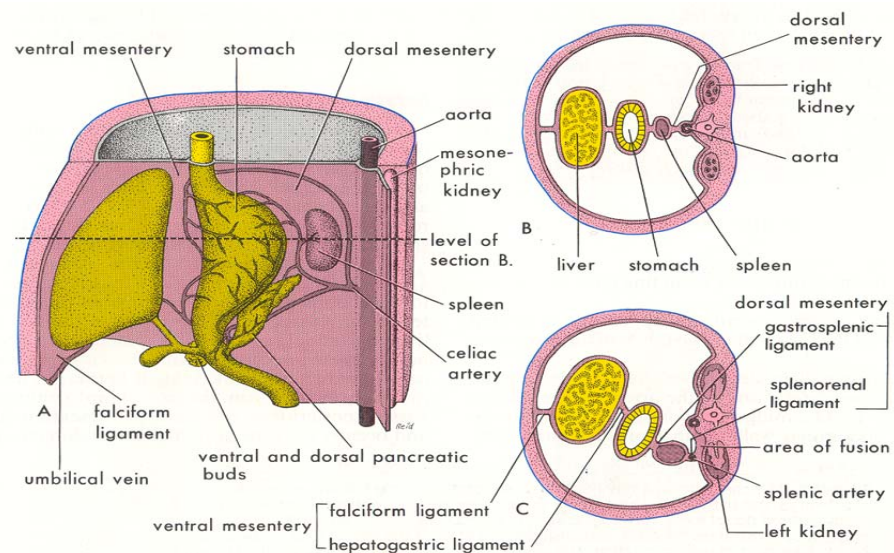
The developing liver divides the ventral mesogastrium into smaller parts. The derivatives of the ventral mesogastrium are:

1. The *Falciform lig. of the liver* (in the lower edge of it the remnant of the umbilical vein forms the *round lig. of the liver*).
2. The *Hepatogastric lig.*
- 3-4. right + left *coronary ligaments* of the liver and *left triangular lig.* of the liver.

#### B) ventral mesoduodenum

The *ventral mesoduodenum* gives rise to the *Hepatoduodenal lig.*

**Lesser omentum** is formed by the Hepatogastric and Hepatoduodenal ligaments.



→ **Dorsal Mesentery:**

The posterior peritoneal duplication develops in the whole length of the gut tube.

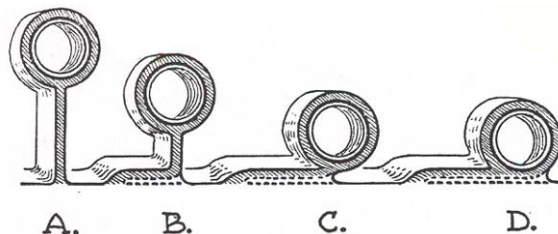
**A) Dorsal Mesogastrium**

The developing spleen divides it into:

1. *Gastrosplenic lig.*
2. *Splenorenal lig. / Phrenicosplenic lig. / Pancreaticosplenic lig.*

The distal part of the duodenum (between the sup. flexure and the duodenojejunal flexure) is originally *Intraperitoneal* (Fig. A), but later becomes retroperitoneal (secondary retroperitoneal organ).

During the fetal development the *ascending and descending colon* as well as the upper 1/3 of the *rectum* fuse with the posterior abdominal wall (Fig. B, C, D), only their anterior part remains peritoneum-covered. These organs can be called **Semi-Intraperitoneal** organs (Fig.D).



**B) From the rest of the dorsal mesentery develop:**

1. **The greater omentum** (4 layers). It later fuses with the 2 layers of the transverse Mesocolon → *Gastrocolic lig.* (6 layers)
2. **The mesentery of the small intestines:** fan shaped double fold of peritoneum that suspends the jejunum and ileum from the posterior abdominal wall and transmits nerve and blood vessels to and from the small intestine. Forms a root that extends from the **Duodenojejunal flexure** to the R. iliac fossa (R. Sacroiliac joint) and is about 15 cm long, contains the superior mesenteric and intestinal vessels, nerves and lymphatics
3. **The Mesoappendix:** connects the appendix to the mesentery of ileum, contains appendicular vessels.
4. **The Transverse Mesocolon:** connects the posterior surface of the transverse colon to the posterior abdominal wall (pancreas), fuses with the greater omentum to form the Gastrocolic ligament. Contains the middle colic vessels/nerves/lymph
5. **The Mesosigmoidium:** peritoneal fold that connects the sigmoid colon to the pelvic wall and contains the sigmoid vessels. Line of attachment has a  $\wedge$ -shape.

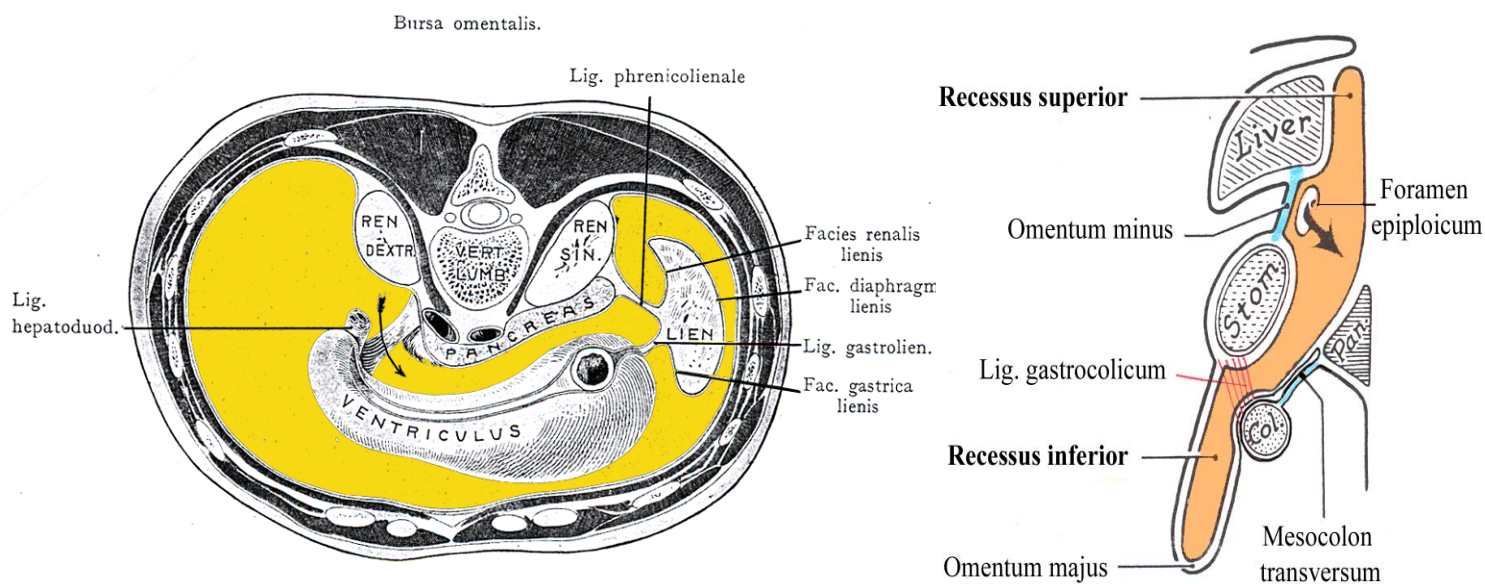
## Subdivisions of peritoneal cavity

### → Omental bursa (lesser sac):

An extensive *sac-like cavity* that lies posterior to the stomach, lesser omentum and adjacent structures. An irregular space, *closed* except for its communication with the *greater sac* (See later) through the **Epiploic (Omental, Winslow) foramen**

It has 3 recesses:

- **Superior recess:** behind the stomach's lesser omentum, and L-lobe of liver.
- **Inferior (Omental) recess:** behind stomach, extending into the layers of greater omentum. Exists *only in the fetal life*, between the 2 duplications of the greater omentum, later this will be *obliterated* (due to the fusion of the membranes of the greater omentum).
- **Splenic recess:** extends left to the hilum of spleen, Bordered by the Gastrosplenic and Phrenicosplenic (Splenorenal) ligaments.



### Parts of the Omental bursa:

#### 1) **Epiploic (Omental, Winslow) foramen:**

Admits two fingers, can be located by running a finger along gallbladder to the free edge of lesser omentum.

#### **Borders of the Epiploic foramen:**

- Anterior: *Hepatoduodenal ligament. (Portal triad)*
- Posterior: *Hepatorenal ligament + Inferior vena cava*
- Superior: *Liver (caudate lobe)*
- Inferior: *superior or first part of duodenum + Duodenorenal ligament*

#### 2) **Vestibule:** between the *Epiploic foramen* and the *gastropancreatic fold*



3) **Gastropancreatic fold (or diaphragma omentale)**: the narrowest part of the omental bursa, separating the *vestibule* from the *cavity proper*. This fold is formed by the *left gastric a.* and the *coronary vein of the stomach* (running behind the peritoneum)

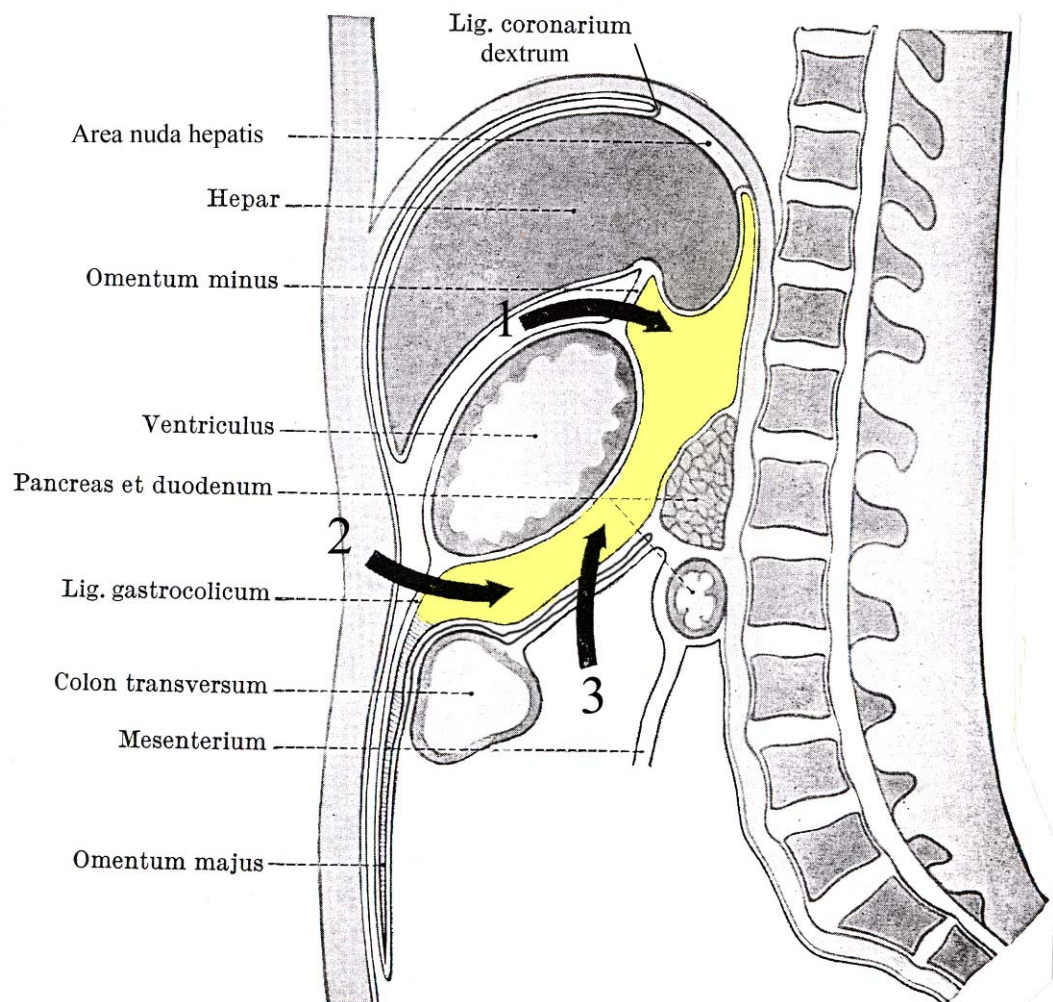
4) **Cavity proper of the omental bursa:**

*Borders:*

- Anterior: *Hepatogastric ligament, stomach, Gastrocolic ligament*
- Inferior: *Transverse colon + Transverse Mesocolon*
- Posterior: *posterior parietal peritoneum + pancreas*

→ **Possible surgical entrances into the omental bursa** (e.g. for pancreas surgery):

- 1) Via the *lesser omentum*
- 2) Via the *greater omentum*
- 3) Via the *Transverse Mesocolon*



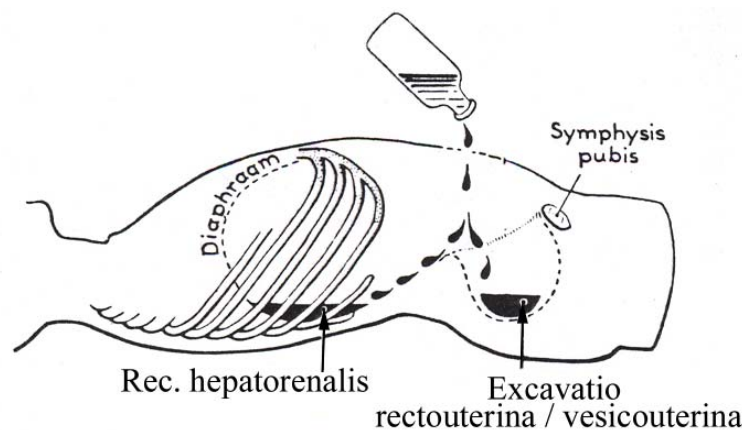
→ **Greater sac (peritoneal cavity):**

Extends across the entire breadth of the abdomen from the diaphragm until the pelvic floor, the interior of the peritoneal sac, normally only a potential *space between the parietal and visceral layers of the peritoneum* (filled with some peritoneal fluid for lubrication – allowing movements of abdominal organs)

*In males* totally closed sac, *in females* the peritoneal cavity communicates with the outer world via the *oviduct – uterus - vagina*.

It has recesses, which can be potential sites of *internal herniae*:

- **Subphrenic (Suprahepatic, Hepaticophrenic) recess:** separated into R. and L. by the falciform ligament. It is the highest point of peritoneal cavity (air, gases can be accumulated here in case of e.g. stomach perforation)
- **Subhepatic recess or Hepatorenal (Morrison's pouch):** deep peritoneal pocket between the liver anteriorly and the R. Kidney & suprarenal gland posteriorly. Communicates with lesser sac via the *Epiploic foramen* and *R. Paracolic gutter*
- **Paracolic recess (gutter):** lays laterally to *ascending colon* (R. Paracolic gutter) and laterally to the *descending colon* (L. Paracolic gutter)
- **Rectovesical pouch** – in males
- **Rectouterine pouch (Douglas):** the deepest point of the female abdominal (peritoneal) cavity. Blood (e.g. from extrauterine gravidity!), inflammation fluid can be accumulated here. The *Douglas pouch* can be examined without opening the abdominal wall: by Douglas puncture (via the post. fornix of vagina).
- **Vesicouterine pouch** – in female
- **Duodenojejunal, Sup. + Inf. Ileocecal, retrocecal, Paracolic recesses, intersigmoid, iliac Subfascial recess.**



## Omentum

A *Double layered* extension or fold of the peritoneum that passes *from the stomach* and proximal part of the duodenum, to adjacent organs/ structures in abdominal cavity.

### 1. **Lesser omentum:**

Connects the lesser curvature of stomach and the proximal part of the duodenum to the liver (*fissure of ligamentum Venosum & Porta hepatica* respectively)

Consists of **Hepatogastric** and **Hepatoduodenal** ligaments and forms the anterior wall of the lesser sac of peritoneal cavity. Acts as a route for the **Left and Right gastric vessels**, which run between the 2 layers along lesser curvature and anastomose there

Has a free margin containing the **portal triad** (inside the Hepatoduodenal ligament)

### 2. **Greater omentum:**

Hangs down like an apron from the *greater curvature of the stomach* and the proximal part of the duodenum. After descending, it folds back and attaches to the anterior surface of the *transverse colon and its mesentery*.

Derived from the *embryonic dorsal mesentery*, transmits the **R. and L. Gastroepiploic** vessels along the greater curvature.

Consist of:

- **Gastrolineal (Gastrosplenic) ligament:** extends from the L. portion of greater curvature of stomach to the hilum of spleen. Contains the **Short gastric vessels** and **L. Gastroepiploic vessels**.
- **Lienorenal (Splenorenal) (Phrenicosplenic) ligament:** runs from hilum of spleen to L. kidney. Contains **Splenic vessels** and the **Tail of pancreas**.
- **Gastrophrenic ligament:** from upper part of greater curvature of stomach to the diaphragm, holds the cardia in position
- **Gastrocolic ligament:** from greater curvature of stomach to the transverse colon.

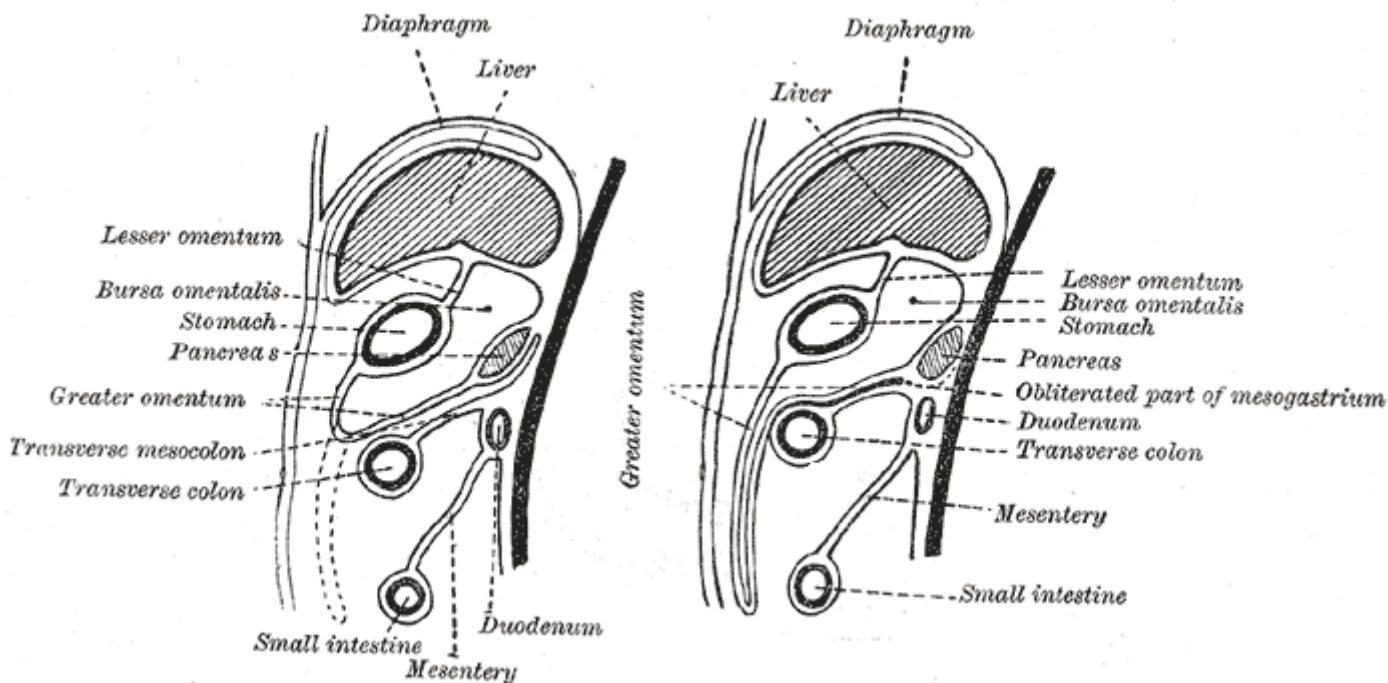
→ **Peritoneal ligaments:** a double layer of peritoneum that connects an organ with another organ or the abdominal wall.

1. **Phrenicocolic ligament (Sustentaculum lienis):** runs from the *L. colic flexure* to the *diaphragm*, form the splenic nest which holds the spleen in position
2. **Falciform ligament:** sickle shaped peritoneal fold, connects the liver to the diaphragm and anterior abdominal wall. Contains the *round ligament of liver (Teres)* + *Paraumbilical vein*, which connects portal vein with subcutaneous veins in umbilical region.
3. **Teres ligament of liver: (Round ligament)** lies in the free margin of Falciform ligament and ascends from umbilicus to the inferior (visceral) surface of liver, lying in a fissure that forms the *L. boundary* of the *quadrate lobe* of the liver. *Remnant of L. umbilical vein (The R. umbilical: obliterated during embryonic period)*.
4. **Coronary ligament of the liver:** peritoneal reflection from diaphragmatic surface of liver onto diaphragm and encloses a triangular area of the R. lobe → *Bare area of liver*. Has a *R. and L. triangular ligaments (as extensions on both sides of liver)*
5. **Ligamentum Venosum:** fibrous remnant of *ductus venosus*. Lies in a fissure on the inferior surface of liver → boundary for the *caudate lobe* of liver.
6. **Hepatoduodenal ligament** (see above)

7. **Hepatogastric ligament** (see above)
8. **Gastrolineal (Gastrosplenic) ligament** (see above)
9. **Lienorenal (Splenorenal) (Phrenicosplenic) ligament** (see above)
10. **Gastrophrenic ligament** (see above)
11. **Gastrocolic ligament** (see above)
12. **Root of the mesentery, mesentery:** contains the Superior mesenteric vessels, lymph nodes and nerve branches

→ **Peritoneal folds:** peritoneal reflections with free edges

1. **Umbilical folds:** discussed previously.
2. **Rectouterine fold:** from cervix of uterus, along side of rectum to posterior pelvic wall, forming *Rectouterine pouch of Douglas (in females)*
3. **Ileocecal fold:** terminal ileum to cecum.



\*\*\*Note:

We have to distinguish the Paracolic gutters from the peritoneal gutters or recesses:

→ **The Paracolic gutters:** are spaces between the ascending / descending colons and the lateral abdominal wall, we have 2:

- *Right Paracolic gutter*
- *Left Paracolic gutter*

→ **The paramesenteric (mesentero-colic) gutters or recesses:** are two spaces (or recesses) in the abdominal cavity between the colon and the root of the mesentery. We have 2 paramesenteric gutters;

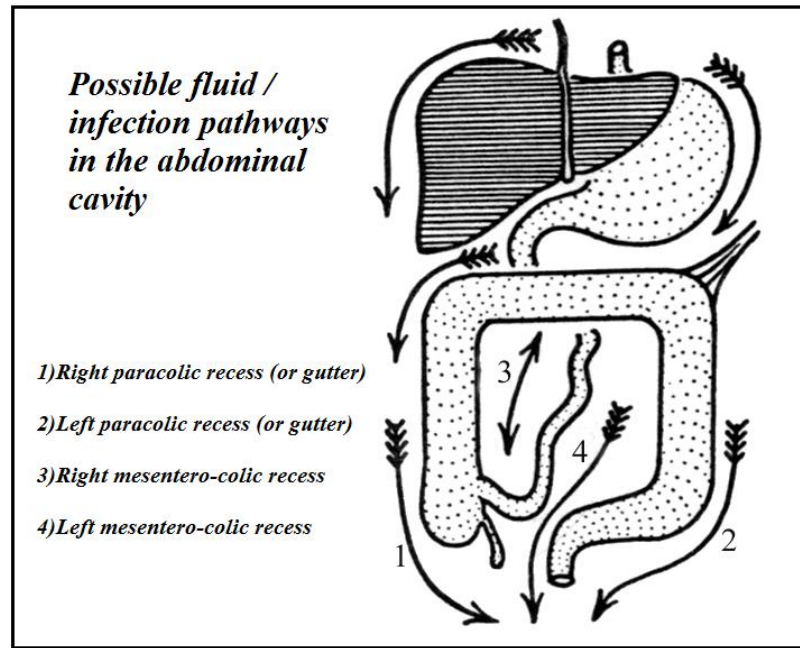
- *Right paramesenteric gutter*, defined by:
  - Ascending colon or cecum laterally
  - Transverse colon superiorly
  - Root of mesentery medially



- *Left paramesenteric gutter*, defined by:
  - Root of mesentery medially
  - Descending colon laterally

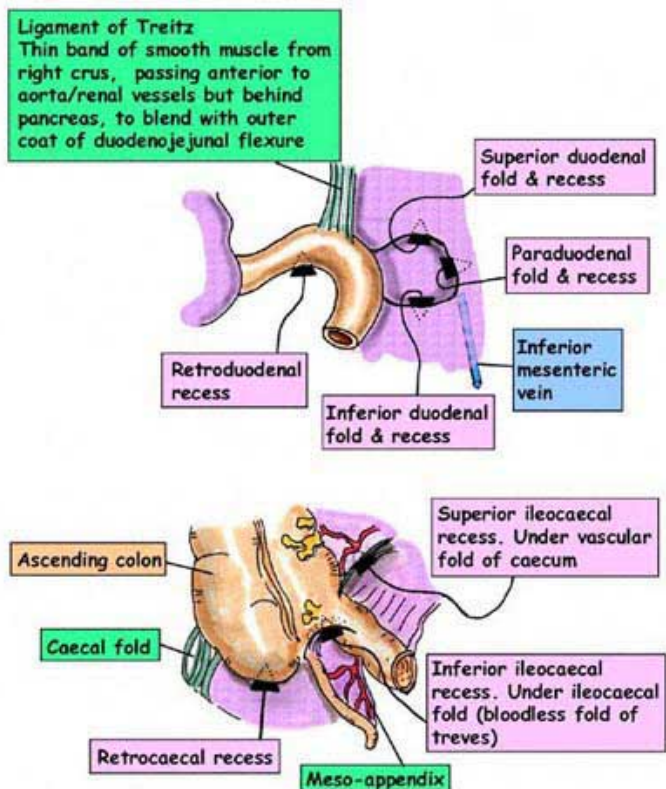
\*\* These gutters are *clinically important* because they allow a passage for infectious fluids from different compartments of the abdomen. For example:

Fluid from an infected appendix can track up the right Paracolic gutter to the Hepatorenal recess.



### PERITONEAL RECESSES

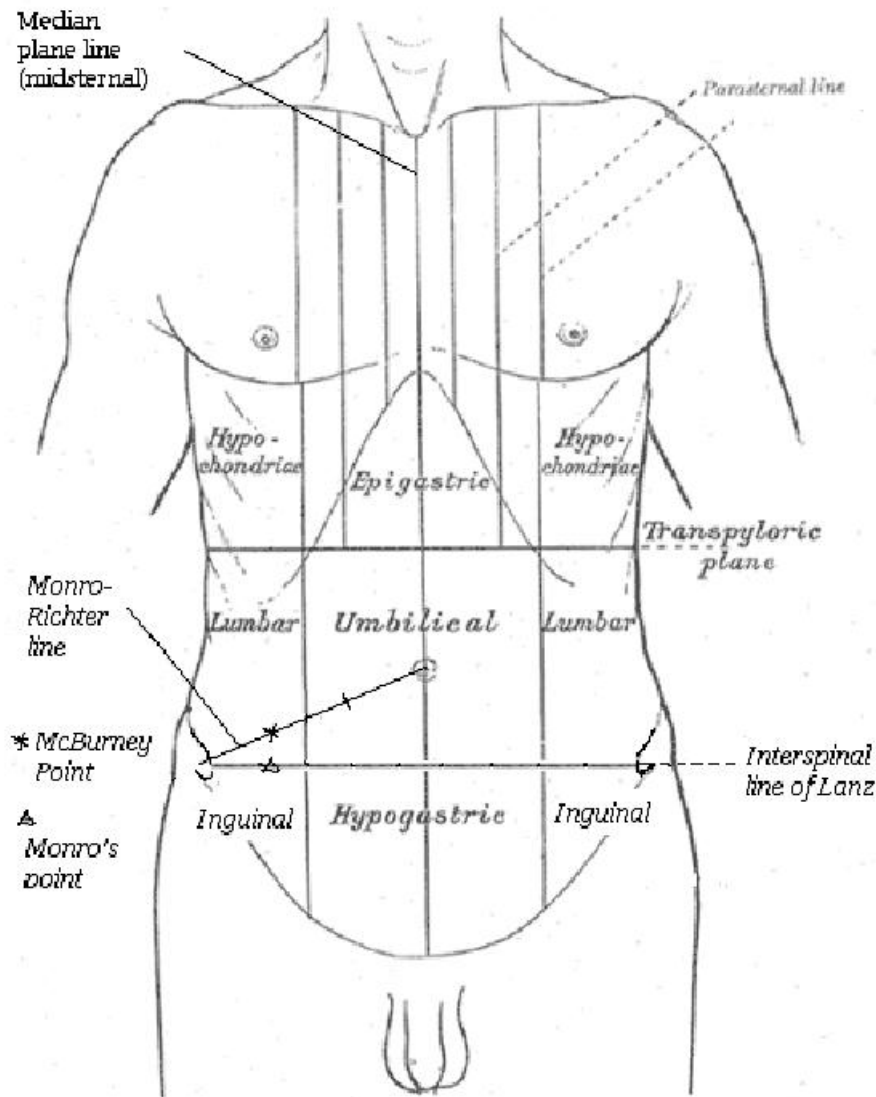
There are a number of small recesses, as shown below, that are potential areas for internal herniation of bowel or other structures





## ***Body surface divisions and important lines:***

### ***Body surface divisions***



- ***Monro-Richter line:*** a line passing from the umbilicus to the anterior superior iliac spine (ASIS)
- ***McBurney point:*** Located on the lateral 1/3 of the Monro-Richter line, marks the approximate location of the base of appendix for appendectomy incision and a sign of acute appendicitis if tenderness occurs in that area
- ***Opposite McBurney point:*** same location of McBurney's point, but on the left lumbar region (left side of the body). This point is used for puncture and drainage of the peritoneal cavity
- ***Monro point:*** in some sources, located on the interspinal line of Lanz, just going vertically below the McBurney's point. Also used for approximation of the location of appendix in appendectomy operation

→ It can be divided into simpler 4 quadrants by 2 lines:

- ***Median plane line***
- ***Transumbilical line*** (horizontal line through the umbilicus)

After intersection, both lines will give: RUQ, RLQ, LUQ, LLQ.

***Right upper quadrant (RUQ):***

Liver, R lobe (RH)  
Gall bladder (RH) can palpate it at X  
Stomach: pylorus (E)  
Duodenum: parts 1-3 (E)  
Pancreas: head (E)  
R suprarenal gland  
R kidney  
R colic (hepatic) flexure (RH)  
Ascending colon: superior part (RL)  
Transverse colon, R half

***Right lower quadrant (RLQ):***

Cecum  
Vermiform appendix  
Most of ileum  
Ascending colon (inferior part)  
R. ovary  
R. uterine tube  
R. ureter (abdominal part)  
R. spermatic cord  
Uterus (if enlarged)  
Urinary bladder (if full)

***Left upper quadrant (LUQ):***

Liver: L lobe  
Spleen  
Stomach  
Jejunum + proximal part of ileum  
Pancreas (body + tail)  
L kidney  
L suprarenal gland  
L colic (splenic) flexure  
Transverse colon (L. half)  
Descending colon (superior part)

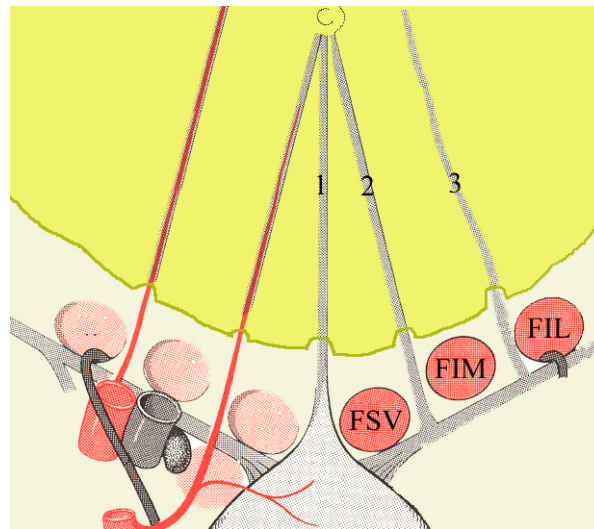
***Left lower quadrant (LLQ):***

Sigmoid colon  
Descending colon (inferior part)  
L. ovary  
L. uterine tube  
L. ureter (abdominal part)  
L. spermatic cord  
Uterus (if enlarged)  
Urinary bladder (if full)

### ***Internal surface of the anterolateral abdominal wall, preperitoneal space:***

The Preperitoneal space is located between the anterior parietal peritoneum and the transversalis fascia. Structures situated here cause **5 folds** on the parietal peritoneum:

4. **Median umbilical fold (unpaired):** extends from the apex of urinary bladder to the umbilicus and covers the median umbilical ligament, remnant of the urachus, which joined the apex of fetal bladder to the umbilicus
5. **Medial umbilical folds (paired):** lateral to median umbilical fold, cover the medial umbilical ligaments, formed by the occluded parts of *umbilical arteries* (remnants)
6. **Lateral umbilical folds (paired):** lateral to medial umbilical folds, cover the *inferior epigastric vessels* → can cause severe bleeding in case of cut



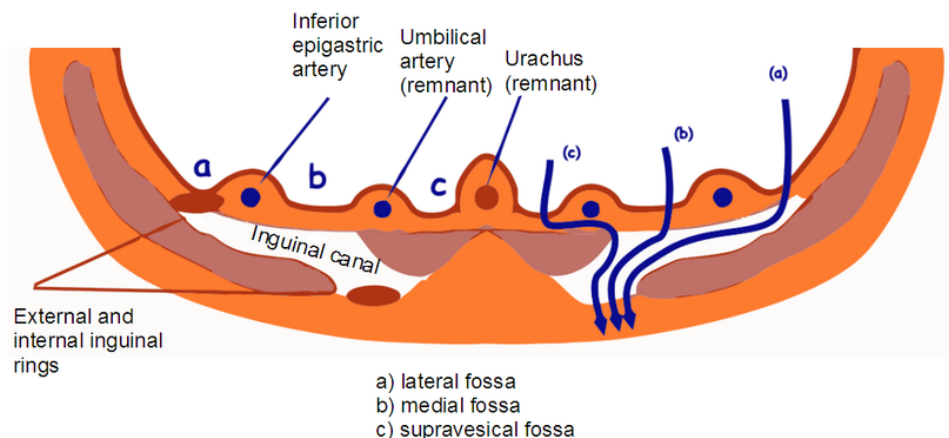
→ FSV: *Supravesical fossa*

→ FIM: *Medial inguinal fossa* (at the level of the superficial inguinal ring), contains the *Hasselbach trigone* (inguinal triangle):

The triangular area in the lower abdominal wall bounded inferiorly by the inguinal ligament, the border of the rectus abdominis medially and the inferior epigastric vessels (lateral umbilical fold) laterally. It is the site of *direct inguinal hernia*.

→ FIL: *Lateral inguinal fossa*, lateral to lateral umbilical folds, including the deep inguinal rings, potential site for *indirect hernia*

**\*\*** Going downwards, the preperitoneal space gets wider: supravesical and prostatic spaces (and prevesical space of **Retzius**). Via the subarcuate hiatus the deep dorsal vein of



penis enters the vesical venous plexus.

\*\* The *supraumbilical* part of the internal surface has a saggitally oriented peritoneal reflection: the falciform ligament extending to the liver, and enclosing the round ligament of liver (Teres ligament) + Paraumbilical vessels in its free edge.

## ***Hernia:***

A condition in which a portion of the intestines protrudes through a weak spot of the body

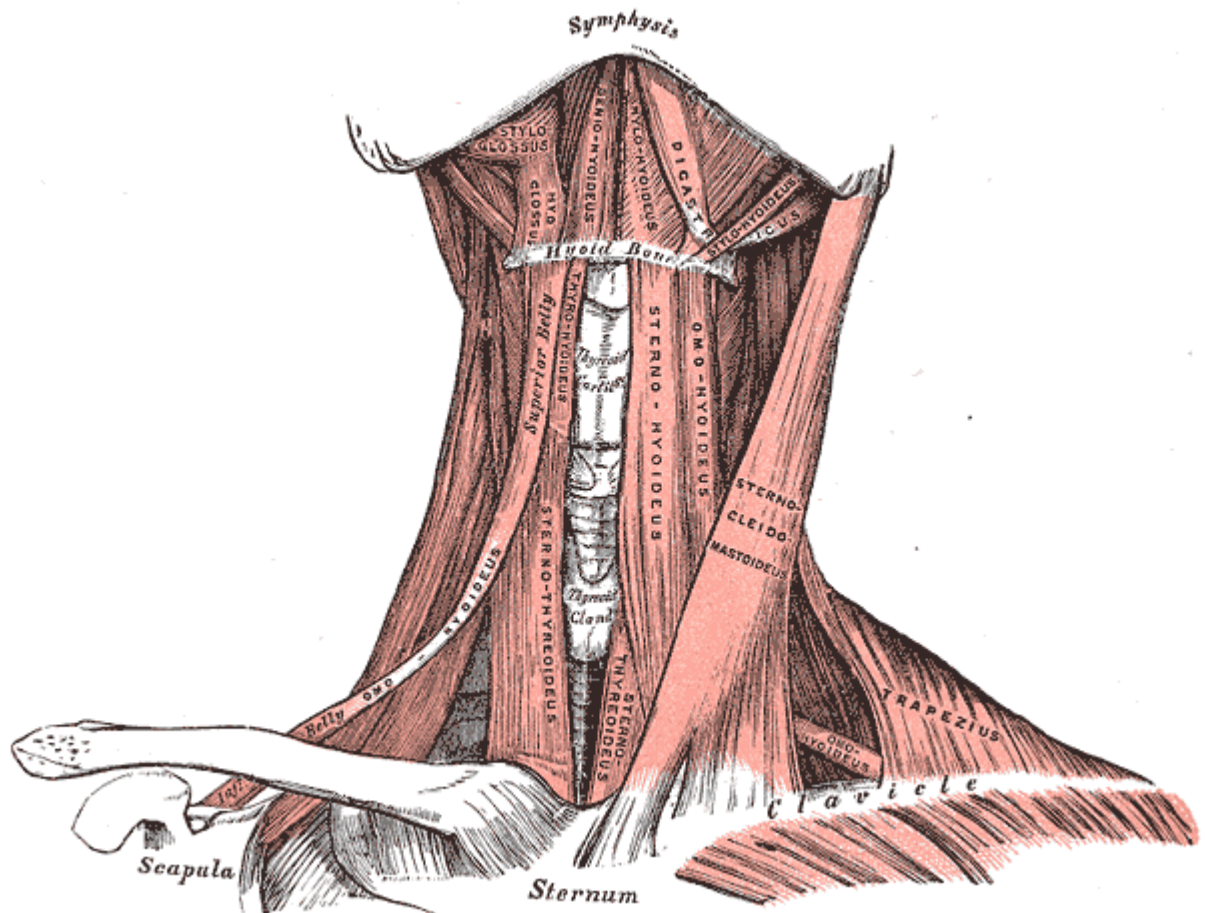
### ***1. Abdominal hernias:***

- a. *Umbilical hernia*: common in newborns, through the weak umbilical ring. It can be acquired in pregnant women or obese people
- b. *Epigastric hernia*: through the Linea alba in the epigastric region, between the xiphoid process and umbilicus in midline
- c. *Spigelian hernia*: along the semilunar lines, in obese people > 40 years
- d. *Incusional hernia*: through a surgical incision
- e. *Lumbocostal hernia*: between diaphragms lumbar and costal parts, lateral to the Arcuate ligament → becomes the Bochdalek foramen hernia (pleuroperitoneal)
- f. *Lumbar hernia*:
  1. *Grynfelt triangle*: bordered by the 12<sup>th</sup> rib + Serratus posterior inferior muscle + internal oblique muscle + Quadratus lumborum
  2. *Petit triangle*: bordered by the Latissimus Dorsi, external oblique muscles and Iliac crest (Check dorsal regions, back)
- g. *Femoral hernia*: through femoral canal, discussed in the Subinguinal hiatus
- h. *Hiatal hernia*: pars of the stomach passing into the mediastinum through the esophageal hiatus of the diaphragm. Can be paraesophageal (rolling) or sliding

### ***2. Inguinal hernias:***

- a. ***Indirect inguinal hernia***: a congenital condition, associated with the presence of patent processus vaginalis, and covered by peritoneum and coverings of the spermatic cord. The hernia passes through the Deep inguinal ring, inguinal canal and the superficial ring → descends into the scrotum (if entire stalk of the processus vaginalis persists)  
Hernia lays lateral to the inferior epigastric vessels (lateral umbilical fold).
- b. ***Direct inguinal hernia***: an acquired condition, associated with weakness in the abdominal wall (posterior wall of the inguinal canal, lateral to the Falx inguinalis, and has a sac that is formed by the peritoneum and sometimes the trasversalis fascia)  
Lays medial to the lateral umbilical fold, in the medial inguinal fossa, through the Hasselbach (inguinal) triangle, and ***doesn't*** descend to the scrotum, but can protrude forward to the superficial inguinal ring

# *Fascia & triangles of the neck, Nuchal region*





## ***Fascia of the neck***

### **A. *Cervical subcutaneous tissue (Superficial cervical fascia) & Platysma:***

A layer of fatty connective tissue that lies between the dermis of the skin and the investing layer of the deep cervical fascia. It's thin anteriorly & contains cutaneous nerves, blood and lymph vessels, superficial lymph nodes and adipose tissue.

The ***Platysma muscle*** is found antero-laterally, it's broad and thin sheet facial muscle.

- *Origin:* the subcutaneous layer and fascia covering the Pectoralis major and deltoid at level of first or second rib
- *Insertion:* on the lower border of mandible, Risorius and Platysma of opposite side
- *Action:* depress the lower lip, forms ridges in skin of neck and upper chest when jaws are “clenched”, denoting stress, anger. (used for stretching the anteriolateral neck for shaving in men)
- *Nerve supply:* cervical branch of facial n. (CN-VII), as the *superficial cervical ansa*, anastomosing with the sensory transverse cervical nerves coming from the cervical plexus
- *Features:* deep to the Platysma, we find the external jugular vein descending from the angle of mandible until the middle of clavicle

### **B. *Deep cervical fascia:***

Consists of 3 fascial layers or sheaths: *Investing (superficial lamina)*, *pretracheal (middle lamina)* & *prevertebral (deep lamina)*, that support the neck viscera, muscles, vessels & deep lymph nodes. They form natural separation between tissues in surgery and limit spread of abscesses resulting from infections. It allows also slipperiness for structures in the neck to move smoothly without difficulty during head movements and swallowing.

#### **1. *Investing layer of deep cervical fascia (superficial lamina):***

Surrounds the entire neck deep to the skin and encloses the *trapezius* & *sternocleidomastoid* muscles.

*Superiorly*, the investing layer of deep cervical fascia is connected to:

- Superior Nuchal line of occipital bone
- Mastoid process of temporal bone
- Zygomatic arches
- Inferior border of mandible
- Hyoid bone
- Spinous process of the cervical vertebrae

It encloses the *submandibular gland*, forms the *fibrous capsule of the parotid gland*, and thickened to form the *stylomandibular ligament*

*Posteriorly* is continuous with the periosteum of C7 spinous process and with the *Nuchal ligament*.

The *inferior* attachment of the investing layer is:

- Manubrium
- Clavicles
- Acromions
- Spines of scapulae

At the attachment on the manubrium, the investing layer divides into anterior & posterior layers to form the ***supra-sternal space***, which contains:

- anterior jugular veins
- jugular venous arch
- fat
- deep lymph nodes

**2. *Pretracheal layer of deep cervical fascia (middle lamina):***

Thin layer, limited to the anterior of the neck, extending inferiorly from the hyoid bone into the thorax, blending with the fibrous pericardium, it has:

- *Muscular part*: enclosing the infrahyoid muscles (including the Omohyoid laterally)
- *Visceral part*: enclosing the thyroid gland, trachea & esophagus, continuous posteriorly & superiorly with the buccopharyngeal fascia of the pharynx

*Laterally* the pretracheal fascia blends with the *carotid sheaths*.

In the region of hyoid bone, it thickens to form the *trochlea* through which intermediate tendon of digastric muscle passes

**3. *Prevertebral layer of deep cervical fascia (deep lamina):***

Forms a tubular sheath for the vertebral column and muscles associated with it (longus capitis, longus colli anteriorly) (scalene laterally) (deep cervical muscles posteriorly)

*Superiorly* it's fixed to the *cranial base*

*Inferiorly* it blends with the *endothoracic fascia* peripherally, and the *anterior longitudinal ligament* centrally at *Th-3 vertebra* level

*Laterally* it extends as the *axillary sheath*, enclosing the axillary vessels & brachial plexus

→ ***Carotid sheath***: tubular fascial investment that extends from the cranial base to the root of the neck. *Anteriorly* it blends with the *investing & pretracheal layers* of fascia and *posteriorly* with the *prevertebral* layer. It contains:

- Common & internal carotid arteries
- Internal jugular vein
- Vagus nerve (CN-X)
- Deep cervical nodes
- Carotid sinus nerve of Hering (branch of glossopharyngeal nerve CN-IX)
- Sympathetic nerve fibers (carotid periarterial plexuses)

## ***Peripharyngeal space***

The space, filled with loose areolar tissue, around the pharynx; it is divided into two portions, *Parapharyngeal space* and *retrovisceral space*.

### ***1) Parapharyngeal space (Lateral pharyngeal space, Pharyngomaxillary space) (PPS):***

The PPS is a potential space in the shape of an *inverted pyramid* with its base at the skull base and its apex at the greater horn of the hyoid bone. Various fascial layers within the PPS result in further compartmentalization of this complex anatomic region.

The ***superior*** limit of the PPS is the base of skull, namely the sphenoid and temporal bones. It includes in this area the jugular and hypoglossal foramina and the foramen lacerum (through which the internal carotid artery passes)

The ***medial*** wall of the PPS is formed superiorly by the pharyngobasilar fascia & the superior pharyngeal constrictor (Naso- & Oropharynx)

The ***posterior-lateral*** limit of the PPS consists of the prevertebral fascia & paravertebral musculature

The ***posterior-medial*** limit of the PPS is the retrovisceral space & deep lobe of the parotid gland

The ***anterior*** limit is formed by the Pterygomandibular raphe

The ***inferior*** limit of the PPS is represented by the junction of the posterior belly of the digastric muscle and the greater horn of the hyoid bone.

The ***lateral*** wall of the PPS is formed by, from anterior to posterior, the medial Pterygoid fascia (Covering the masticatory space, containing the masticatory muscles), ramus of the mandible, Retromandibular portion of the deep lobe of the parotid, and *below the level of mandible*, the posterior belly of the digastric muscle.

Using the tensor veli-styloid fascia (runs between the tensor veli palatini muscle, styloid process and styloid musculature) as separating layer the parapharyngeal space can be subdivided into two compartments:

- *Prestyloid compartment*: found anterolaterally
- *Poststyloid compartment*: found posteriomediaally (including the carotid sheath & space)

***Contents*** of this space: in its *anterior* part the following muscles: *Stylopharyngeus*, *Styloglossus*, and *Stylohyoid*. In its *posterior* part we can find the *Glossopharyngeal*, *Vagus*, *accessory*, *hypoglossal* nerves, the *cervical sympathetic trunk*, the *internal Carotid artery*, the *Internal Jugular vein*, and of course, some lymph nodes.

## 2) *The Retrovisceral space:*

The retrovisceral space is divided into the *retropharyngeal space* and the *danger space* by the *alar fascia*.

The *alar fascia* is considered in most of the sources as part of the prevertebral fascia, as a thin layer attaching to the midline of the buccopharyngeal fascia from the cranium to the level of C7 vertebra, from this attachment it extends laterally and terminate in the carotid sheath.

**\*\*** The retrovisceral space is of particular clinical importance because it is a main route by which *oropharyngeal infections* can spread into the *mediastinum*.

### a) *The retropharyngeal space:*

A virtual space behind the pharynx, bordered by:

- Anteriorly: the pretracheal fascia (buccopharyngeal fascia)
- Posteriorly: the alar fascia
- Superiorly: base of the skull
- Laterally: the carotid sheath

It extends from the base of skull superiorly *until the level of the Th-4* vertebra level, where the pretracheal & alar fasciae fuse together (Separated from the mediastinum)

It is divided by the midline attachment of the alar fascia into the buccopharyngeal fascia to *right & left spaces*, making infections occurring in this space look unilateral  
The retropharyngeal space contains:

- *Adipose tissue*
- ***The retropharyngeal lymph nodes***: three groups of lymph nodes, one median and two lateral, they receive lymph from the nasopharynx, the auditory tube, and the atlanto-occipital and atlantoaxial joints

### b) *The danger space:*

A virtual space found anterior to the prevertebral fascia (covering the cervical spine & musculature) and posterior to the alar fascia, so it's behind the retropharyngeal space.

This space is limited superiorly by the base of the skull, but *communicates freely* with the *posterior mediastinum* inferiorly, reaching the level of the diaphragm, allowing a cranio-caudal spread of infection (hence the name), and what makes it more dangerous is that this space has no midline raphe (As in the retropharyngeal space), so the infections can easily spread to either side.

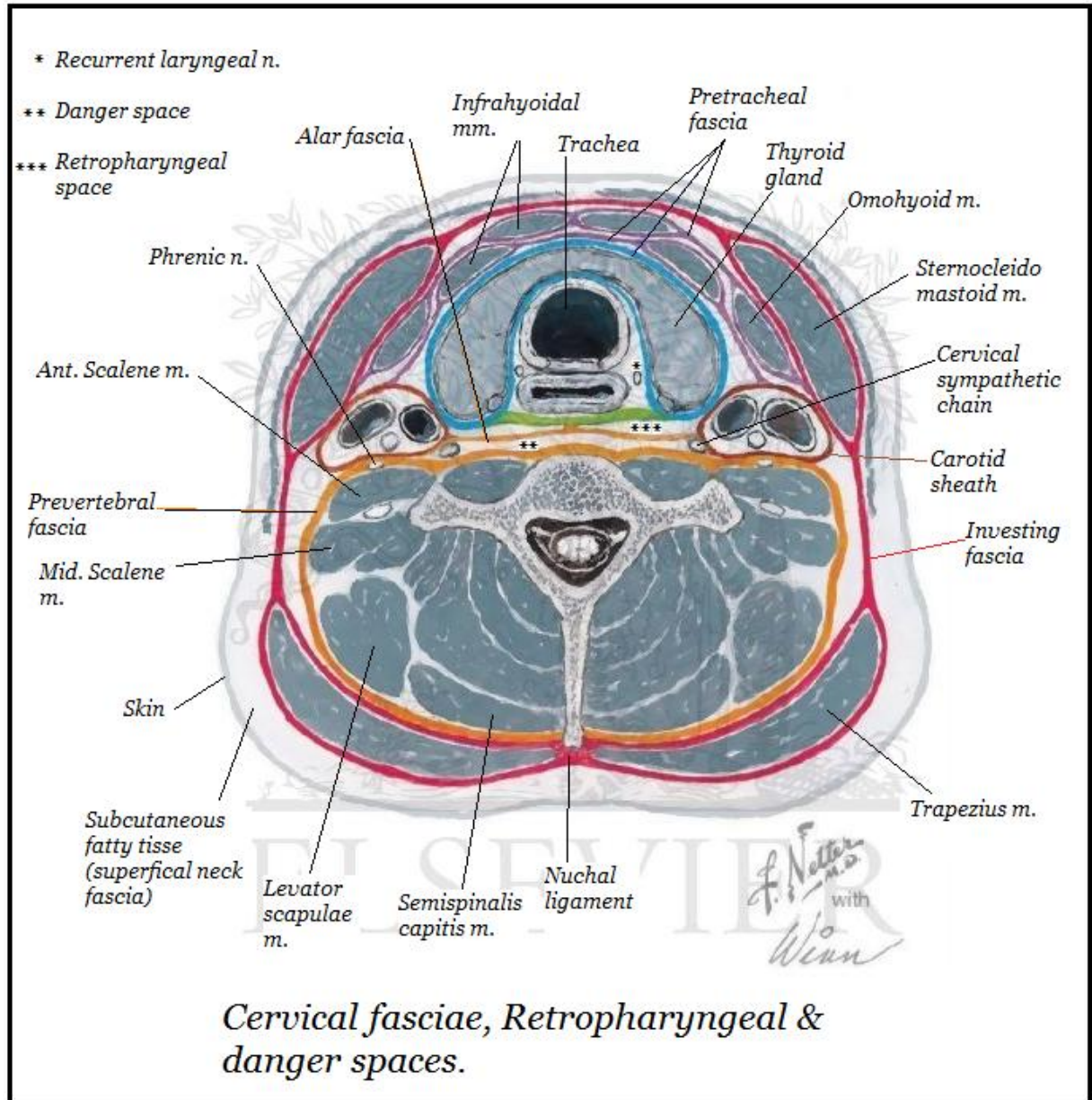
### **\*\* Clinical importance of the retrovisceral space, example:**

This is most commonly seen in children. The infection begins in the *lymphatic tissue of Waldeyers ring* and spreads to the *retropharyngeal lymph nodes*, causing adenitis. These nodes may become *suppurative* and eventually cause *retropharyngeal cellulites* or a *retropharyngeal abscess*.

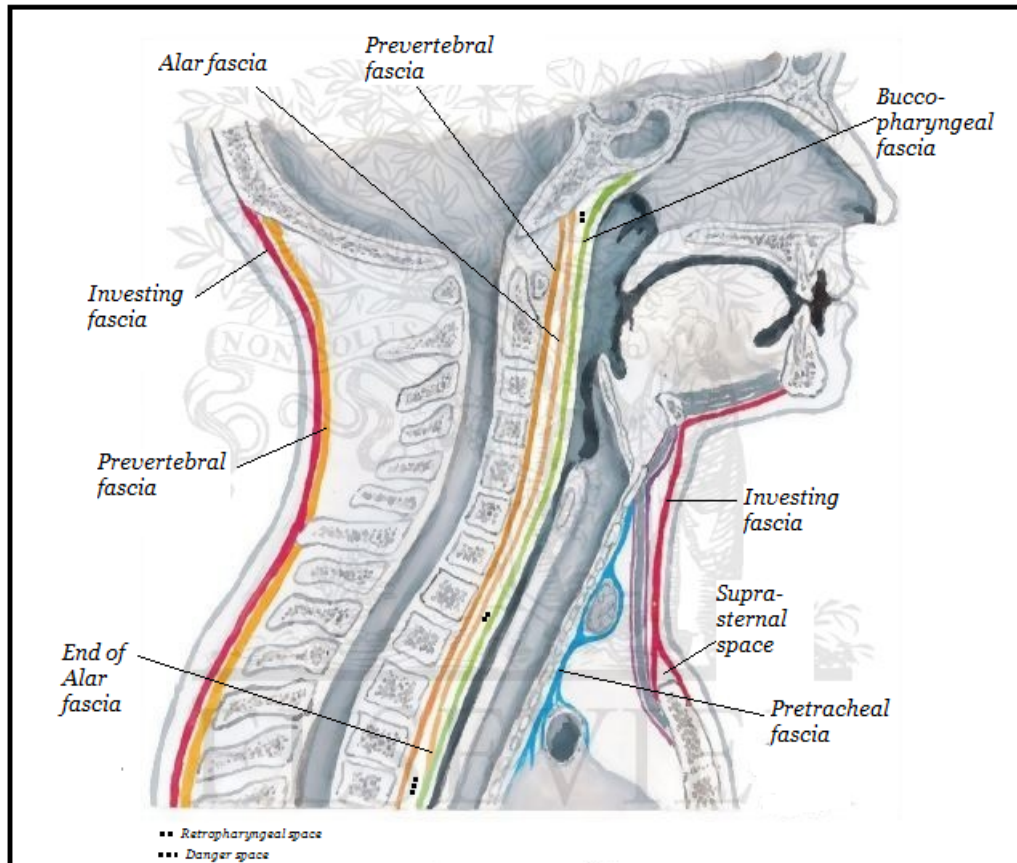
*Pus* from this abscess can either cross the *alar fascia* reaching the *danger space*, and from there *to the posterior mediastinum* infecting the structures found there, or it can reach laterally to the *prestyloid compartment of the parapharyngeal space* and damage the cranial nerves (IX, X, XI, XII) located there.

\* **Radiographically**, pathology within this danger space *cannot* be distinguished from retropharyngeal space pathology.

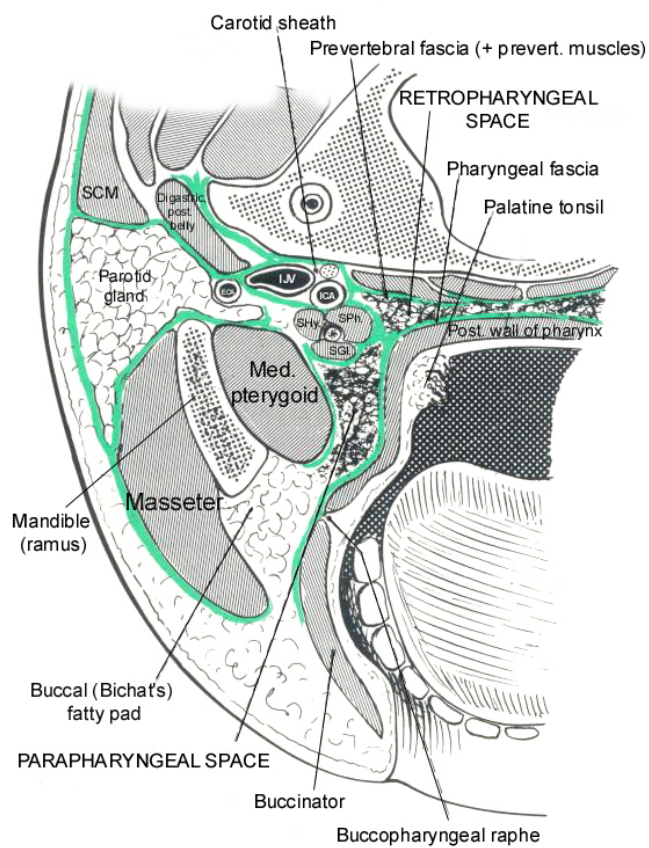
\*\* Some sources indicate that the retrovisceral space is the same as the retropharyngeal space







### PARA- and RETROPHARYNGEAL SPACES



## ***Cervical Regions/Triangles and their contents:***

### ***A) Sternocleidomastoidal region:***

The area over the SCM muscle, we can find there:

- Superior part of external jugular vein
- Transverse cervical & greater auricular nerves
- ***Lesser supraclavicular fossa:*** a space between the two heads of origin of the sternocleidomastoid muscle. It contains:
  - Inferior part of the internal jugular vein

### ***B) Posterior cervical region (Nuchal region):***

The back of neck, including trapezius muscles & the suboccipital region laying deep in the superior part of this region (check dorsal regions for description of this region)

### ***C) Lateral cervical region (posterior triangle of neck):***

We find here:

- ***Occipital (Omotrapius) triangle:*** a triangle of the neck bounded by the trapezius, the sternocleidomastoid, and the omohyoid muscles. It contains:
  - external jugular vein
  - transverse cervical artery
  - spinal accessory nerve
  - trunks of brachial nerve plexus
  - posterior branches of cervical nerve plexus
- ***Omoclavicular (Supraclavicular, Subclavian) triangle:*** the triangle bounded by the clavicle, the omohyoid muscle, and the sternocleidomastoid muscle. It contains:
  - Subclavian artery & vein (vein not always)
  - Suprascapular artery
  - Suprascapular lymph nodes (on the left side, its called Signal or Virchow lymph node, that can have metastasis of gastric cc.)

***\*\* The term greater supraclavicular fossa*** was formerly considered a synonym for the *omoclavicular triangle*, today the Terminologica Anatomica still reserves this term for the *surface feature* that overlies the omoclavicular triangle: a depressed area above the middle of the clavicle, lateral to the sternocleidomastoid.

### ***\*\* The Scalenotracheal fossa:***

The deep part of the supraclavicular region, between the scalenus anterior muscle and the trachea. Inside, there are longitudinal and transverse structures:

- The ***longitudinal*** structures are:
  - Structures of the *carotid sheath* and those parallel to it
  - *Middle ganglion of the cervical sympathetic trunk*, from the middle cervical ganglion, fibers surrounding the subclavian artery are called the *ansa subclavia*

- The **transverse** structures are branches of the **thoracic part the subclavian artery** passing through the fossa:
  - *Thyrocervical* trunk and its branches (mainly the inferior thyroid artery)
  - *The inferior thyroid artery* goes to the thyroid gland and crosses a nerve in the scalenotracheal fossa; the *inferior laryngeal nerve* (or recurrent laryngeal nerve)
  - *The vertebral artery*, which ascends through the transverse foramen of the cervical vertebrae
  - *The internal thoracic artery*, descending to run parallel to sternum on the interior of the thoracic cavity

\*\* The recurrent laryngeal nerve comes from the thoracic cavity. The left recurrent laryngeal nerve hooks around the aortic arch and the right nerve goes around the right subclavian artery because both develop from the 4th branchial arteries.

\*\* From a surgical point of view, this crossing is very important because if a surgeon operates on the thyroid gland, he has to ligate first the superior and inferior thyroid arteries to cut the gland. If the surgeon cuts this nerve, the patient will have a big problem, because the inferior laryngeal nerve innervates all the intrinsic muscles of the larynx (except the Cricothyroid), and the voice of the patient will be affected (Husky voice)

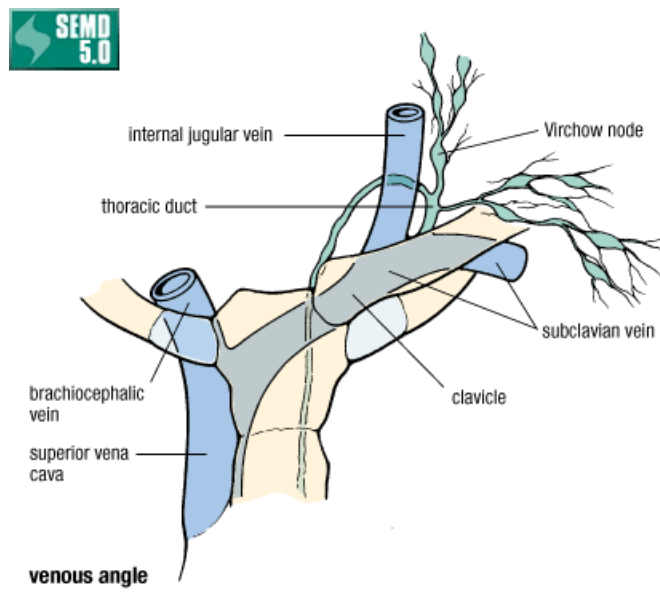
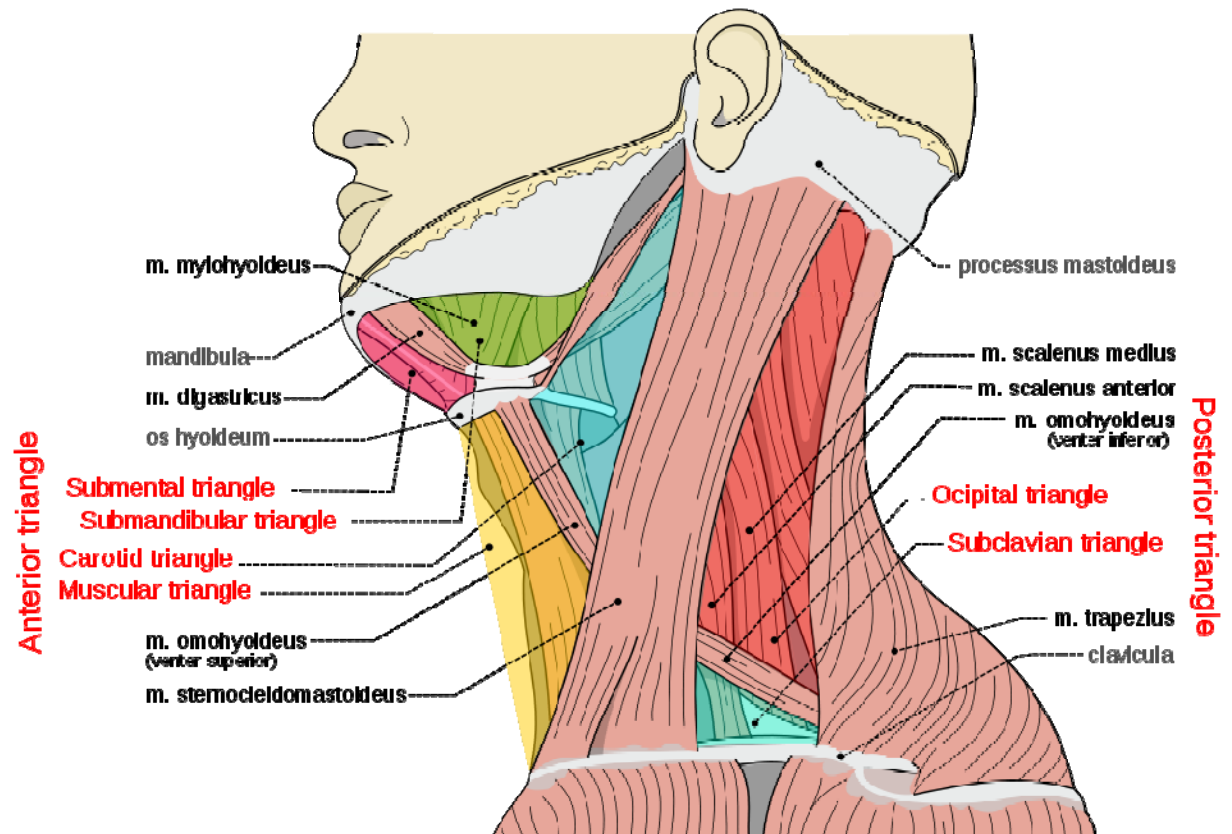
#### **D) Anterior cervical region (anterior triangle of neck):**

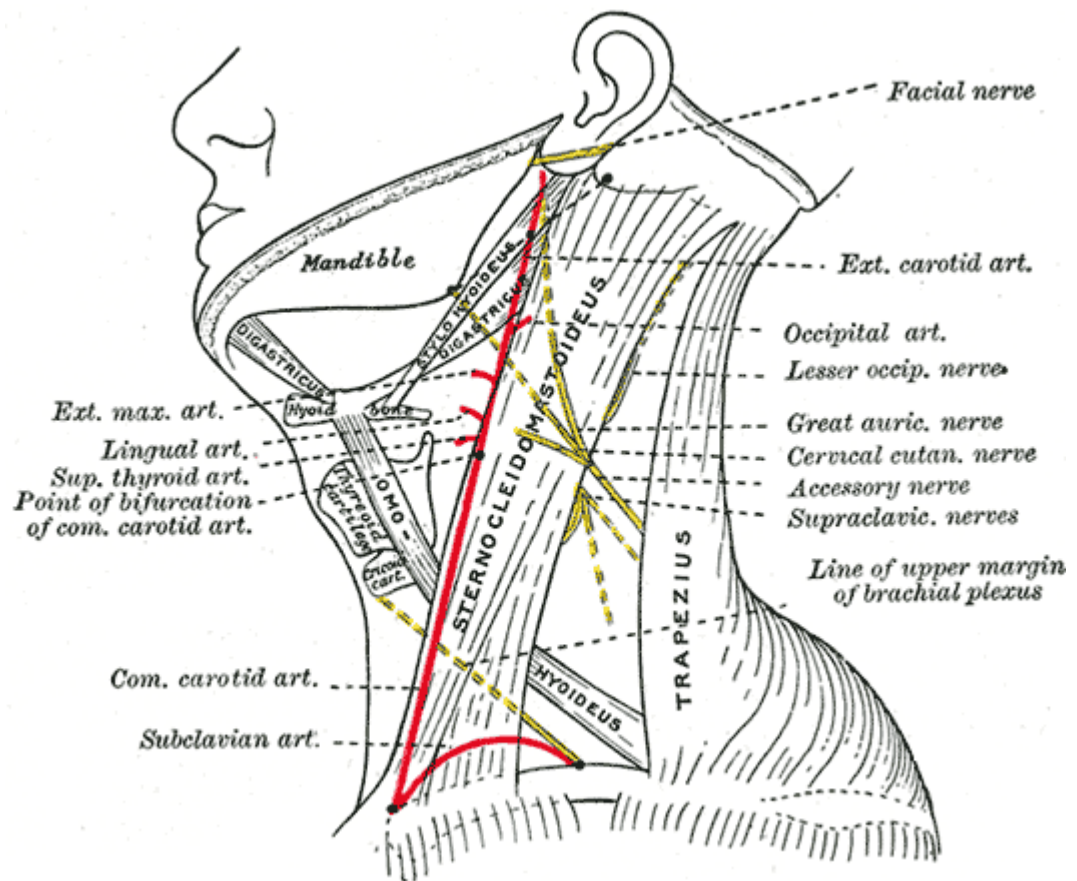
The area of the neck bounded by the mandible, the anterior border of the sternocleidomastoid muscle, and the anterior midline of the neck, it is subdivided into:

- **Submandibular (digastric triangle):** the triangle of the neck bounded by the mandible and the two bellies of the digastric muscle, it contains:
  - Submandibular gland
  - Submandibular lymph nodes
  - Mylohyoid nerve
  - Part of facial artery & vein
- **Submental triangle:** a triangle bounded by the anterior belly of the digastric muscles, the hyoid bone, and the midline; the mylohyoid muscle forms its floor. It contains:
  - Submental lymph nodes
  - Small veins, that unite to form the anterior jugular veins
- **Carotid triangle (Gerdy hyoid fossa):** a space bounded by the superior belly of the omohyoid muscle, anterior border of the sternocleidomastoid, and posterior belly of the digastric. It contains:
  - Carotid sheath with its contents
  - Hypoglossal nerve
  - Superior root of ansa cervicalis
  - Spinal accessory nerve
  - Thyroid gland
  - Larynx
  - Pharynx
  - Deep cervical lymph nodes

- **Muscular (omotracheal) triangle:** the triangle bounded by the sternocleidomastoid muscle, the superior belly of the omohyoid muscle, and the anterior midline of the neck, it contains:
  - Infrahyoidal muscles (Sternothyroid, Sternohyoid, thyrohyoid)
  - Thyroid gland

\*\* The 2 muscular trigons are called *middle cervical region*





### ***The Nuchal Region:***

- Borders:**
- Superior: External Occipital Protuberance
  - Lateral: A line along the side of Mastoid process
  - Inferior: Vertebral Prominence (7<sup>th</sup> Cervical)
  - Medial: a vertical line From the External Occipital Protuberance to the Vertebral Prominence

***Skin Innervation:***

- Lesser Occipital nerve (C2/C3, Cervical plexus)
- Greater Occipital nerve (Dorsal branch of C2)
- Tertiary Occipital nerve (Dorsal branch of C3)

Beneath the skin, the *superficial nuchal fascia*, covers the *trapezius* medially and laterally the *splenius capitis*.

If you remove the fascia, you will see the *trapezius*. Beneath the *trapezius*, lies the *splenius capitis* muscle, a V-shaped muscle originating from the spinous processes of the lower cervical and upper thoracic vertebrae and ascending to the superior nuchal line of the occipital bone. The lateral part of the *splenius capitis* is called the *splenius cervicis* because lateral fibers do not ascend to the skull, but they terminate on the transverse processes of the cervical vertebrae.



Cutting the splenius capitis and cervicis, you will see the *semispinalis capitis muscle*, an upsidedown V-shaped muscle (like the greek delta) originating from the transverse processes of the upper thoracic and lower cervical vertebrae and inserting to the superior nuchal line, medial to the splenius capitis. *The semispinalis and splenius capitis make the rounded shape to the back of the neck.*

The great occipital nerve and the tertiary occipital nerve can be seen coming out from the deep through these muscles, becoming superficial, and turning onto the head.

Together with the lesser occipital nerve, we have the occipital artery (a little deeper).

Deep to the semispinalis capitis, the deep nuchal fascia covers the suboccipital triangle and the semispinalis cervicis muscle

### → **The Suboccipital Triangle:**

#### **Borders:**

- **Lateral:** *Obliquus capitis superior* (*origin*, transverse process of atlas; *insertion*, lateral third of inferior Nuchal line)
- **Infeior:** *Obliquus capitis inferior* (*origin*, spinous process of axis; *insertion*, transverse process of the atlas, note that despite its name, it has to attachment to the skull)
- **Superior:**
  - *Rectus Capitis Posterior Major* (*origin*, spinous process of axis; *insertion*, middle of inferior Nuchal line of occipital bone)
  - *Rectus Capitis Posterior Minor* (medial to Major) (*origin*, from posterior tubercle of atlas; *insertion*, medial third of inferior Nuchal line of occipital bone)
- **Roof:** *Deep Nuchal Fascia*
- **Flooor:** *Posterior Atlantococcipital membrane*

#### **Contents:**

- **Vertebral Artery:** Located on the posterior arch of Atlas (can be palpated) in the vertebral suclus. On the atlas, it runs a little lateral, and then turns medially. after, it pierces through the *posterior atlantooccipital membrane* and enters the skull to supply the brain
- **Suboccipital nerve:** dorsal ramus of C1, passing through the suboccipital triangle and sending branches to the Rectus capitis posterior major and minor, Obliquus capitis superior and inferior, Rectus capitis Lateralis, and semispinalis capitis. its generally considered to have only motor fibers, but the suboccipital nerve receives sensory fibers for proprioception via a communicating branch from the second cervical spinal nerve

At the lateral border of the region, you can see the Sternocleidomastoid muscle (arising from the *Manubrium of Sternum* and medial third of the clavicle and inserting to the mastoid process of the temporal bone and the occipital bone)

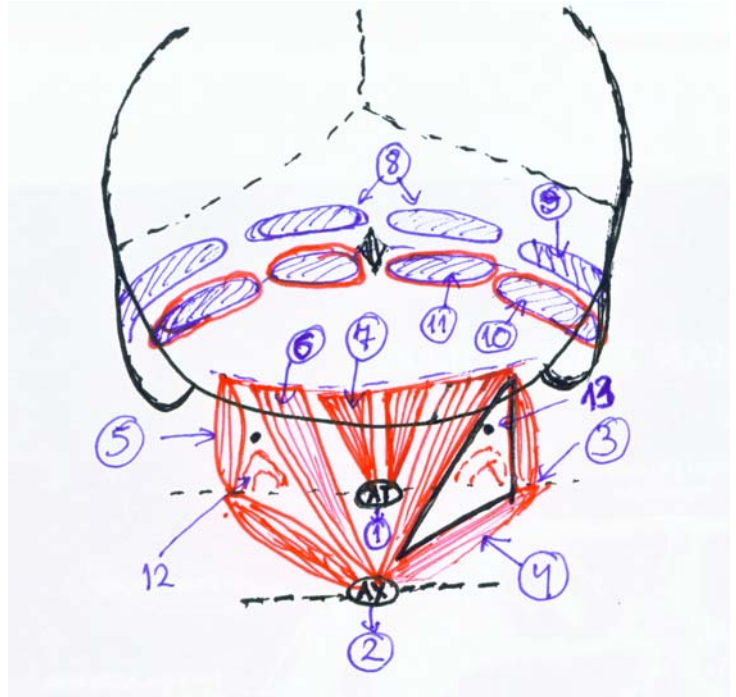
**\*\* The important thing about this region is to know the layers, so let's summarize up the layers from most superficially to the deepest:**

- 1) Skin
- 2) Subcutaneous fat
- 3) Superficial nuchal fascia
- 4) Trapezius muscle (Descending fibers, inverted V-shape)

- 5) Splenius Capitis muscle (V-shaped)
- 6) Semispinalis Capitis muscle (inverted V-Shape, or vertical || )
- 7) Deep Nuchal fascia
- 8) Suboccipital Triangle
- 9) Posterior Atlantooccipital membrane
- 10) Semi Spinalis Cervicis muscle

\* Drawing By Dr. Hanics:

- 1- Atlas (Posterior Tubercle)
- 2- Axis (Spinous Process)
- 3- Atlas (Transverse Process)
- 4- Obliquus capitis inf. M.
- 5- Obliquus capitis sup. M.
- 6- Rectus capitis post. Major m.
- 7- Rectus capitis post minor m.
- 8- Trapezium m.
- 9- Sternocleidomastoid m.
- 10- Splenius capitis m.



## ***The External Carotid Artery***

The external carotid artery begins the upper border of the thyroid cartilage, and, taking a slightly curved course, passes upward and forward, and then inclines backward to the space behind the neck of the mandible, where it divides into the superficial temporal and maxillary arteries. It rapidly diminishes in size in its course up the neck. In the child, it is somewhat smaller than the internal carotid; but in the adult, the two vessels are of nearly equal size. At its origin, this artery is more superficial, and placed nearer the middle line than the internal carotid, and is contained within the carotid triangle.

### **→ Relations:**

The external carotid artery is *covered* by the skin, superficial fascia, Platysma, deep fascia, and anterior margin of the Sternocleidomastoid. It is crossed by the hypoglossal nerve, by the lingual, common facial, and superior thyroid veins; and by the Digastric and Stylohyoid.

Higher up it passes deeply into the substance of the parotid gland, where it lies deep to the facial nerve and the junction of the temporal and internal maxillary veins. *Medial* to it are the hyoid bone, the wall of the pharynx, the superior laryngeal nerve, and a portion of the parotid gland. *Lateral* to it, in the lower part of its course, is the internal carotid artery. *Posterior* to it, near its origin, is the superior laryngeal nerve; and higher up, it is separated from the internal carotid by the Styloglossus and Stylopharyngeus, the glossopharyngeal nerve, the pharyngeal branch of the Vagus, and part of the parotid gland.

→ **Branches:** The branches of the external carotid artery may be divided into four sets:

1. *Anterior:* Superior Thyroid, Lingual, Facial artery
2. *Posterior:* Occipital, Posterior Auricular, Sternocleidomastoidal (not always)
3. *Ascending:* Ascending Pharyngeal
4. *Terminal:* Superficial Temporal, Maxillary

**A) Anterior**

1. **Superior thyroid artery:** *arises* from the external carotid artery just below the level of the greater horn of the hyoid bone and ends in the thyroid gland. Besides the arteries distributed to the muscles and to the thyroid gland, the branches of the superior thyroid artery are: *Infrahyoid, Superior Laryngeal, Sternocleidomastoidal and Cricothyroid.*

2. **Lingual artery:** *arises* from the external carotid between the superior thyroid and facial artery; it first runs obliquely upward and medial toward to the greater horn of the hyoid bone; it then curves downward and forward, forming a loop which is crossed by the hypoglossal nerve, (Pirigoff trigone) and passing beneath the Digastric and Stylohyoid it runs horizontally forward, beneath the Hyoglossus, (Beclard trigone) and finally, ascending almost perpendicularly to the tongue, turns forward on its lower surface as far as the tip giving the Suprahyoidal, dorsal lingual branches and sublingual artery.

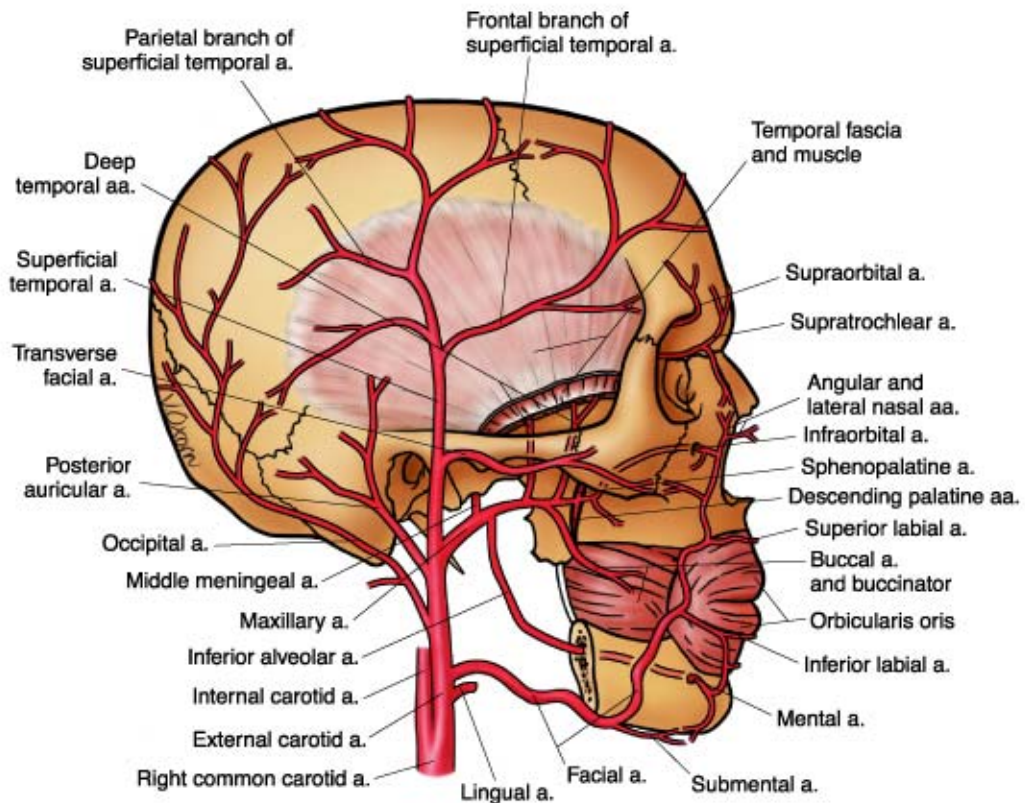
3. **Facial artery:** *arises* in the carotid triangle a little above the lingual artery and, sheltered by the ramus of the mandible, passes obliquely up beneath the Digastric and Stylohyoid, over which it arches to enter a groove on the posterior surface of the Submandibular gland. It then curves upward over the body of the mandible at the antero-inferior angle of the Messeter; passes forward and upward across the cheek to the angle of the mouth, then ascends along the side of the nose, and ends at the medial commissure of the eye, under the name of the **angular artery**. It gives the ascending palatine, tonsillar, glandular, submental, inferior labial, superior labial, messeteric, buccal and lateral nasal branches before terminating as the angular artery.

**\*\* Sometimes the lingual and facial arteries both arise in a common trunk called the *Linguofacial trunk*.**

**B) Posterior**

1. **Occipital artery:** *arises* from the posterior part of the external carotid, opposite the facial artery, near the lower margin of the posterior belly of the Digastric, and ends in the posterior part of the scalp giving auricular, mastoidal, occipital and meningeal branches.

2. **Posterior auricular artery:** it's small and *arises* from above the Digastric and Stylohyoid, opposite the apex of the styloid process. It ascends, under the parotid gland, on the styloid process of the temporal bone, to the groove between the cartilage of the ear and the mastoid process, and then it divides into its auricular and occipital branches.



**FIGURE 8.26 External carotid artery and its branches. Lateral view.**

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### C) Ascending

**Ascending pharyngeal artery:** the smallest branch of the external carotid is a long, deeply seated in the neck, beneath the other branches of the external carotid and under the Stylopharyngeus. It *arises* from the back part of the external carotid, and ascends vertically between the internal carotid and the side of the pharynx, to the under surface of the base of the skull, lying on the Longus capitis.

### D) Terminal

1. **Superficial temporal artery:** the smaller of the two terminal branches of the external carotid, appears, from its direction, to be the continuation of that vessel. It begins in the substance of the parotid gland, behind the neck of the mandible, and crosses over the posterior root of the zygomatic process of the temporal bone; about 5 cm. above this process it divides into two branches, a *frontal* and a *parietal*.

2. **Maxillary artery:** the larger of the two terminal branches of the external carotid, *arises* behind the neck of the mandible, and is at first imbedded in the substance of the parotid gland; it passes forward between the ramus of the mandible and the *sphenomandibular* ligament, and then runs to the pterygopalatine fossa. It supplies the deep structures of the face, see details later

## ***The Maxillary artery (Internal maxillary artery)***

This vessel is the larger of the 2 terminal branches of the external carotid artery (other branch being the superficial temporal a.).

It arises posterior to the neck of the condylar process of the mandible, within the parotid gland, and passes anteriorly deep to the neck, traversing the infratemporal fossa. It passes superficial to the lateral Pterygoid muscle and then disappears in the infratemporal fossa.

→ *The maxillary artery is divided into 3 parts in relation to the lateral Pterygoid muscle.*

*The 1<sup>st</sup> part* passes deep to the neck of the mandible between the bone and sphenomandibular ligament and runs anteriorly lateral to the inferior alveolar n. to reach the border of the lateral Pterygoid.

*The 2<sup>nd</sup> part* angles anteromedial between the 2 heads of the lateral Pterygoid and anterior and posterior divisions of the Mandibular nerve.

*The 3<sup>rd</sup> part* leaves the lateral Pterygoid to enter the pterygopalatine fossa where it terminates as branches that accompany the branches of the maxillary division of the trigeminal nerve.

### ***A) Branches of the 1<sup>st</sup> part of maxillary artery:***

***1. Deep auricular:*** to the external acoustic meatus

***2. Anterior tympanic:*** to the tympanic membrane

***3. Middle meningeal:***

The most important artery of the dura matter, it passes through the foramen spinosum into the middle cranial fossa. During its extra-dural course it divides into an anterior and posterior branch. This artery is usually surrounded near its origin by 2 branches of a cutaneous nerve, the *auricotemporal* n. (branch of Mandibular n. V3)

***4. Accessory meningeal arteries:*** to the cranial cavity

***5. Inferior alveolar:***

Gives off the mylohyoid branch, before entering the Mandibular canal along with the inferior alveolar n.

This artery supplies the teeth, bone and soft tissues of the lower jaw. Finally as the mental artery it traverses the mental foramen to run to the skin of the chin.

### ***B) Branches of the 2<sup>nd</sup> part of maxillary artery:*** all are branches to the muscles of mastication,

***1. Massetric***

***2. Deep*** anterior and posterior ***temporal*** arteries.

***3. Pterygoid*** branches

***4. Buccal:*** to the cheek and buccal mucous membrane.



**C) Branches of the 3<sup>rd</sup> part of the maxillary artery:** these branches arise just before it enters the pterygopalatine fossa,

**1. Superior posterior alveolar:** enters the maxilla at the maxillary tuberosity, and supplies the molar teeth and gums. It forms an anastomosis with the infraorbital branch

**2. Infraorbital:** which passes through the inferior orbital fissure to the orbit and leaves it through the infraorbital canal and foramen, on its way the infraorbital a. gives off the *anterior superior and middle superior alveolar arteries* which supply the incisor, canine and premolar teeth and related gums.

**3. Descending palatine:** reaches the hard palate through the pterygopalatine canal. It emerges from the greater palatine foramen, runs forward in groove on the medial side of the alveolar border of the hard palate to the direction of incisive canal, where it anastomosis with the nasopalatine artery.

Branches are distributed to the gums, palatine glands and the mucous membrane of the roof of the mouth. In the pterygopalatine canal it gives off twigs of branches that descend in the lesser palatine canals to supply to soft palate and palatine tonsils, anastomosing with the ascending palatine a.

\*\* (some sources state the descending palatine a. branches into the great and lesser palatine arteries, while other sources equate the descending palatine to the greater palatine artery)

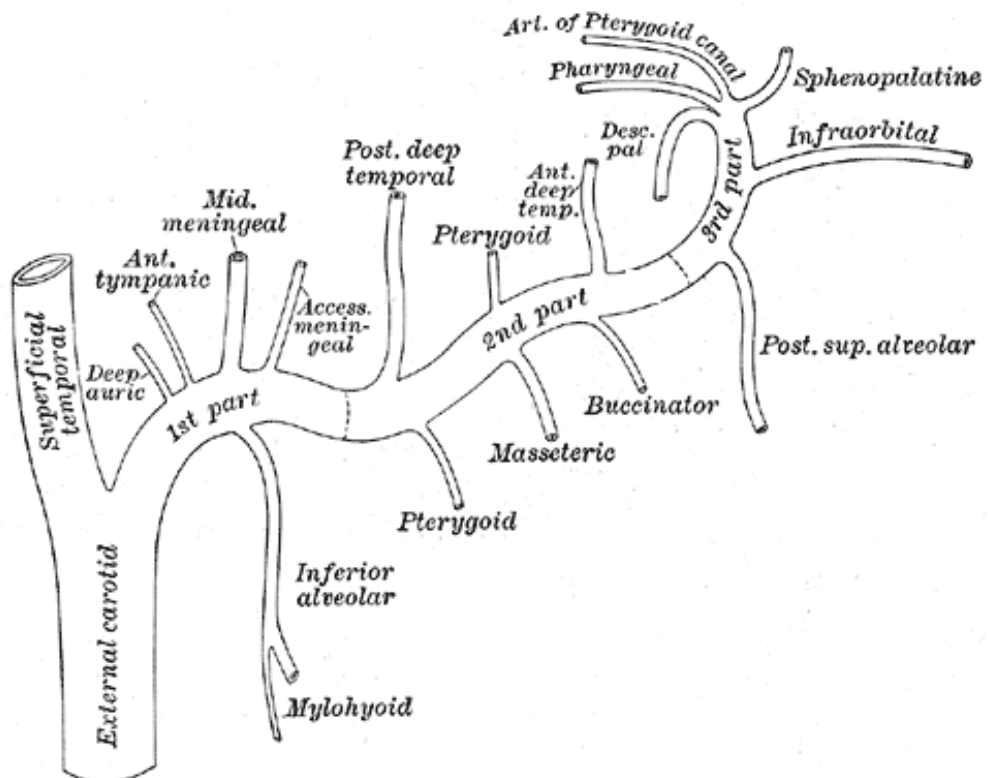
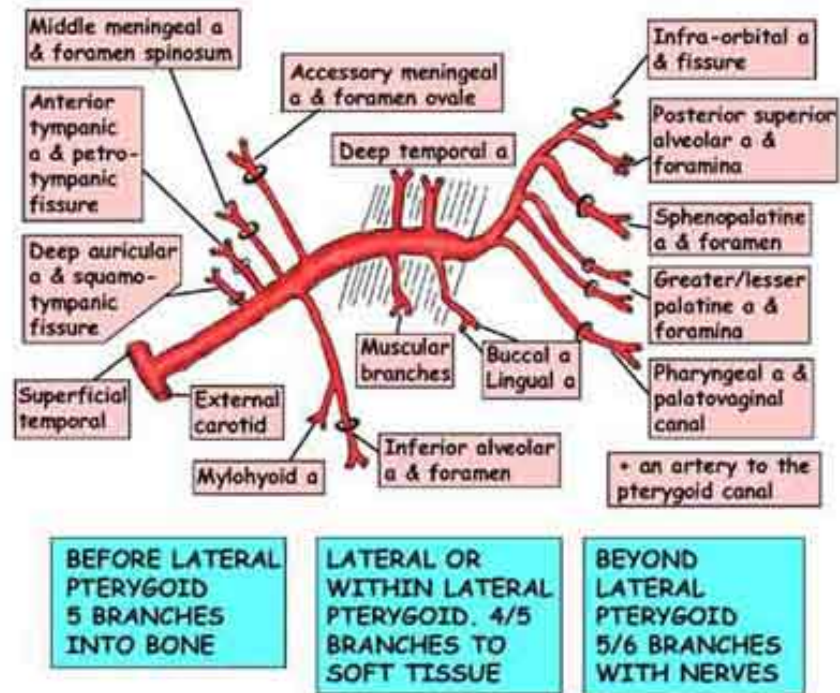
**4. Artery of Pterygoid canal:** runs in this canal to reach the upper pharynx and Eustachian tube

**5. Pharyngeal artery:** runs through the *Palatovaginal canal*, on the undersurface of the vaginal process of the sphenoid bone, which is a furrow that is converted into a canal by the Sphenoidal process of the palatine bone; it transmits the pharyngeal branch of the maxillary artery and the pharyngeal nerve from the pterygopalatine ganglion.

**6. Sphenopalatine artery:** enters the nasal cavity through the nasopalatine foramen, and supplies the lateral, posterior and medial walls via the posterior lateral nasal arteries

## MAXILLARY ARTERY

In infratemporal fossa, either within or lateral to the superficial head of lateral pterygoid muscle. This muscle is shown below



## ***Facial artery (External maxillary artery):***

An anterior branch of the external carotid artery that supplies structures in the face, The facial artery arises in the *carotid triangle* from the external carotid artery a little above the lingual artery and, sheltered by the ramus of the mandible, passes obliquely up beneath the *digastric and stylohyoid muscles*, over which it arches to enter a groove on the posterior surface of the *submandibular gland*.

It then curves upward over the *body of the mandible* at the antero-inferior angle of the masseter; passes forward and upward across the cheek to the angle of the mouth, then ascends along the side of the nose, and ends at the *medial commissure of the eye*, under the name of the ***angular artery***.

\*\* This vessel, both in the neck and on the face, is remarkably *tortuous*: in the former situation, to accommodate itself to the movements of the *pharynx in deglutition*; and in the latter, to the movements of the *mandible, lips, and cheeks*.

### ***Branches:***

#### ***A. Cervical branches:***

1. ***Ascending palatine artery:*** arises close to the origin of facial, and passes between the Styloglossus & Stylopharyngeus to the side of the pharynx up the base of skull between the superior constrictor of pharynx & medial Pterygoid muscle. It supplies:
  - a. *Lateral walls of pharynx*
  - b. *Palatine tonsils*
  - c. *Auditory tubes*
  - d. *Soft palate*
2. ***Tonsillar branch:*** primary blood supply to *palatine tonsil*, with extensive anastomoses with other tonsillar arteries
3. ***Glandular branches:*** branches of facial artery to the *submandibular gland*
4. ***Submental artery:*** largest cervical branch that runs forward upon the mylohyoid, just below the mandible, and beneath the digastric muscle. It supplies the mylohyoid muscle, submandibular and sublingual glands, and structures of lower lip & anastomoses with the inferior labial, mental branch of inferior dental (alveolar) and sublingual artery.

#### ***B. Facial branches:***

1. ***Inferior labial artery:*** arises near the angle of mouth, penetrates the Orbicularis Oris muscle and runs in a tortuous course along the edge of the lower lip between this muscle and the mucous membrane. It supplies the structures of the *lower lip (labial glands, muscle, and mucous membrane)*
2. ***Masseteric & Buccal branches:*** branches to the muscles of cheek and masseter that anastomose with the same branches of maxillary artery.

3. **Superior labial artery:** also arises near the angle of mouth, penetrates the Orbicularis Oris muscle and runs in a tortuous course along the edge of the upper lip between this muscle and the mucous membrane and supplies:
  - a. *Structures of the upper lip* (labial glands, muscle and mucous membrane)
  - b. The anterior, lower part of nasal septum by a *septal branch*
  - c. Lateral side of the external nose by an *alar branch*
4. **Lateral nasal branch:** branch of facial artery to the *side of the nose* (ala and dorsum); anastomoses with its contralateral partner, as well as the septal and alar branches of the superior labial, the dorsal nasal branch of the ophthalmic, and the infraorbital branch of the maxillary artery
5. **Angular artery:** a *terminal* branch of the facial artery that ascends to the *medial angle (commissure) of the orbit* accompanied by the angular vein. It supplies the muscles and skin of side of nose, and anastomosis with:
  - a. Lateral nasal branches
  - b. Infraorbital artery
  - c. *Dorsal artery of nose (from the ophthalmic artery)*, thereby providing an external-internal carotid arterial anastomosis

